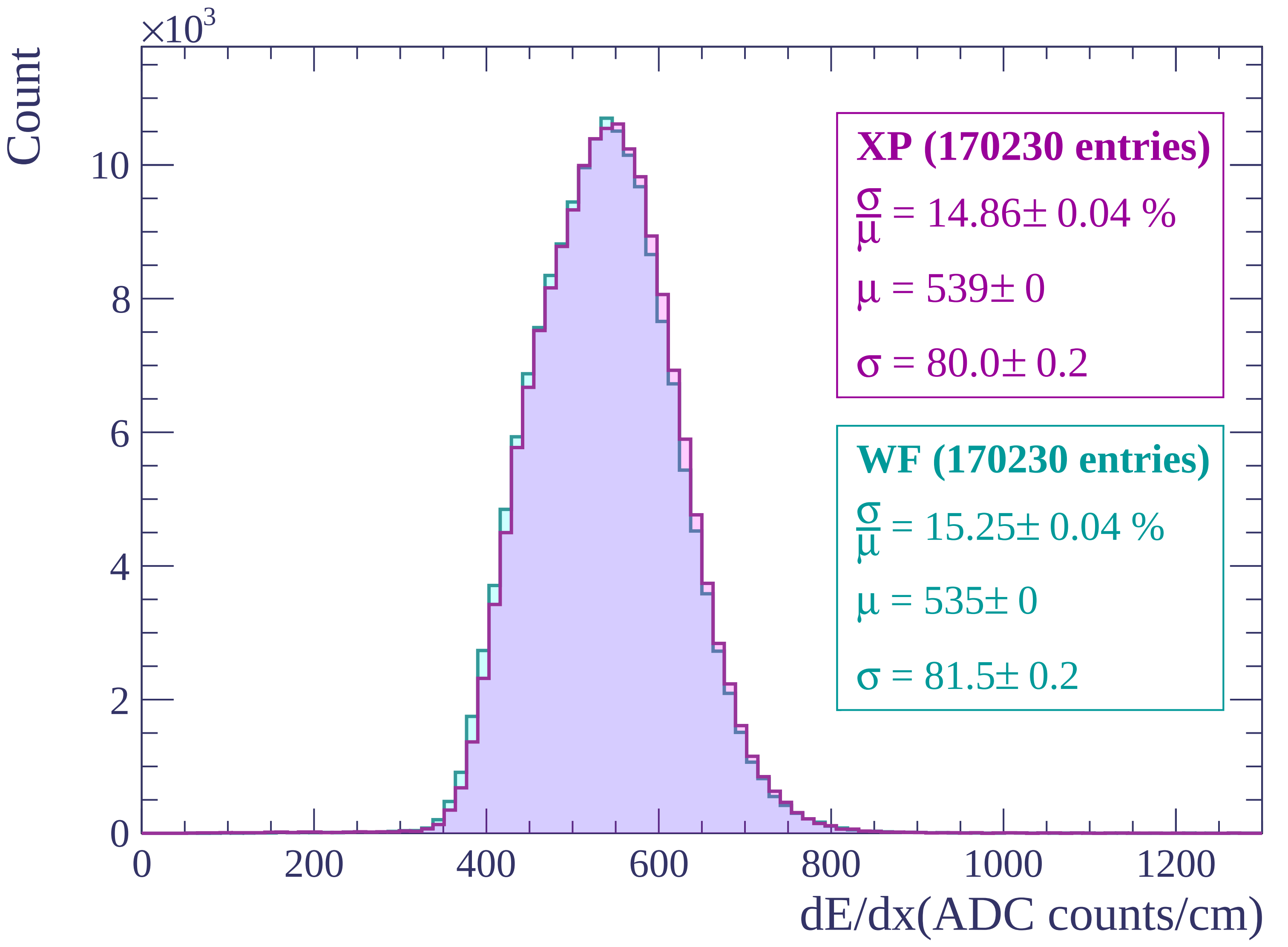
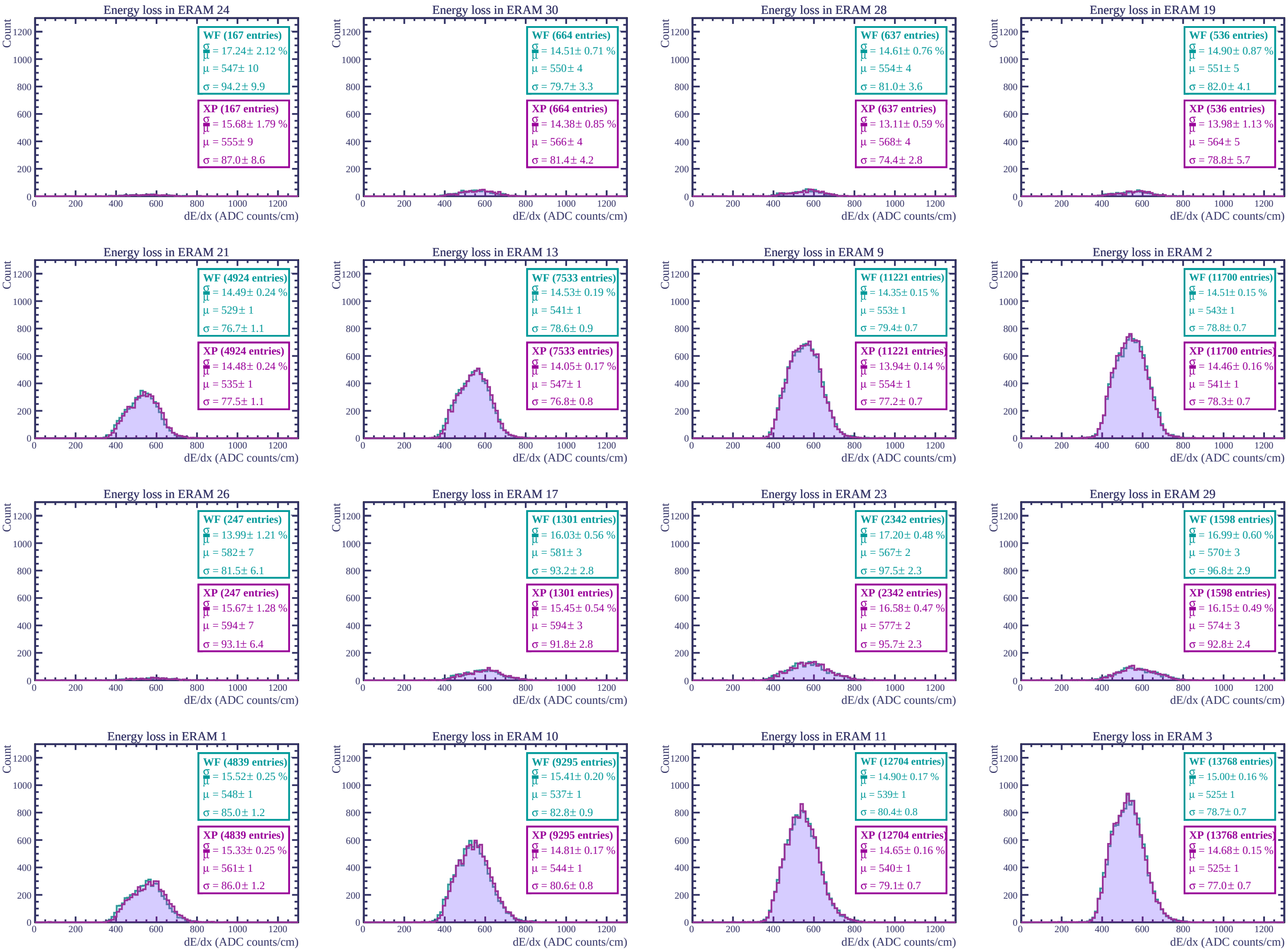
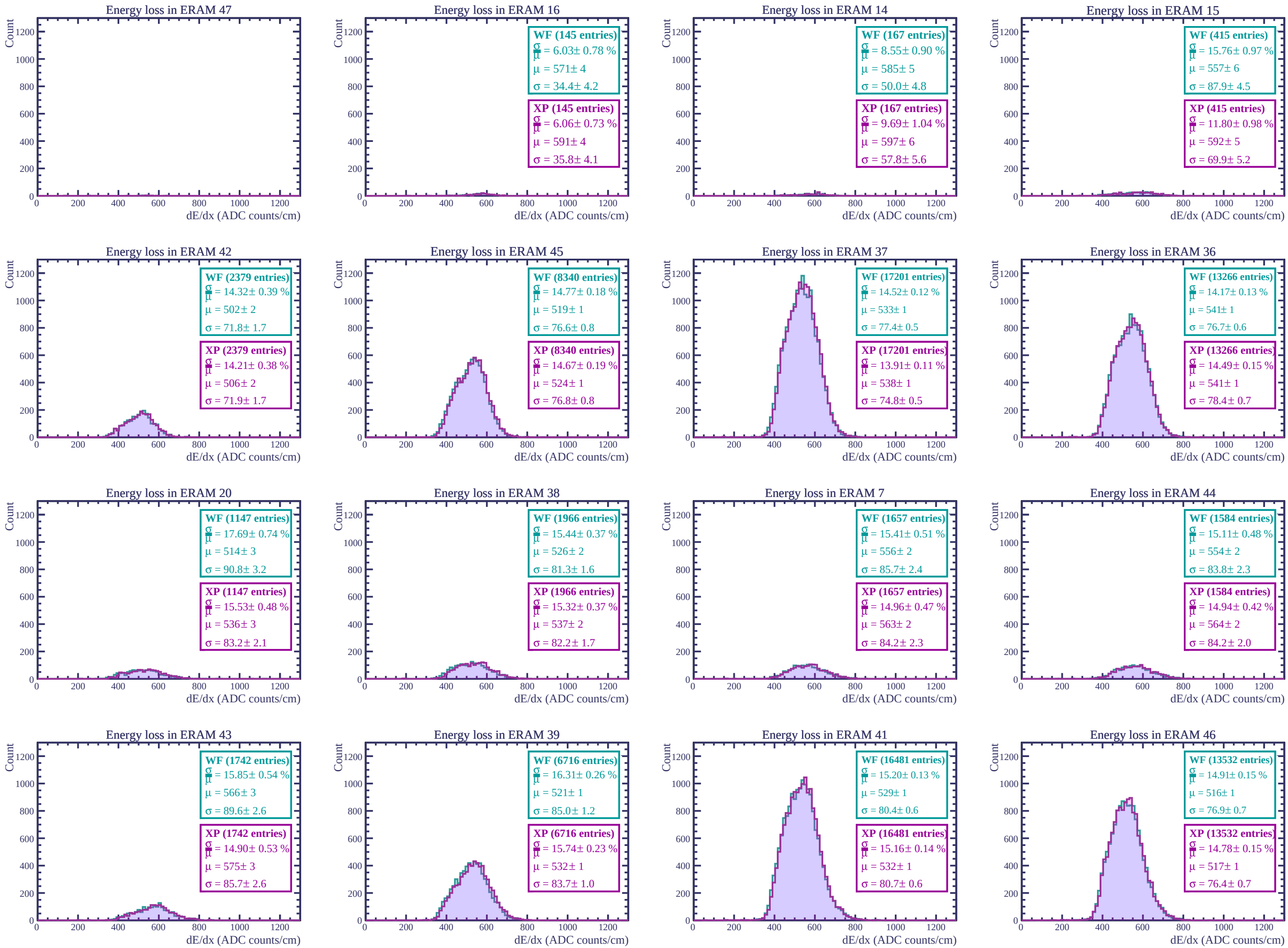
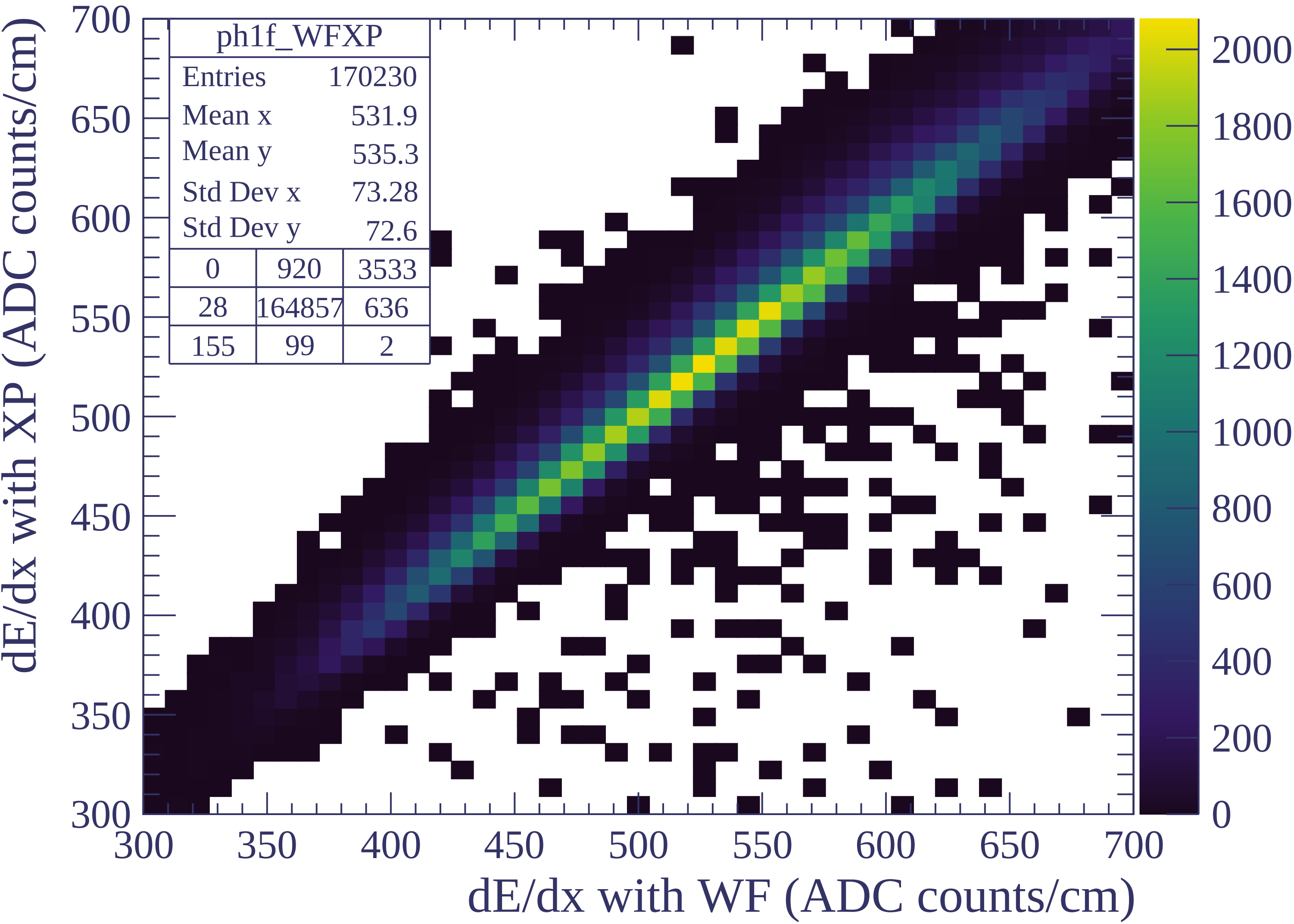

×10³
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (170230 entries)
σ/μ = 14.86± 0.04 %
μ = 539± 0
σ = 80.0± 0.2
WF (170230 entries)
σ/μ = 15.25± 0.04 %
μ = 535± 0
σ = 81.5± 0.2

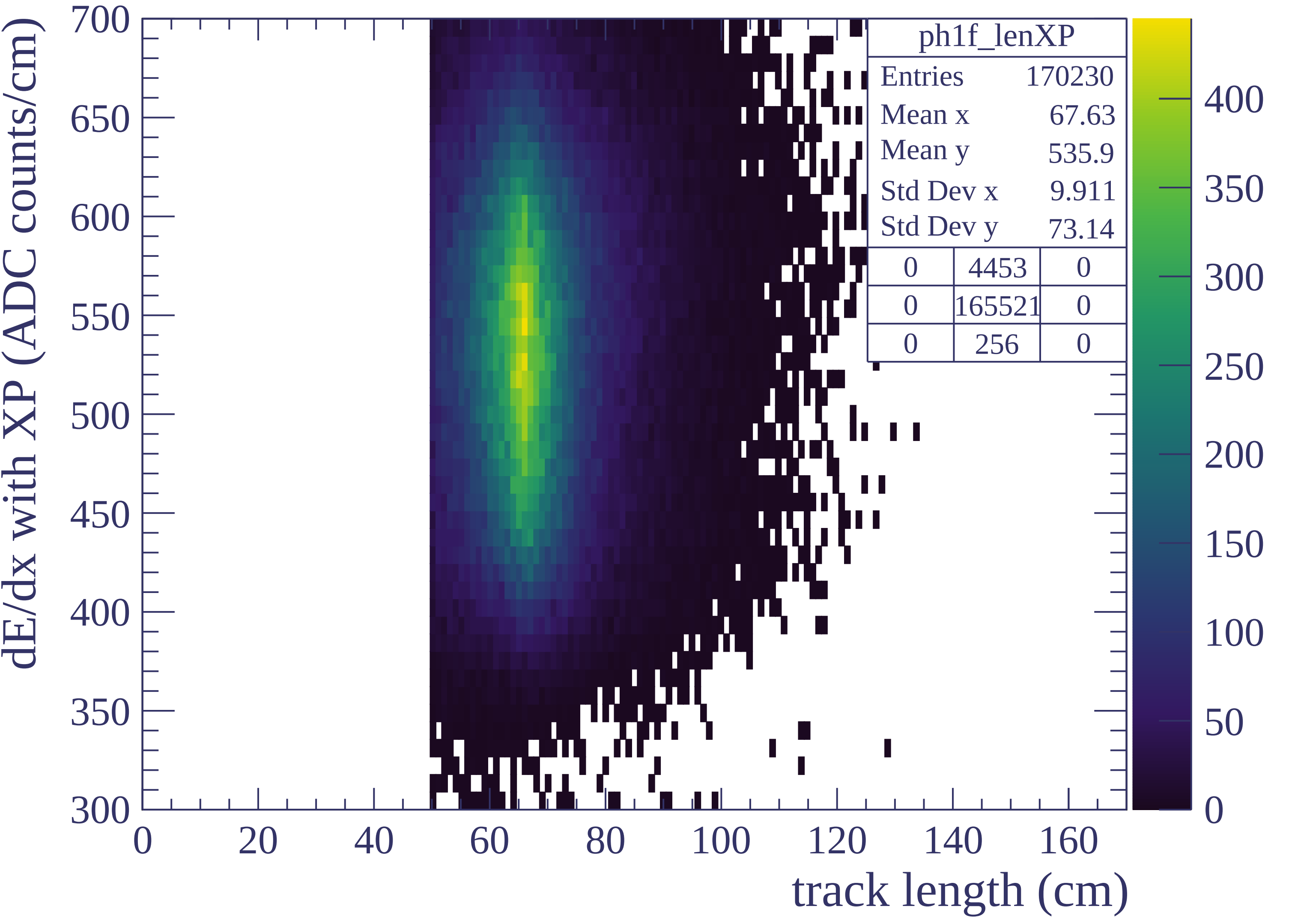


Energy loss in ERAM 24
WF (167 entries)
σ/μ = 17.24± 2.12 %
μ = 547± 10
σ = 94.2± 9.9
XP (167 entries)
σ/μ = 15.68± 1.79 %
μ = 555± 9
σ = 87.0± 8.6
Energy loss in ERAM 30
WF (664 entries)
σ/μ = 14.51± 0.71 %
μ = 550± 4
σ = 79.7± 3.3
XP (664 entries)
σ/μ = 14.38± 0.85 %
μ = 566± 4
σ = 81.4± 4.2
Energy loss in ERAM 28
WF (637 entries)
σ/μ = 14.61± 0.76 %
μ = 554± 4
σ = 81.0± 3.6
XP (637 entries)
σ/μ = 13.11± 0.59 %
μ = 568± 4
σ = 74.4± 2.8
Energy loss in ERAM 19
WF (536 entries)
σ/μ = 14.90± 0.87 %
μ = 551± 5
σ = 82.0± 4.1
XP (536 entries)
σ/μ = 13.98± 1.13 %
μ = 564± 5
σ = 78.8± 5.7
Energy loss in ERAM 21
WF (4924 entries)
σ/μ = 14.49± 0.24 %
μ = 529± 1
σ = 76.7± 1.1
XP (4924 entries)
σ/μ = 14.48± 0.24 %
μ = 535± 1
σ = 77.5± 1.1
Energy loss in ERAM 13
WF (7533 entries)
σ/μ = 14.53± 0.19 %
μ = 541± 1
σ = 78.6± 0.9
XP (7533 entries)
σ/μ = 14.05± 0.17 %
μ = 547± 1
σ = 76.8± 0.8
Energy loss in ERAM 9
WF (11221 entries)
σ/μ = 14.35± 0.15 %
μ = 553± 1
σ = 79.4± 0.7
XP (11221 entries)
σ/μ = 13.94± 0.14 %
μ = 554± 1
σ = 77.2± 0.7
Energy loss in ERAM 2
WF (11700 entries)
σ/μ = 14.51± 0.15 %
μ = 543± 1
σ = 78.8± 0.7
XP (11700 entries)
σ/μ = 14.46± 0.16 %
μ = 541± 1
σ = 78.3± 0.7
Energy loss in ERAM 26
WF (247 entries)
σ/μ = 13.99± 1.21 %
μ = 582± 7
σ = 81.5± 6.1
XP (247 entries)
σ/μ = 15.67± 1.28 %
μ = 594± 7
σ = 93.1± 6.4
Energy loss in ERAM 17
WF (1301 entries)
σ/μ = 16.03± 0.56 %
μ = 581± 3
σ = 93.2± 2.8
XP (1301 entries)
σ/μ = 15.45± 0.54 %
μ = 594± 3
σ = 91.8± 2.8
Energy loss in ERAM 23
WF (2342 entries)
σ/μ = 17.20± 0.48 %
μ = 567± 2
σ = 97.5± 2.3
XP (2342 entries)
σ/μ = 16.58± 0.47 %
μ = 577± 2
σ = 95.7± 2.3
Energy loss in ERAM 29
WF (1598 entries)
σ/μ = 16.99± 0.60 %
μ = 570± 3
σ = 96.8± 2.9
XP (1598 entries)
σ/μ = 16.15± 0.49 %
μ = 574± 3
σ = 92.8± 2.4
Energy loss in ERAM 1
WF (4839 entries)
σ/μ = 15.52± 0.25 %
μ = 548± 1
σ = 85.0± 1.2
XP (4839 entries)
σ/μ = 15.33± 0.25 %
μ = 561± 1
σ = 86.0± 1.2
Energy loss in ERAM 10
WF (9295 entries)
σ/μ = 15.41± 0.20 %
μ = 537± 1
σ = 82.8± 0.9
XP (9295 entries)
σ/μ = 14.81± 0.17 %
μ = 544± 1
σ = 80.6± 0.8
Energy loss in ERAM 11
WF (12704 entries)
σ/μ = 14.90± 0.17 %
μ = 539± 1
σ = 80.4± 0.8
XP (12704 entries)
σ/μ = 14.65± 0.16 %
μ = 540± 1
σ = 79.1± 0.7
Energy loss in ERAM 3
WF (13768 entries)
σ/μ = 15.00± 0.16 %
μ = 525± 1
σ = 78.7± 0.7
XP (13768 entries)
σ/μ = 14.68± 0.15 %
μ = 525± 1
σ = 77.0± 0.7
Count
dE/dx (ADC counts/cm)

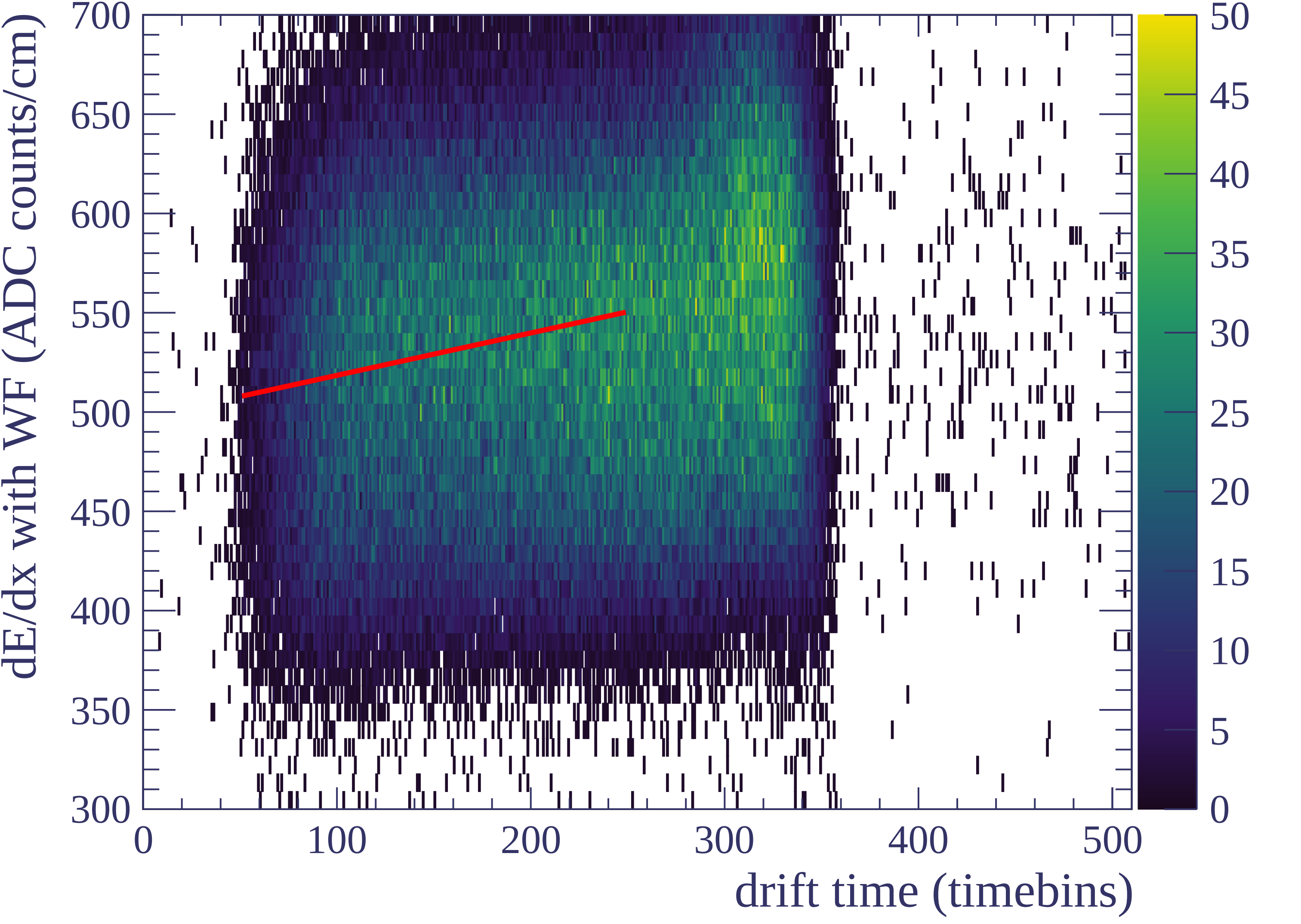

Energy loss in ERAM 47
Energy loss in ERAM 16
WF (145 entries)
σ/μ = 6.03± 0.78 %
μ = 571± 4
σ = 34.4± 4.2
XP (145 entries)
σ/μ = 6.06± 0.73 %
μ = 591± 4
σ = 35.8± 4.1
Energy loss in ERAM 14
WF (167 entries)
σ/μ = 8.55± 0.90 %
μ = 585± 5
σ = 50.0± 4.8
XP (167 entries)
σ/μ = 9.69± 1.04 %
μ = 597± 6
σ = 57.8± 5.6
Energy loss in ERAM 15
WF (415 entries)
σ/μ = 15.76± 0.97 %
μ = 557± 6
σ = 87.9± 4.5
XP (415 entries)
σ/μ = 11.80± 0.98 %
μ = 592± 5
σ = 69.9± 5.2
Energy loss in ERAM 42
WF (2379 entries)
σ/μ = 14.32± 0.39 %
μ = 502± 2
σ = 71.8± 1.7
XP (2379 entries)
σ/μ = 14.21± 0.38 %
μ = 506± 2
σ = 71.9± 1.7
Energy loss in ERAM 45
WF (8340 entries)
σ/μ = 14.77± 0.18 %
μ = 519± 1
σ = 76.6± 0.8
XP (8340 entries)
σ/μ = 14.67± 0.19 %
μ = 524± 1
σ = 76.8± 0.8
Energy loss in ERAM 37
WF (17201 entries)
σ/μ = 14.52± 0.12 %
μ = 533± 1
σ = 77.4± 0.5
XP (17201 entries)
σ/μ = 13.91± 0.11 %
μ = 538± 1
σ = 74.8± 0.5
Energy loss in ERAM 36
WF (13266 entries)
σ/μ = 14.17± 0.13 %
μ = 541± 1
σ = 76.7± 0.6
XP (13266 entries)
σ/μ = 14.49± 0.15 %
μ = 541± 1
σ = 78.4± 0.7
Energy loss in ERAM 20
WF (1147 entries)
σ/μ = 17.69± 0.74 %
μ = 514± 3
σ = 90.8± 3.2
XP (1147 entries)
σ/μ = 15.53± 0.48 %
μ = 536± 3
σ = 83.2± 2.1
Energy loss in ERAM 38
WF (1966 entries)
σ/μ = 15.44± 0.37 %
μ = 526± 2
σ = 81.3± 1.6
XP (1966 entries)
σ/μ = 15.32± 0.37 %
μ = 537± 2
σ = 82.2± 1.7
Energy loss in ERAM 7
WF (1657 entries)
σ/μ = 15.41± 0.51 %
μ = 556± 2
σ = 85.7± 2.4
XP (1657 entries)
σ/μ = 14.96± 0.47 %
μ = 563± 2
σ = 84.2± 2.3
Energy loss in ERAM 44
WF (1584 entries)
σ/μ = 15.11± 0.48 %
μ = 554± 2
σ = 83.8± 2.3
XP (1584 entries)
σ/μ = 14.94± 0.42 %
μ = 564± 2
σ = 84.2± 2.0
Energy loss in ERAM 43
WF (1742 entries)
σ/μ = 15.85± 0.54 %
μ = 566± 3
σ = 89.6± 2.6
XP (1742 entries)
σ/μ = 14.90± 0.53 %
μ = 575± 3
σ = 85.7± 2.6
Energy loss in ERAM 39
WF (6716 entries)
σ/μ = 16.31± 0.26 %
μ = 521± 1
σ = 85.0± 1.2
XP (6716 entries)
σ/μ = 15.74± 0.23 %
μ = 532± 1
σ = 83.7± 1.0
Energy loss in ERAM 41
WF (16481 entries)
σ/μ = 15.20± 0.13 %
μ = 529± 1
σ = 80.4± 0.6
XP (16481 entries)
σ/μ = 15.16± 0.14 %
μ = 532± 1
σ = 80.7± 0.6
Energy loss in ERAM 46
WF (13532 entries)
σ/μ = 14.91± 0.15 %
μ = 516± 1
σ = 76.9± 0.7
XP (13532 entries)
σ/μ = 14.78± 0.15 %
μ = 517± 1
σ = 76.4± 0.7
Count
dE/dx (ADC counts/cm)

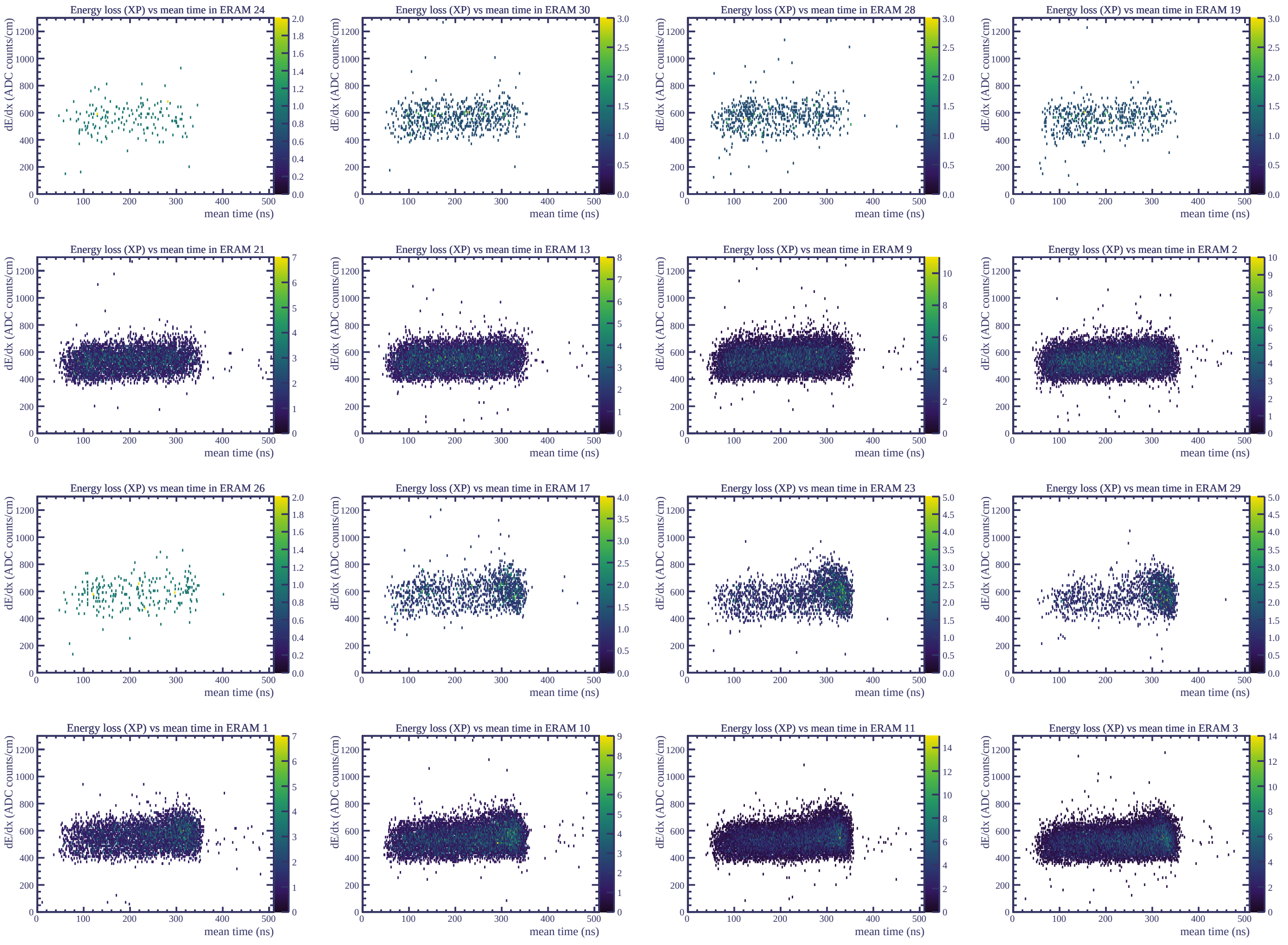


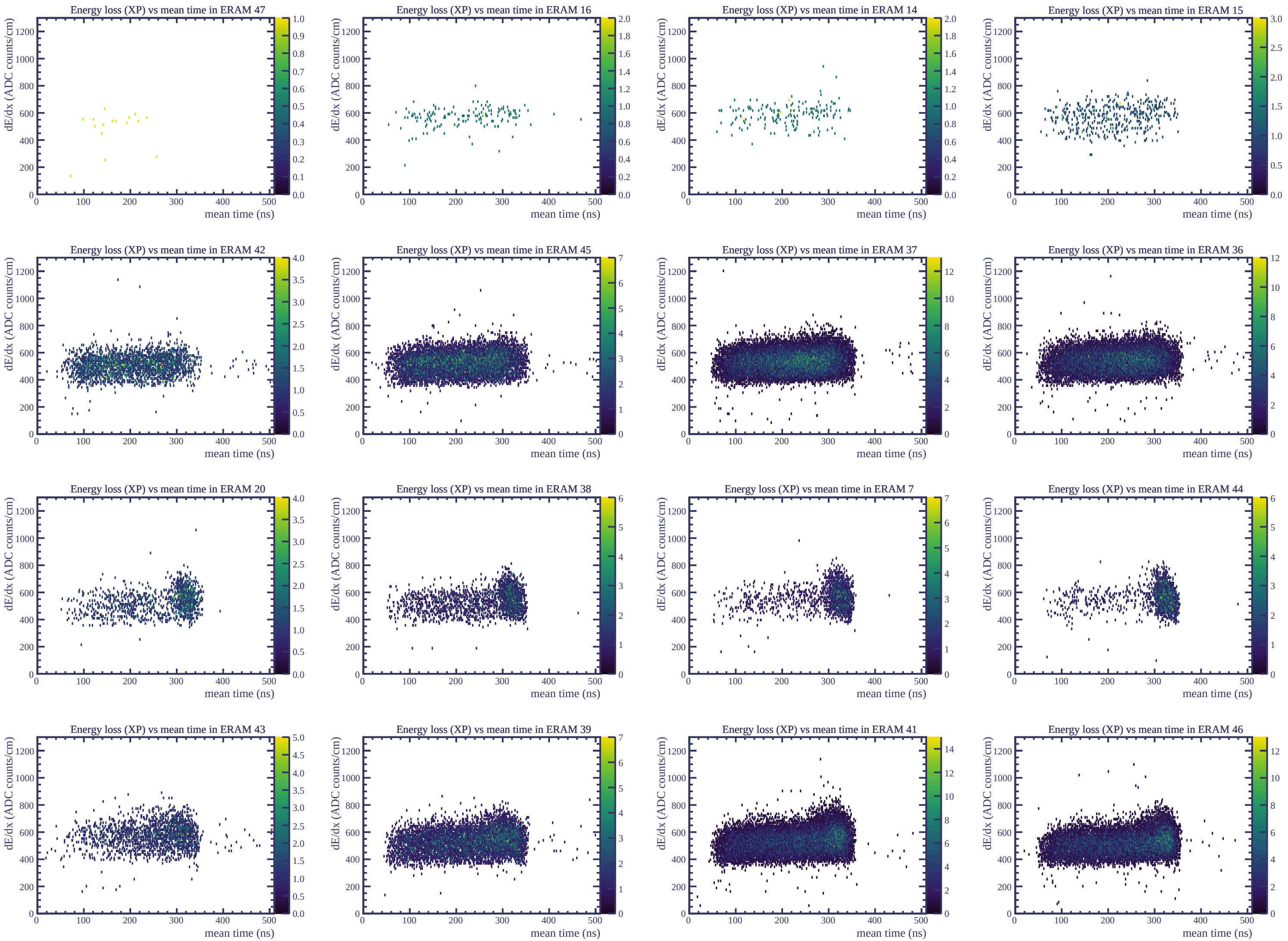
ph1f_WFXP
Entries 170230
Mean x 531.9
Mean y 535.3
Std Dev x 73.28
Std Dev y 72.6
0 920 3533
28 164857 636
155 99 2
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
300
350
400
450
500
550
600
650
700
0
200
400
600
800
1000
1200
1400
1600
1800
2000

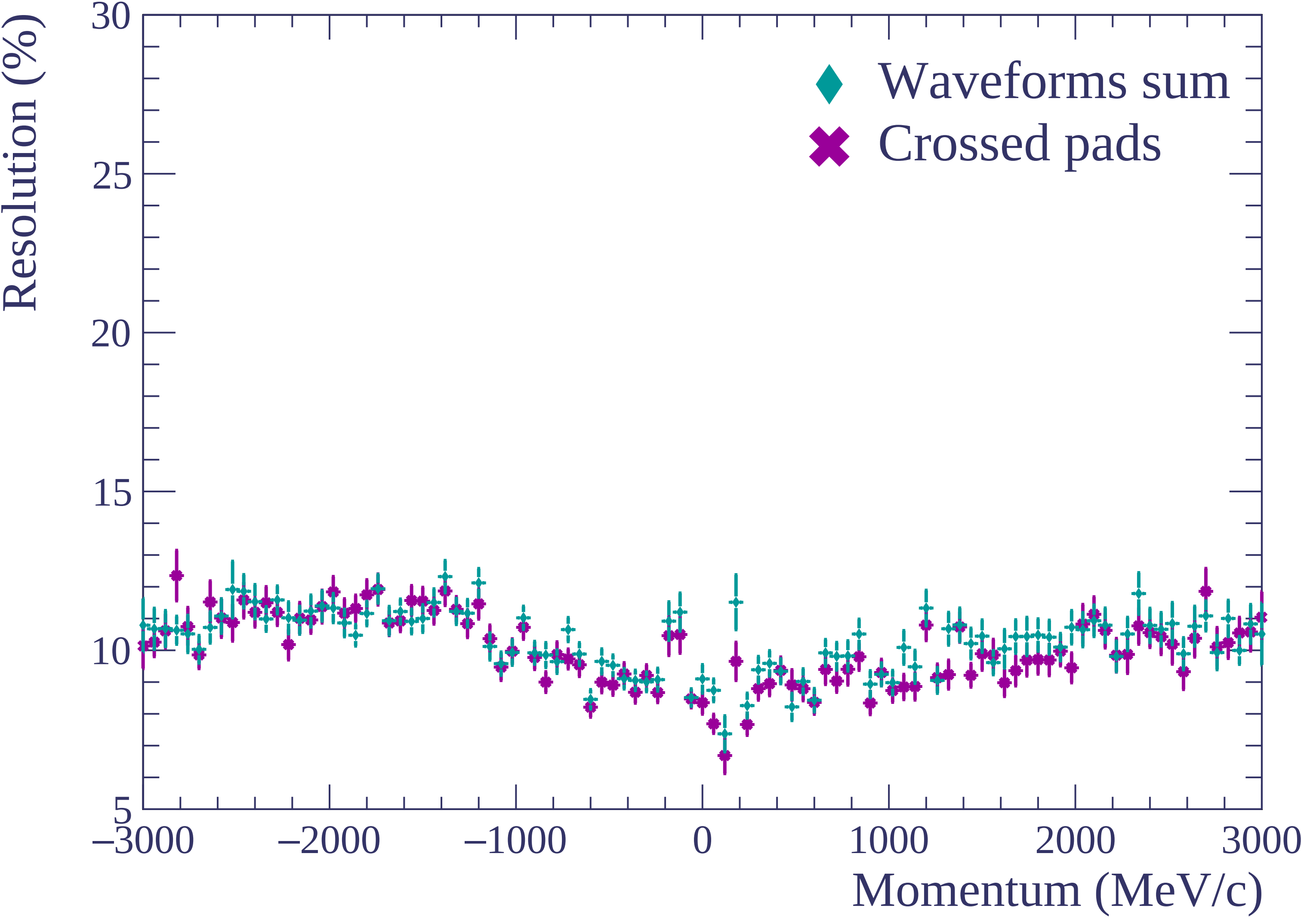


ph1f_lenXP
Entries 170230
Mean x 67.63
Mean y 535.9
Std Dev x 9.911
Std Dev y 73.14
0 4453 0
0 165521 0
0 256 0
dE/dx with XP (ADC counts/cm)
track length (cm)
700
650
600
550
500
450
400
350
300
0
20
40
60
80
100
120
140
160
400
350
300
250
200
150
100
50
0

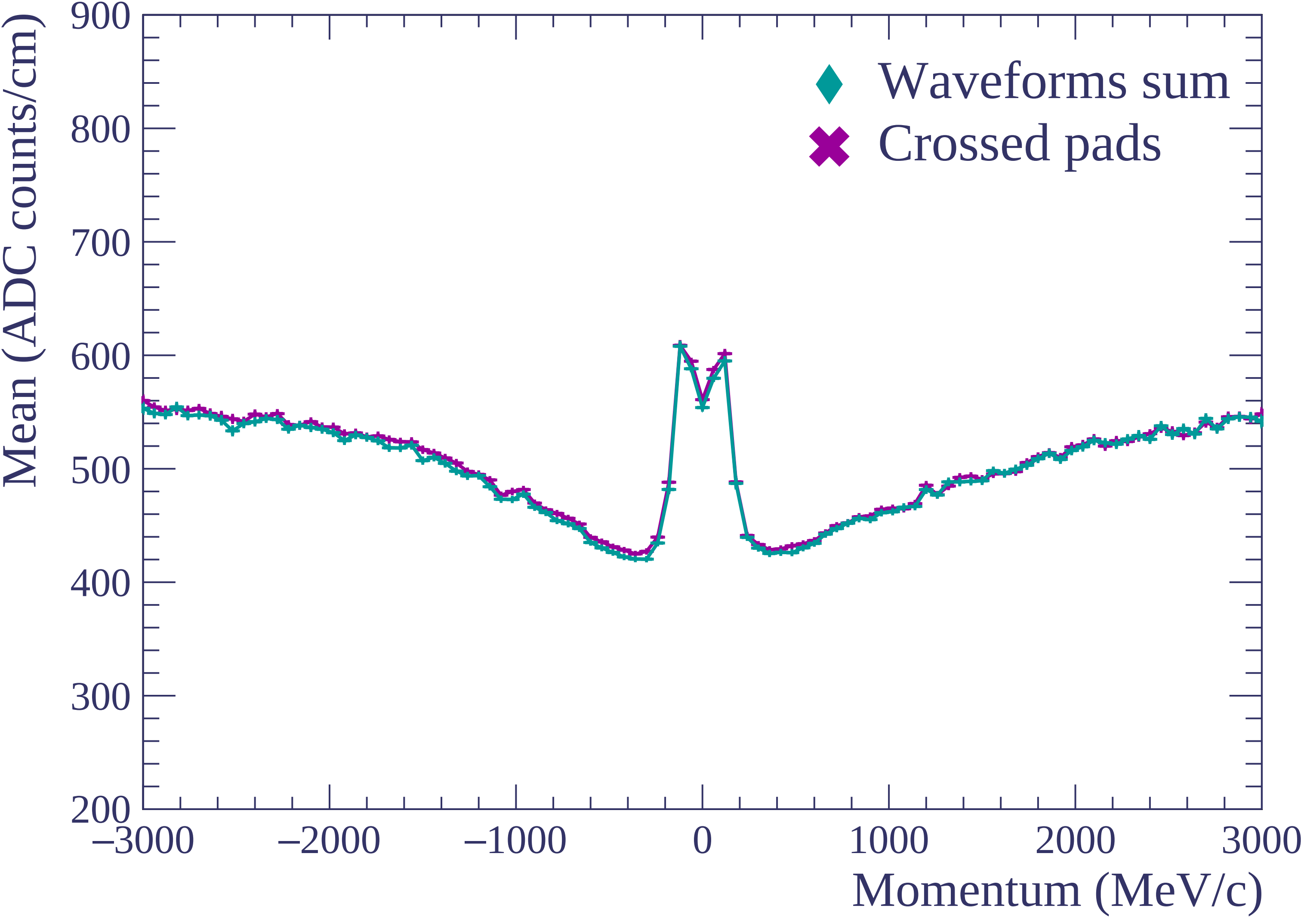

dE/dx with WF (ADC counts/cm)
drift time (timebins)
700
650
600
550
500
450
400
350
300
0
100
200
300
400
500
50
45
40
35
30
25
20
15
10
5
0


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

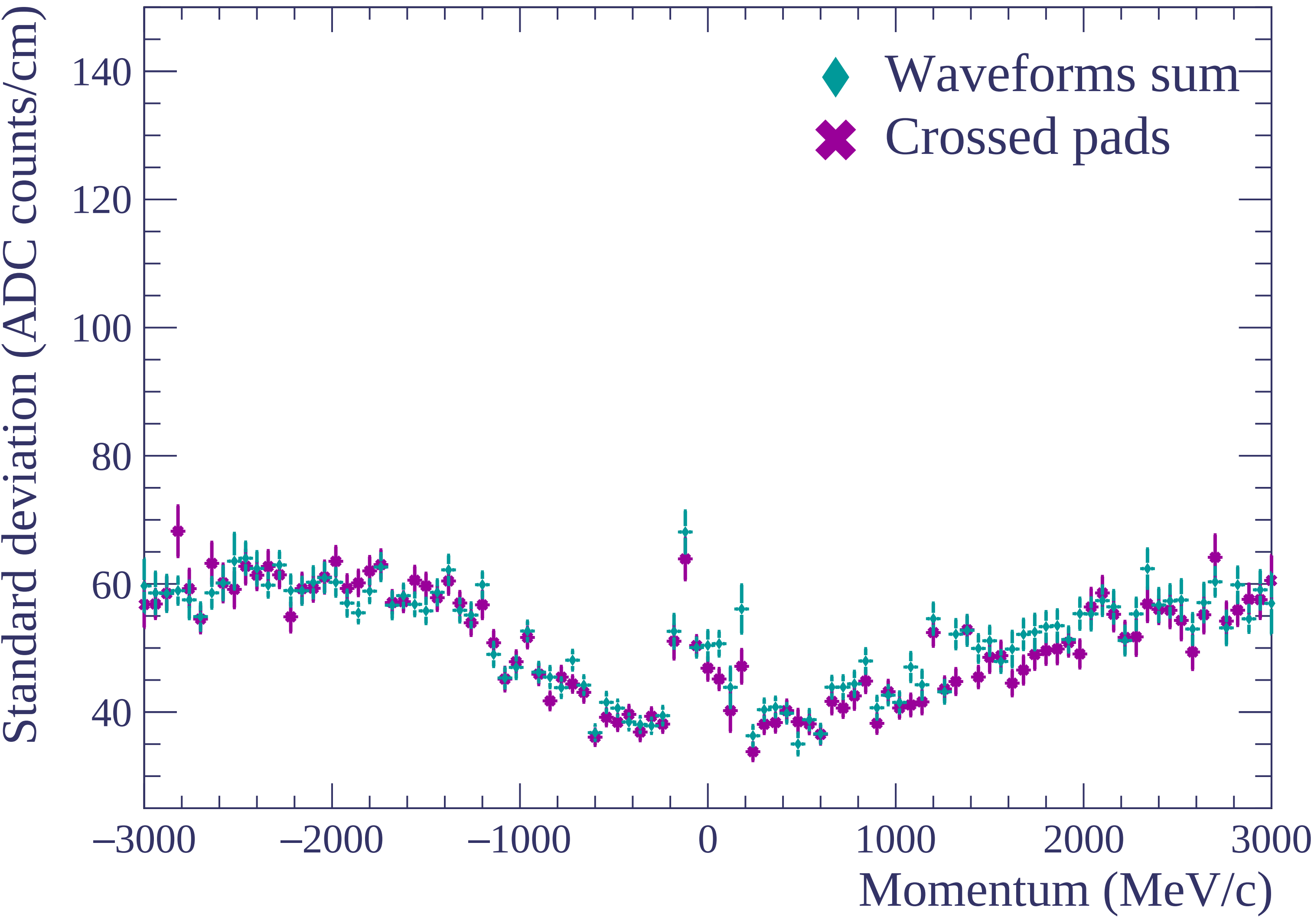

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)



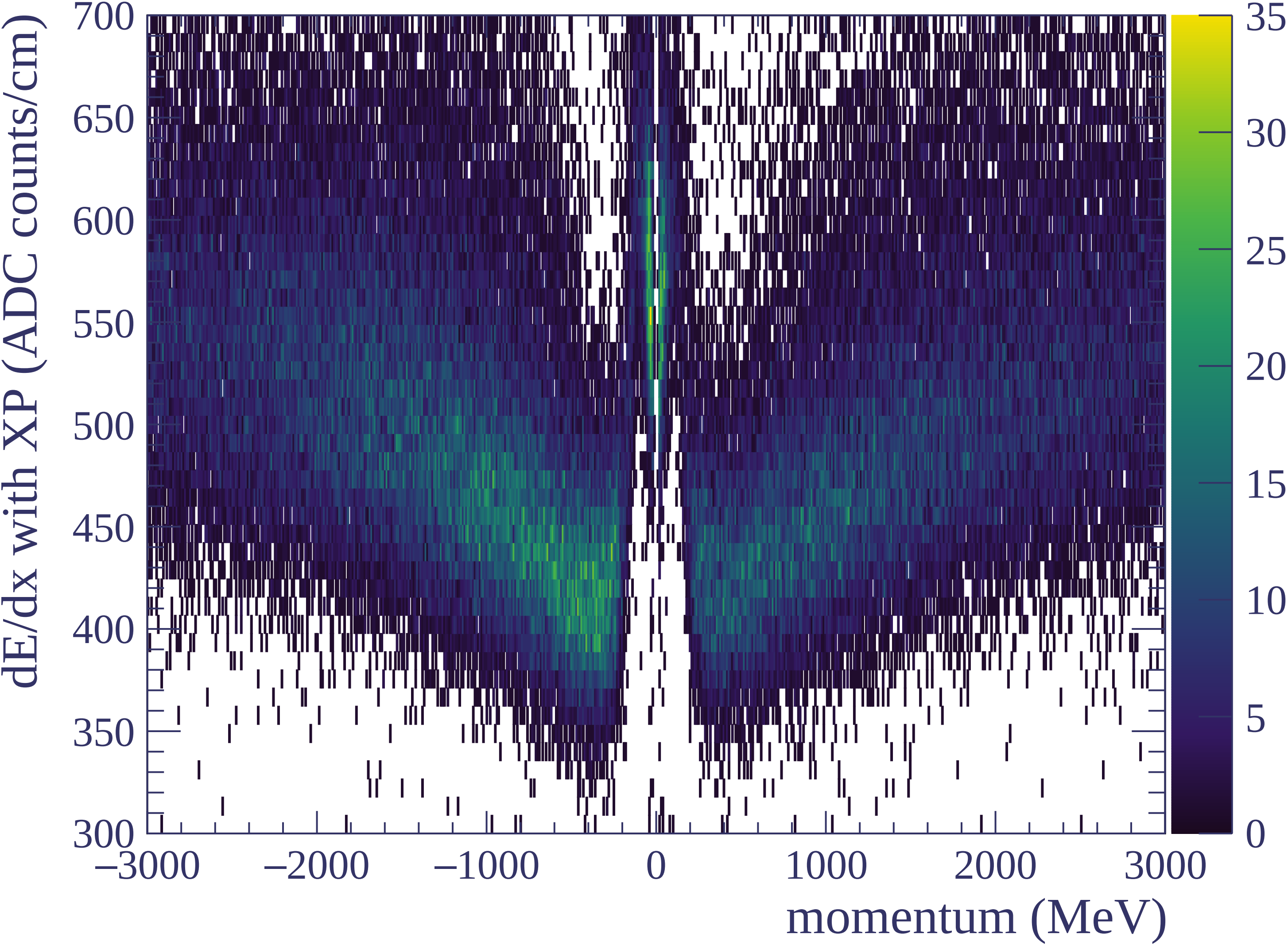
Resolution (%)
30
25
20
15
10
5
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



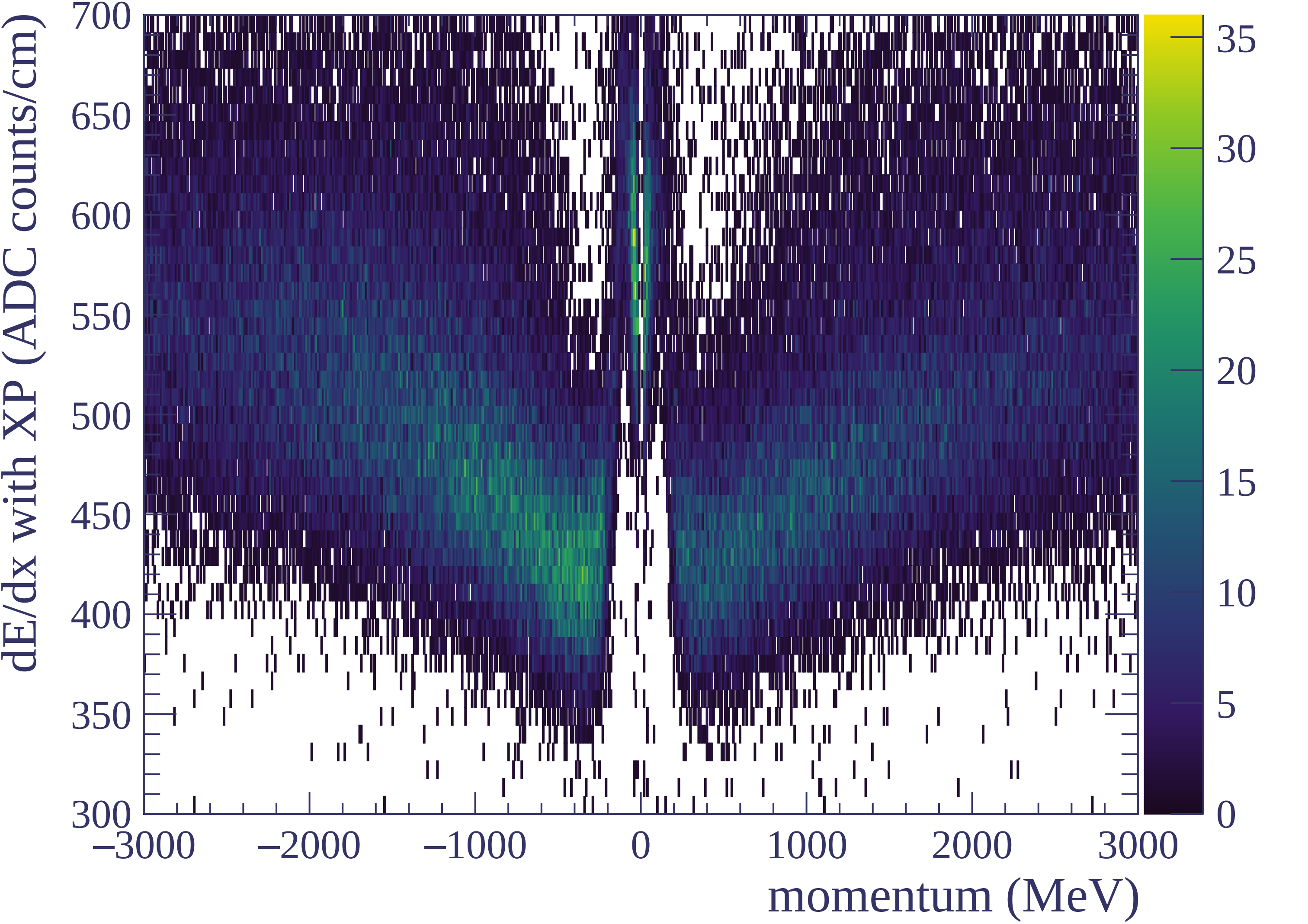
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Momentum (MeV/c)
900
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000



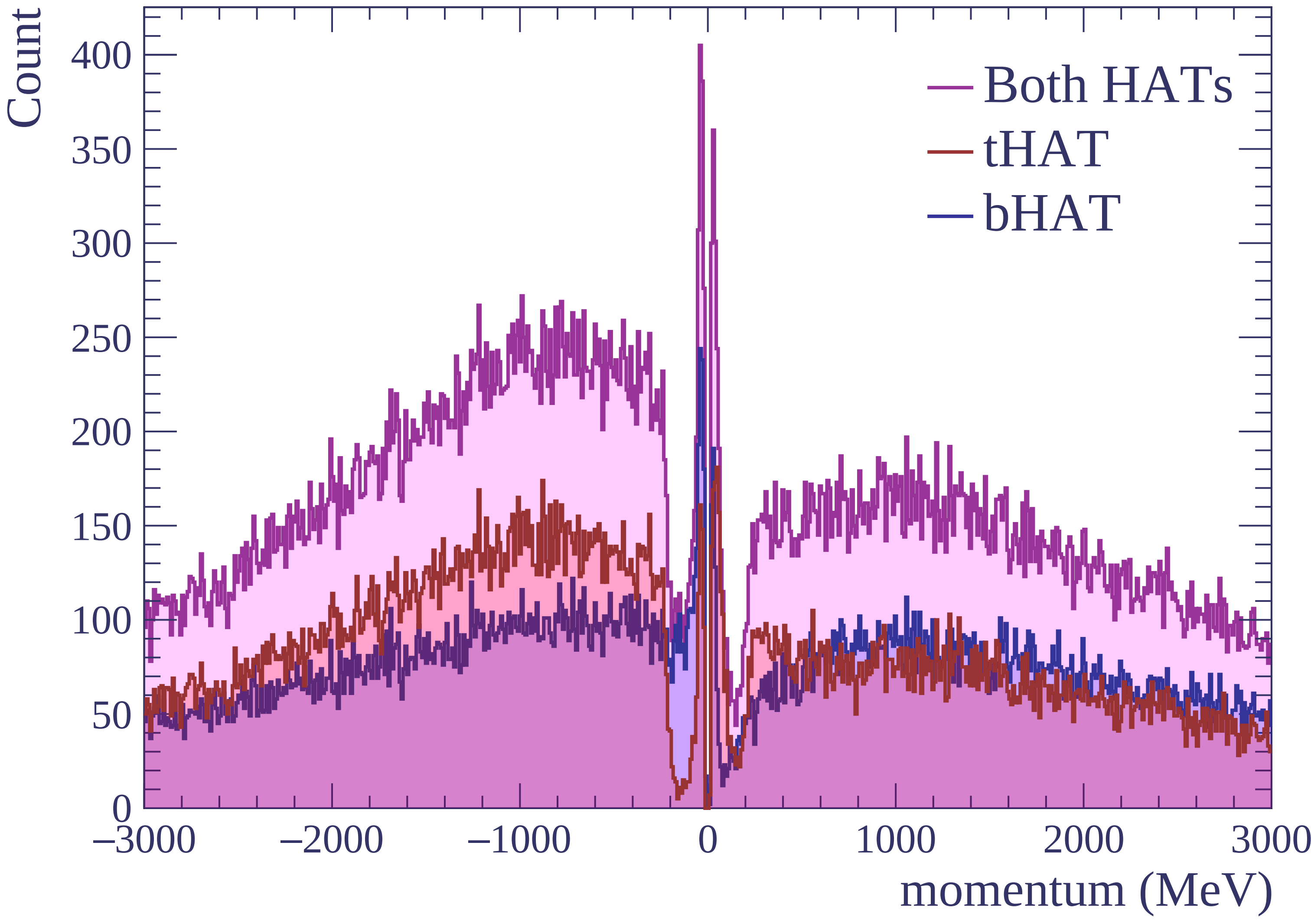
Standard deviation (ADC counts/cm)
140
120
100
80
60
40
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

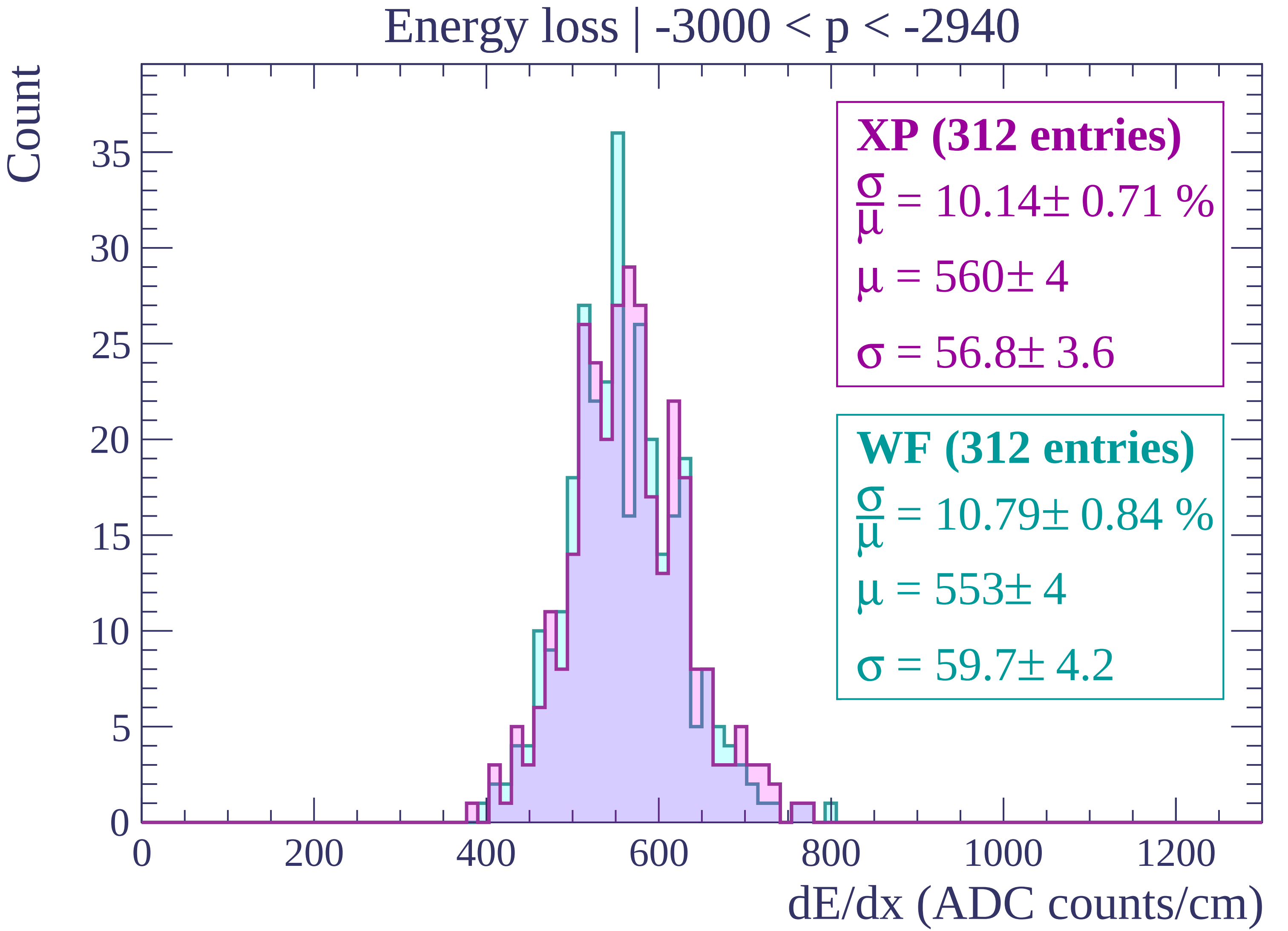


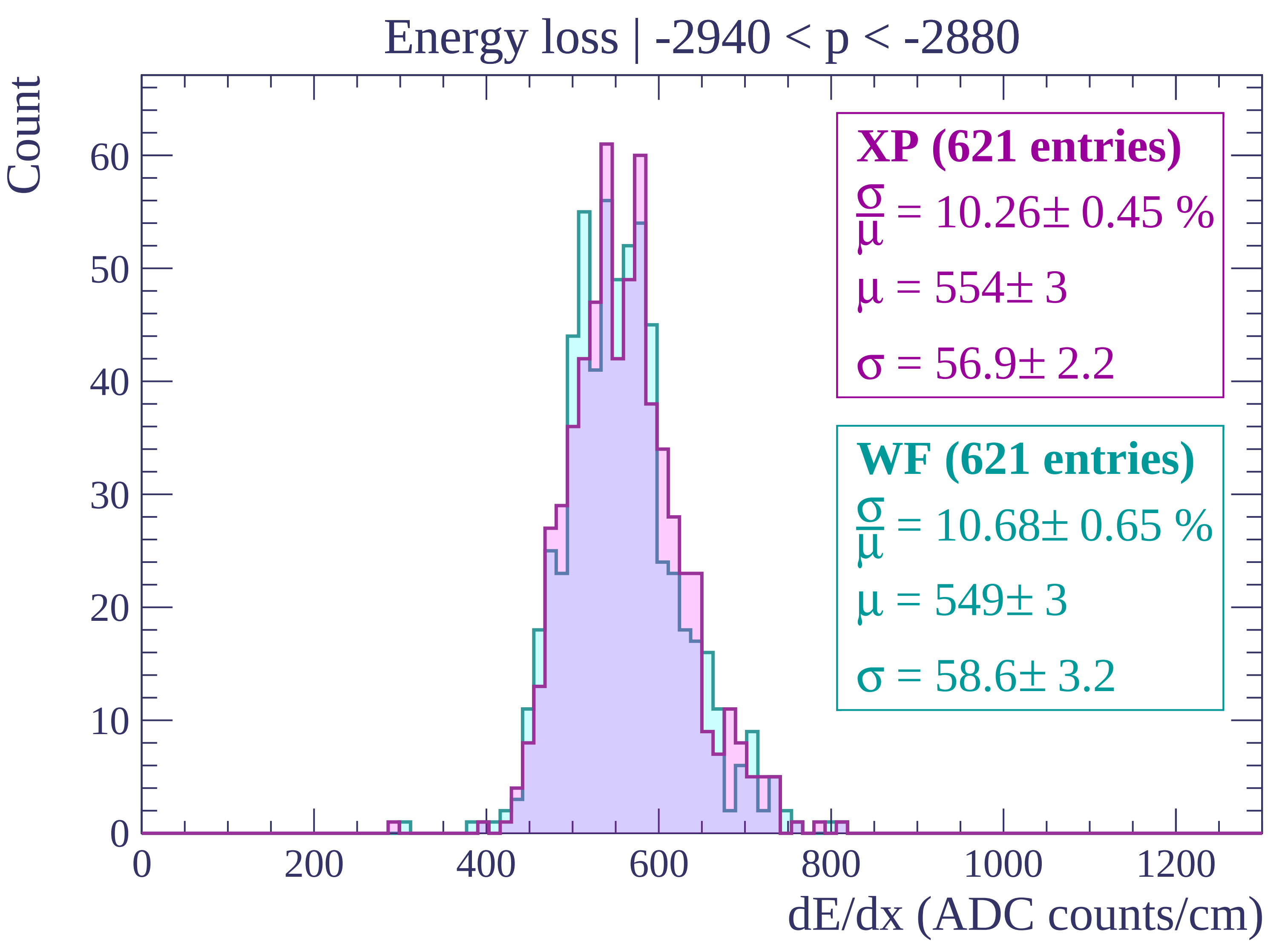
dE/dx with XP (ADC counts/cm)
700
650
600
550
500
450
400
350
300
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
35
30
25
20
15
10
5
0

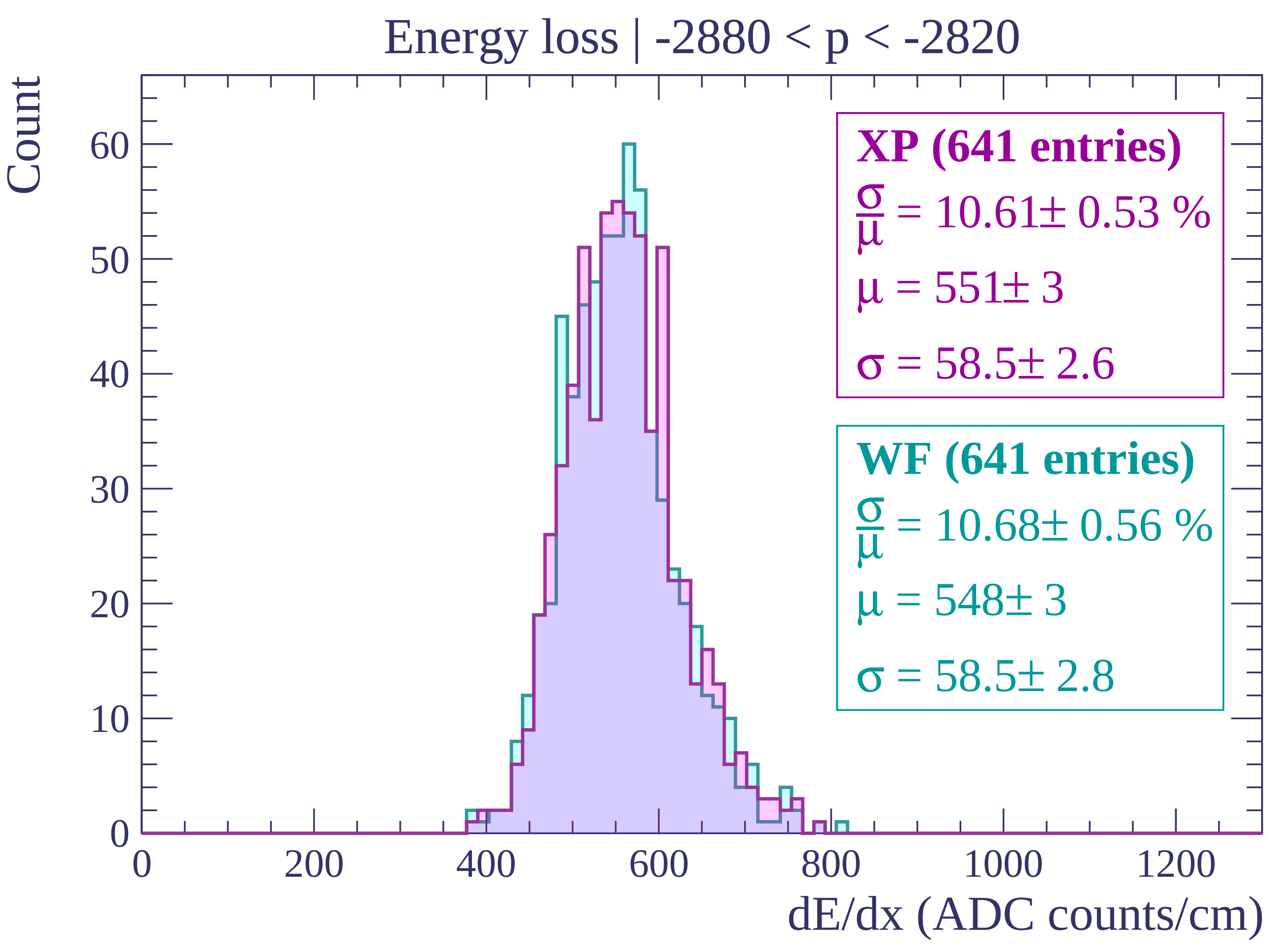


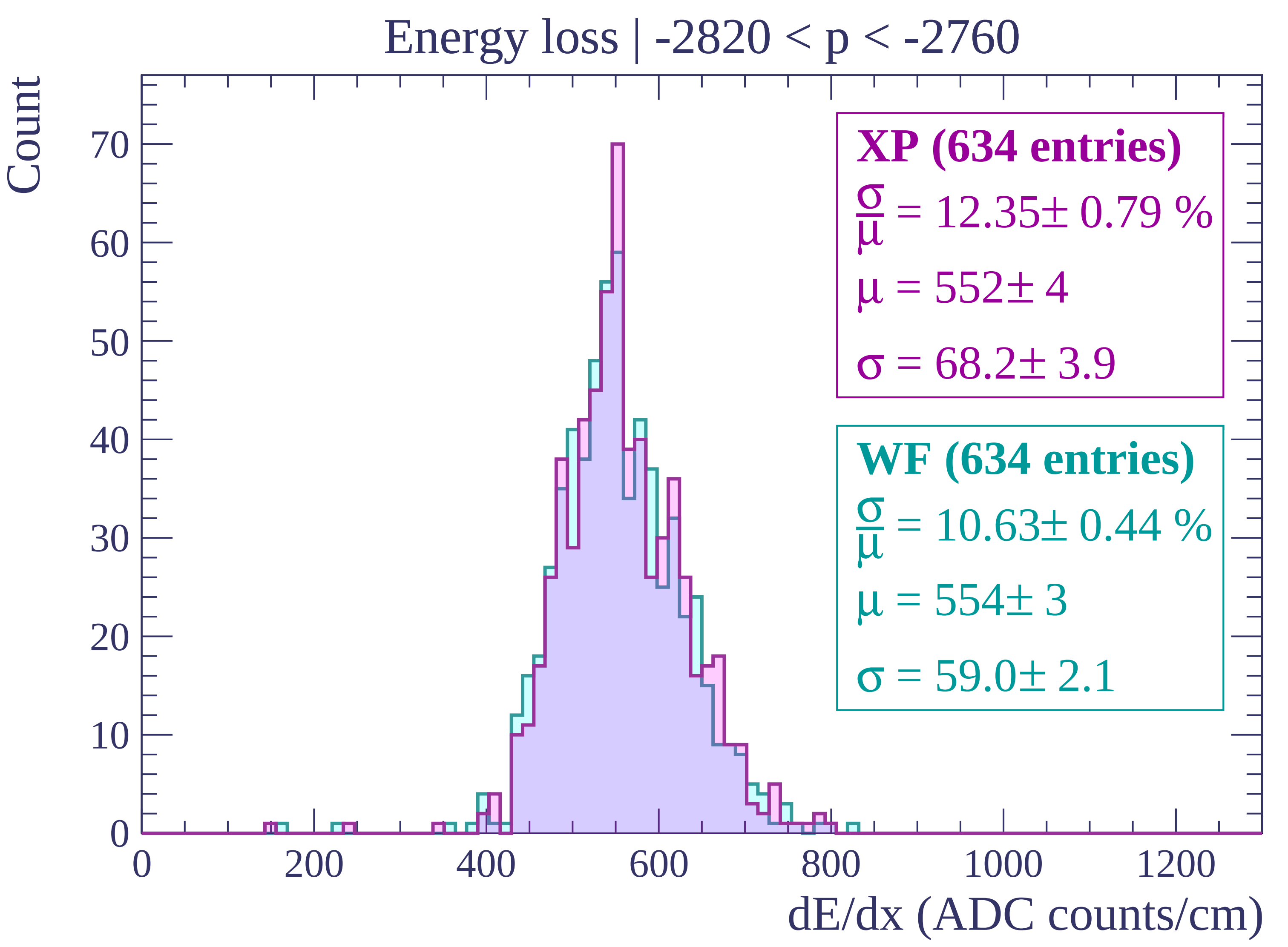
dE/dx with XP (ADC counts/cm)
700
650
600
550
500
450
400
350
300
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
35
30
25
20
15
10
5
0


Count
400
350
300
250
200
150
100
50
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

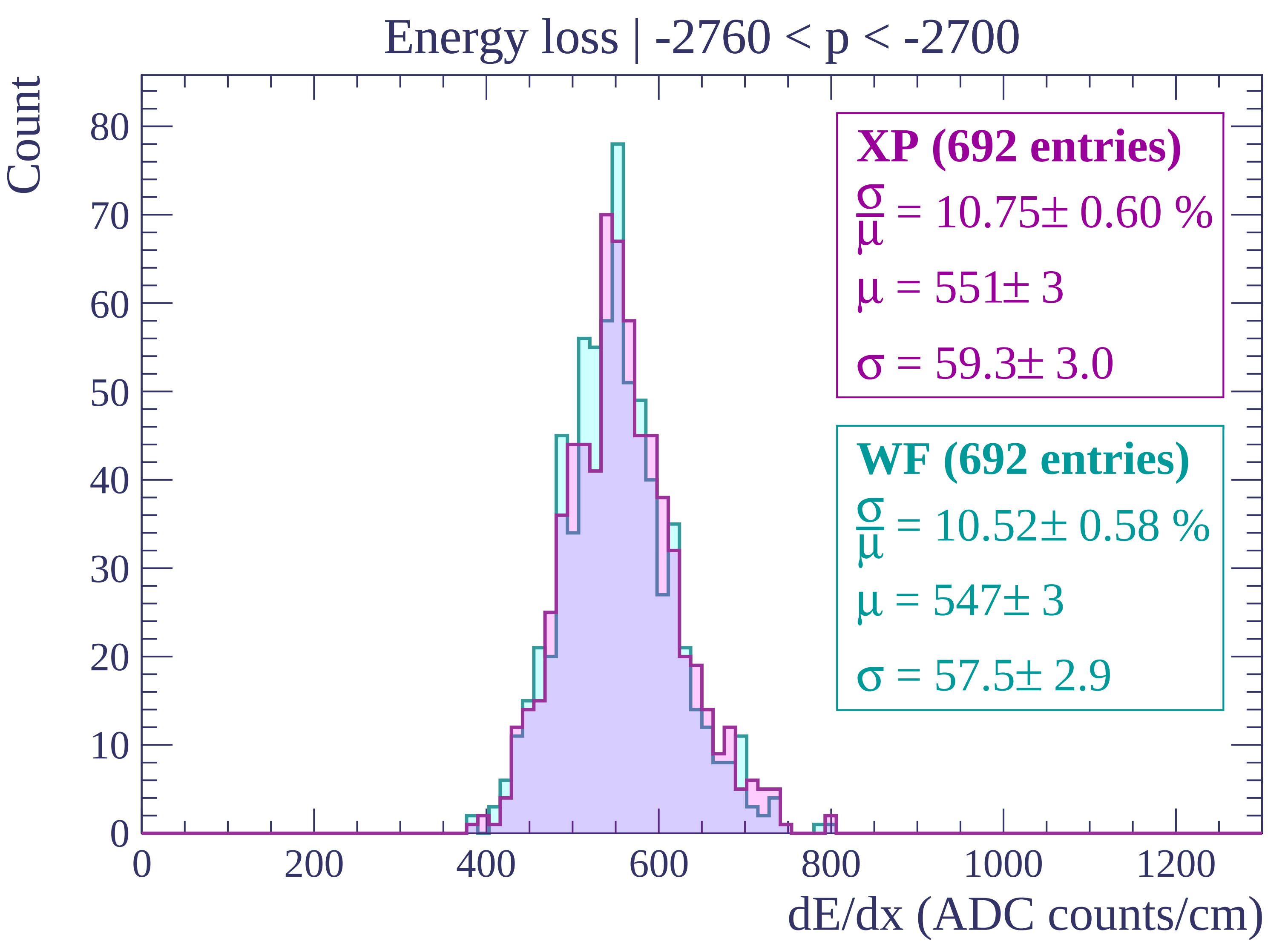

Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
XP (312 entries)
σ/μ = 10.14± 0.71 %
μ = 560± 4
σ = 56.8± 3.6
WF (312 entries)
σ/μ = 10.79± 0.84 %
μ = 553± 4
σ = 59.7± 4.2

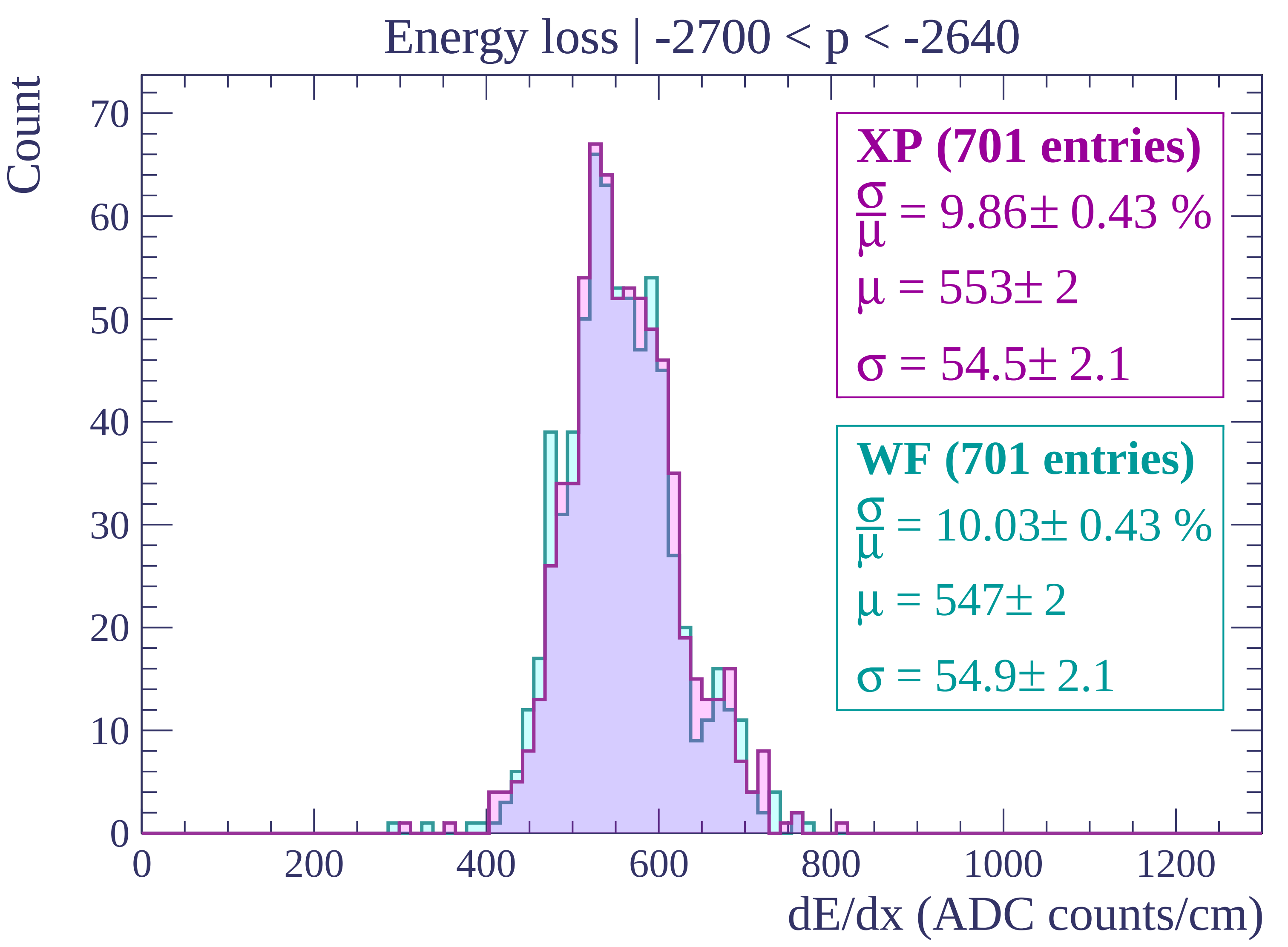

Energy loss | -2940 < p < -2880
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (621 entries)
σ/μ = 10.26± 0.45 %
μ = 554± 3
σ = 56.9± 2.2
WF (621 entries)
σ/μ = 10.68± 0.65 %
μ = 549± 3
σ = 58.6± 3.2


Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP (641 entries)
$\frac{\sigma}{\mu}$ = 10.61± 0.53 %
μ = 551± 3
σ = 58.5± 2.6
WF (641 entries)
$\frac{\sigma}{\mu}$ = 10.68± 0.56 %
μ = 548± 3
σ = 58.5± 2.8

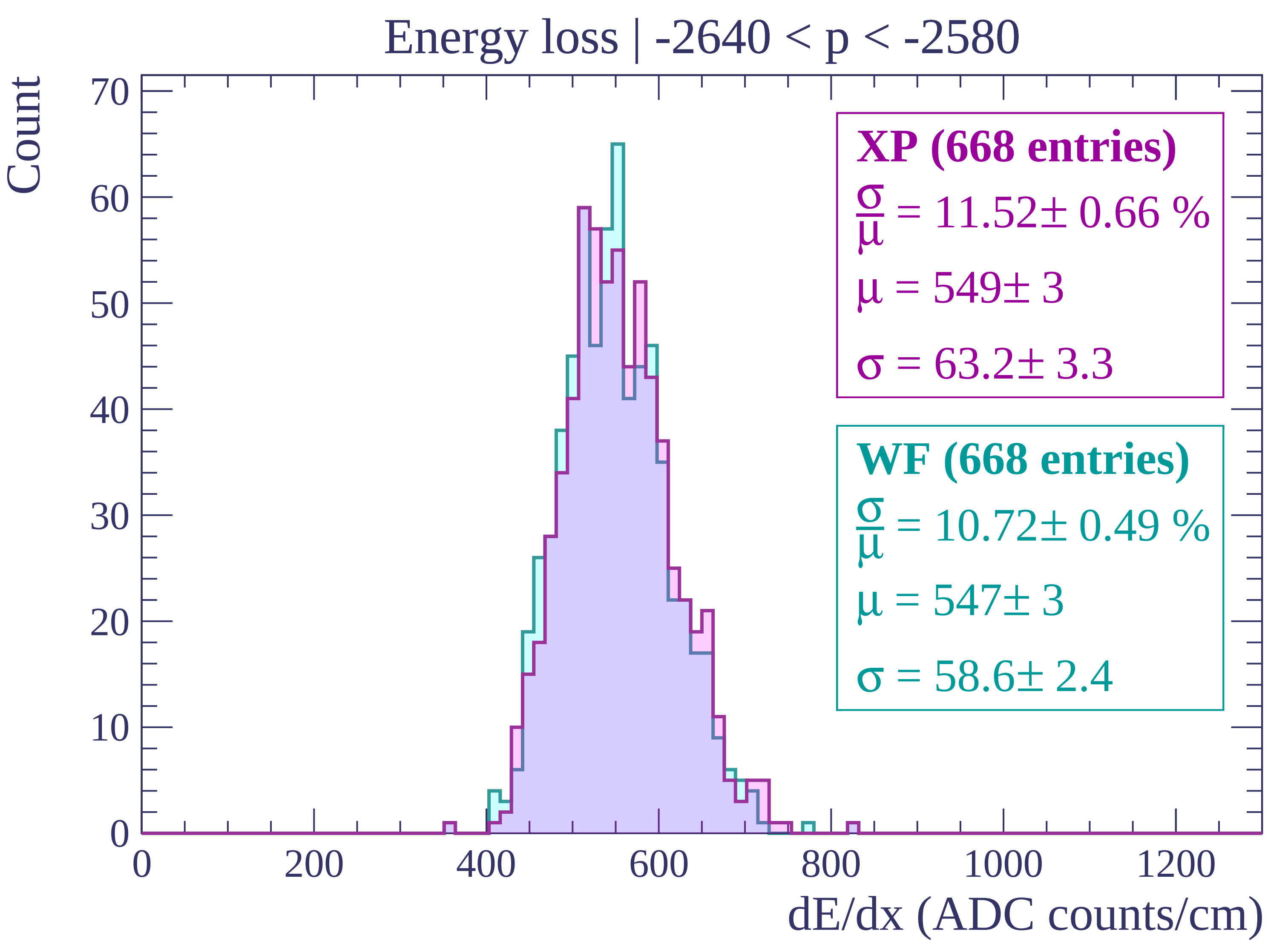


Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (634 entries)
σ/μ = 12.35± 0.79 %
μ = 552± 4
σ = 68.2± 3.9
WF (634 entries)
σ/μ = 10.63± 0.44 %
μ = 554± 3
σ = 59.0± 2.1

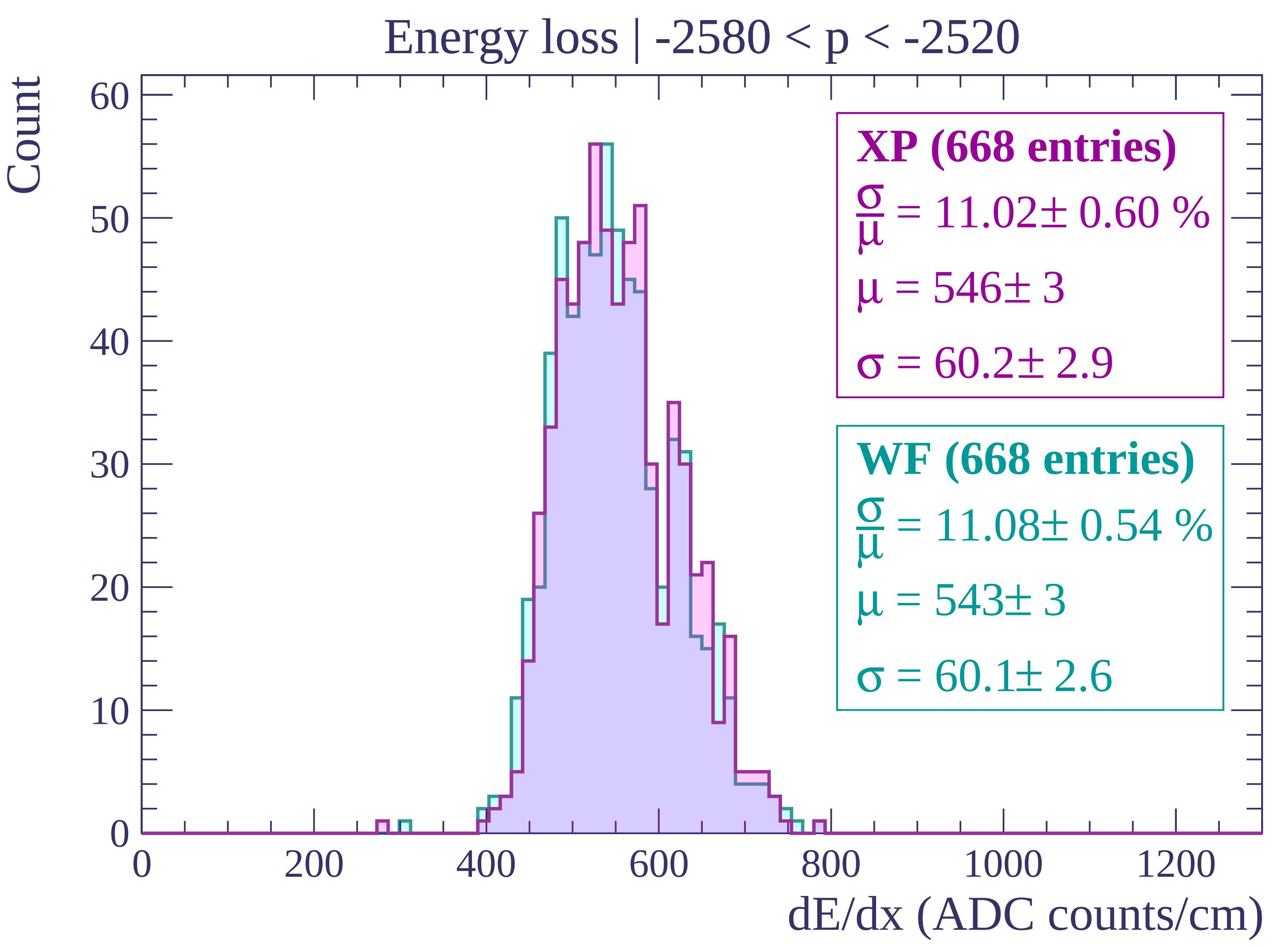

Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP (692 entries)
σ/μ = 10.75± 0.60 %
μ = 551± 3
σ = 59.3± 3.0
WF (692 entries)
σ/μ = 10.52± 0.58 %
μ = 547± 3
σ = 57.5± 2.9



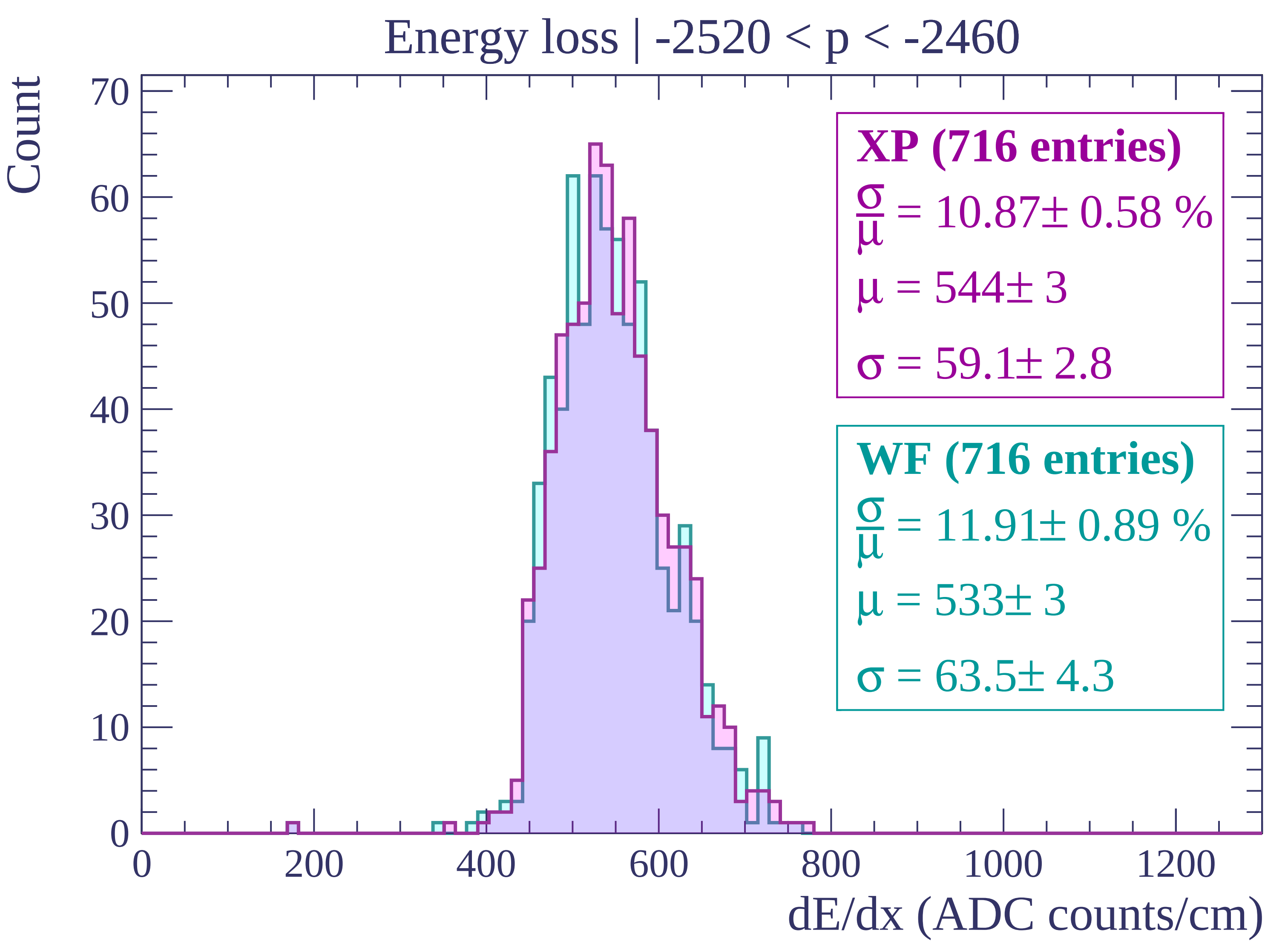
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (701 entries)
$\frac{\sigma}{\mu}$ = 9.86± 0.43 %
μ = 553± 2
σ = 54.5± 2.1
WF (701 entries)
$\frac{\sigma}{\mu}$ = 10.03± 0.43 %
μ = 547± 2
σ = 54.9± 2.1

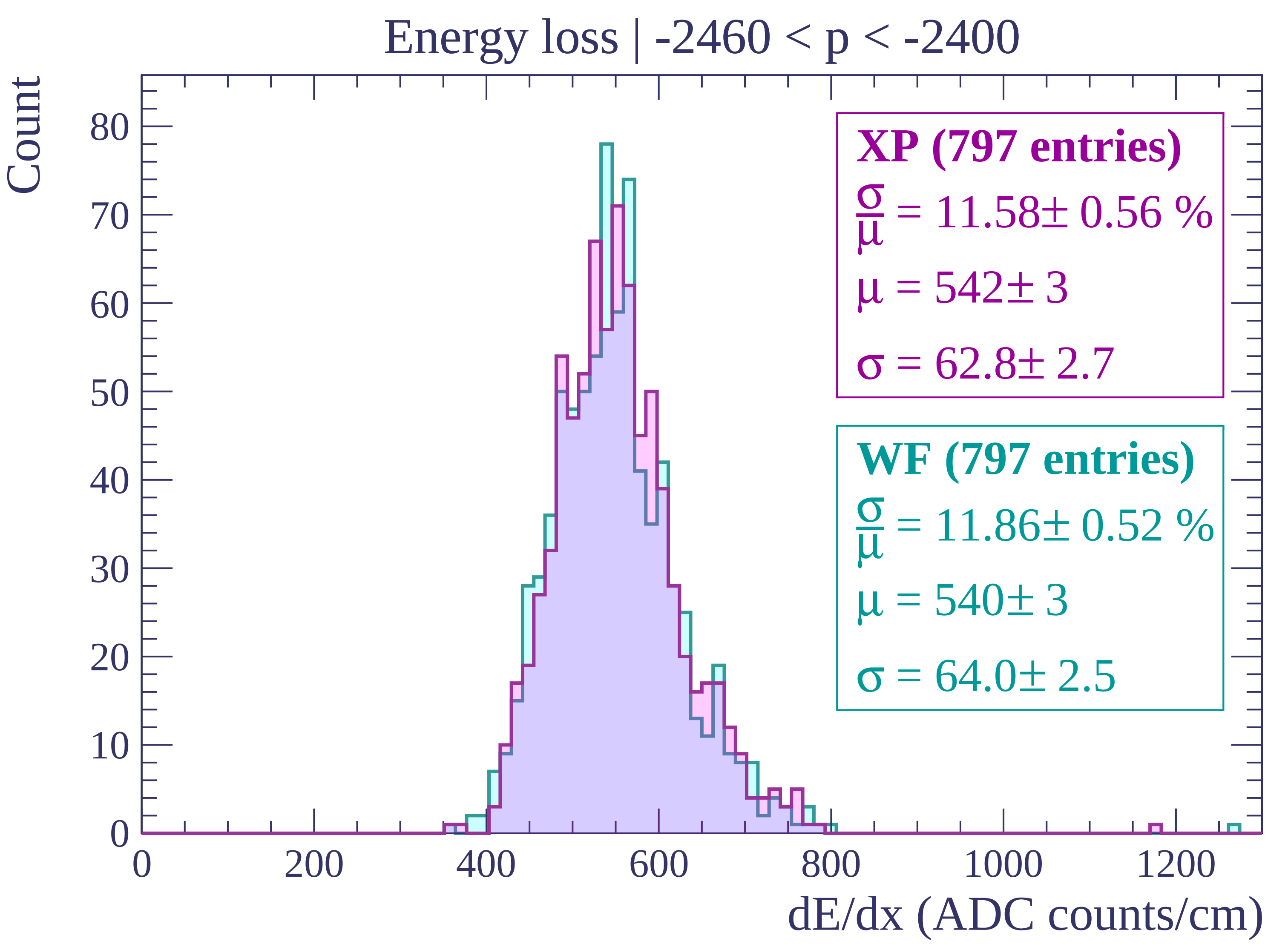


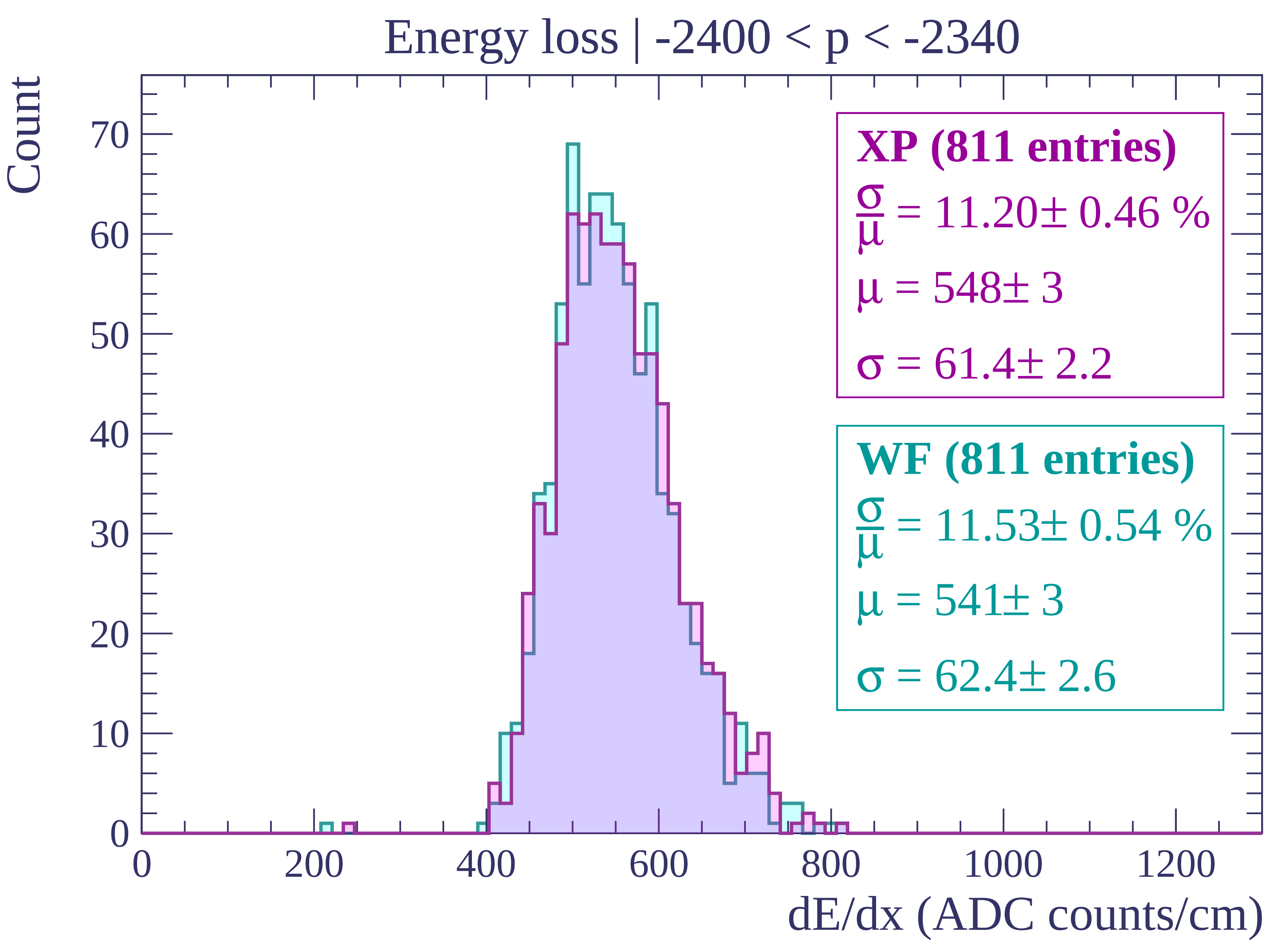
Energy loss | -2640 < p < -2580
Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (668 entries)
σ/μ = 11.52± 0.66 %
μ = 549± 3
σ = 63.2± 3.3
WF (668 entries)
σ/μ = 10.72± 0.49 %
μ = 547± 3
σ = 58.6± 2.4

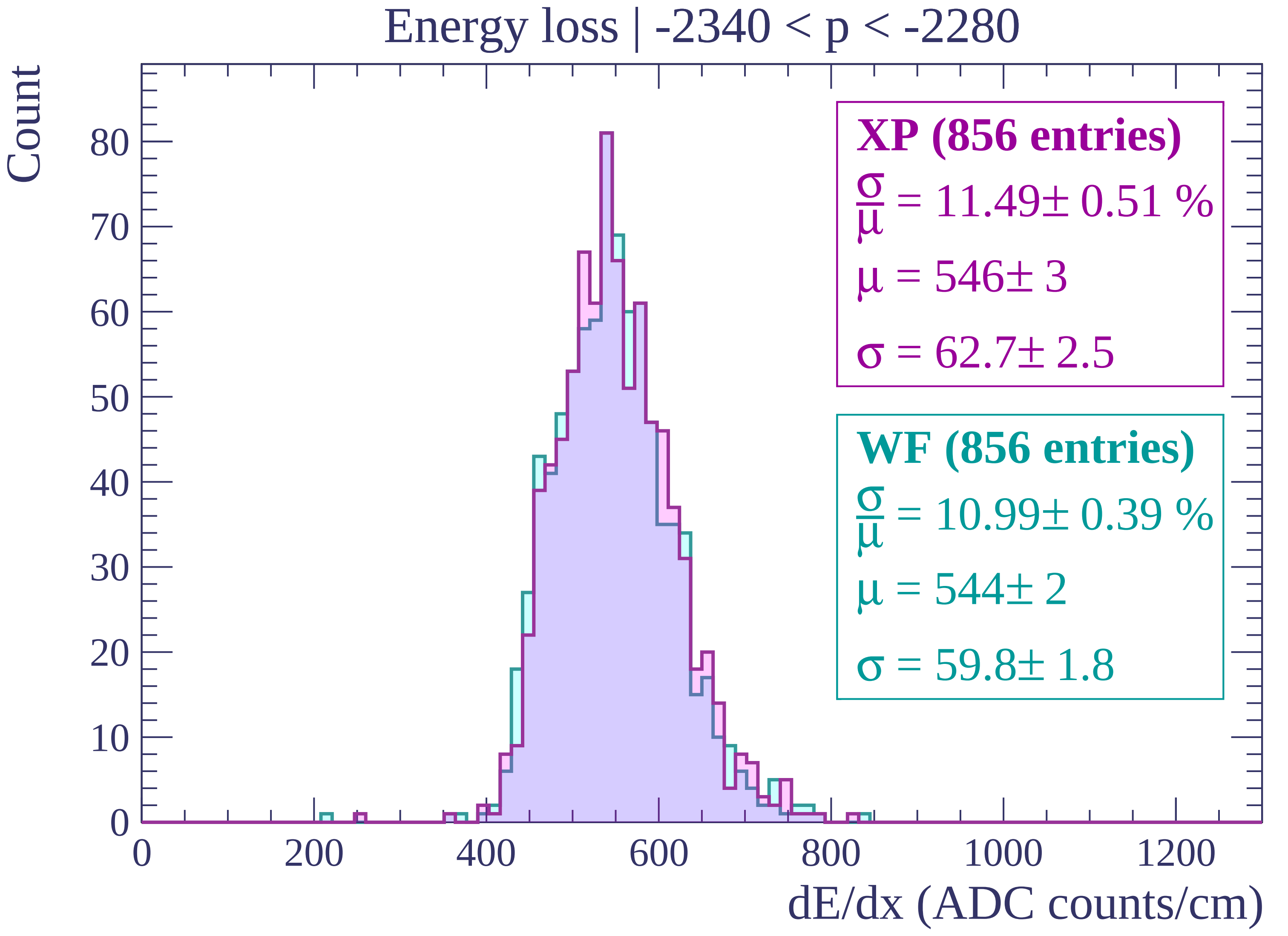


Energy loss | -2580 < p < -2520
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (668 entries)
σ/μ = 11.02± 0.60 %
μ = 546± 3
σ = 60.2± 2.9
WF (668 entries)
σ/μ = 11.08± 0.54 %
μ = 543± 3
σ = 60.1± 2.6

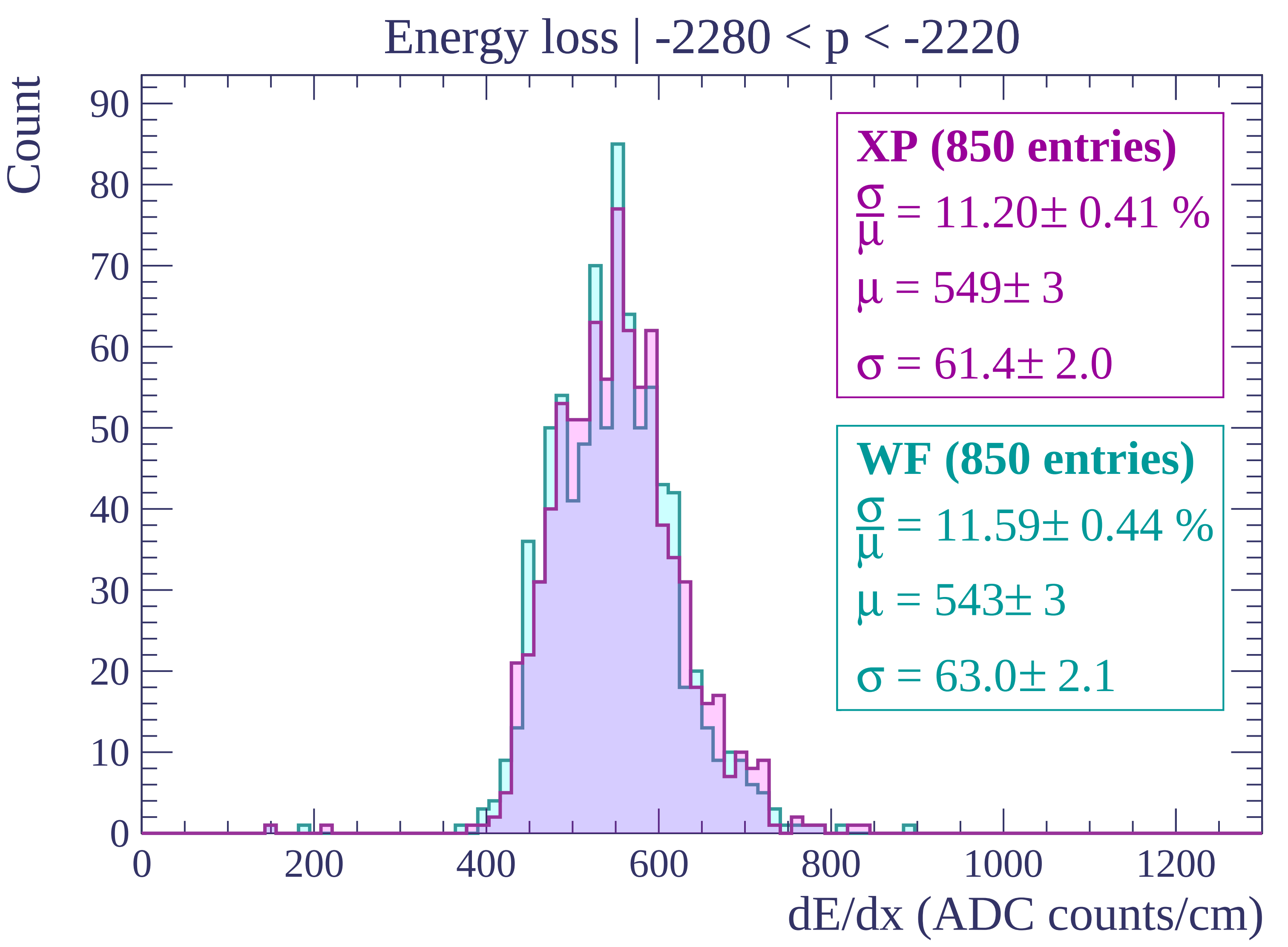

Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
XP (716 entries)
$\frac{\sigma}{\mu}$ = 10.87± 0.58 %
μ = 544± 3
σ = 59.1± 2.8
WF (716 entries)
$\frac{\sigma}{\mu}$ = 11.91± 0.89 %
μ = 533± 3
σ = 63.5± 4.3


Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
XP (797 entries)
σ/μ = 11.58± 0.56 %
μ = 542± 3
σ = 62.8± 2.7
WF (797 entries)
σ/μ = 11.86± 0.52 %
μ = 540± 3
σ = 64.0± 2.5
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

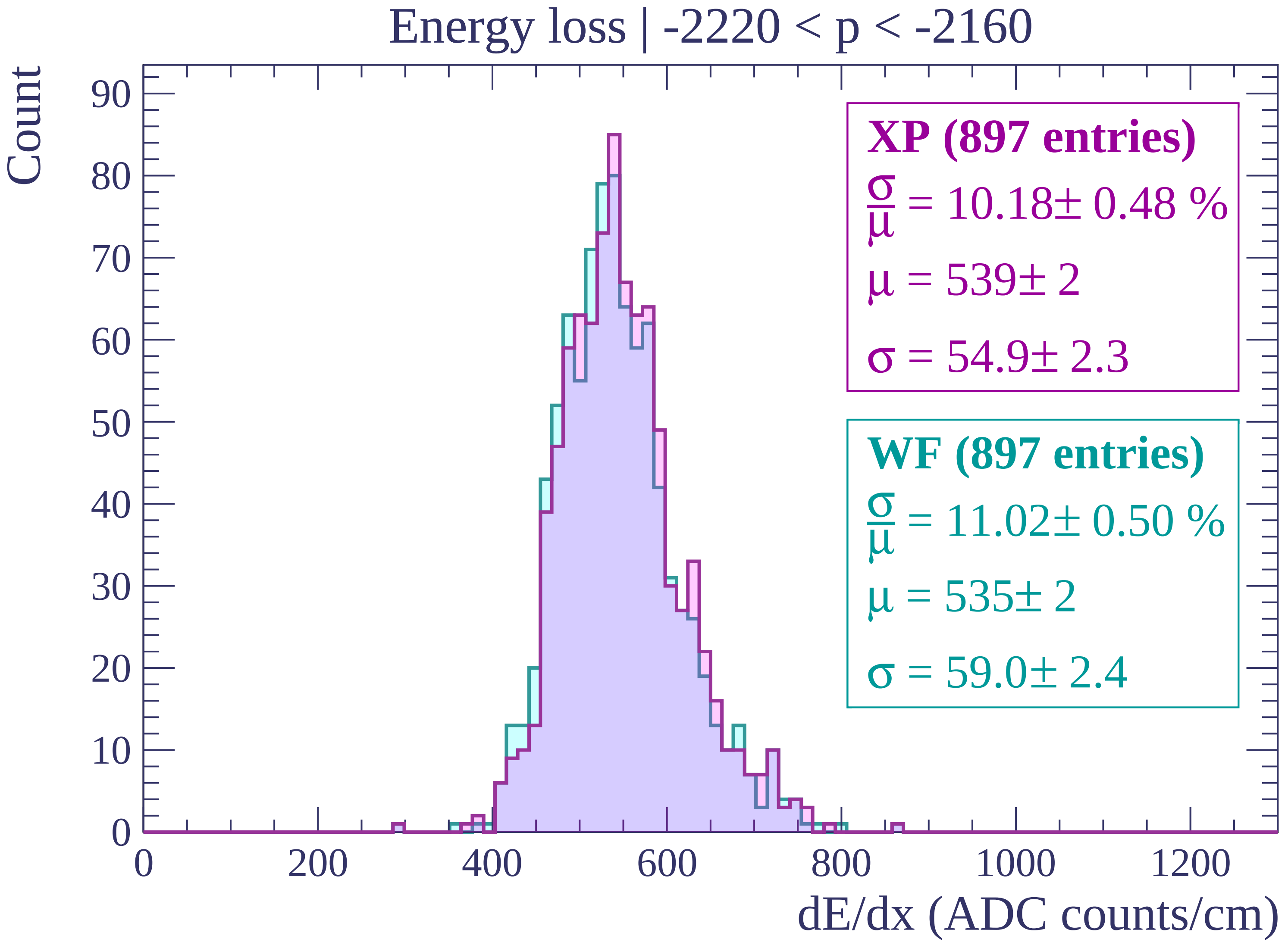

Energy loss | -2400 < p < -2340
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (811 entries)
σ/μ = 11.20± 0.46 %
μ = 548± 3
σ = 61.4± 2.2
WF (811 entries)
σ/μ = 11.53± 0.54 %
μ = 541± 3
σ = 62.4± 2.6

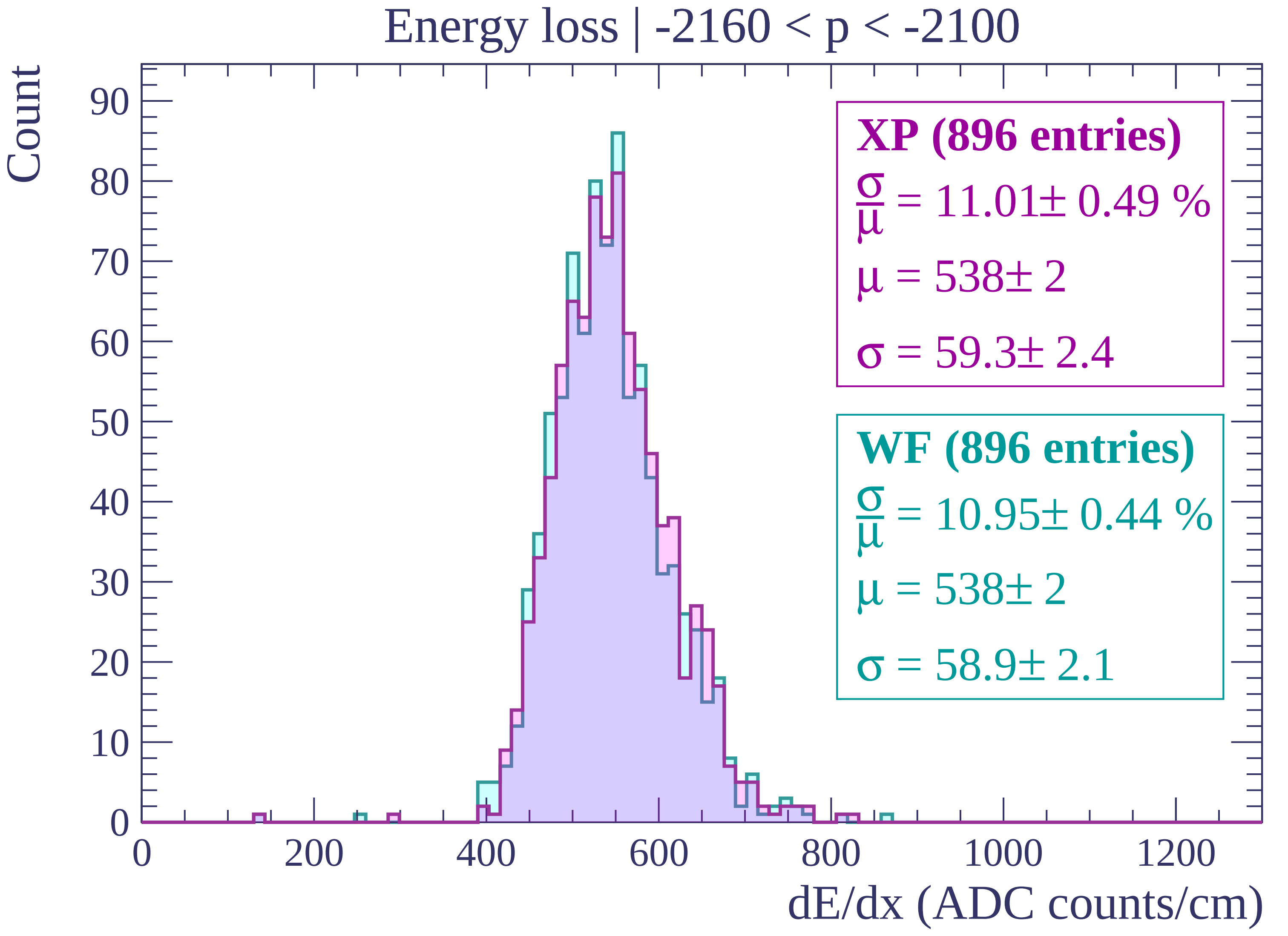


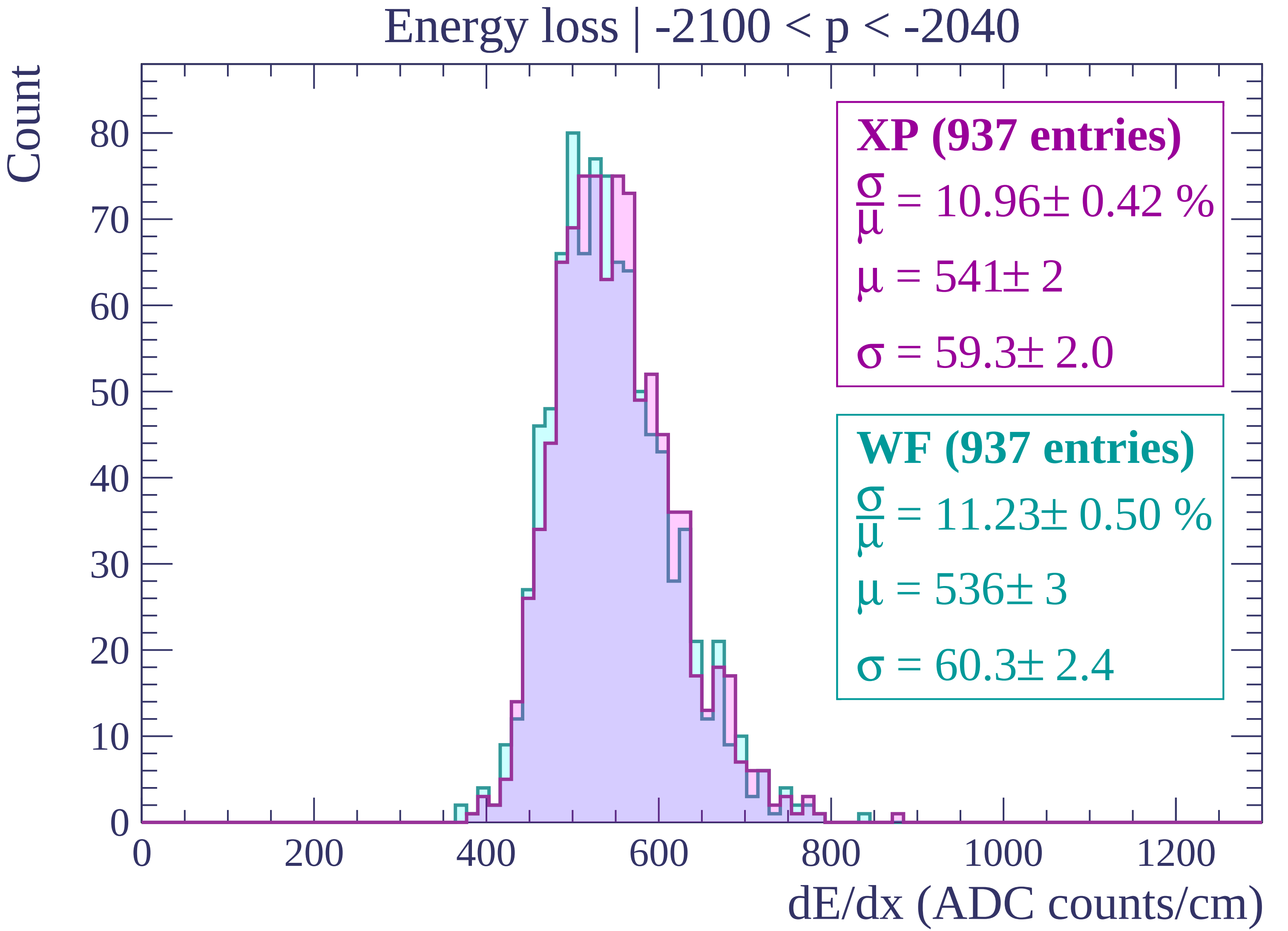
Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (856 entries)
$\frac{\sigma}{\mu}$ = 11.49± 0.51 %
μ = 546± 3
σ = 62.7± 2.5
WF (856 entries)
$\frac{\sigma}{\mu}$ = 10.99± 0.39 %
μ = 544± 2
σ = 59.8± 1.8

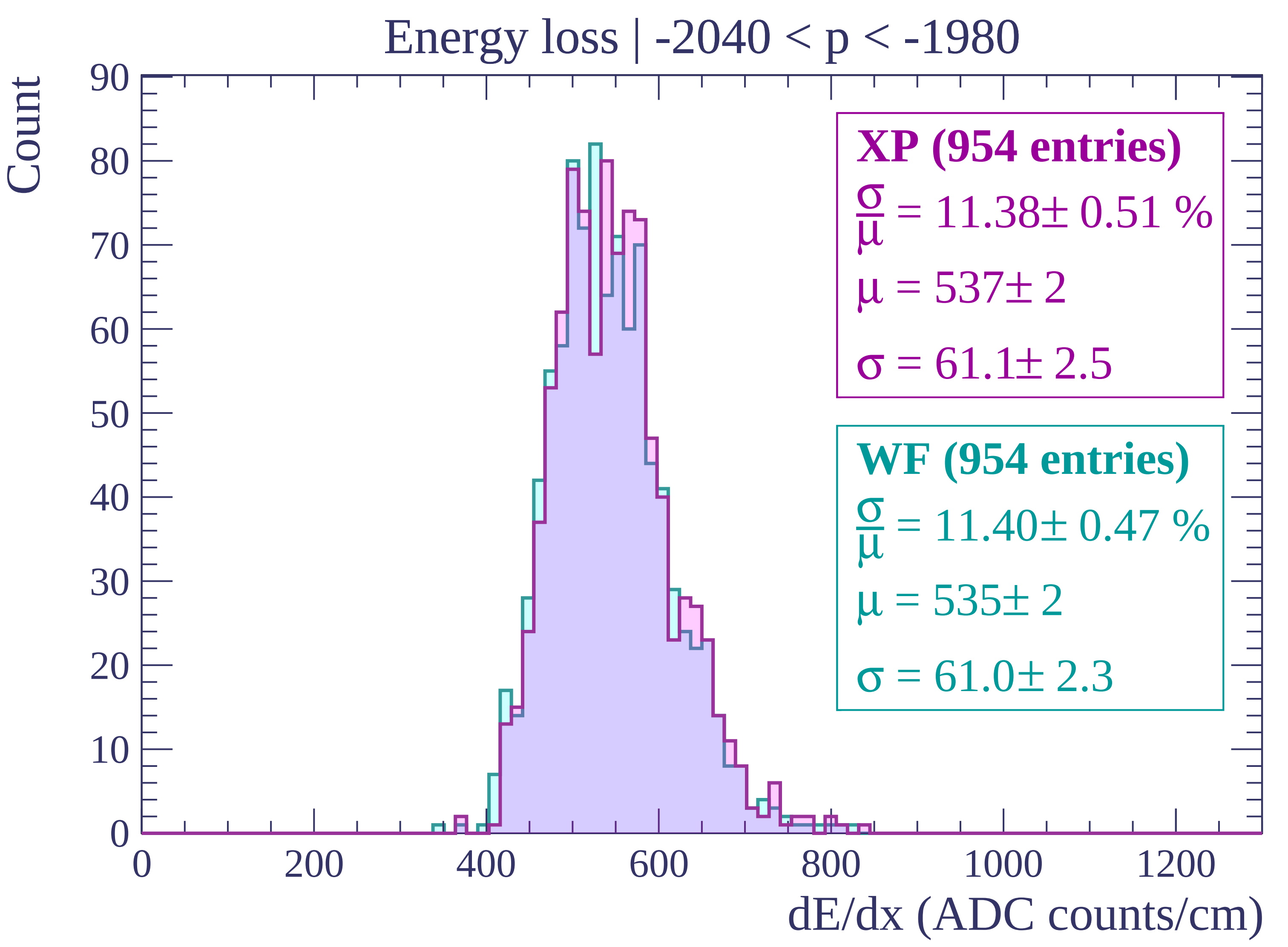


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (850 entries)
σ/μ = 11.20± 0.41 %
μ = 549± 3
σ = 61.4± 2.0
WF (850 entries)
σ/μ = 11.59± 0.44 %
μ = 543± 3
σ = 63.0± 2.1

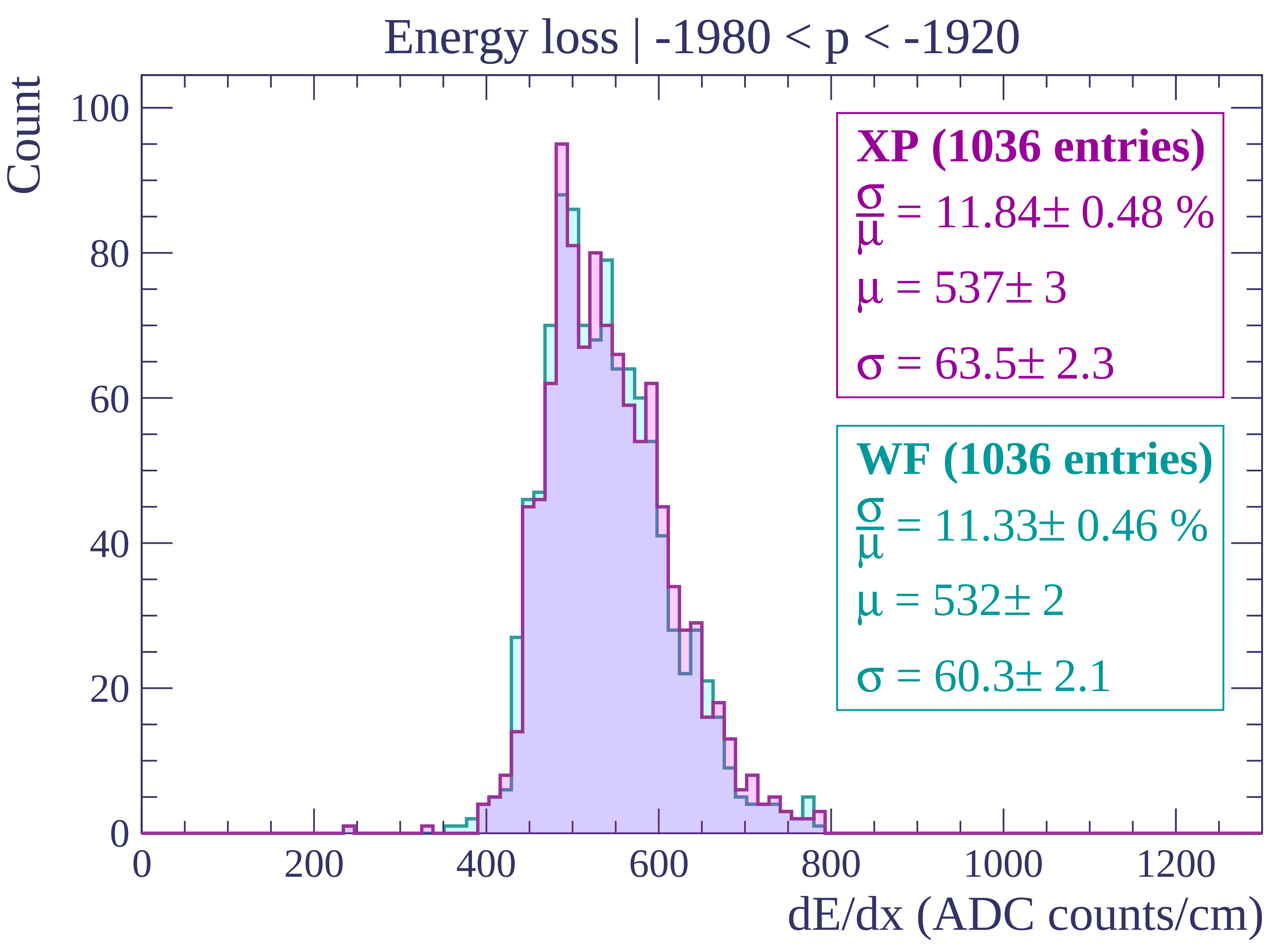

Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
XP (897 entries)
σ/μ = 10.18± 0.48 %
μ = 539± 2
σ = 54.9± 2.3
WF (897 entries)
σ/μ = 11.02± 0.50 %
μ = 535± 2
σ = 59.0± 2.4
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200


Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP (896 entries)
σ/μ = 11.01± 0.49 %
μ = 538± 2
σ = 59.3± 2.4
WF (896 entries)
σ/μ = 10.95± 0.44 %
μ = 538± 2
σ = 58.9± 2.1
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90

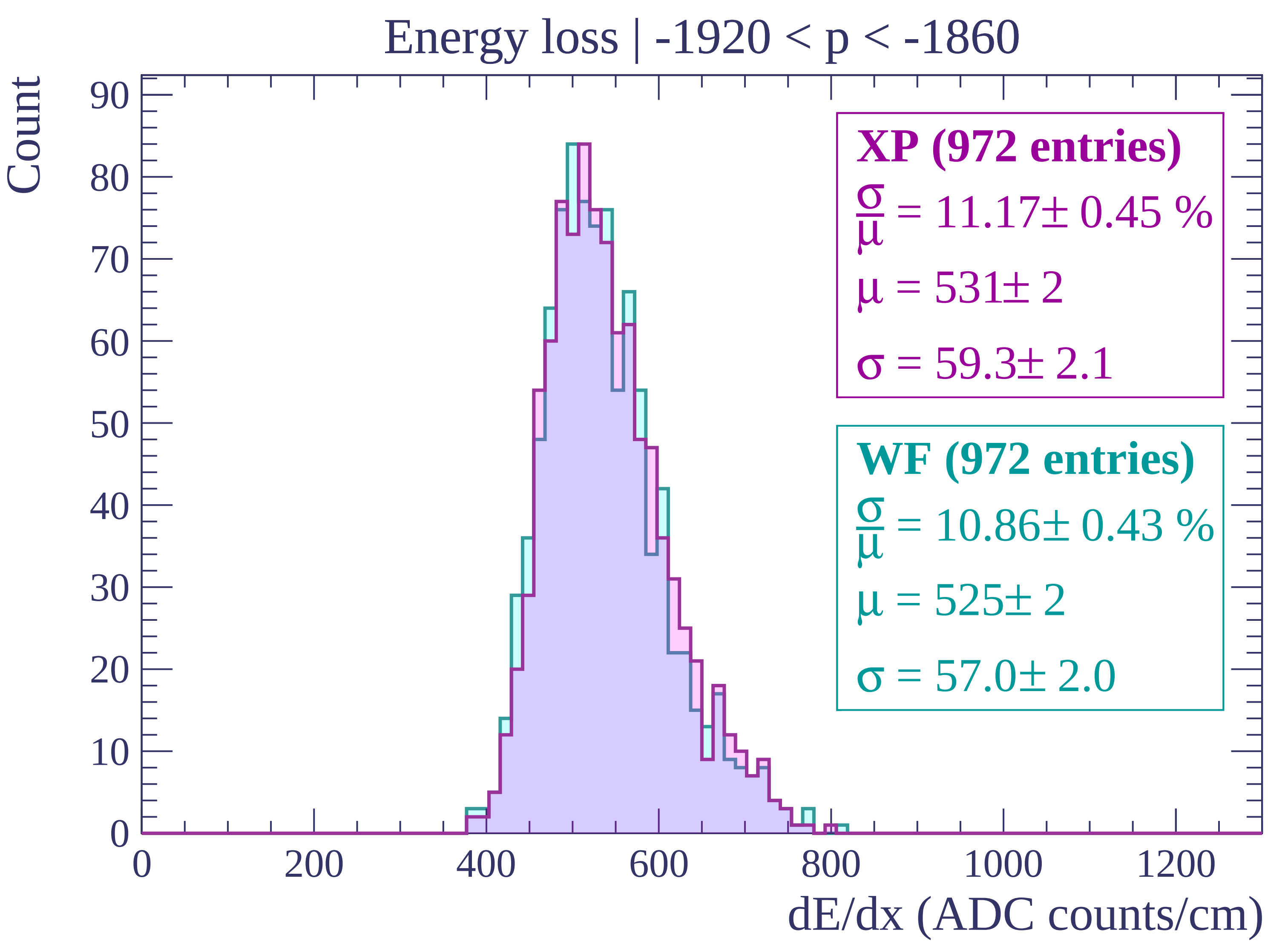

Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (937 entries)
σ/μ = 10.96± 0.42 %
μ = 541± 2
σ = 59.3± 2.0
WF (937 entries)
σ/μ = 11.23± 0.50 %
μ = 536± 3
σ = 60.3± 2.4

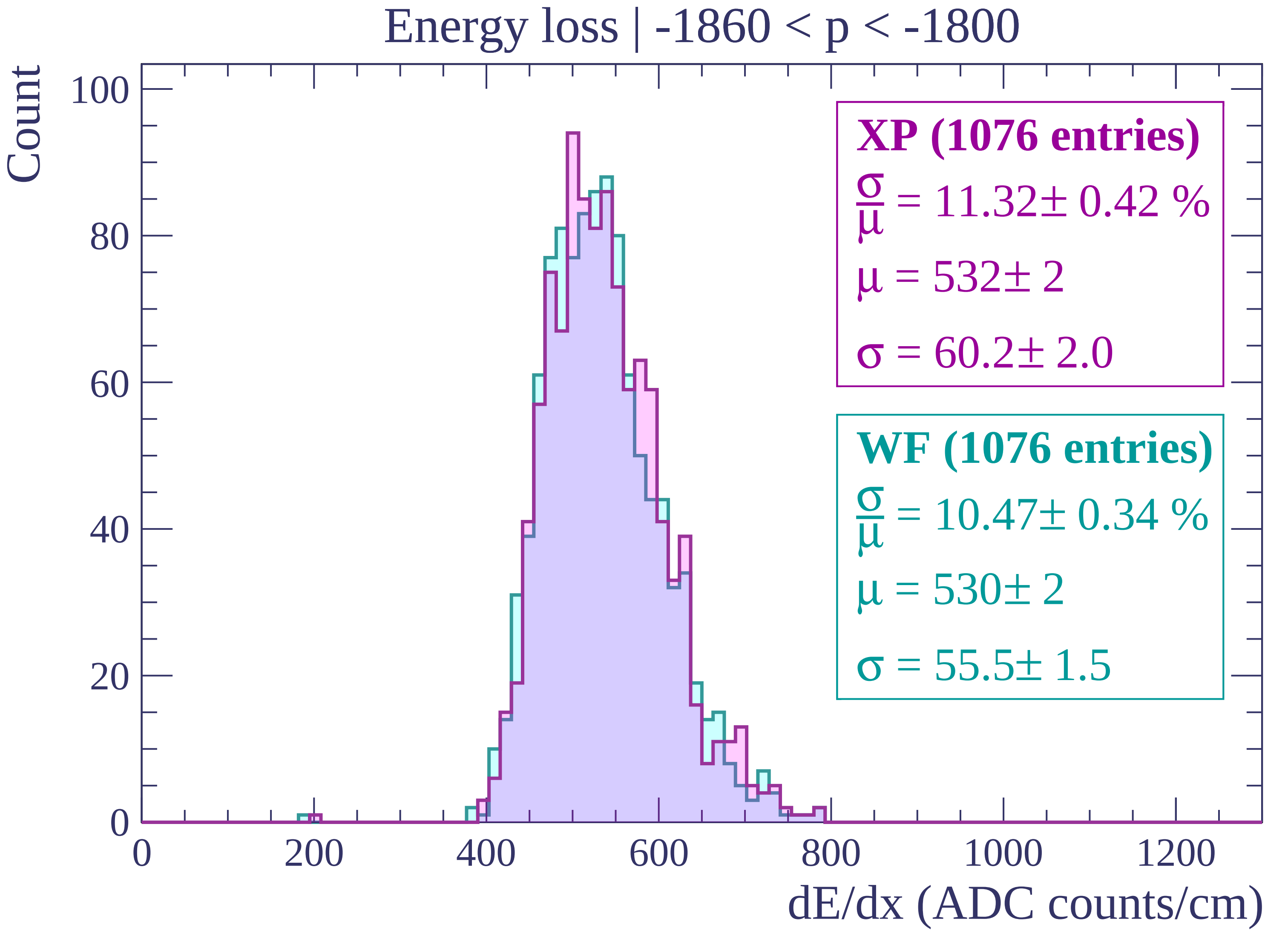


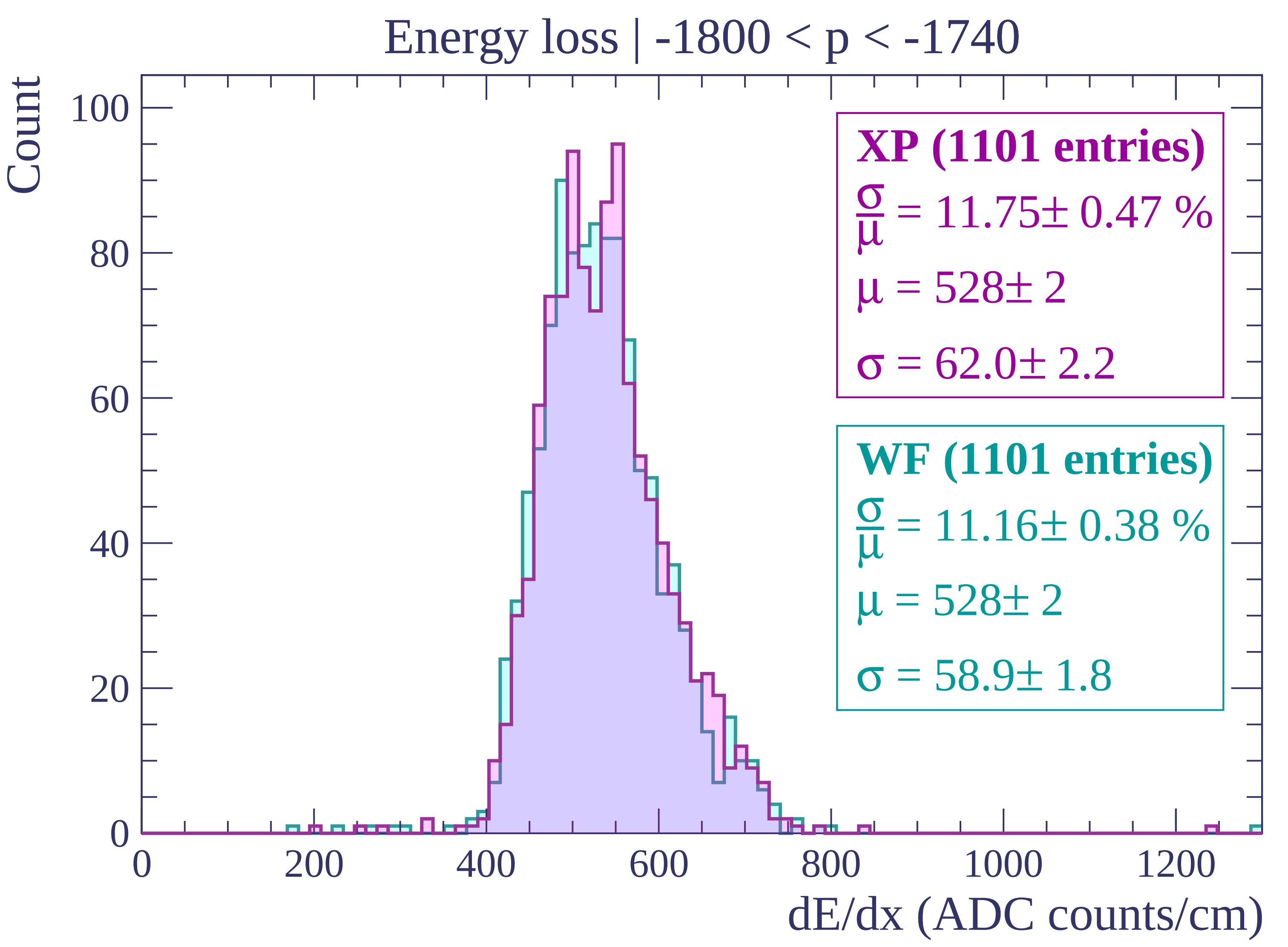
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (954 entries)
$\frac{\sigma}{\mu}$ = 11.38± 0.51 %
μ = 537± 2
σ = 61.1± 2.5
WF (954 entries)
$\frac{\sigma}{\mu}$ = 11.40± 0.47 %
μ = 535± 2
σ = 61.0± 2.3



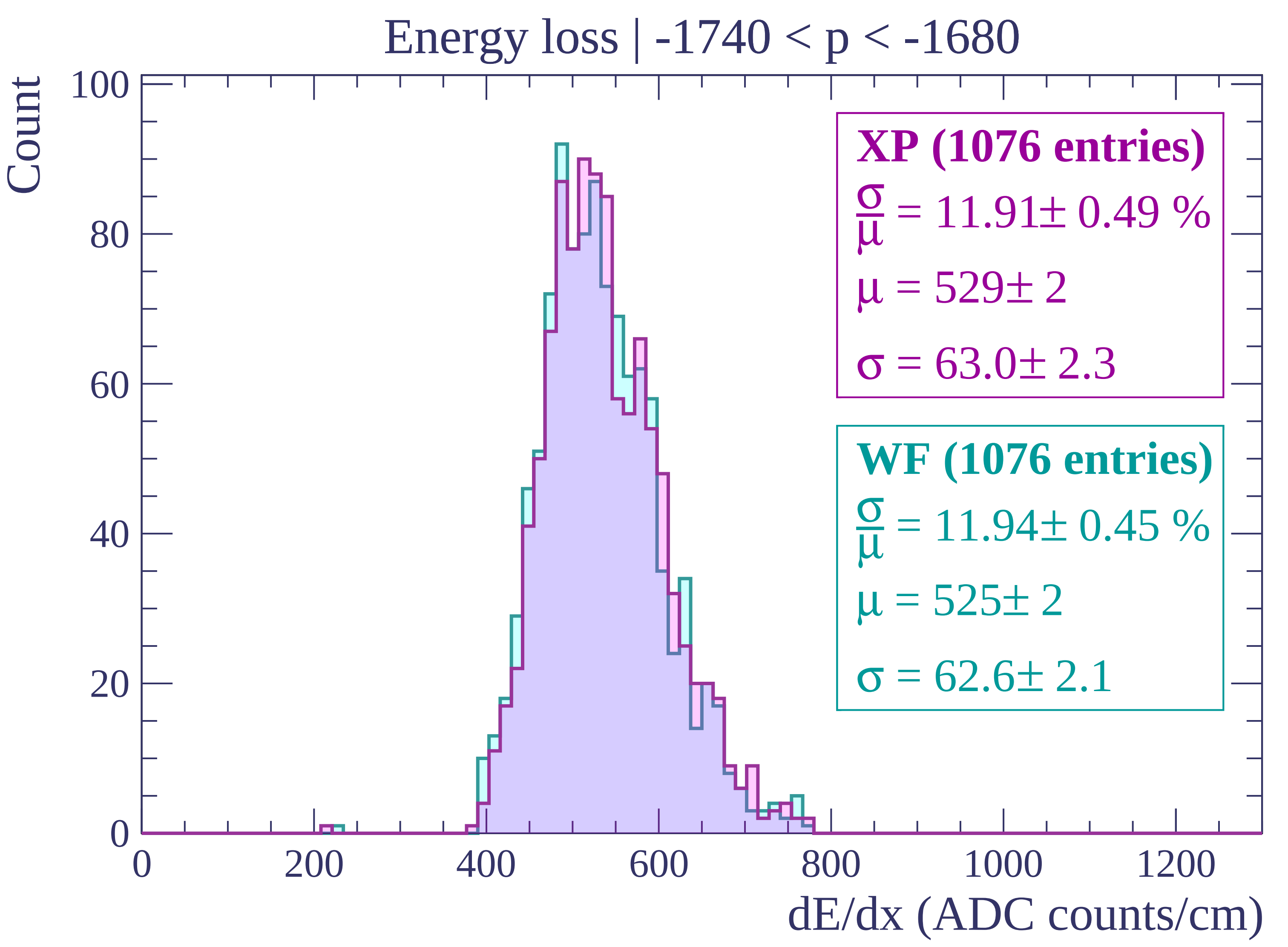
Energy loss | -1980 < p < -1920
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1036 entries)
$\frac{\sigma}{\mu}$ = 11.84± 0.48 %
μ = 537± 3
σ = 63.5± 2.3
WF (1036 entries)
$\frac{\sigma}{\mu}$ = 11.33± 0.46 %
μ = 532± 2
σ = 60.3± 2.1

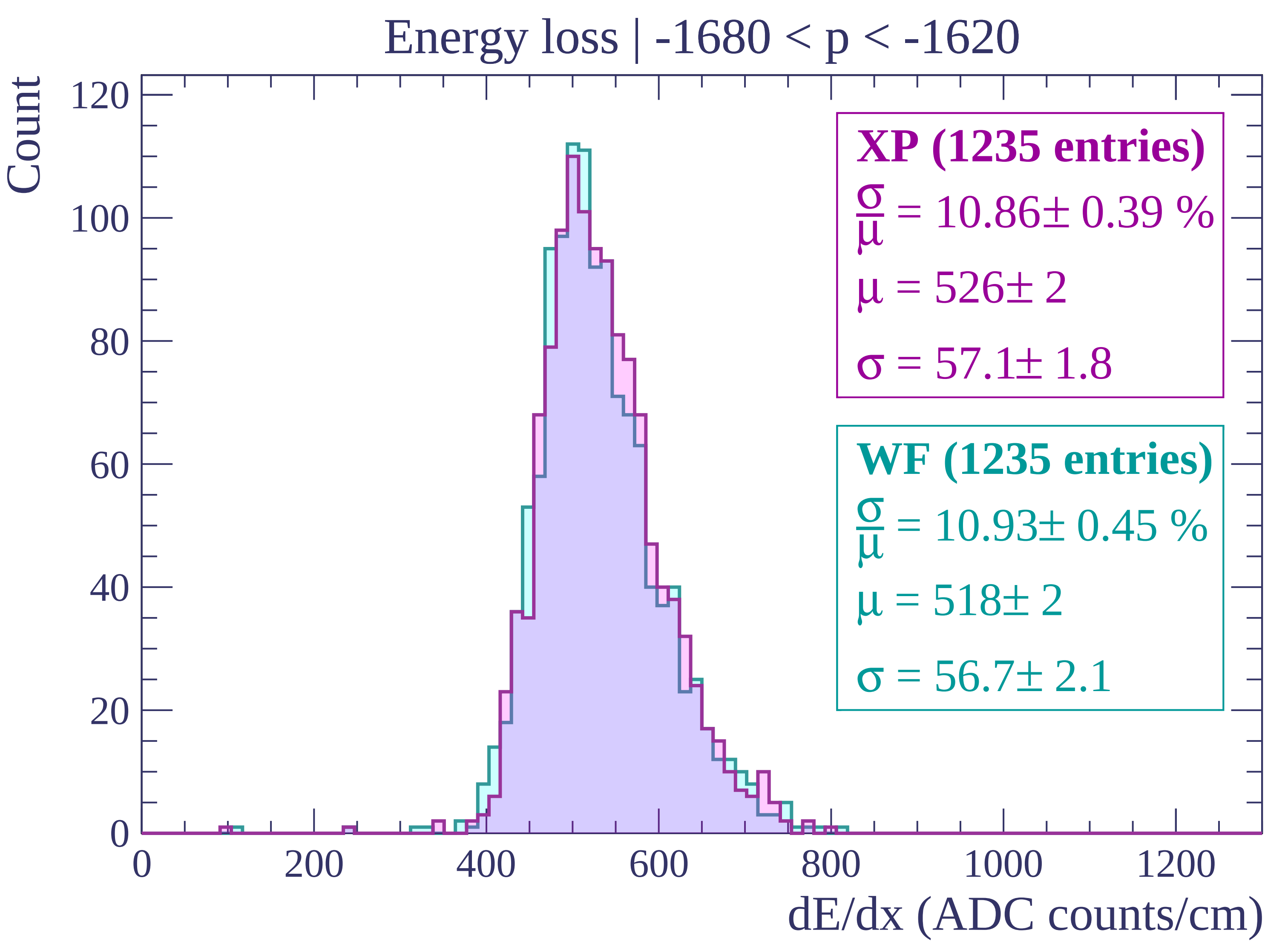

Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
XP (972 entries)
$\frac{\sigma}{\mu}$ = 11.17± 0.45 %
μ = 531± 2
σ = 59.3± 2.1
WF (972 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.43 %
μ = 525± 2
σ = 57.0± 2.0


Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1076 entries)
σ/μ = 11.32± 0.42 %
μ = 532± 2
σ = 60.2± 2.0
WF (1076 entries)
σ/μ = 10.47± 0.34 %
μ = 530± 2
σ = 55.5± 1.5

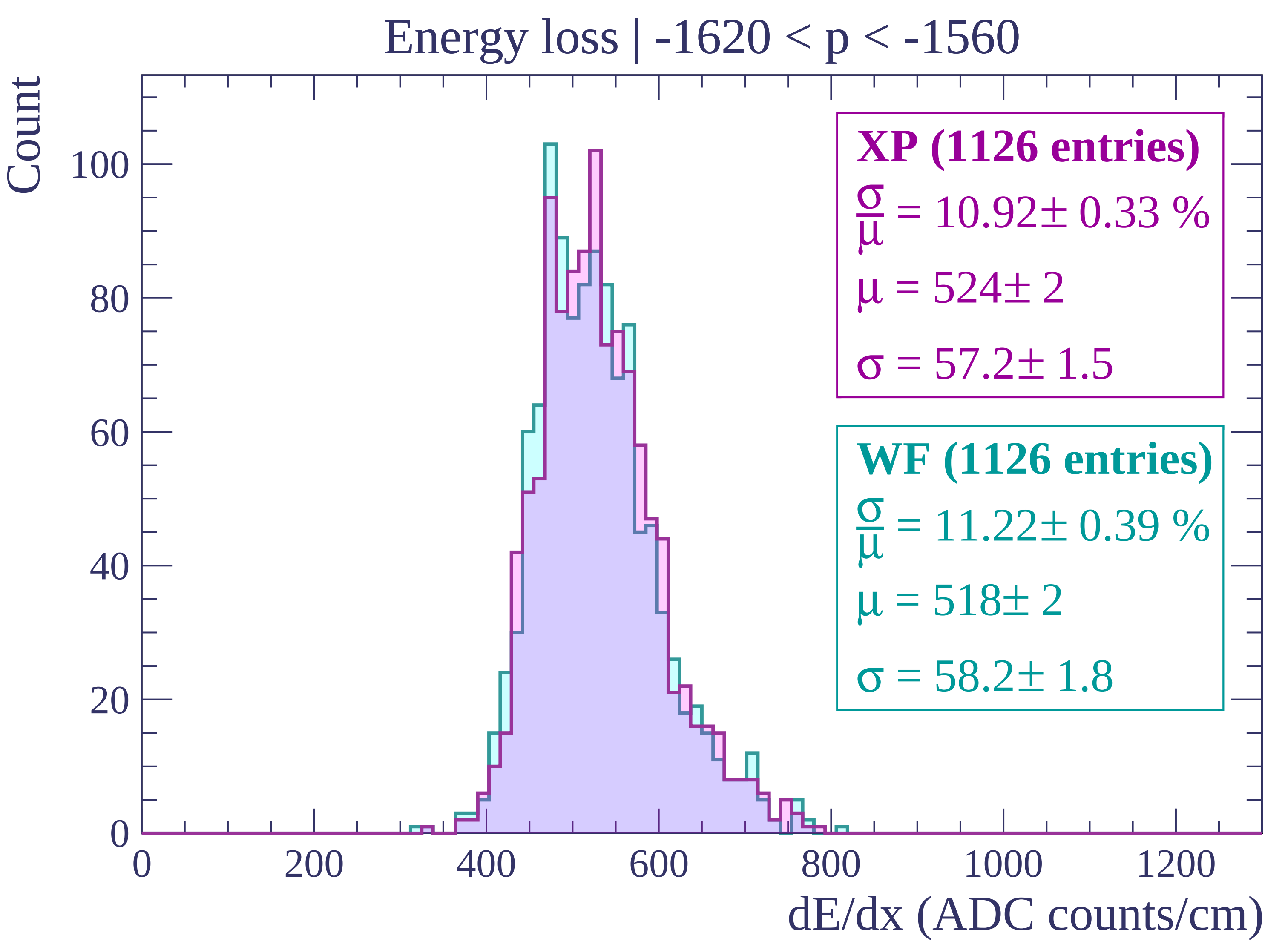


Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (1101 entries)
σ/μ = 11.75± 0.47 %
μ = 528± 2
σ = 62.0± 2.2
WF (1101 entries)
σ/μ = 11.16± 0.38 %
μ = 528± 2
σ = 58.9± 1.8

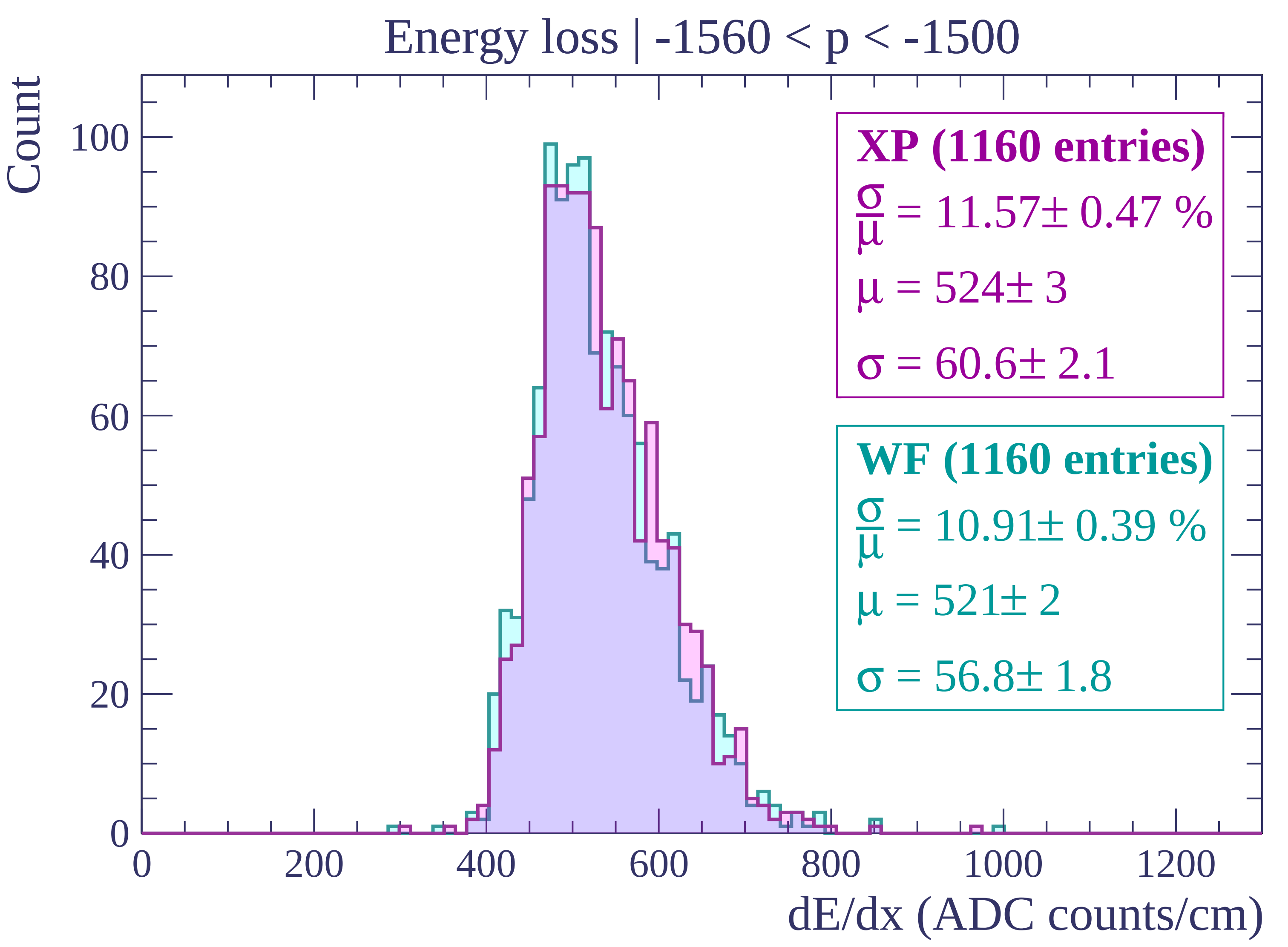

Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP (1076 entries)
$\frac{\sigma}{\mu}$ = 11.91± 0.49 %
μ = 529± 2
σ = 63.0± 2.3
WF (1076 entries)
$\frac{\sigma}{\mu}$ = 11.94± 0.45 %
μ = 525± 2
σ = 62.6± 2.1



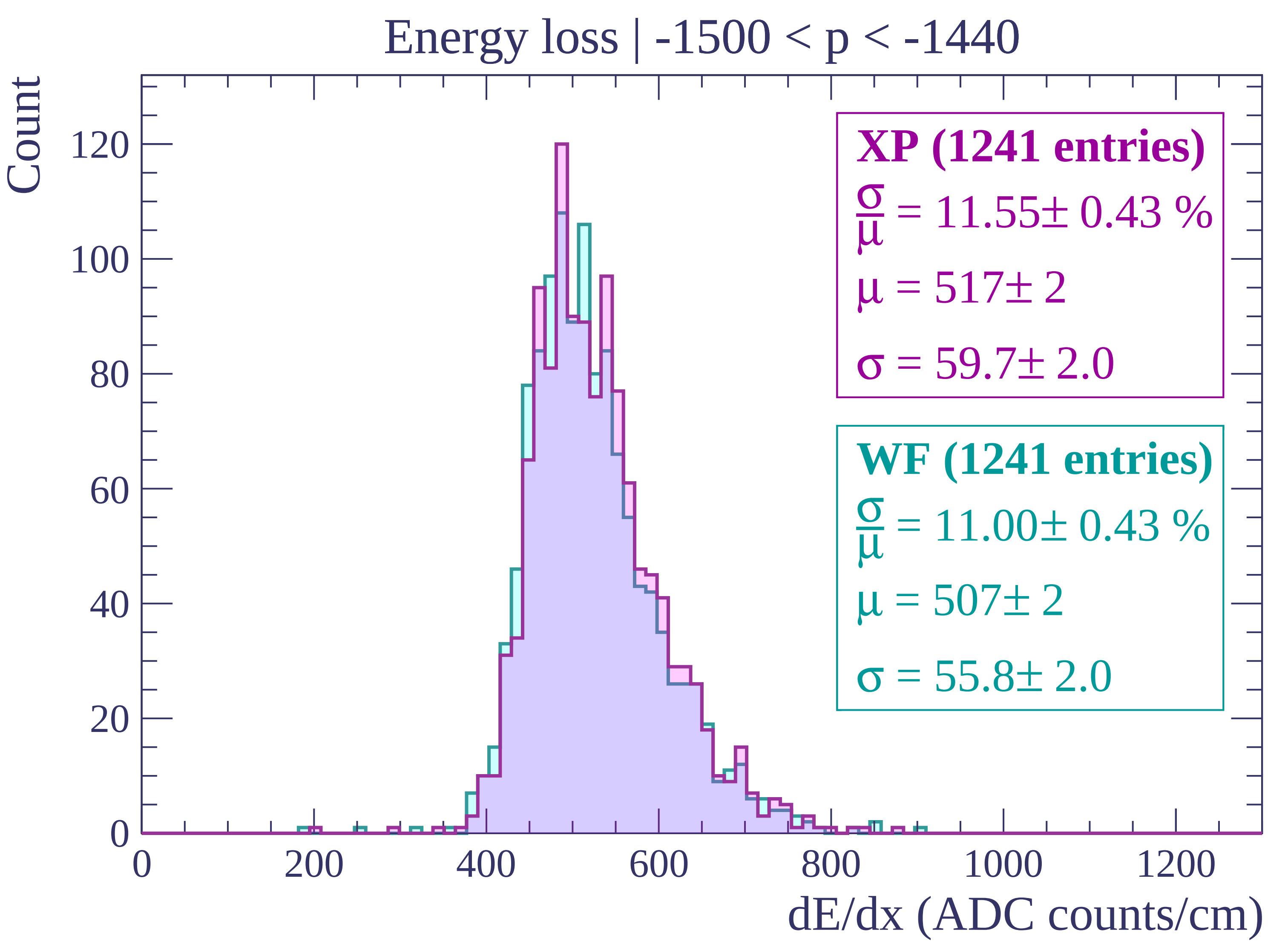
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1235 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.39 %
μ = 526± 2
σ = 57.1± 1.8
WF (1235 entries)
$\frac{\sigma}{\mu}$ = 10.93± 0.45 %
μ = 518± 2
σ = 56.7± 2.1



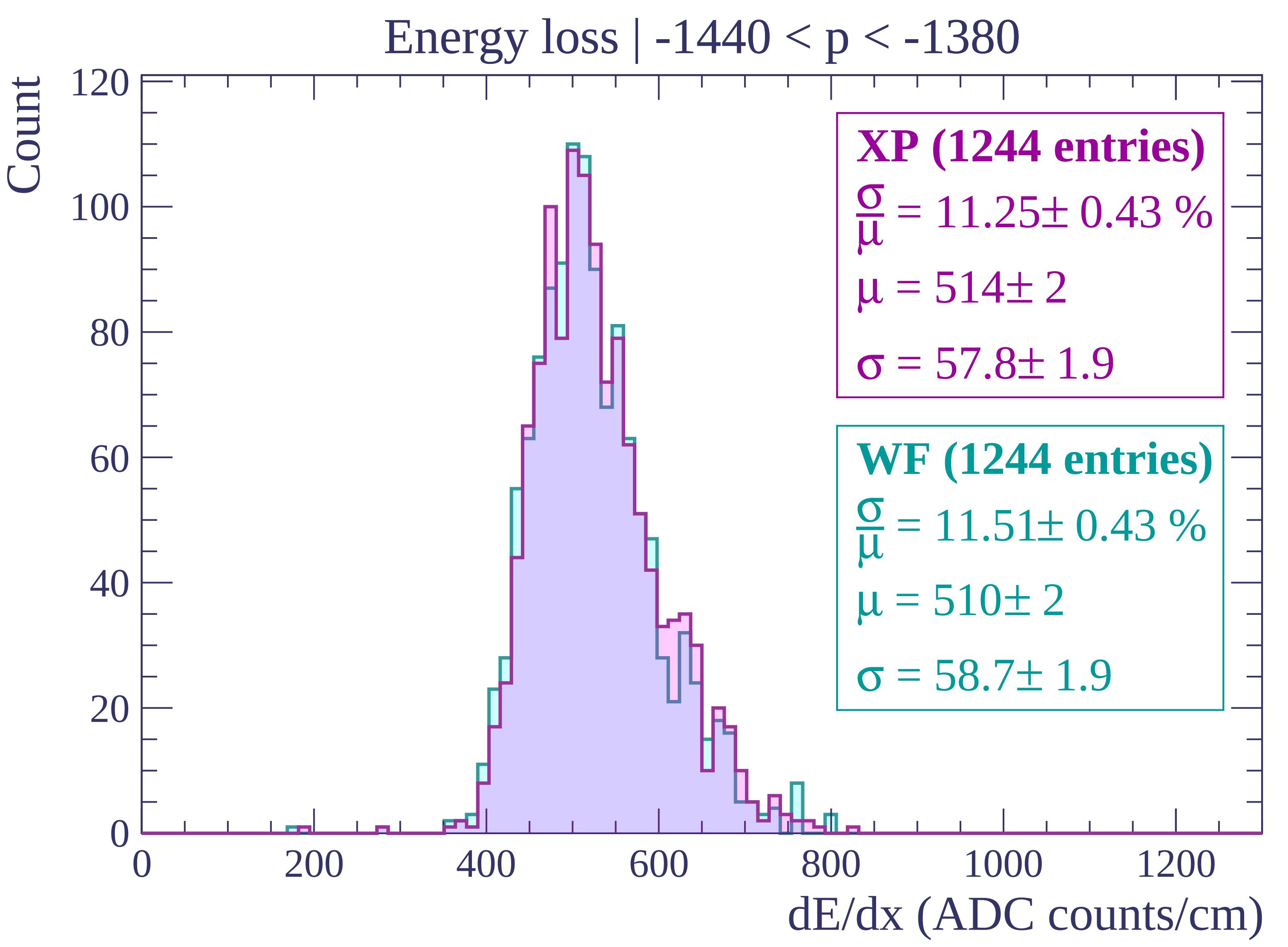
Energy loss | -1620 < p < -1560
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1126 entries)
σ/μ = 10.92± 0.33 %
μ = 524± 2
σ = 57.2± 1.5
WF (1126 entries)
σ/μ = 11.22± 0.39 %
μ = 518± 2
σ = 58.2± 1.8

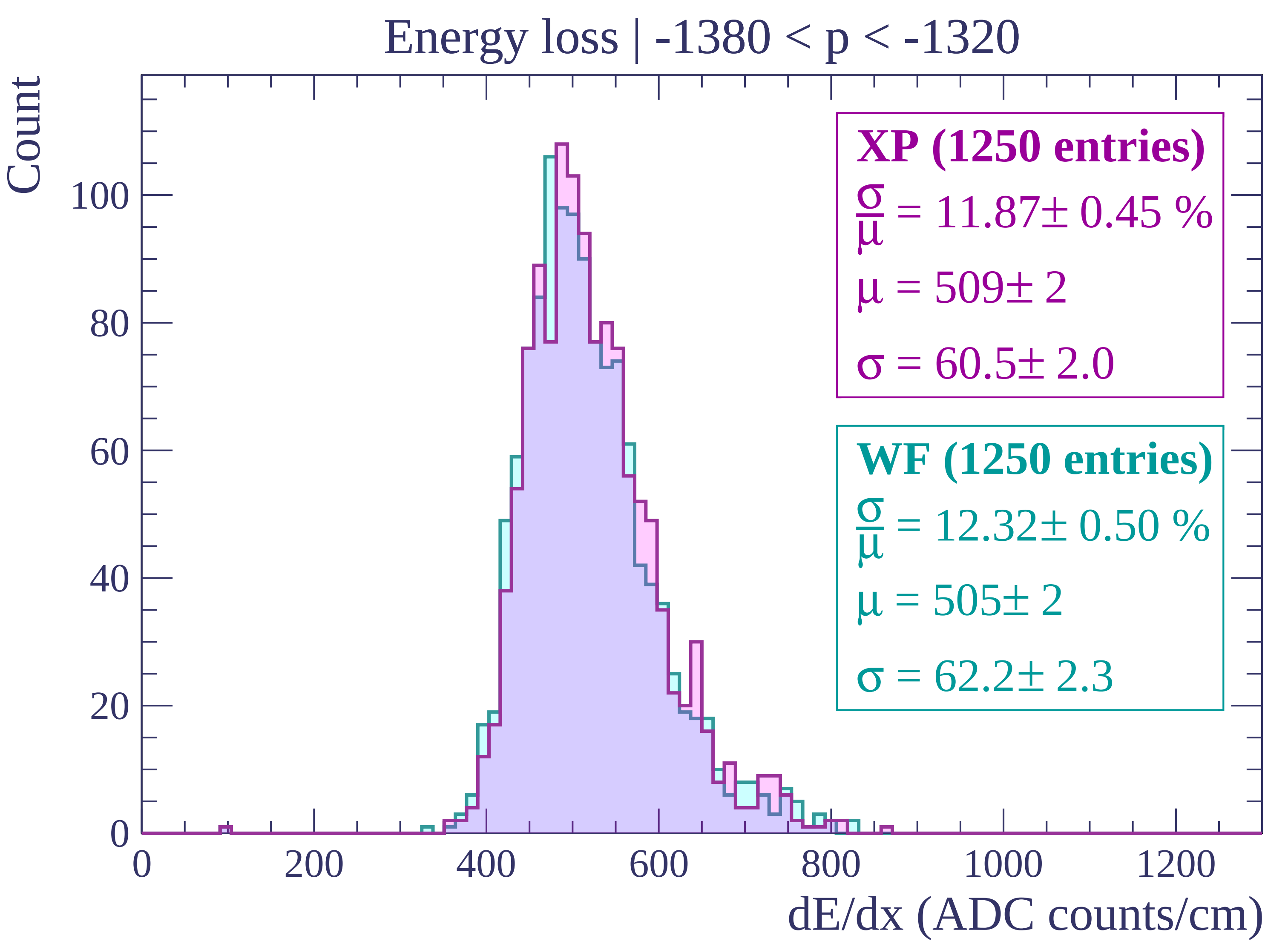


Energy loss | -1560 < p < -1500
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1160 entries)
$\frac{\sigma}{\mu}$ = 11.57± 0.47 %
μ = 524± 3
σ = 60.6± 2.1
WF (1160 entries)
$\frac{\sigma}{\mu}$ = 10.91± 0.39 %
μ = 521± 2
σ = 56.8± 1.8

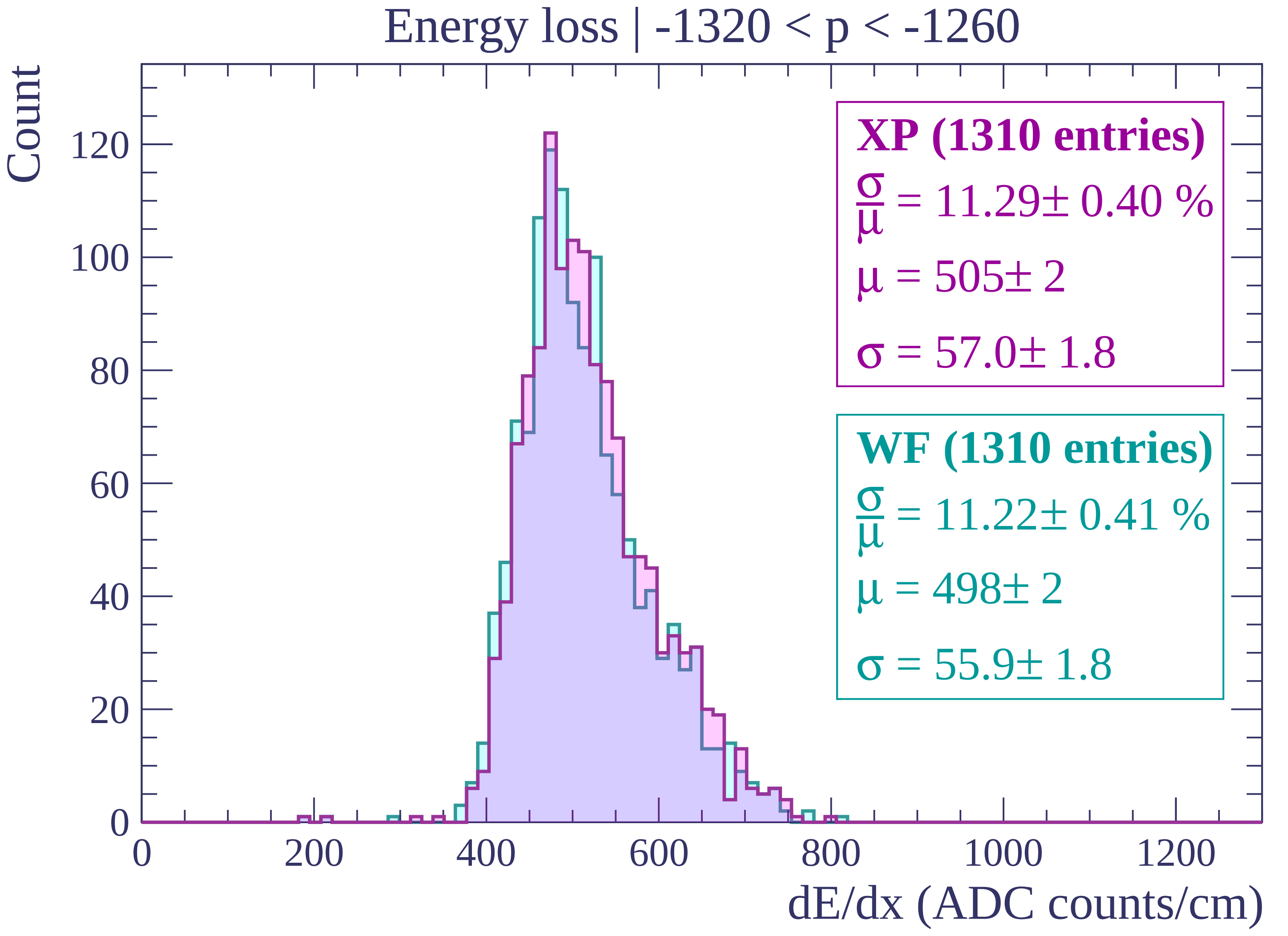


Energy loss | -1500 < p < -1440
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1241 entries)
σ/μ = 11.55± 0.43 %
μ = 517± 2
σ = 59.7± 2.0
WF (1241 entries)
σ/μ = 11.00± 0.43 %
μ = 507± 2
σ = 55.8± 2.0

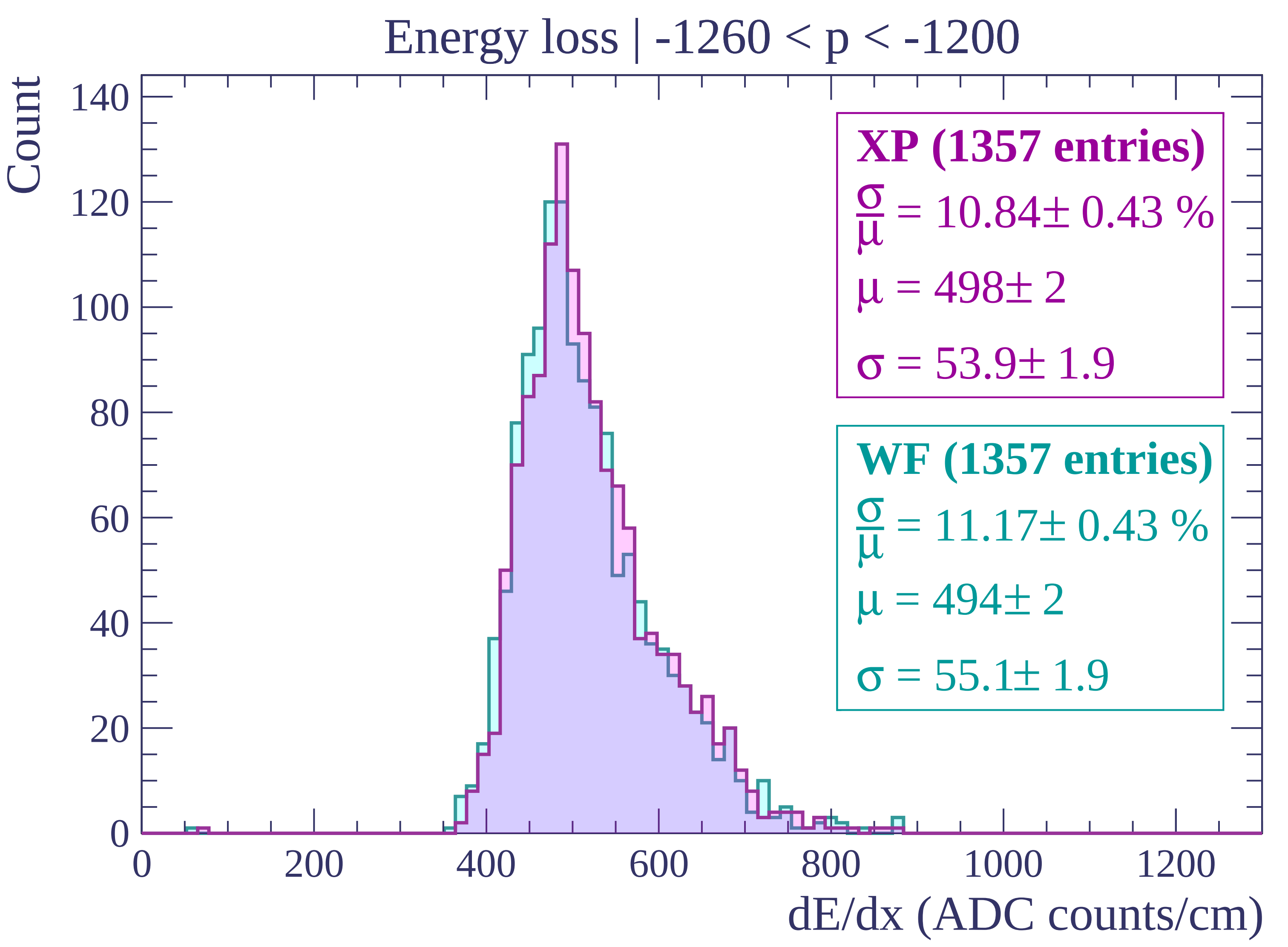

Energy loss | -1440 < p < -1380
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1244 entries)
σ/μ = 11.25± 0.43 %
μ = 514± 2
σ = 57.8± 1.9
WF (1244 entries)
σ/μ = 11.51± 0.43 %
μ = 510± 2
σ = 58.7± 1.9

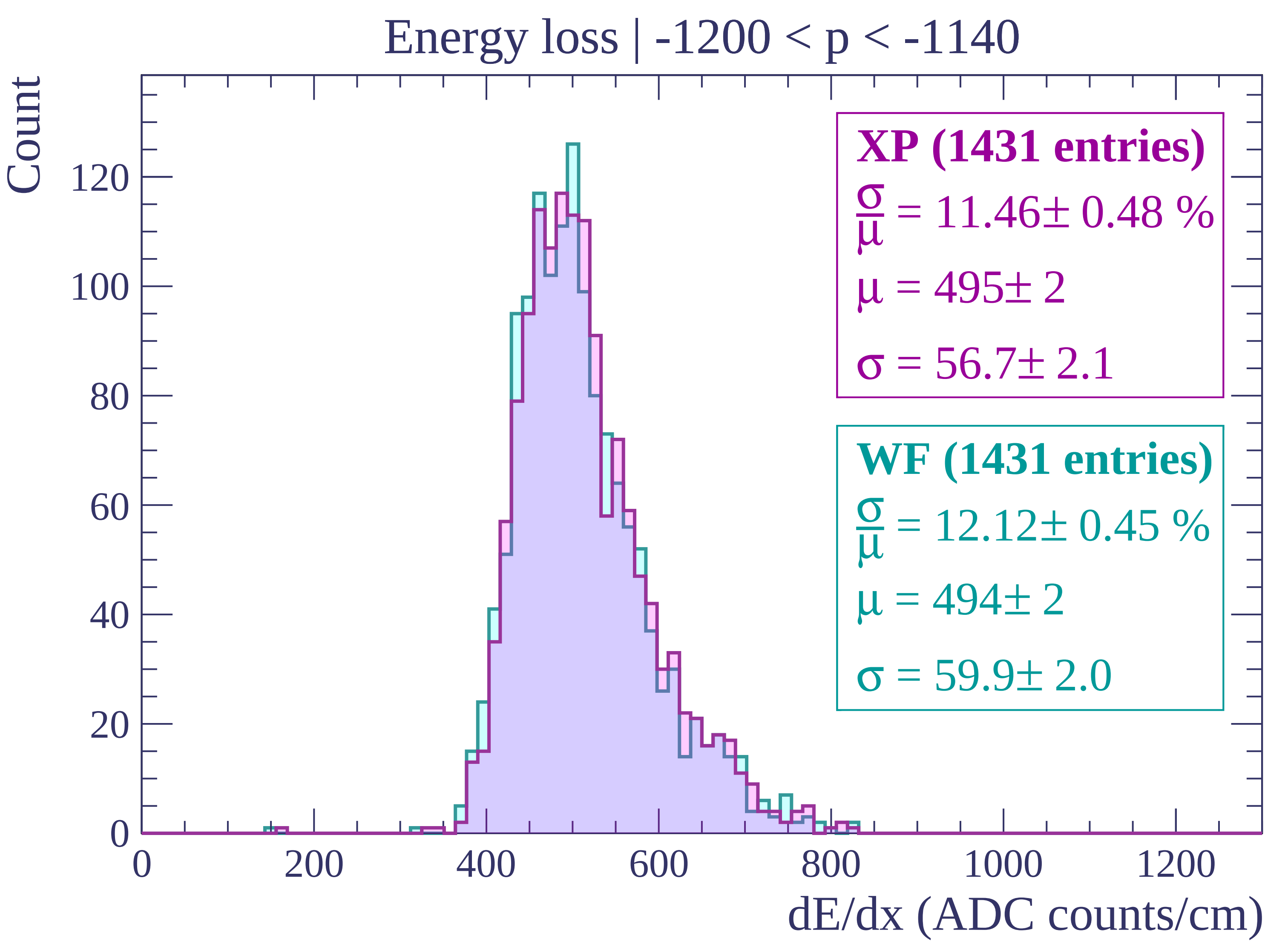


Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1250 entries)
σ/μ = 11.87± 0.45 %
μ = 509± 2
σ = 60.5± 2.0
WF (1250 entries)
σ/μ = 12.32± 0.50 %
μ = 505± 2
σ = 62.2± 2.3

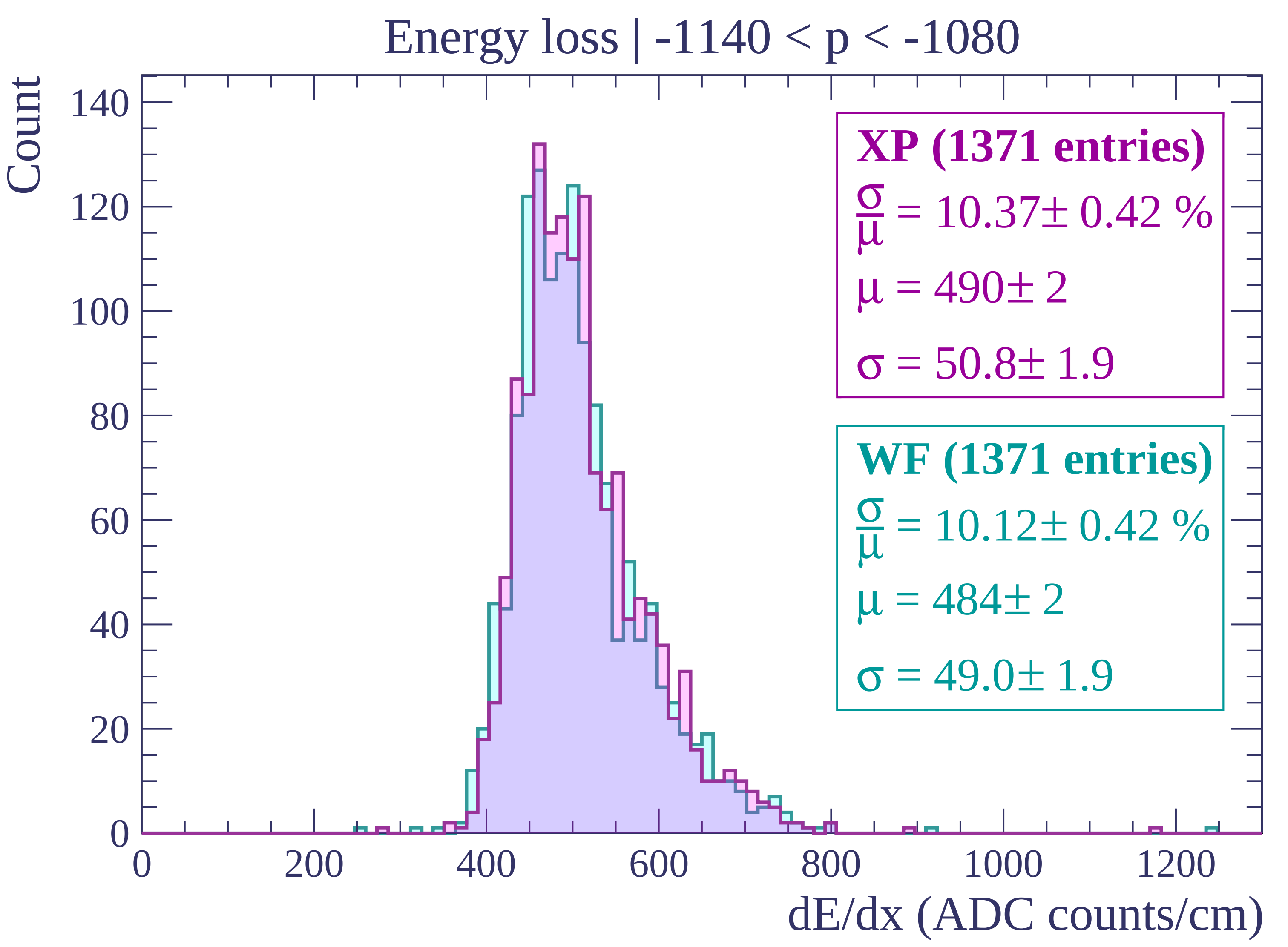


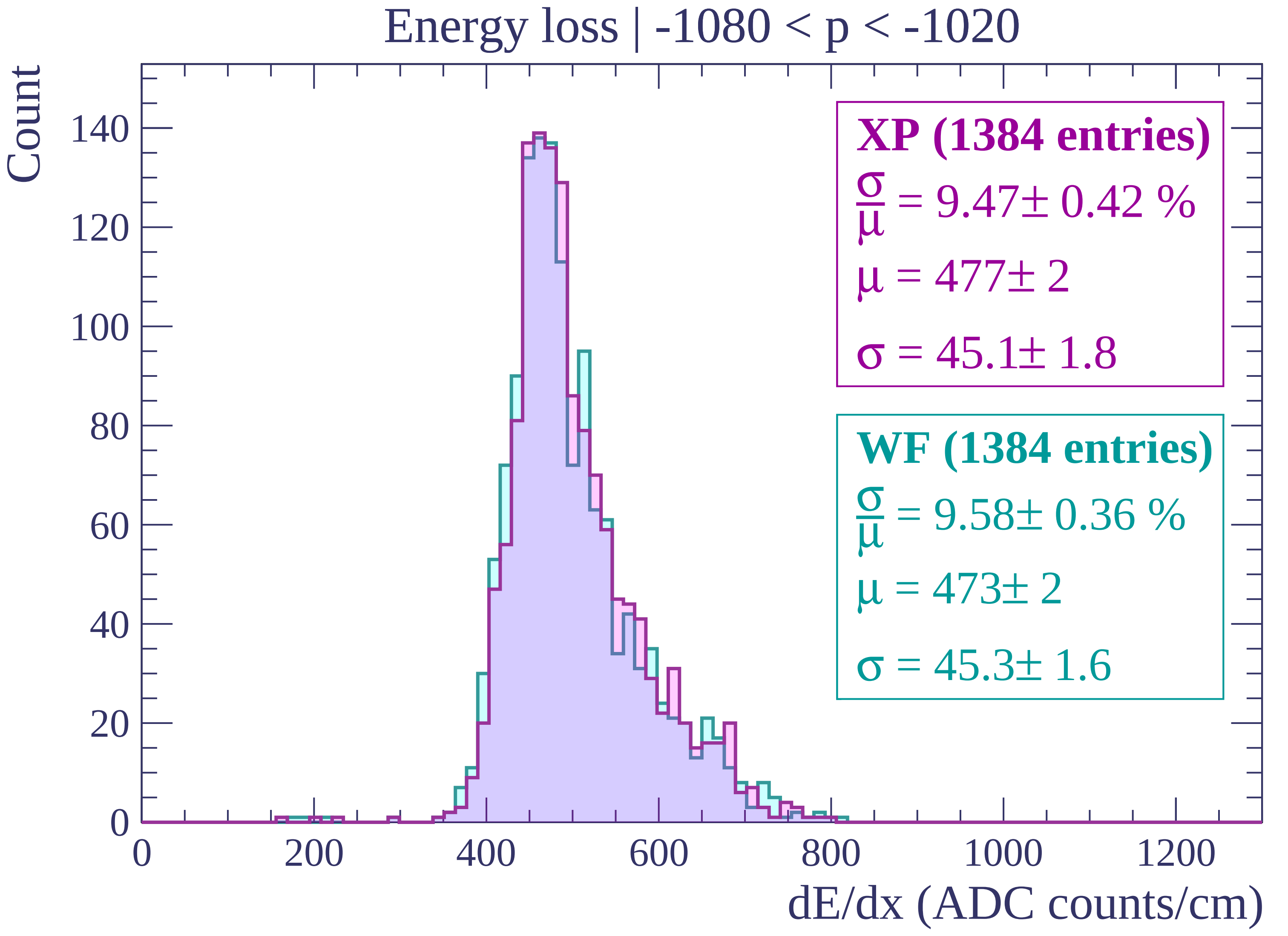
Energy loss | -1320 < p < -1260
Count
XP (1310 entries)
$\frac{\sigma}{\mu}$ = 11.29± 0.40 %
μ = 505± 2
σ = 57.0± 1.8
WF (1310 entries)
$\frac{\sigma}{\mu}$ = 11.22± 0.41 %
μ = 498± 2
σ = 55.9± 1.8
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)


Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
XP (1357 entries)
$\frac{\sigma}{\mu}$ = 10.84± 0.43 %
μ = 498± 2
σ = 53.9± 1.9
WF (1357 entries)
$\frac{\sigma}{\mu}$ = 11.17± 0.43 %
μ = 494± 2
σ = 55.1± 1.9
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

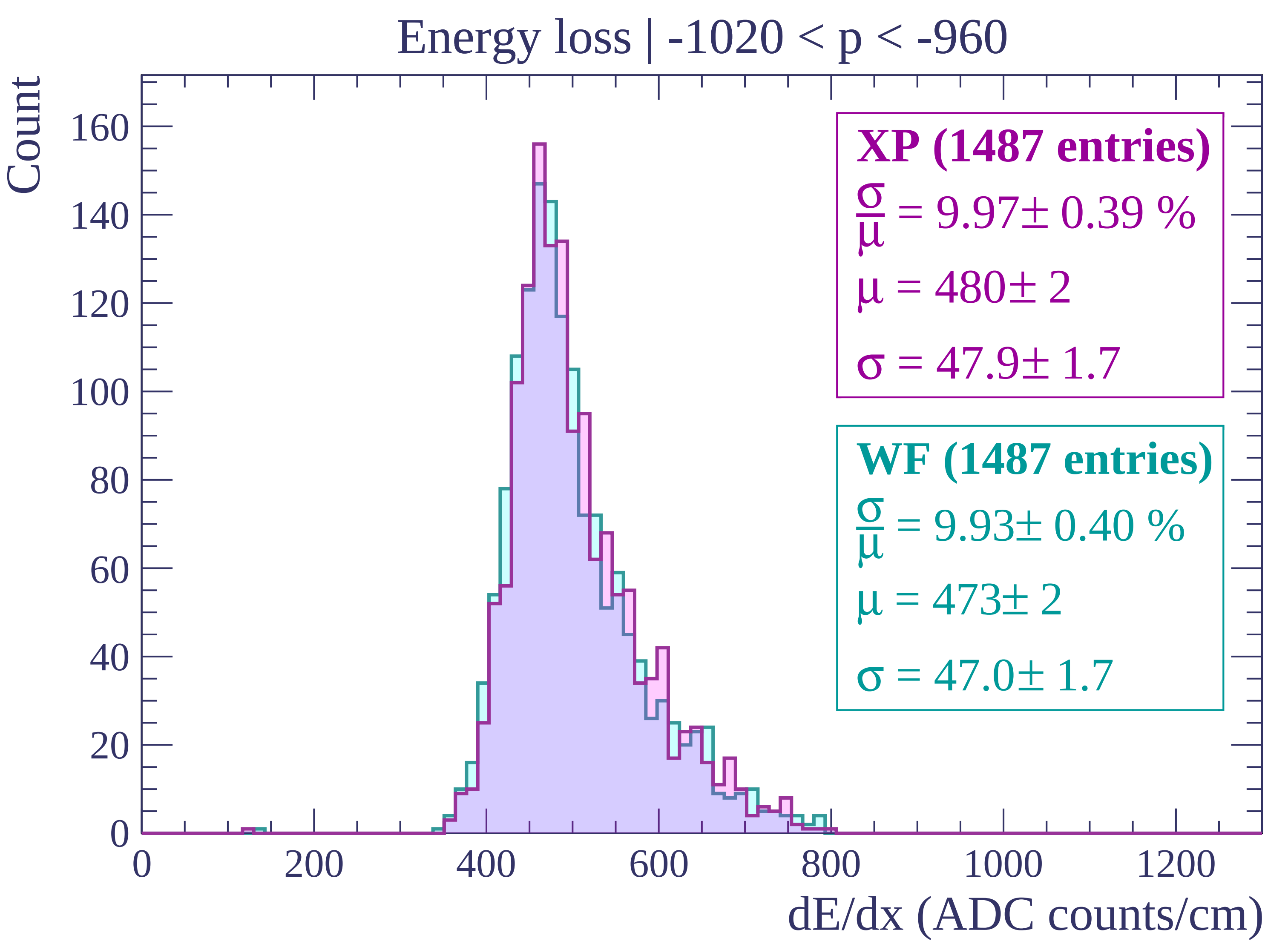


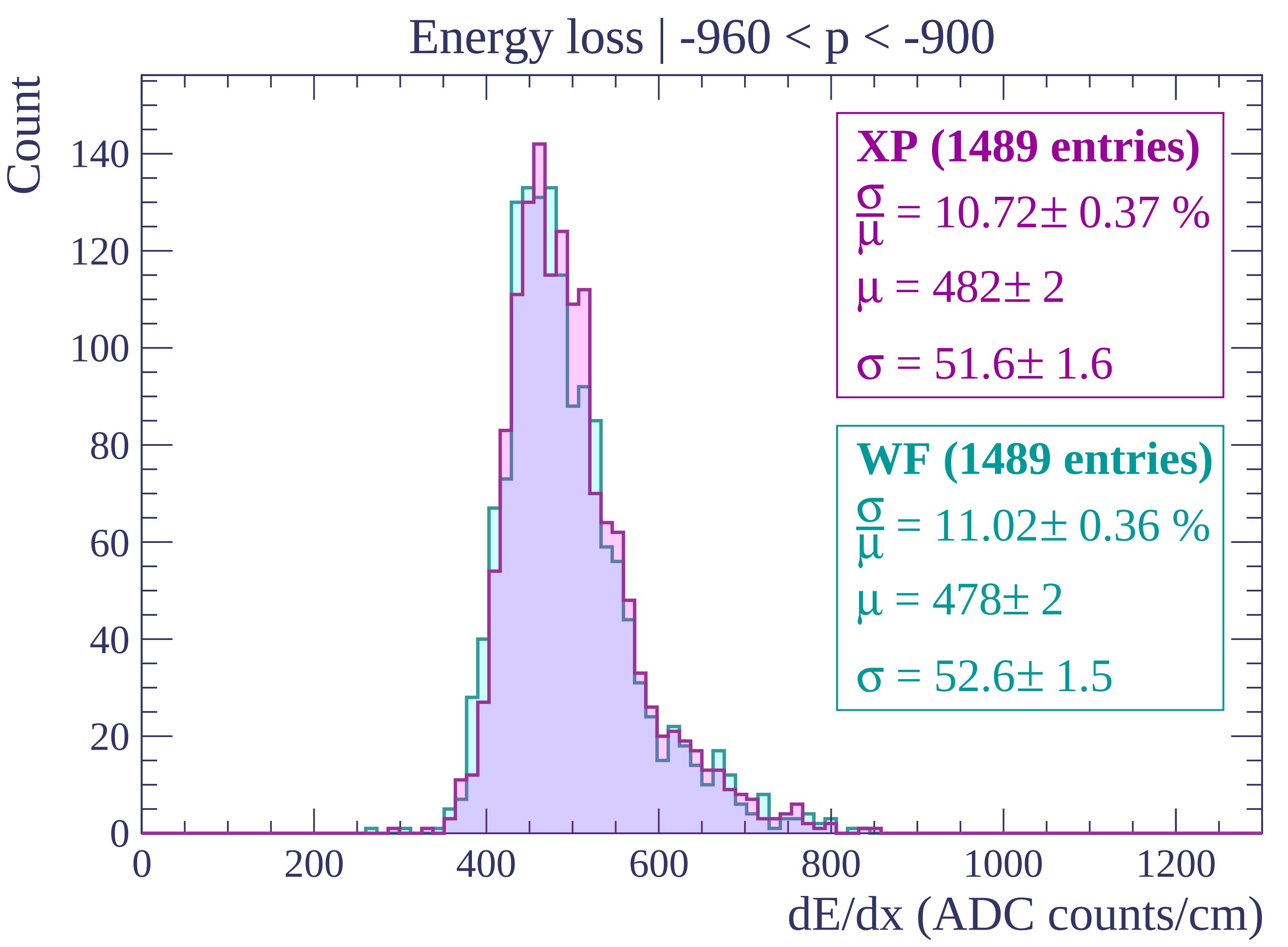
Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1431 entries)
σ/μ = 11.46± 0.48 %
μ = 495± 2
σ = 56.7± 2.1
WF (1431 entries)
σ/μ = 12.12± 0.45 %
μ = 494± 2
σ = 59.9± 2.0


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1371 entries)
σ/μ = 10.37± 0.42 %
μ = 490± 2
σ = 50.8± 1.9
WF (1371 entries)
σ/μ = 10.12± 0.42 %
μ = 484± 2
σ = 49.0± 1.9

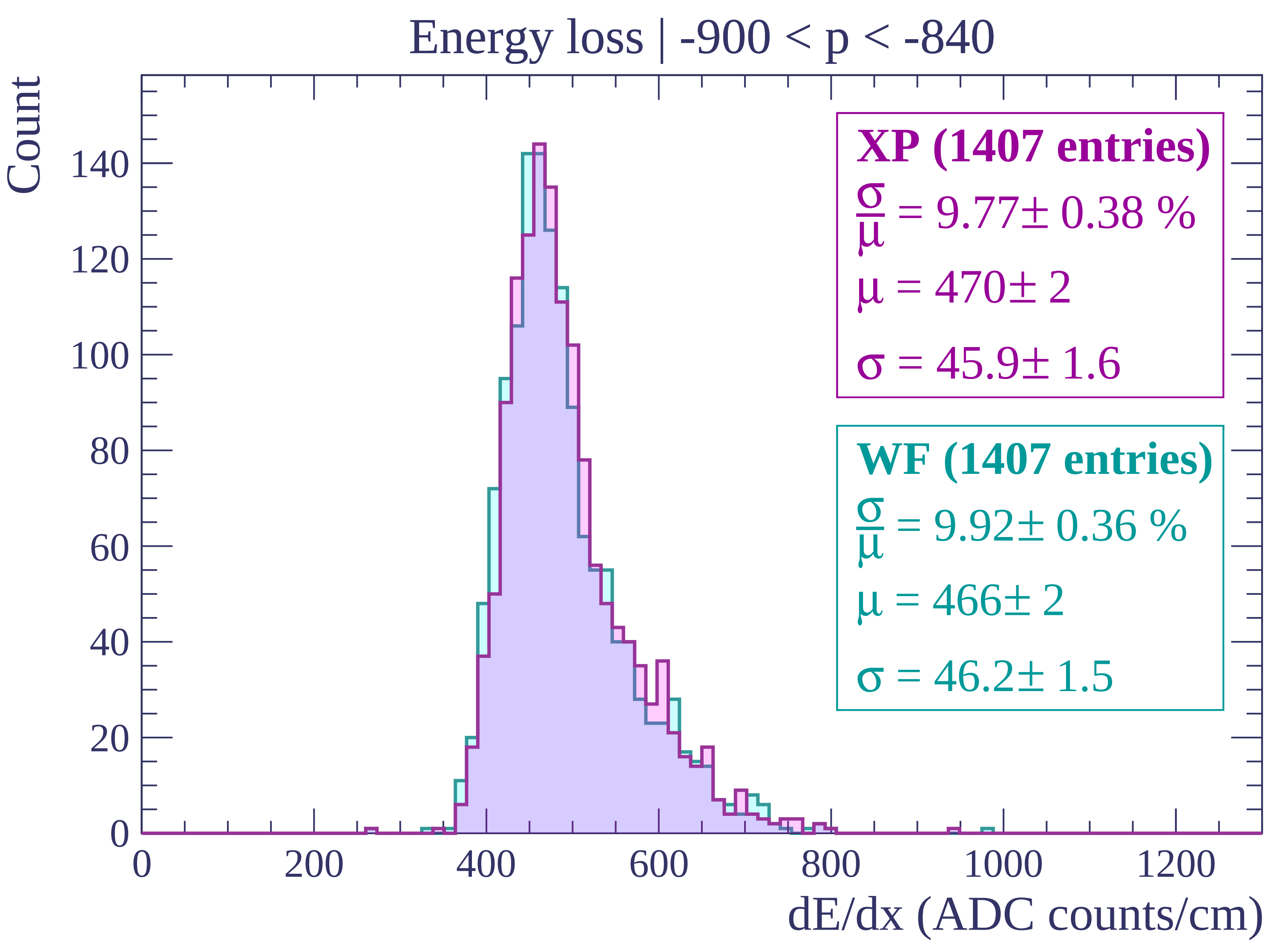


Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
XP (1384 entries)
σ/μ = 9.47± 0.42 %
μ = 477± 2
σ = 45.1± 1.8
WF (1384 entries)
σ/μ = 9.58± 0.36 %
μ = 473± 2
σ = 45.3± 1.6

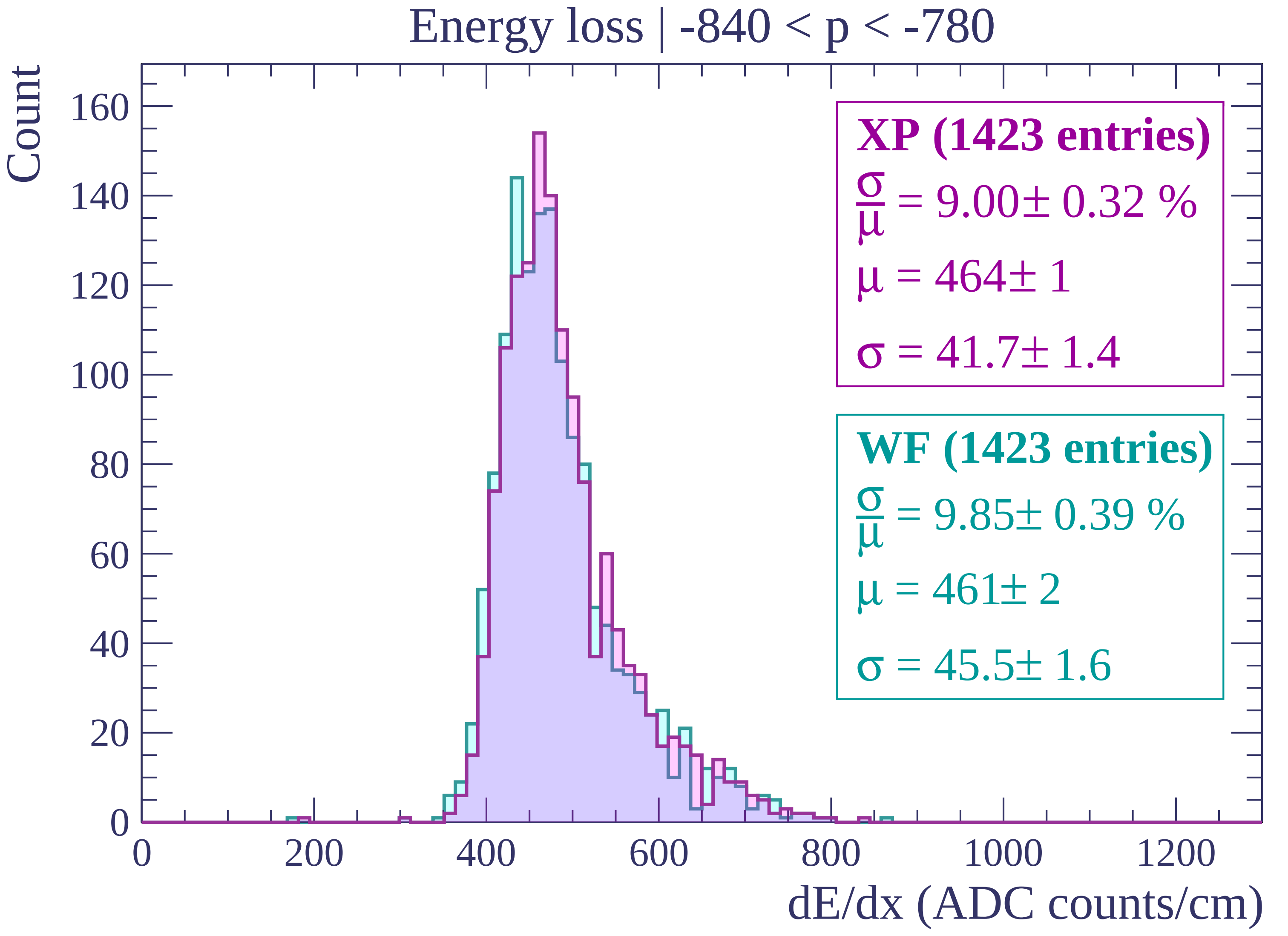

Energy loss | -1020 < p < -960
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1487 entries)
σ/μ = 9.97± 0.39 %
μ = 480± 2
σ = 47.9± 1.7
WF (1487 entries)
σ/μ = 9.93± 0.40 %
μ = 473± 2
σ = 47.0± 1.7



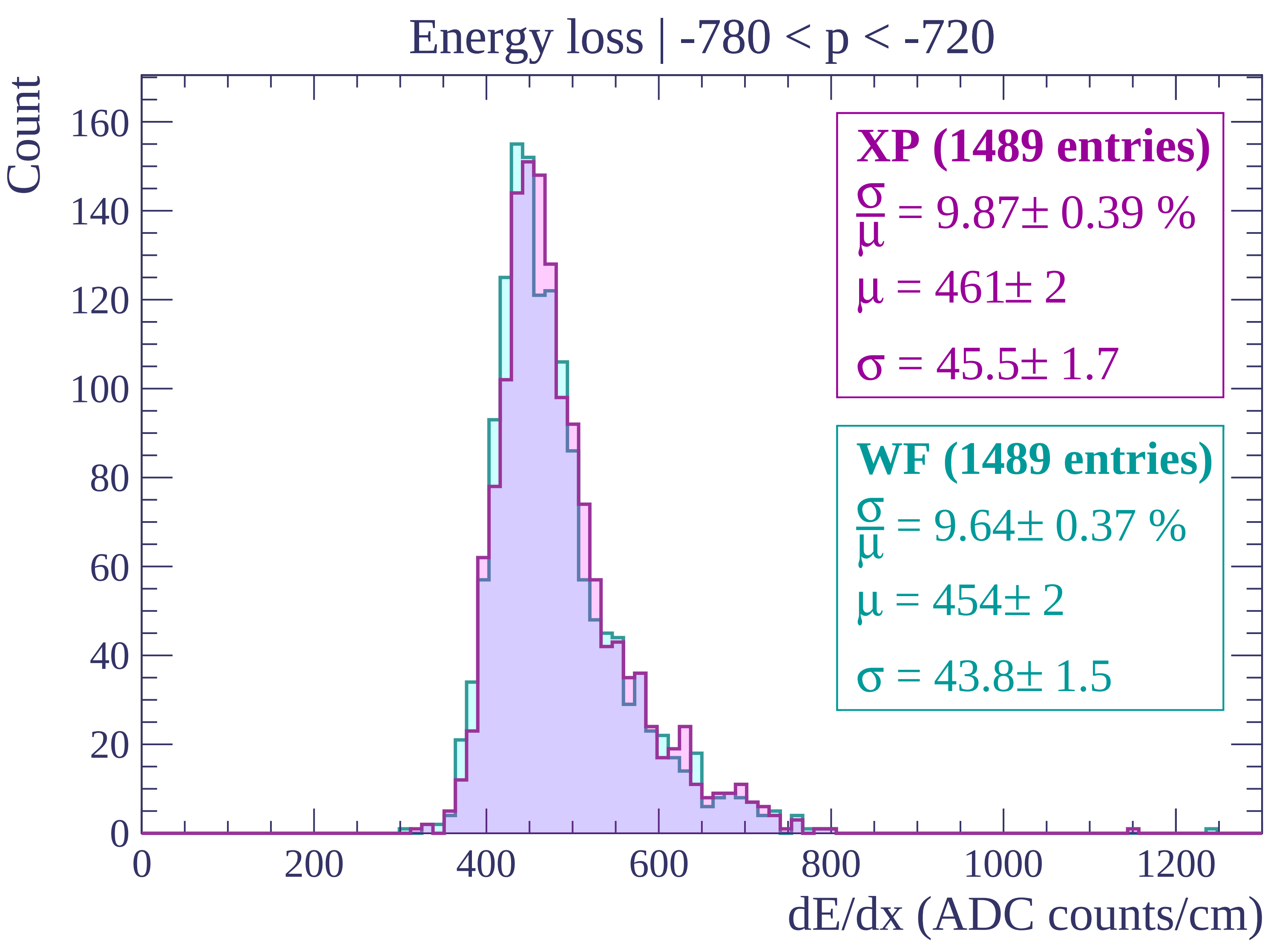
Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1489 entries)
σ/μ = 10.72± 0.37 %
μ = 482± 2
σ = 51.6± 1.6
WF (1489 entries)
σ/μ = 11.02± 0.36 %
μ = 478± 2
σ = 52.6± 1.5

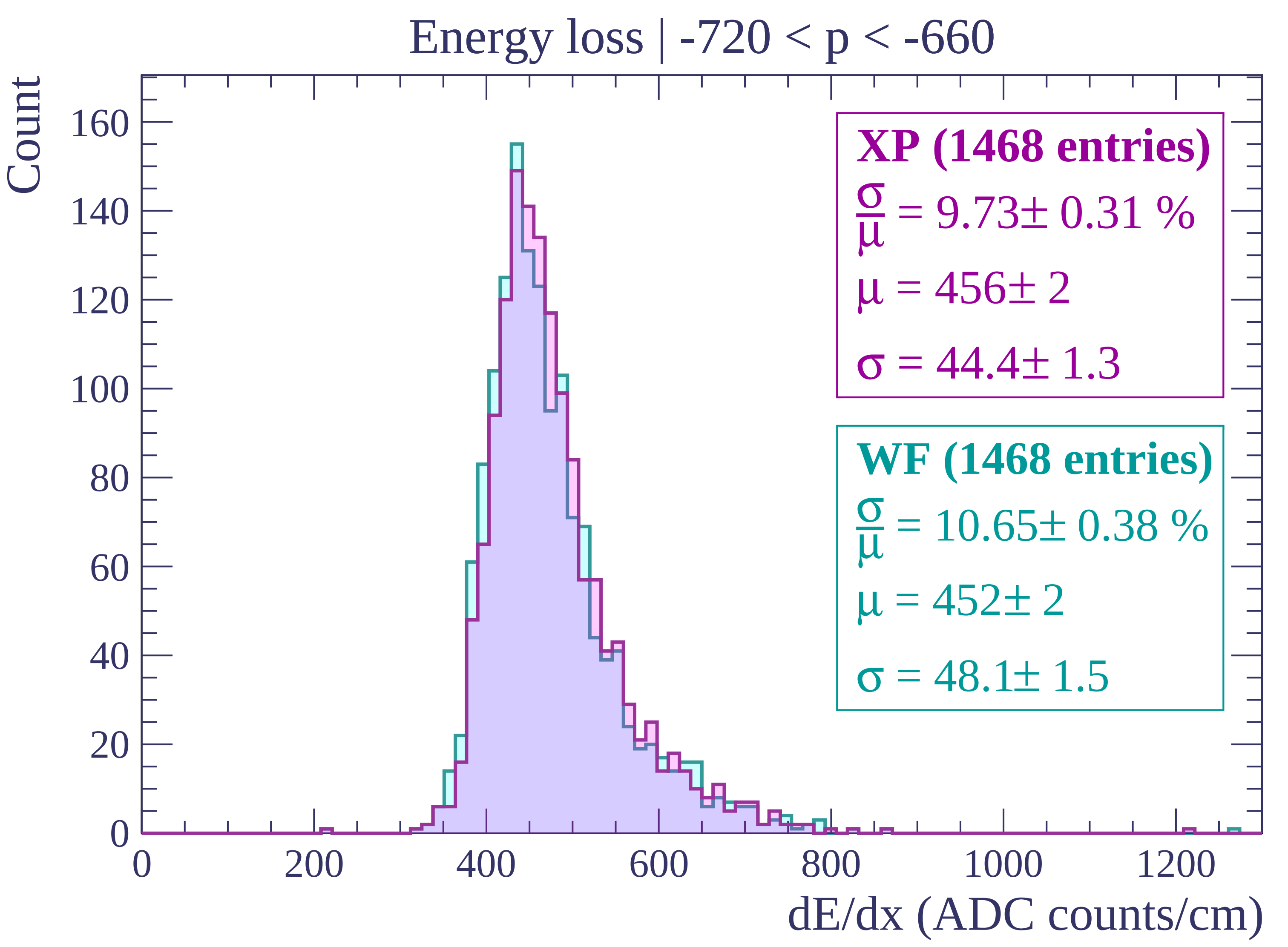


Energy loss | -900 < p < -840
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1407 entries)
$\frac{\sigma}{\mu}$ = 9.77± 0.38 %
μ = 470± 2
σ = 45.9± 1.6
WF (1407 entries)
$\frac{\sigma}{\mu}$ = 9.92± 0.36 %
μ = 466± 2
σ = 46.2± 1.5

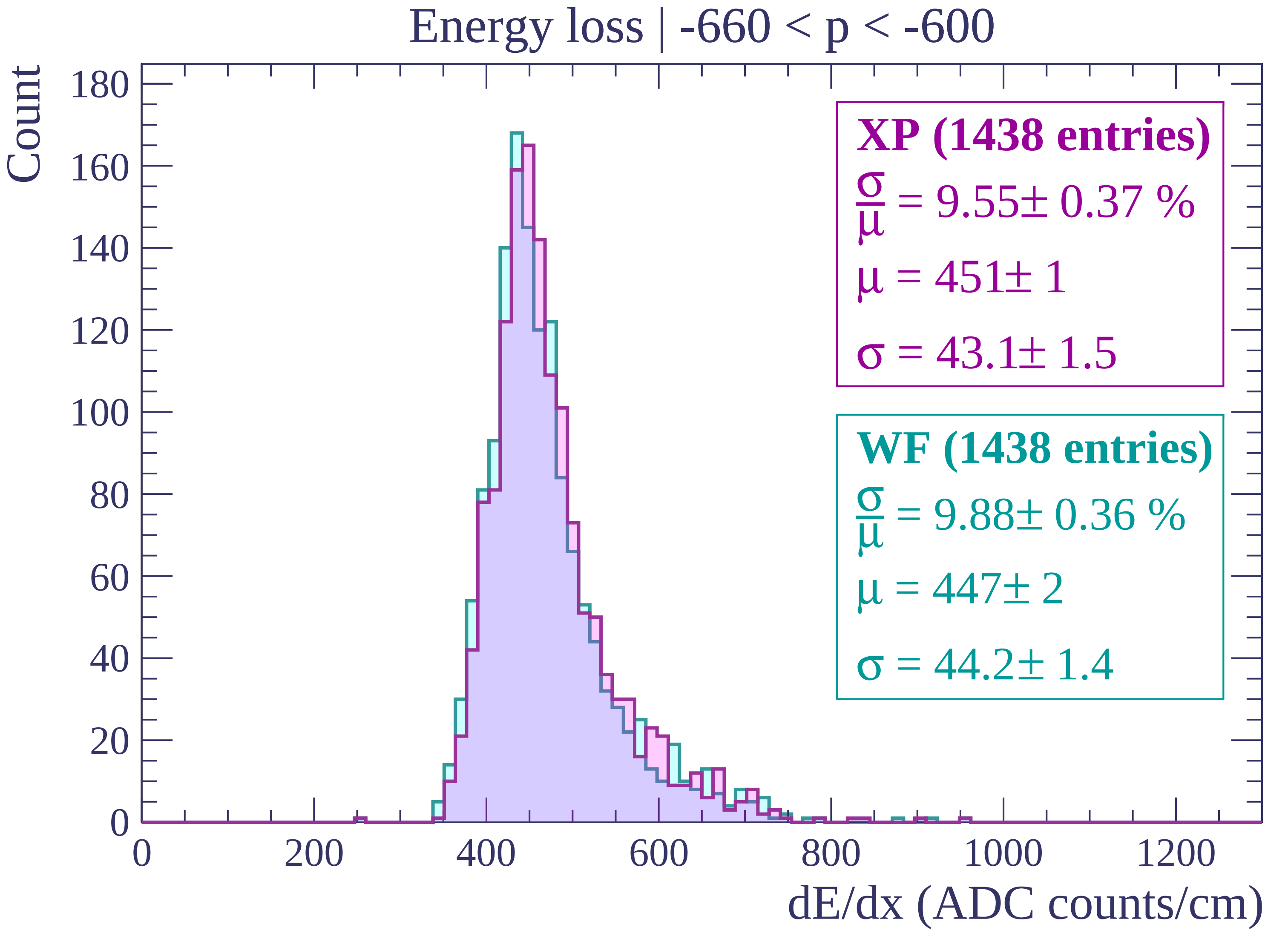


Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
XP (1423 entries)
σ/μ = 9.00± 0.32 %
μ = 464± 1
σ = 41.7± 1.4
WF (1423 entries)
σ/μ = 9.85± 0.39 %
μ = 461± 2
σ = 45.5± 1.6
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200

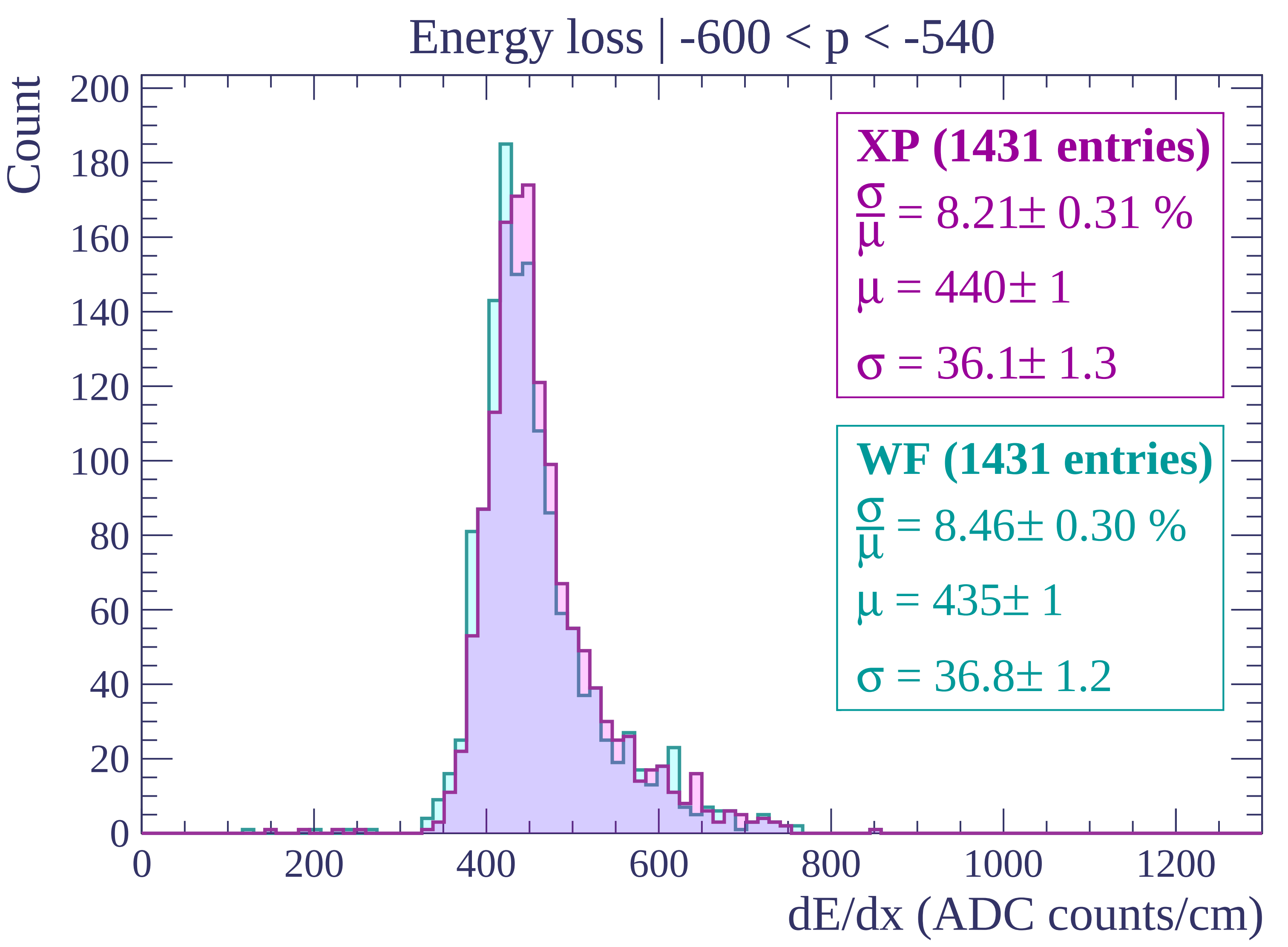

Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
XP (1489 entries)
$\frac{\sigma}{\mu}$ = 9.87± 0.39 %
μ = 461± 2
σ = 45.5± 1.7
WF (1489 entries)
$\frac{\sigma}{\mu}$ = 9.64± 0.37 %
μ = 454± 2
σ = 43.8± 1.5

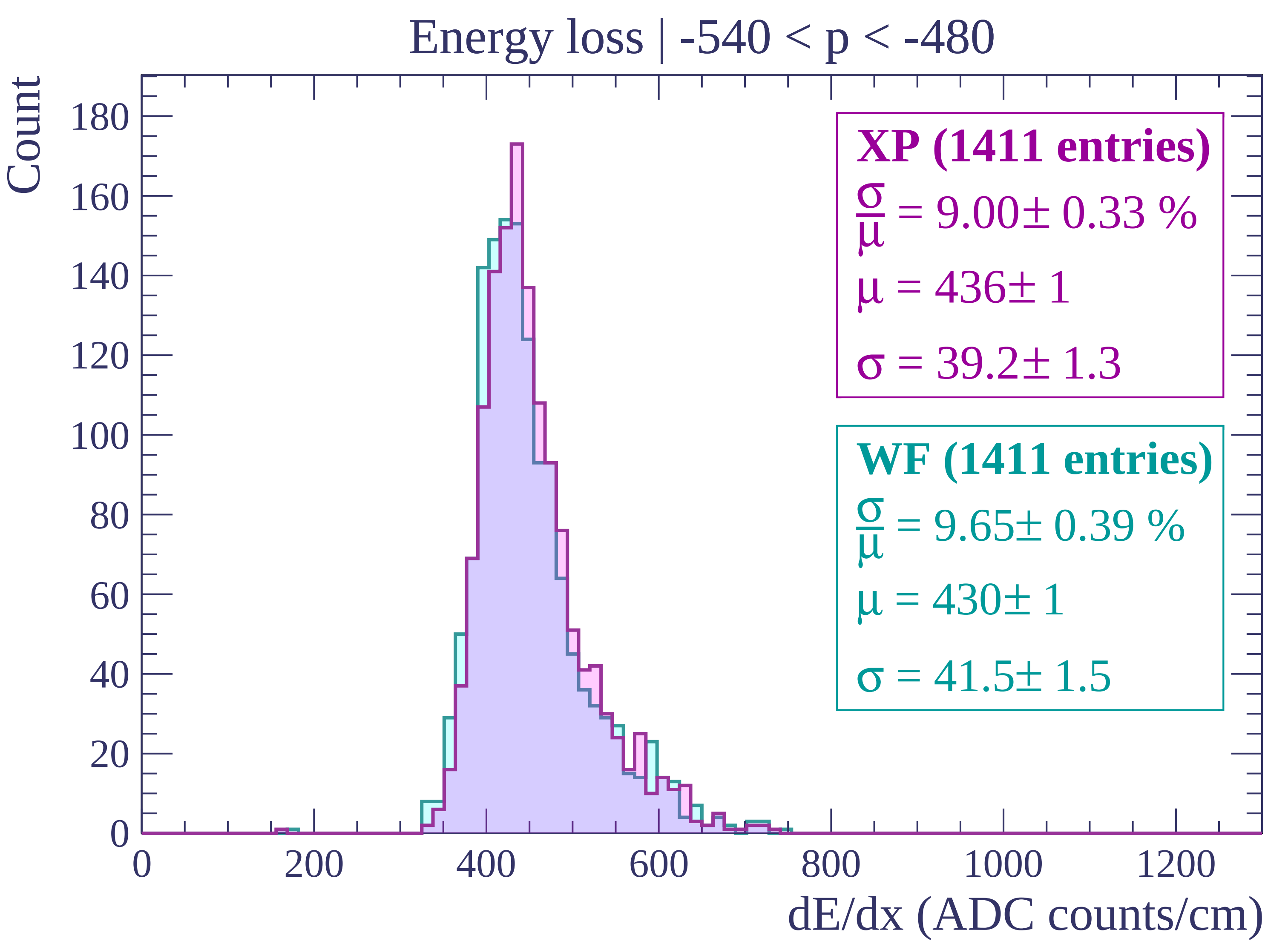


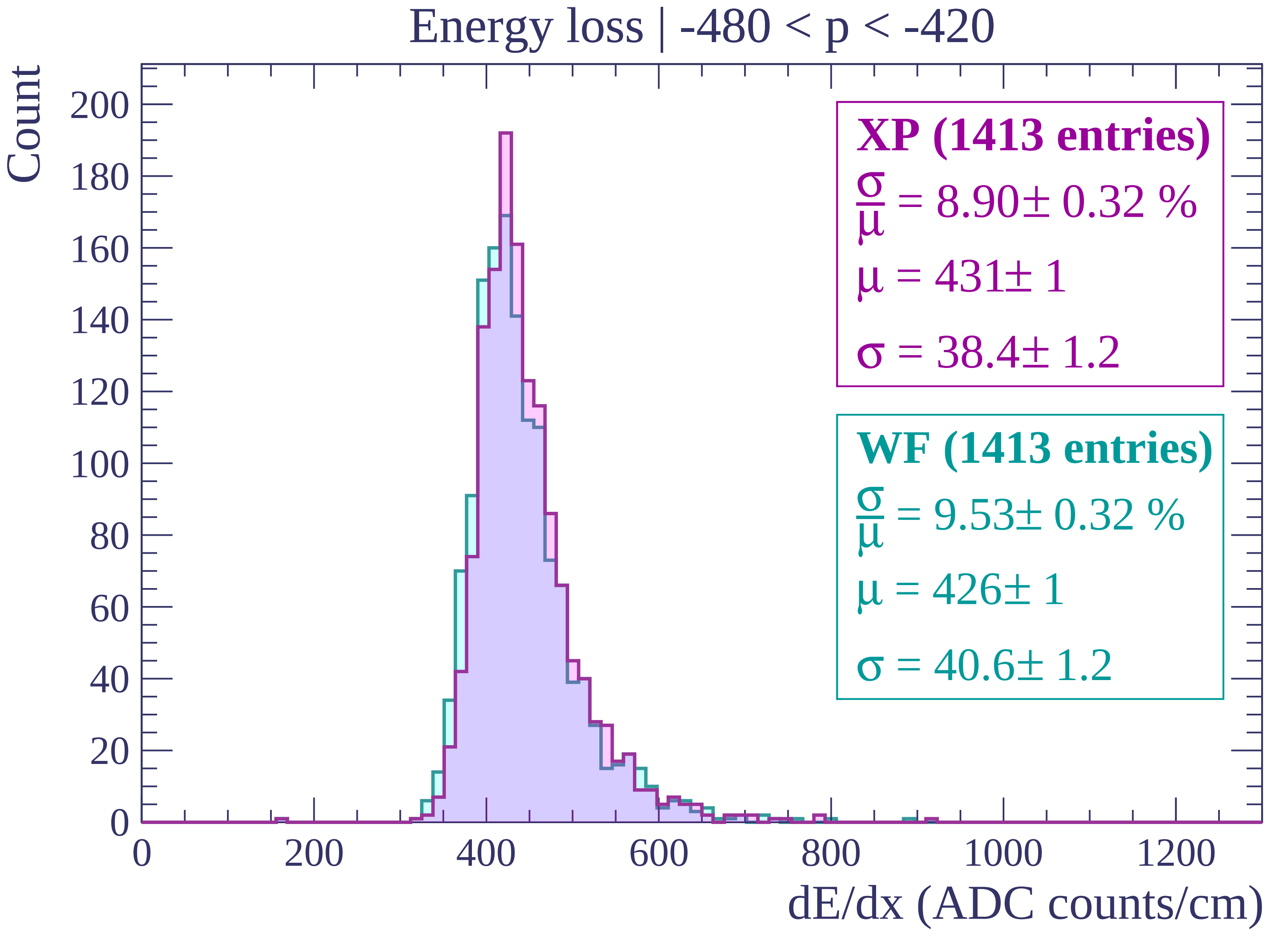
Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
XP (1468 entries)
$\frac{\sigma}{\mu}$ = 9.73± 0.31 %
μ = 456± 2
σ = 44.4± 1.3
WF (1468 entries)
$\frac{\sigma}{\mu}$ = 10.65± 0.38 %
μ = 452± 2
σ = 48.1± 1.5

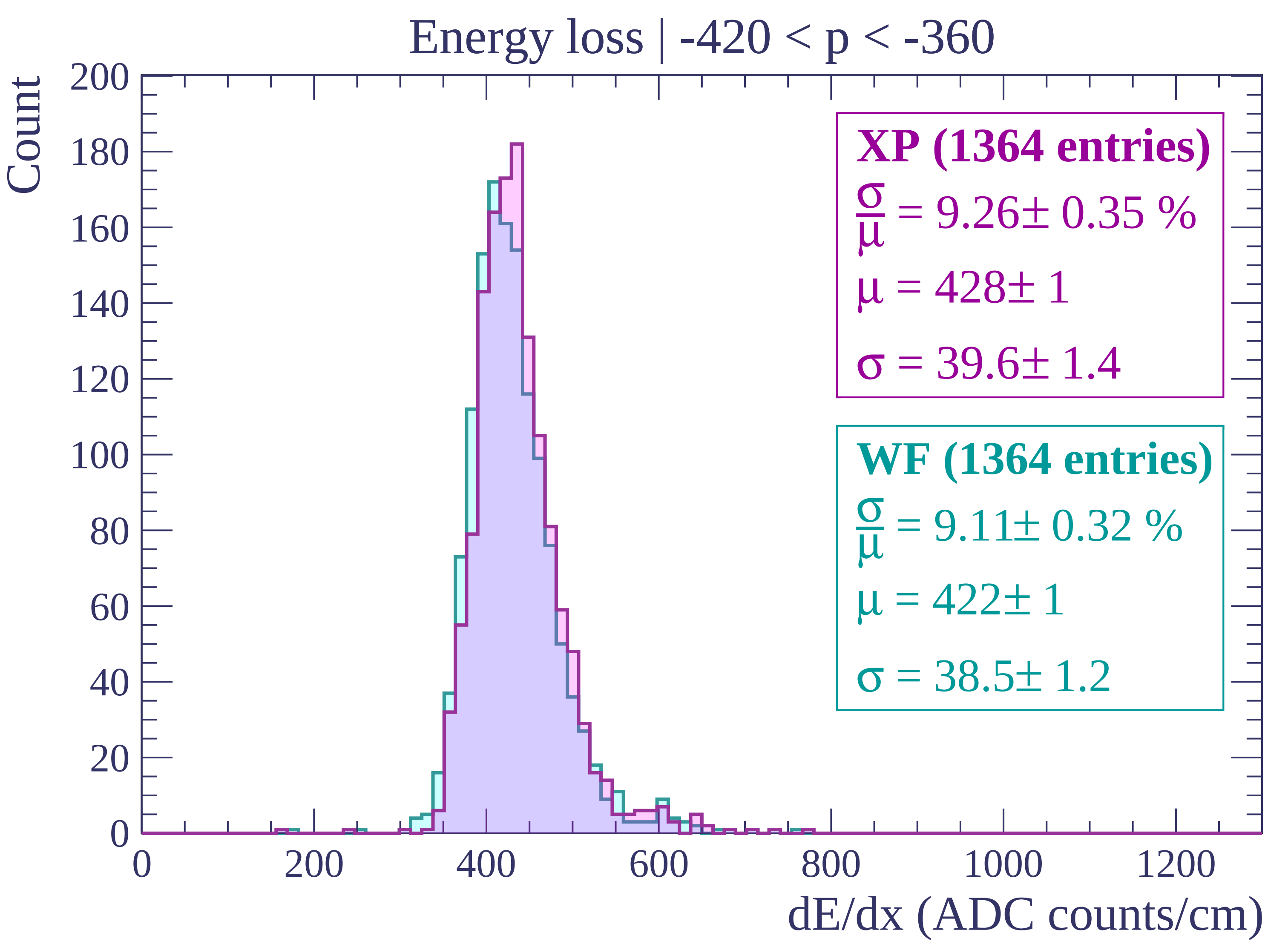


Energy loss | -660 < p < -600
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1438 entries)
σ/μ = 9.55± 0.37 %
μ = 451± 1
σ = 43.1± 1.5
WF (1438 entries)
σ/μ = 9.88± 0.36 %
μ = 447± 2
σ = 44.2± 1.4

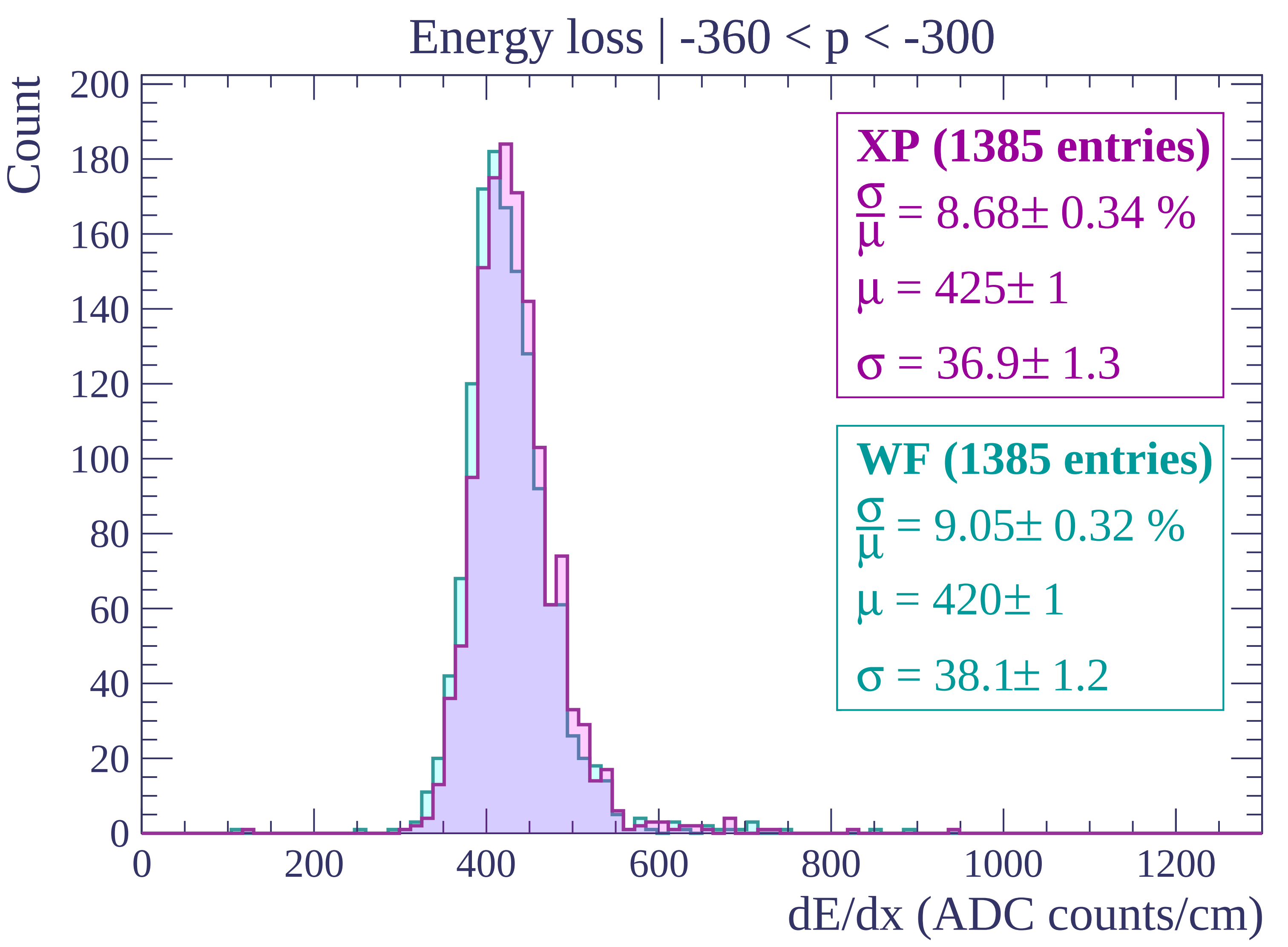

Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
XP (1431 entries)
$\frac{\sigma}{\mu}$ = 8.21± 0.31 %
μ = 440± 1
σ = 36.1± 1.3
WF (1431 entries)
$\frac{\sigma}{\mu}$ = 8.46± 0.30 %
μ = 435± 1
σ = 36.8± 1.2
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200


Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
180
XP (1411 entries)
$\frac{\sigma}{\mu}$ = 9.00± 0.33 %
μ = 436± 1
σ = 39.2± 1.3
WF (1411 entries)
$\frac{\sigma}{\mu}$ = 9.65± 0.39 %
μ = 430± 1
σ = 41.5± 1.5

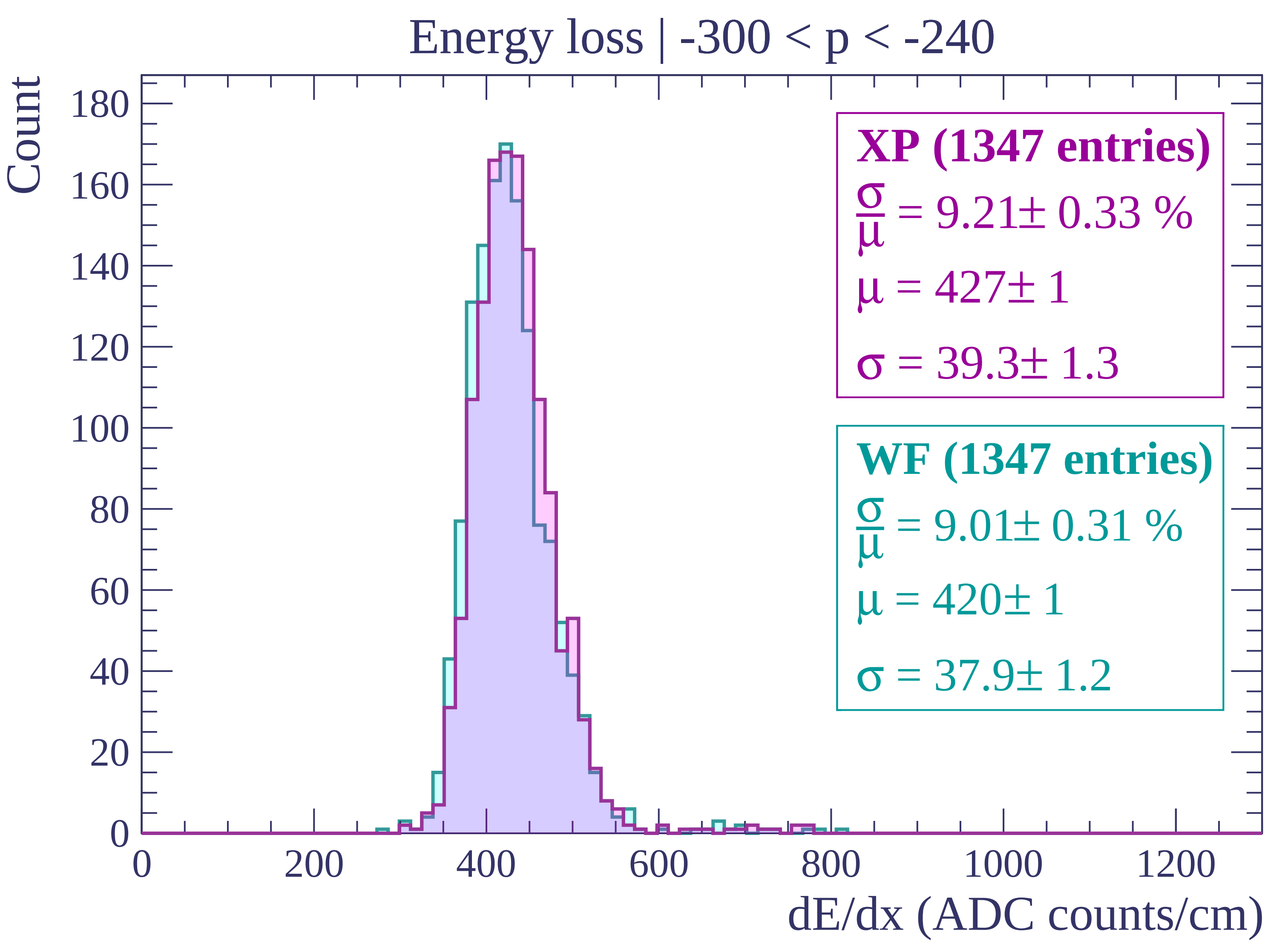

Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1413 entries)
σ/μ = 8.90± 0.32 %
μ = 431± 1
σ = 38.4± 1.2
WF (1413 entries)
σ/μ = 9.53± 0.32 %
μ = 426± 1
σ = 40.6± 1.2



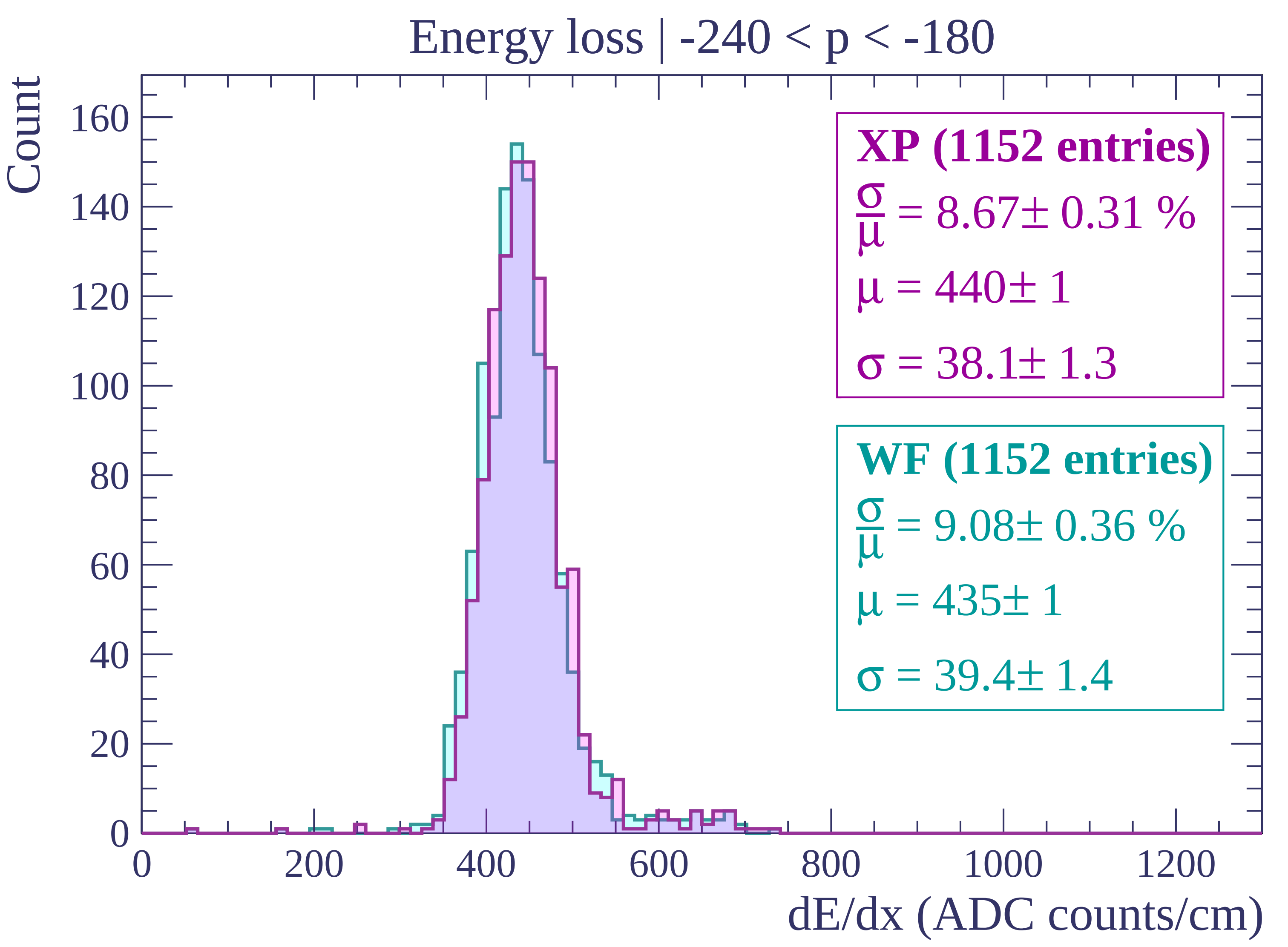
Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
XP (1364 entries)
$\frac{\sigma}{\mu}$ = 9.26± 0.35 %
μ = 428± 1
σ = 39.6± 1.4
WF (1364 entries)
$\frac{\sigma}{\mu}$ = 9.11± 0.32 %
μ = 422± 1
σ = 38.5± 1.2



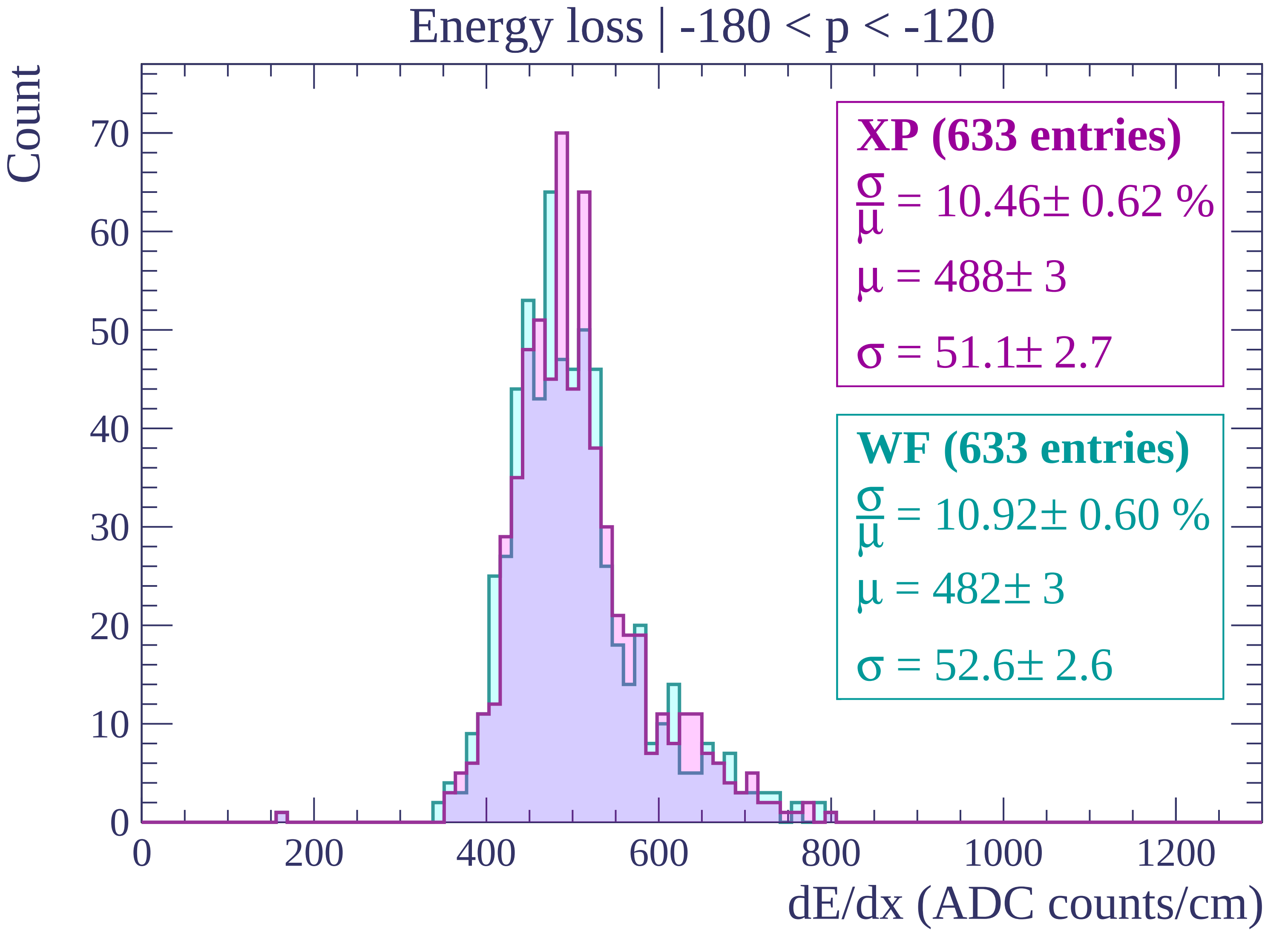
Energy loss | -360 < p < -300
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1385 entries)
σ/μ = 8.68± 0.34 %
μ = 425± 1
σ = 36.9± 1.3
WF (1385 entries)
σ/μ = 9.05± 0.32 %
μ = 420± 1
σ = 38.1± 1.2



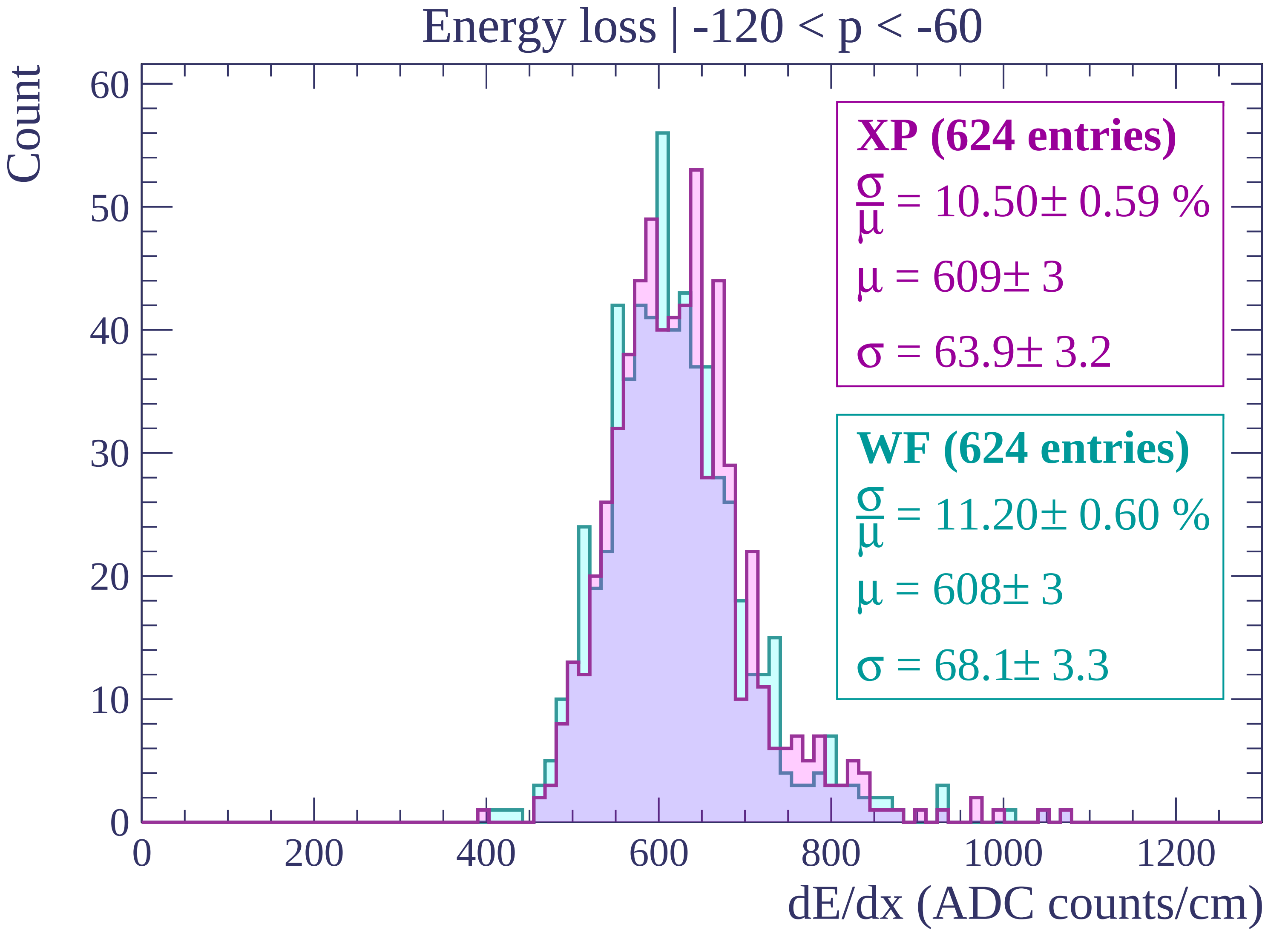
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1347 entries)
σ/μ = 9.21± 0.33 %
μ = 427± 1
σ = 39.3± 1.3
WF (1347 entries)
σ/μ = 9.01± 0.31 %
μ = 420± 1
σ = 37.9± 1.2



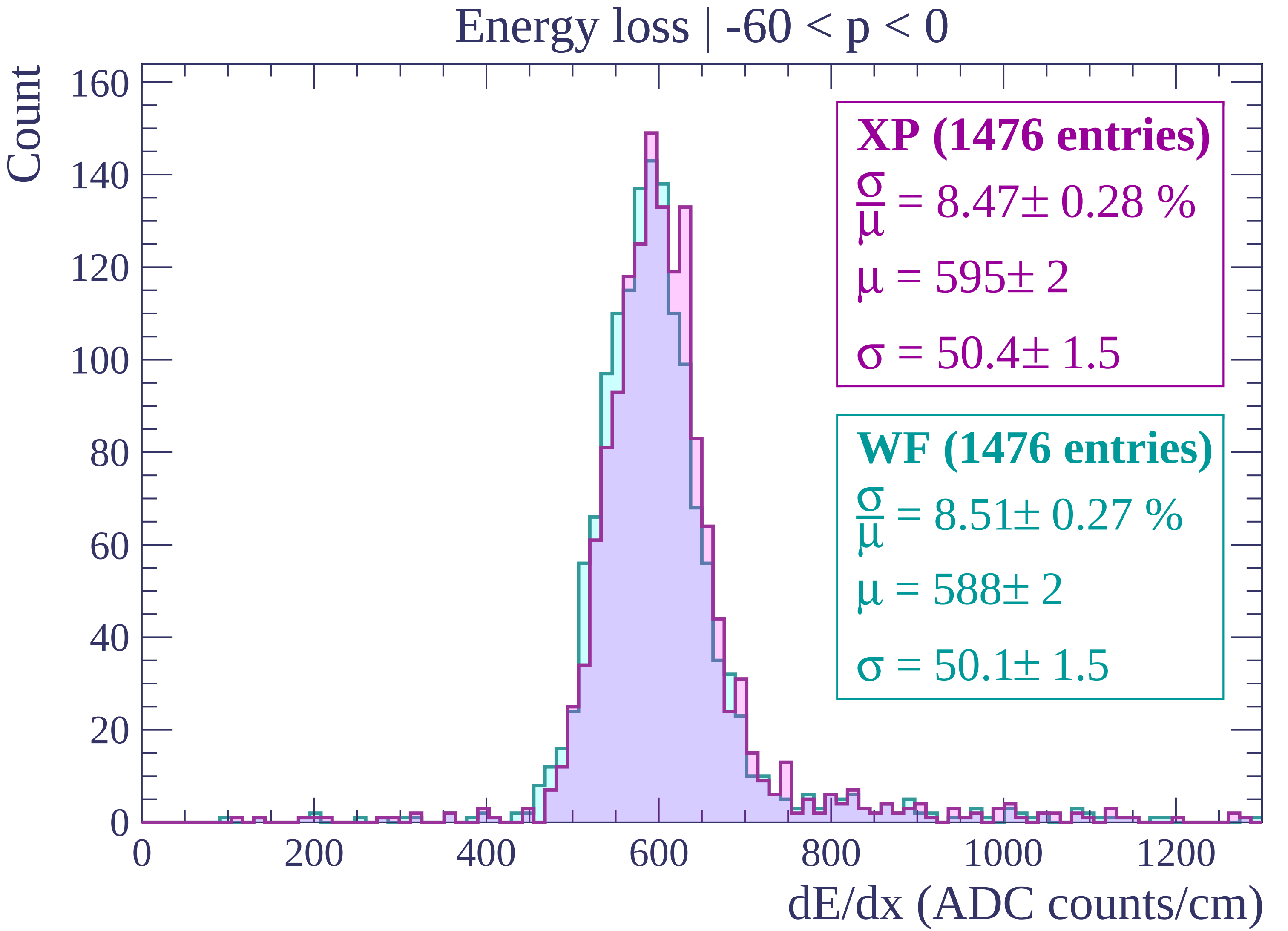
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
XP (1152 entries)
$\frac{\sigma}{\mu}$ = 8.67± 0.31 %
μ = 440± 1
σ = 38.1± 1.3
WF (1152 entries)
$\frac{\sigma}{\mu}$ = 9.08± 0.36 %
μ = 435± 1
σ = 39.4± 1.4



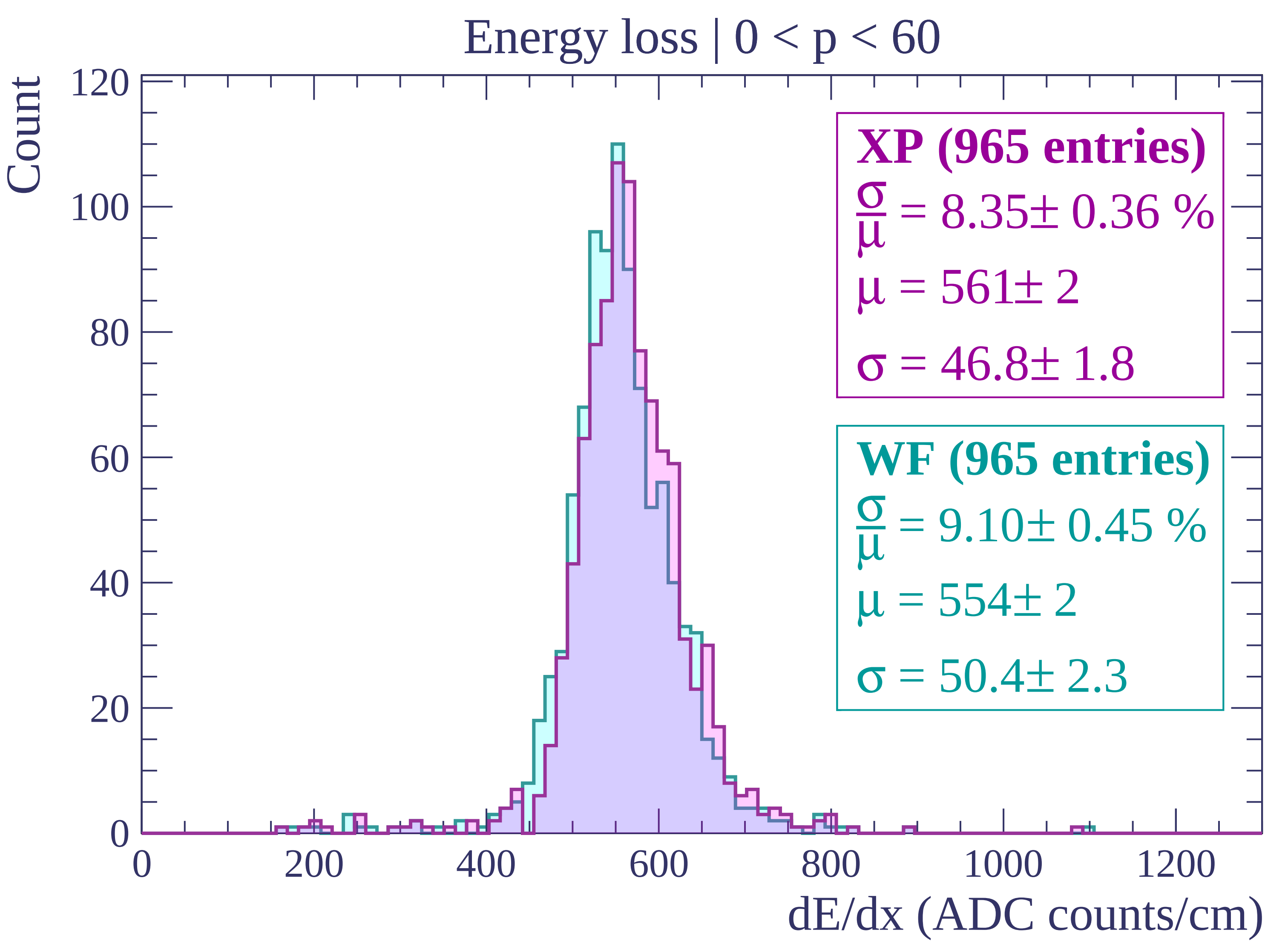
Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (633 entries)
$\frac{\sigma}{\mu}$ = 10.46± 0.62 %
μ = 488± 3
σ = 51.1± 2.7
WF (633 entries)
$\frac{\sigma}{\mu}$ = 10.92± 0.60 %
μ = 482± 3
σ = 52.6± 2.6



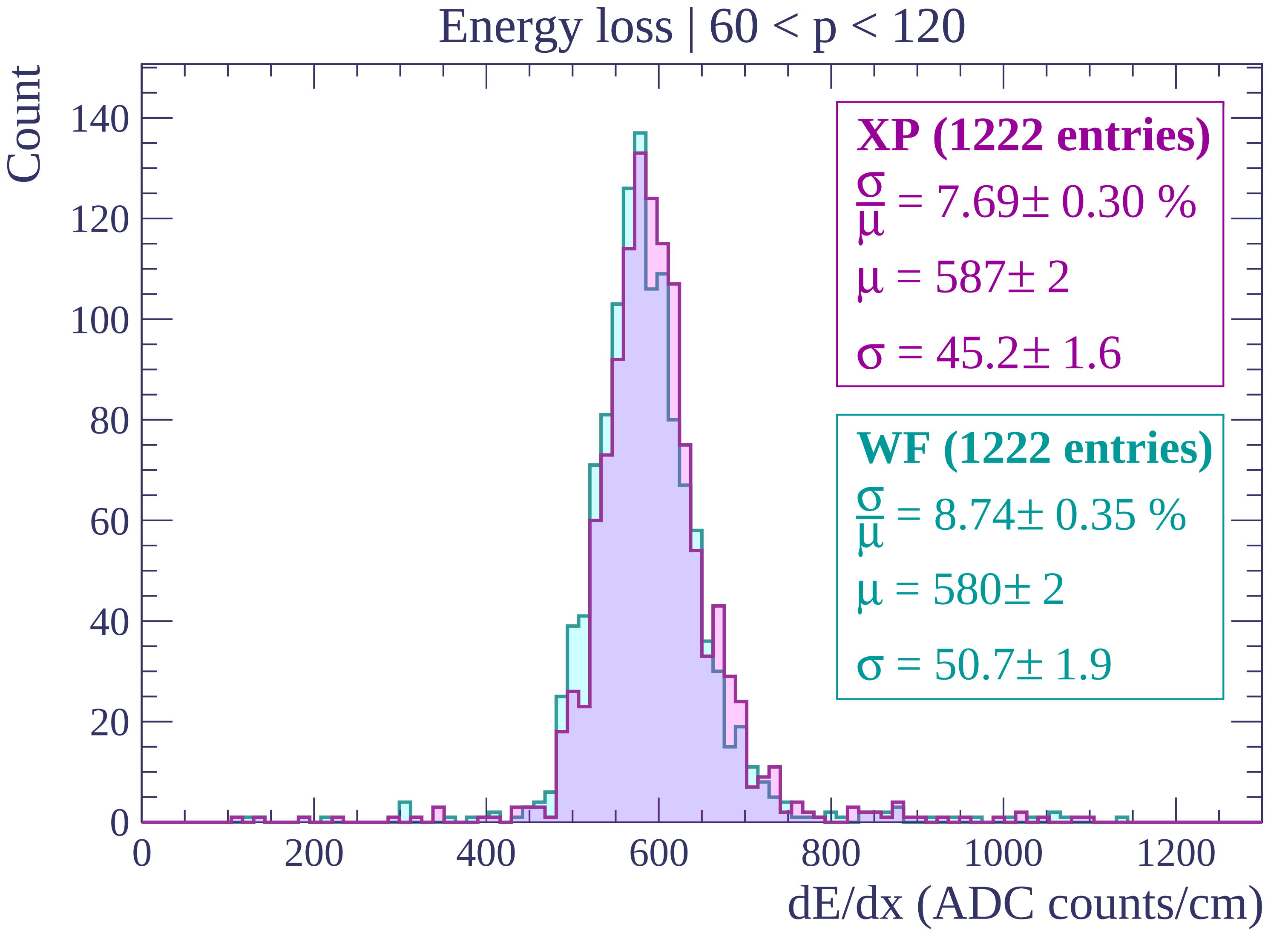
Energy loss | -120 < p < -60
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (624 entries)
$\frac{\sigma}{\mu}$ = 10.50± 0.59 %
μ = 609± 3
σ = 63.9± 3.2
WF (624 entries)
$\frac{\sigma}{\mu}$ = 11.20± 0.60 %
μ = 608± 3
σ = 68.1± 3.3



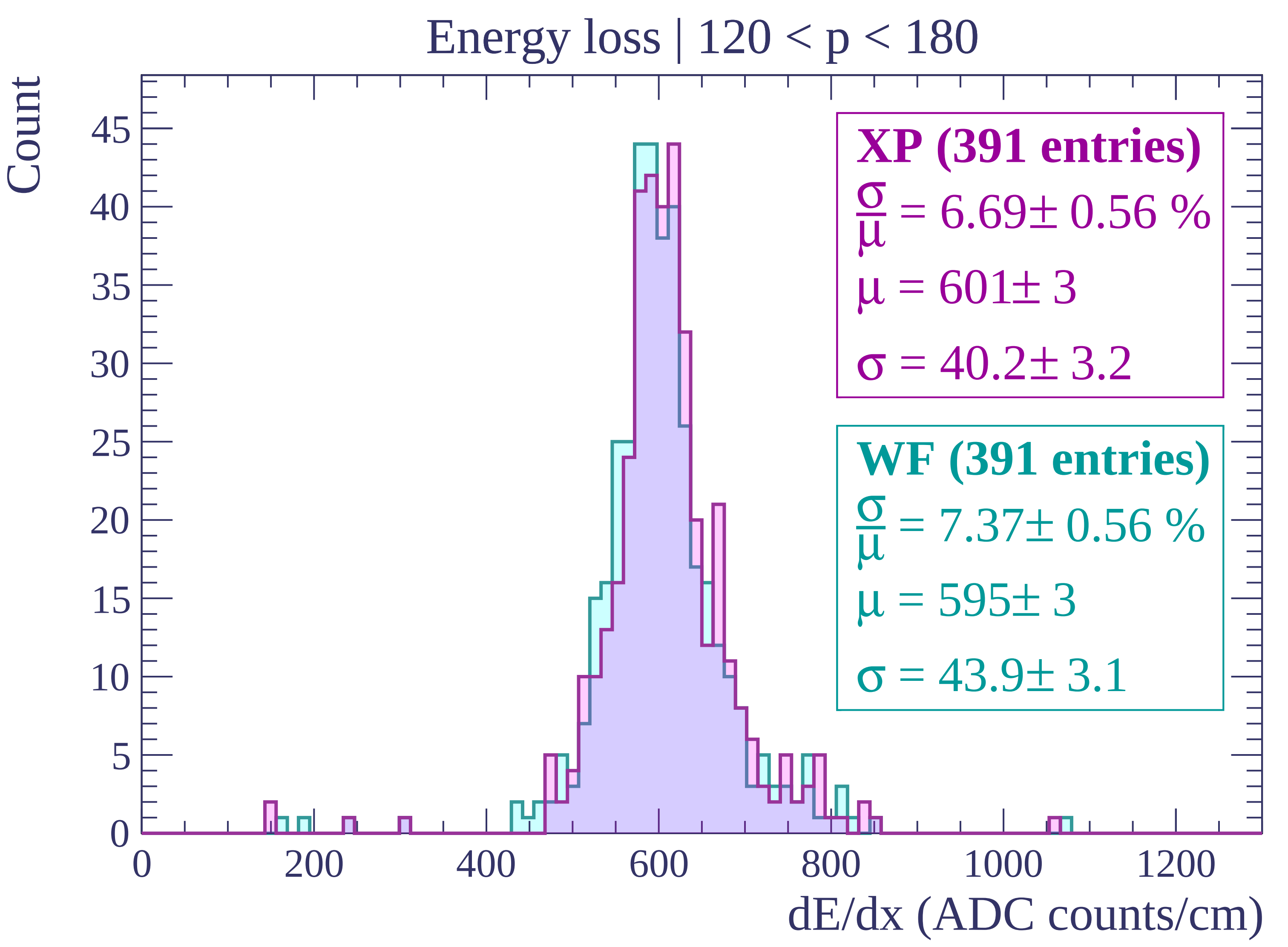
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
XP (1476 entries)
σ/μ = 8.47± 0.28 %
μ = 595± 2
σ = 50.4± 1.5
WF (1476 entries)
σ/μ = 8.51± 0.27 %
μ = 588± 2
σ = 50.1± 1.5
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200

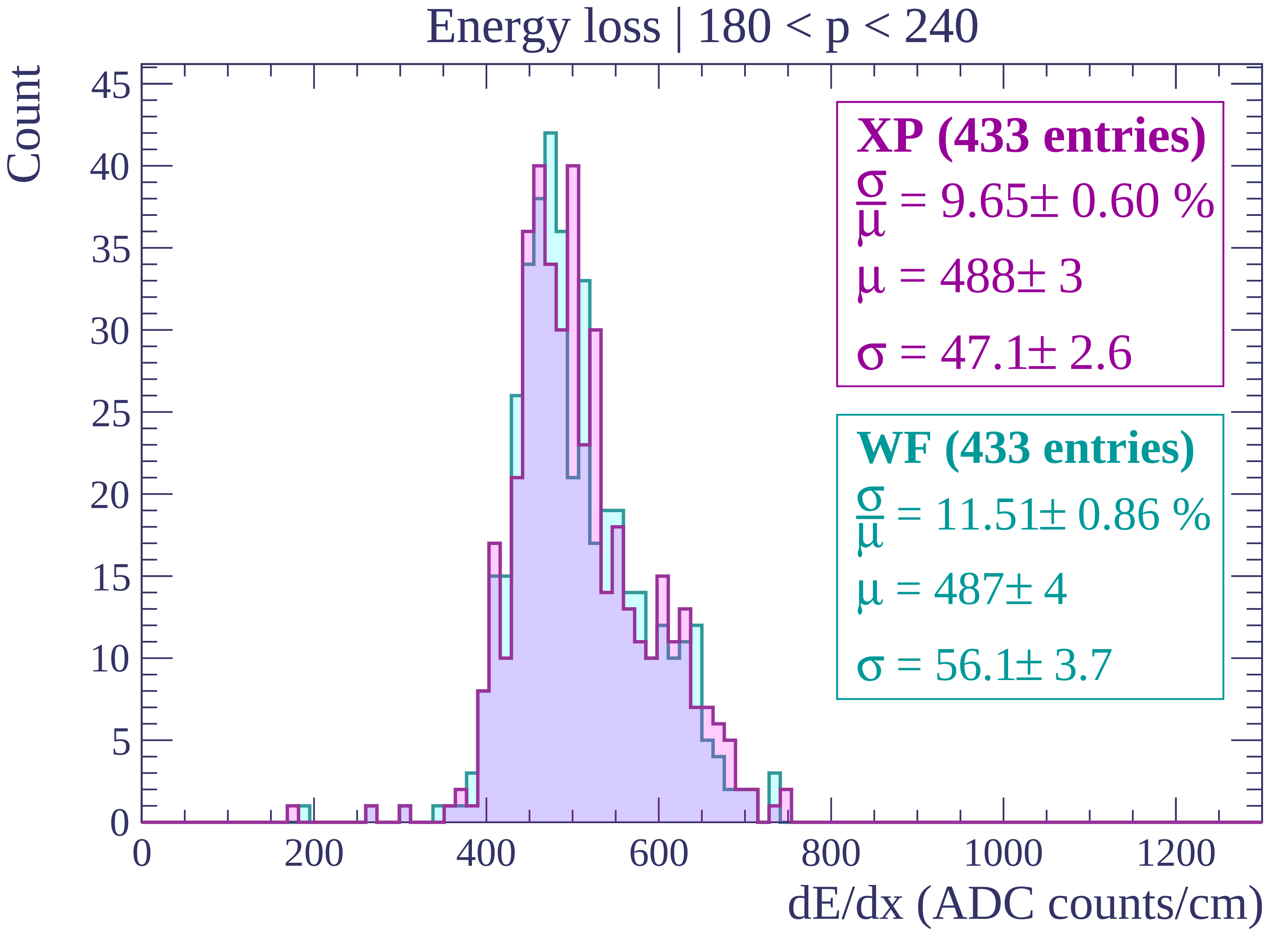


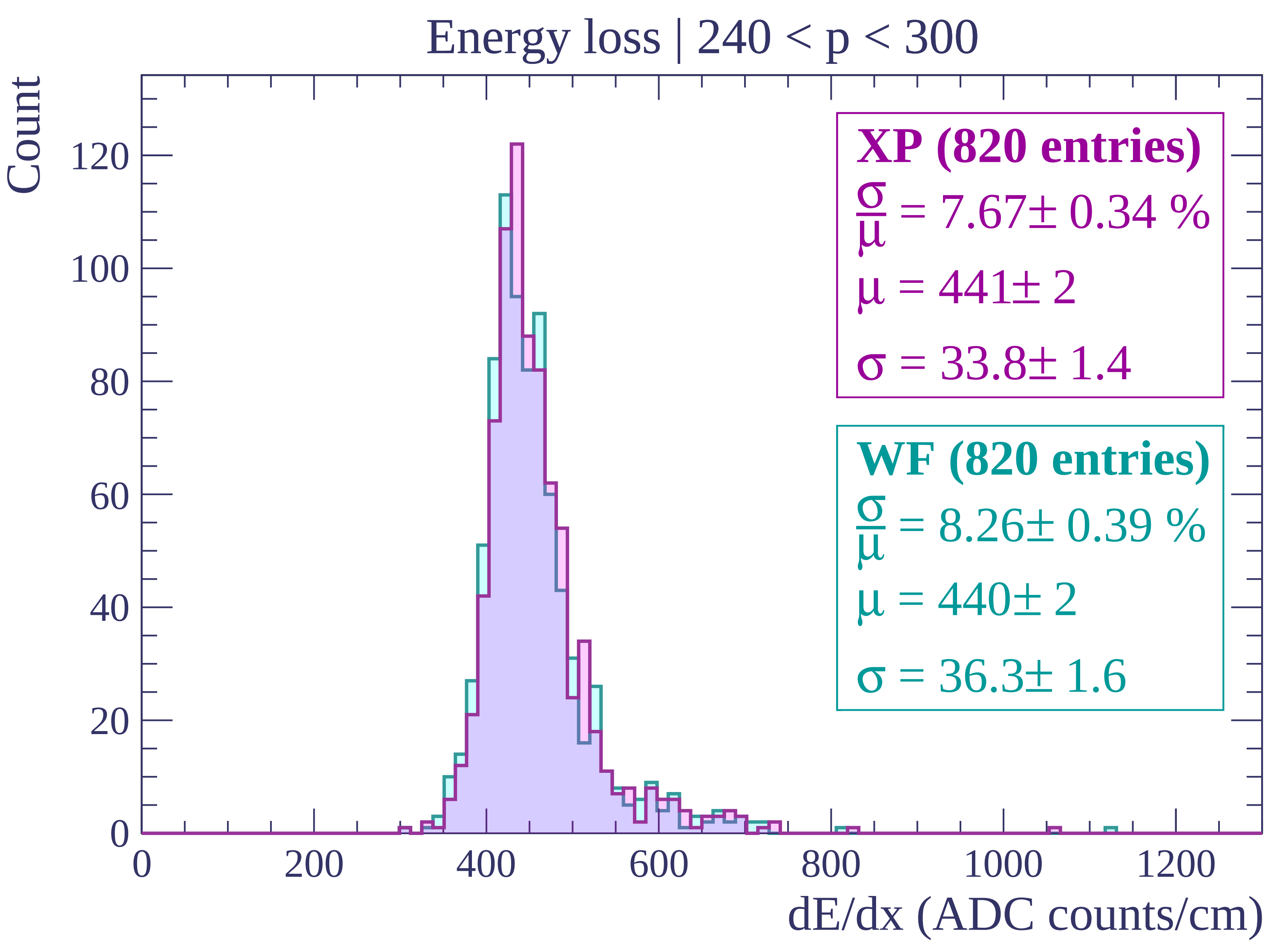
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP (965 entries)
σ/μ = 8.35± 0.36 %
μ = 561± 2
σ = 46.8± 1.8
WF (965 entries)
σ/μ = 9.10± 0.45 %
μ = 554± 2
σ = 50.4± 2.3

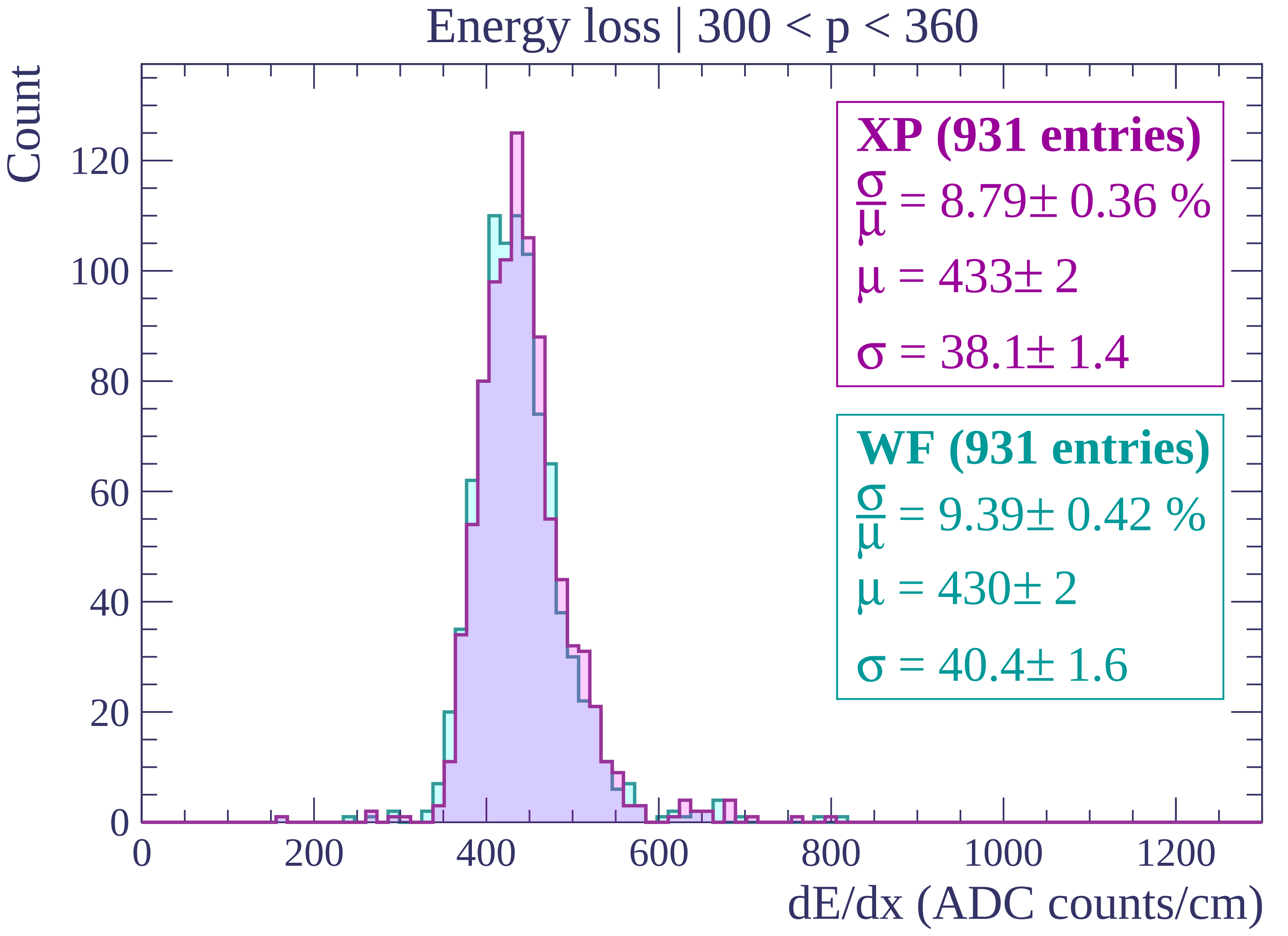


Energy loss | 60 < p < 120
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1222 entries)
σ/μ = 7.69± 0.30 %
μ = 587± 2
σ = 45.2± 1.6
WF (1222 entries)
σ/μ = 8.74± 0.35 %
μ = 580± 2
σ = 50.7± 1.9

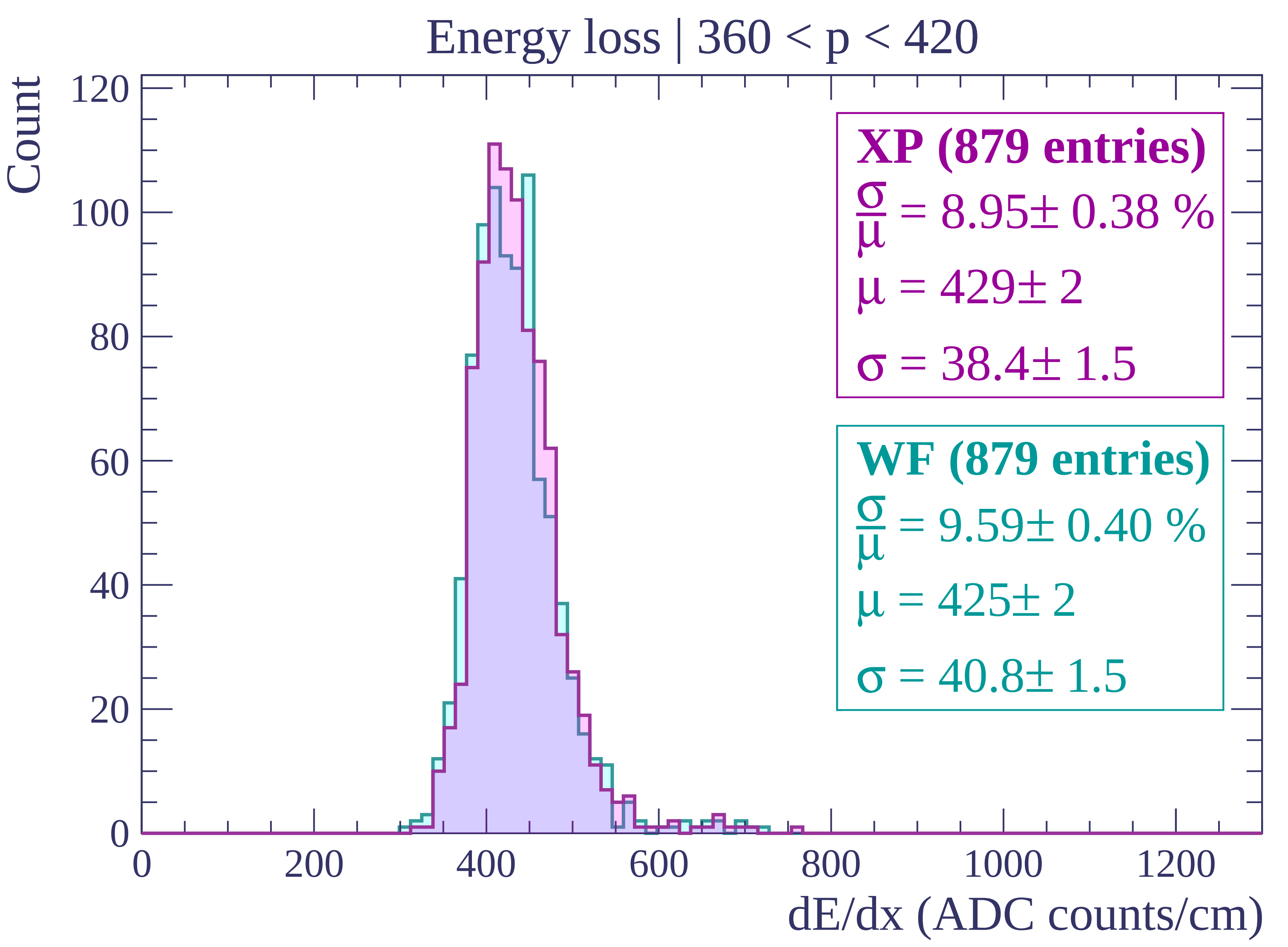

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP (391 entries)
$\frac{\sigma}{\mu}$ = 6.69± 0.56 %
μ = 601± 3
σ = 40.2± 3.2
WF (391 entries)
$\frac{\sigma}{\mu}$ = 7.37± 0.56 %
μ = 595± 3
σ = 43.9± 3.1


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP (433 entries)
σ/μ = 9.65± 0.60 %
μ = 488± 3
σ = 47.1± 2.6
WF (433 entries)
σ/μ = 11.51± 0.86 %
μ = 487± 4
σ = 56.1± 3.7

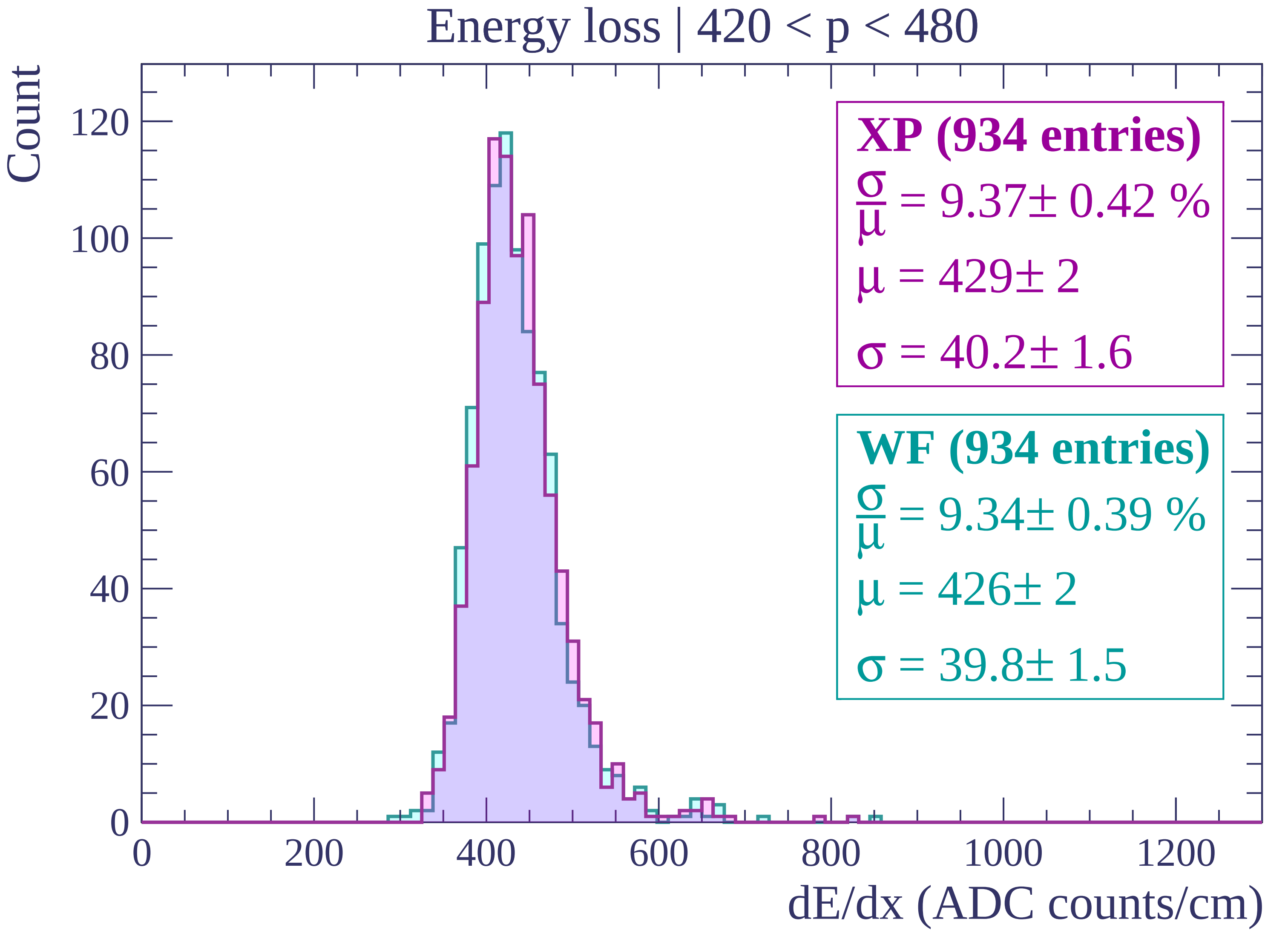

Energy loss | 240 < p < 300
Count
XP (820 entries)
σ/μ = 7.67± 0.34 %
μ = 441± 2
σ = 33.8± 1.4
WF (820 entries)
σ/μ = 8.26± 0.39 %
μ = 440± 2
σ = 36.3± 1.6
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



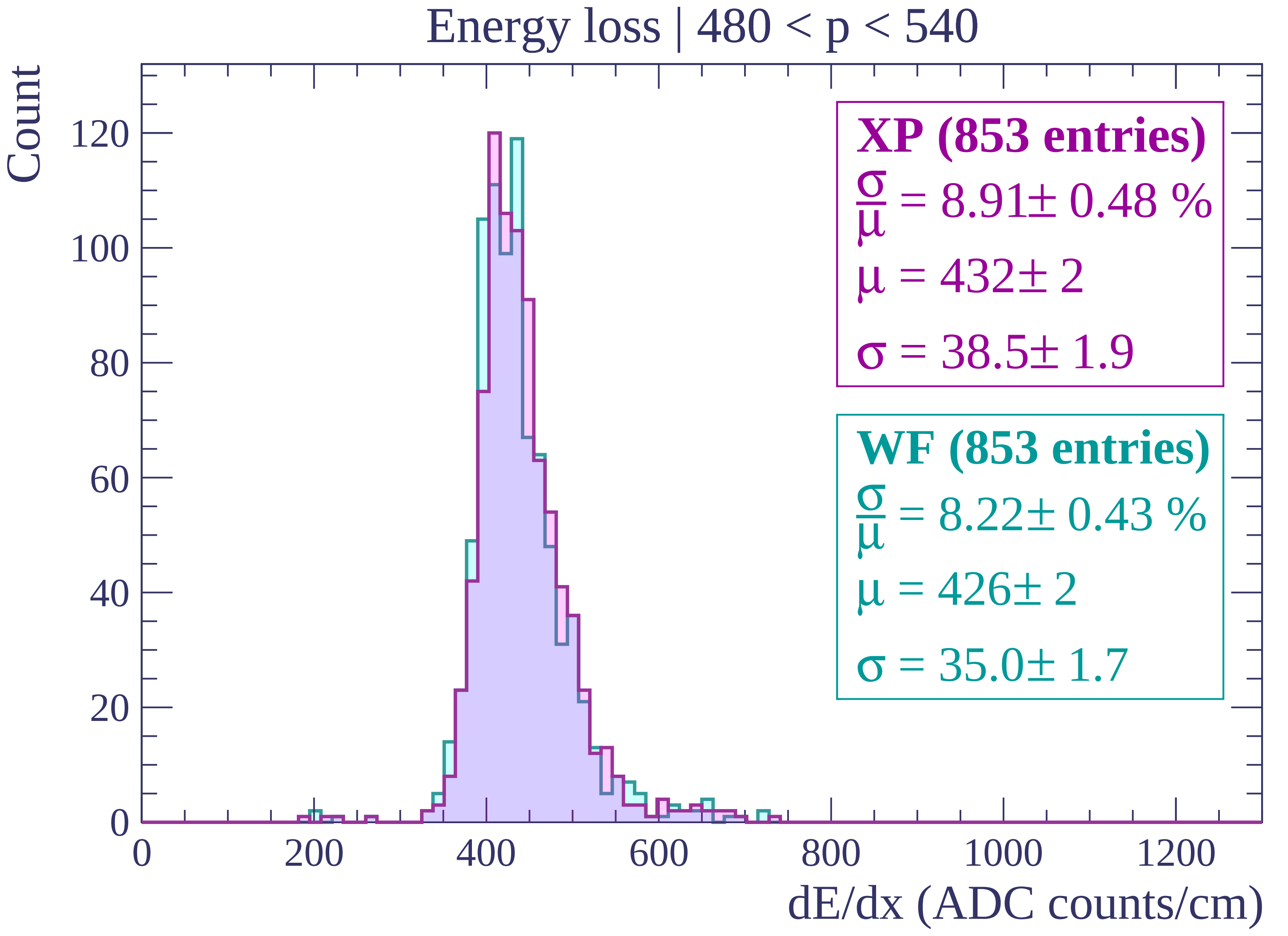
Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
XP (931 entries)
$\frac{\sigma}{\mu}$ = 8.79± 0.36 %
μ = 433± 2
σ = 38.1± 1.4
WF (931 entries)
$\frac{\sigma}{\mu}$ = 9.39± 0.42 %
μ = 430± 2
σ = 40.4± 1.6
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

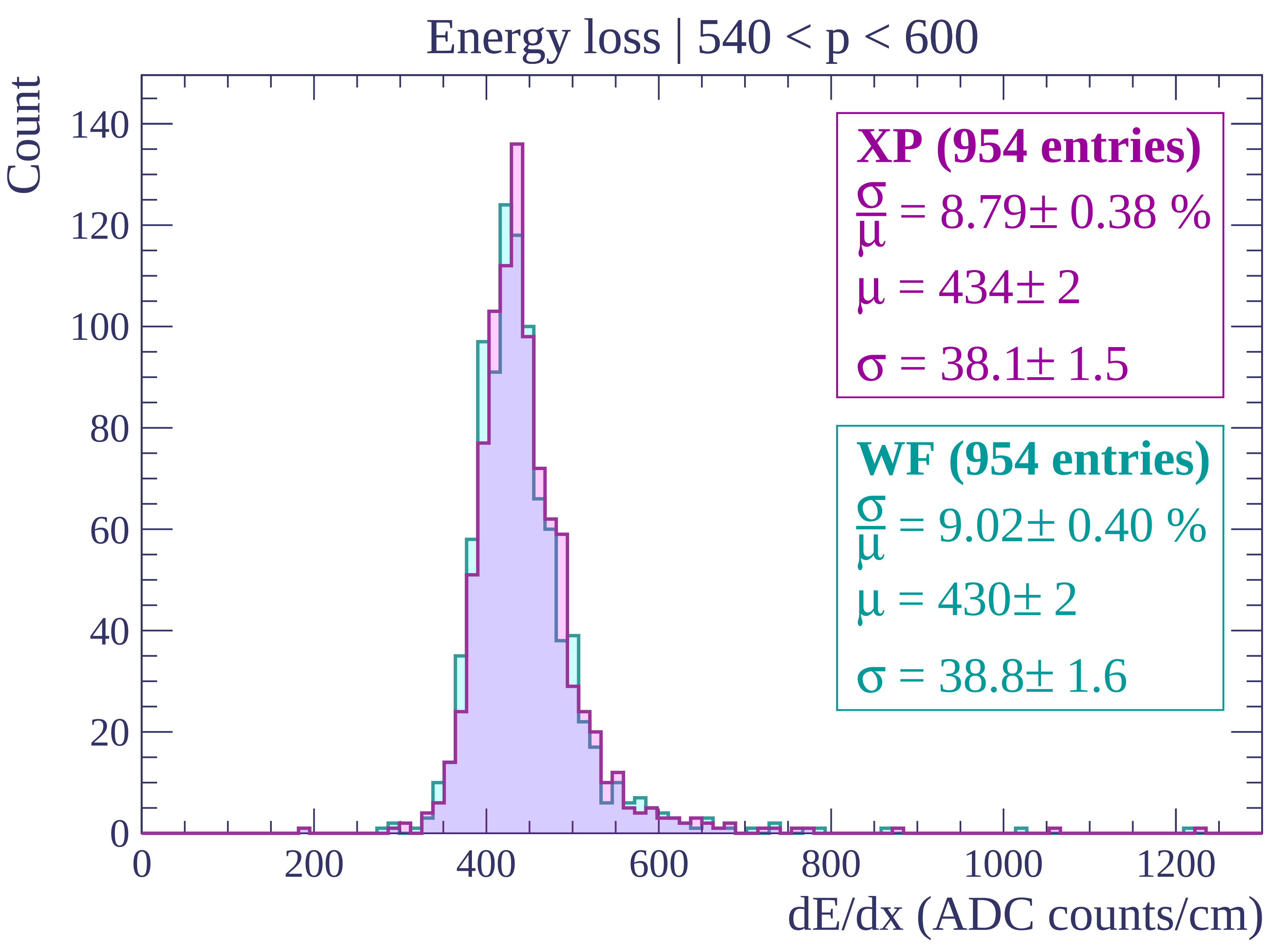


Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (879 entries)
σ/μ = 8.95± 0.38 %
μ = 429± 2
σ = 38.4± 1.5
WF (879 entries)
σ/μ = 9.59± 0.40 %
μ = 425± 2
σ = 40.8± 1.5

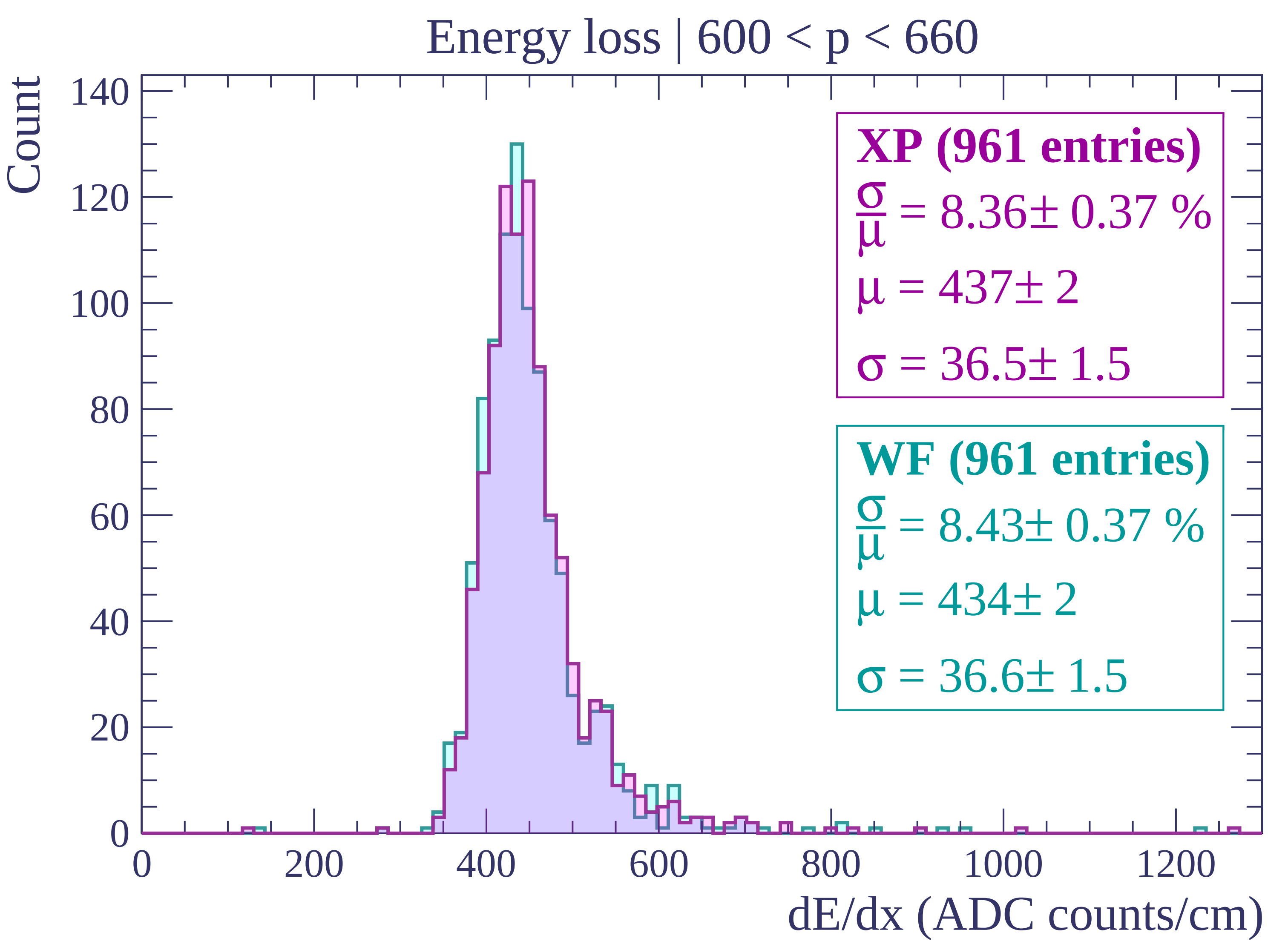


Energy loss | 420 < p < 480
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (934 entries)
σ/μ = 9.37± 0.42 %
μ = 429± 2
σ = 40.2± 1.6
WF (934 entries)
σ/μ = 9.34± 0.39 %
μ = 426± 2
σ = 39.8± 1.5

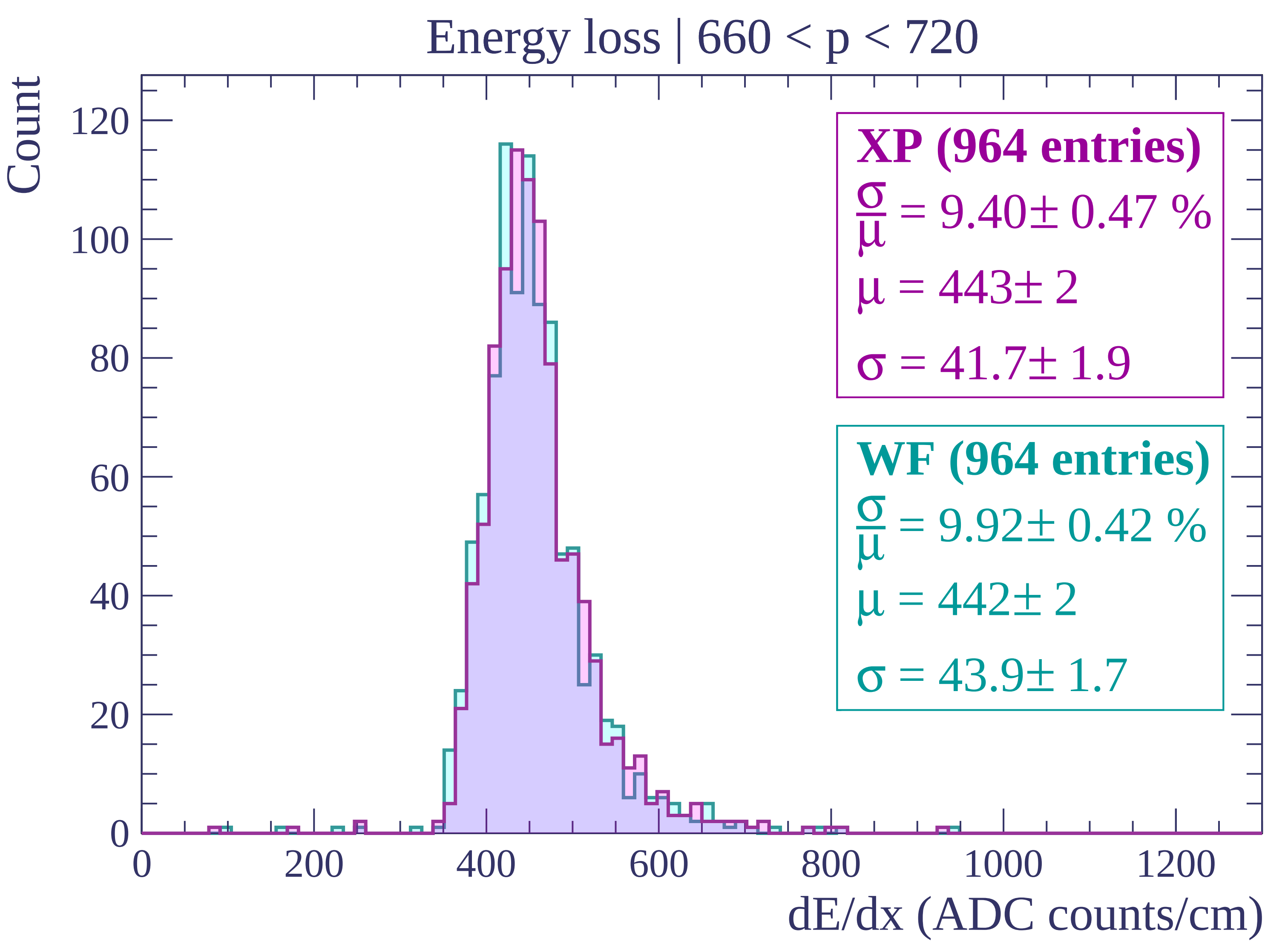

Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
XP (853 entries)
σ/μ = 8.91± 0.48 %
μ = 432± 2
σ = 38.5± 1.9
WF (853 entries)
σ/μ = 8.22± 0.43 %
μ = 426± 2
σ = 35.0± 1.7
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

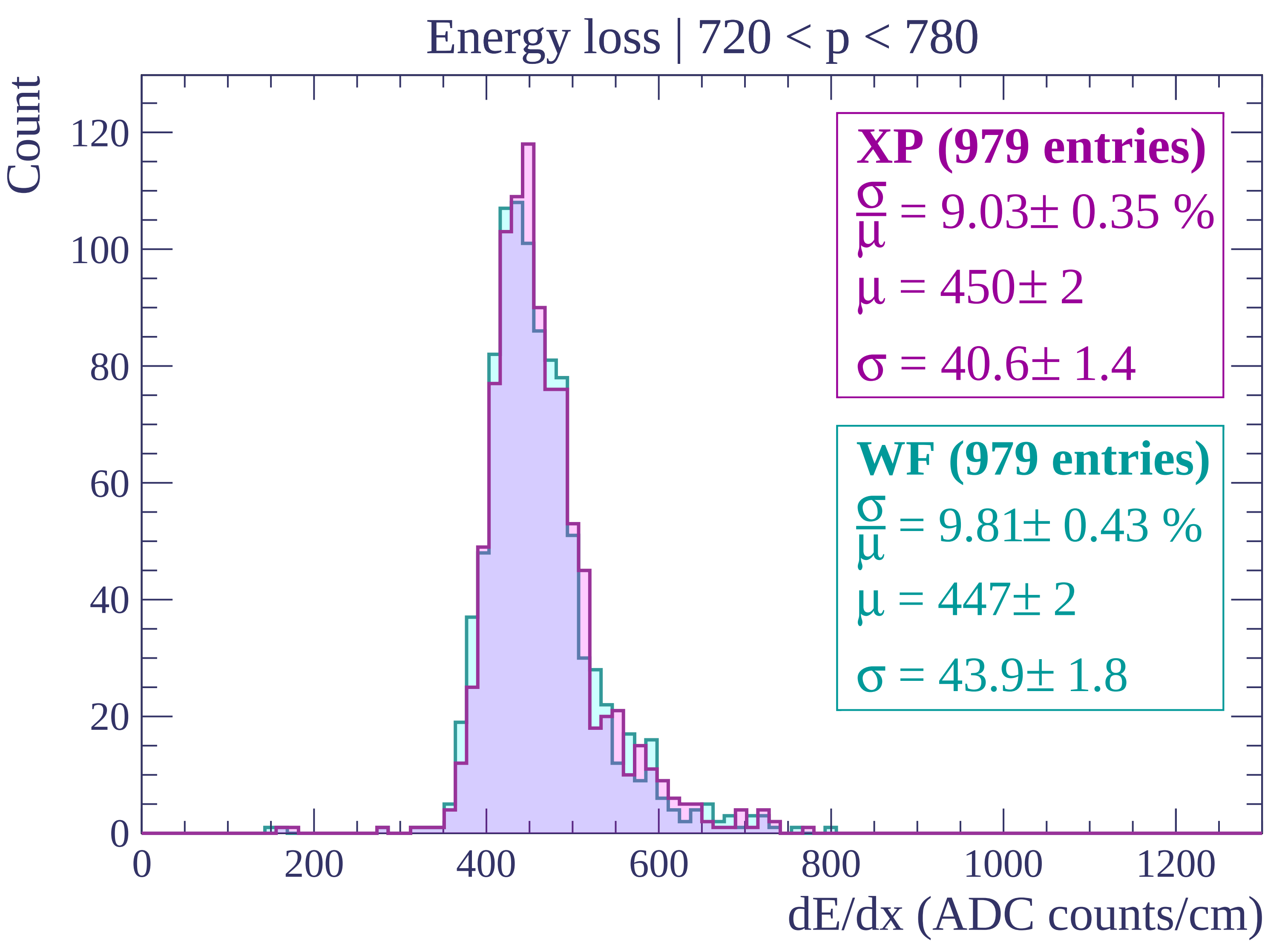


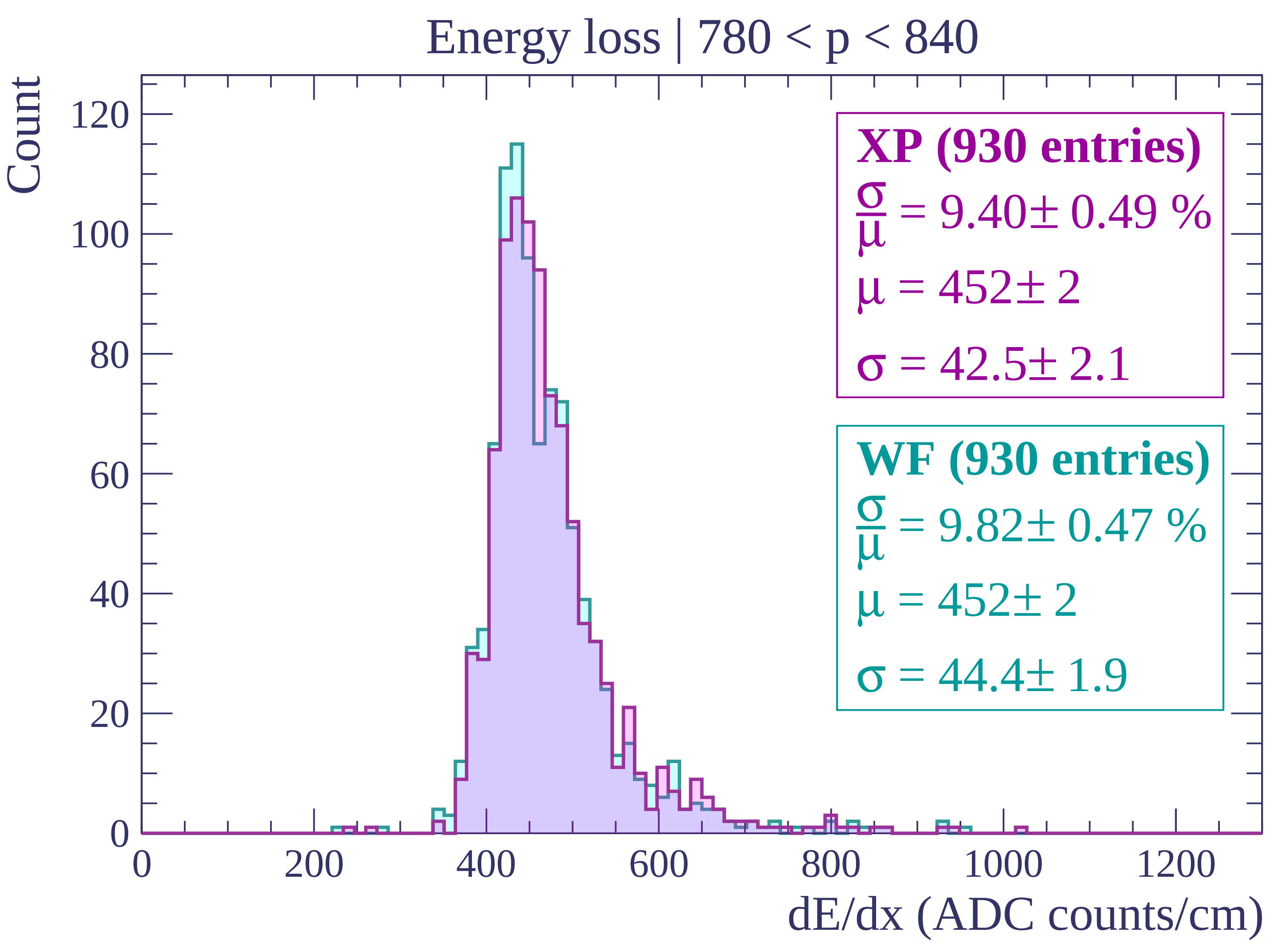
Energy loss | 540 < p < 600
Count
XP (954 entries)
σ/μ = 8.79± 0.38 %
μ = 434± 2
σ = 38.1± 1.5
WF (954 entries)
σ/μ = 9.02± 0.40 %
μ = 430± 2
σ = 38.8± 1.6
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



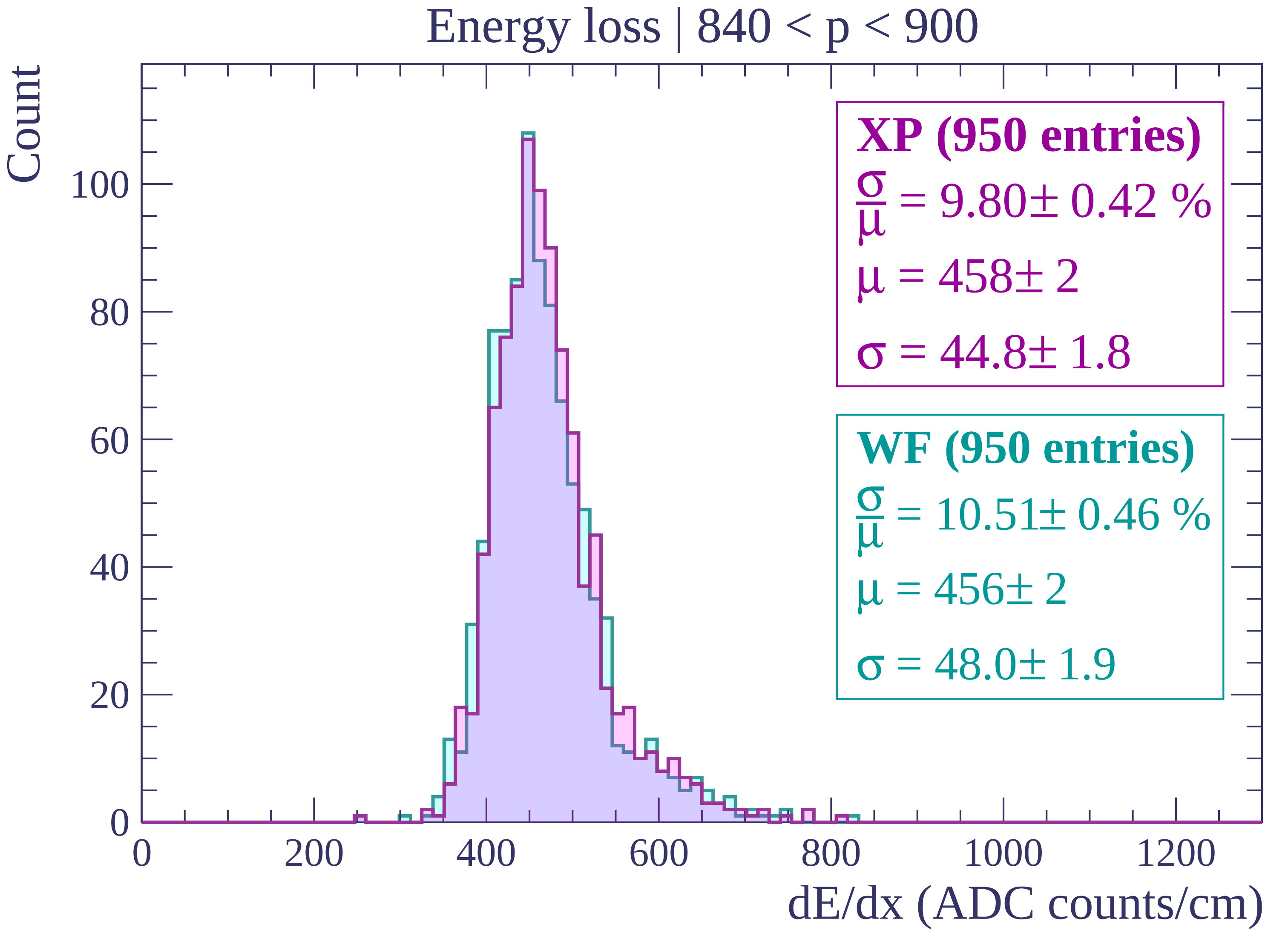
Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
XP (961 entries)
σ/μ = 8.36± 0.37 %
μ = 437± 2
σ = 36.5± 1.5
WF (961 entries)
σ/μ = 8.43± 0.37 %
μ = 434± 2
σ = 36.6± 1.5
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

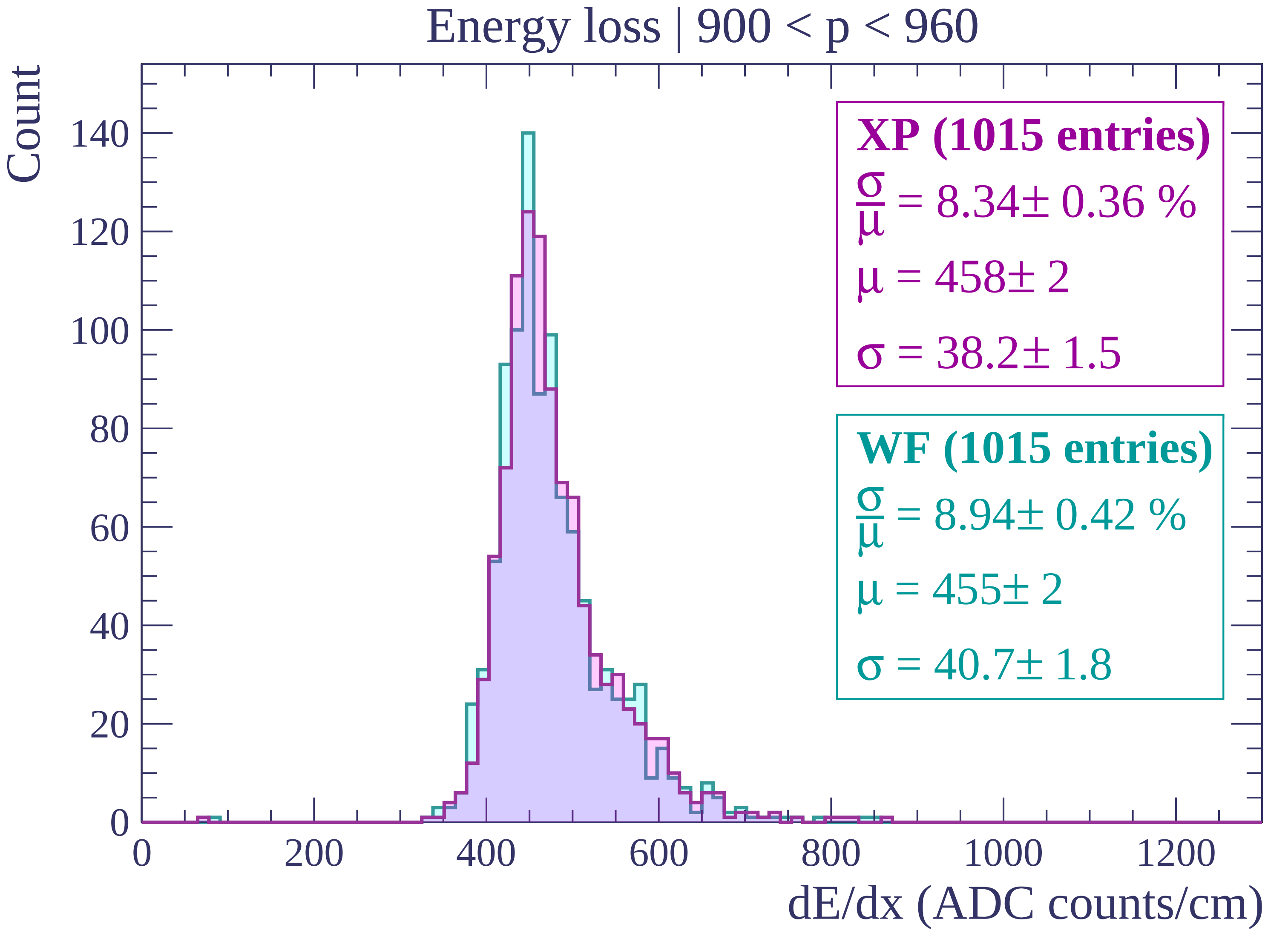

Energy loss | 660 < p < 720
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (964 entries)
σ/μ = 9.40± 0.47 %
μ = 443± 2
σ = 41.7± 1.9
WF (964 entries)
σ/μ = 9.92± 0.42 %
μ = 442± 2
σ = 43.9± 1.7

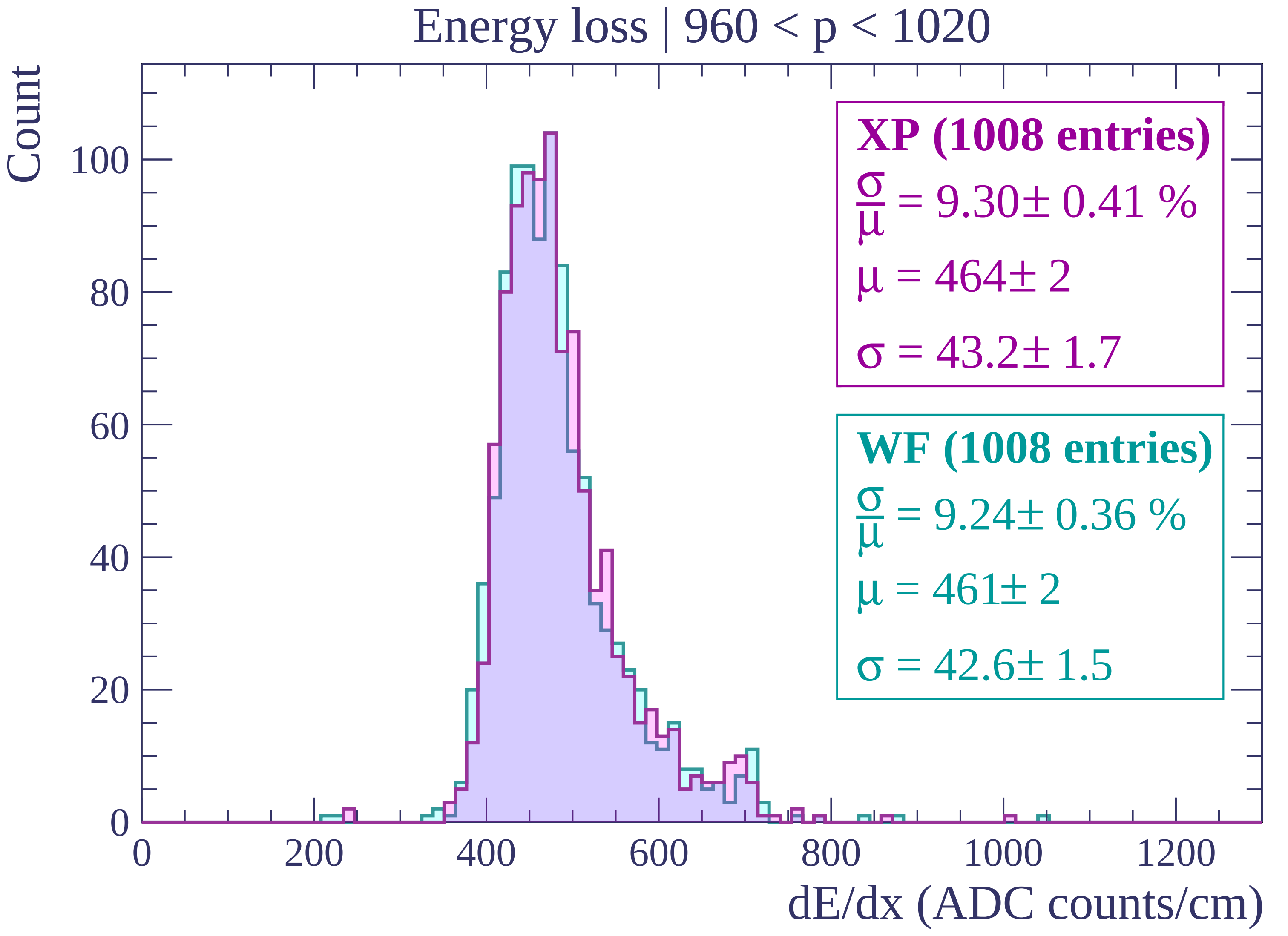

Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (979 entries)
σ/μ = 9.03± 0.35 %
μ = 450± 2
σ = 40.6± 1.4
WF (979 entries)
σ/μ = 9.81± 0.43 %
μ = 447± 2
σ = 43.9± 1.8



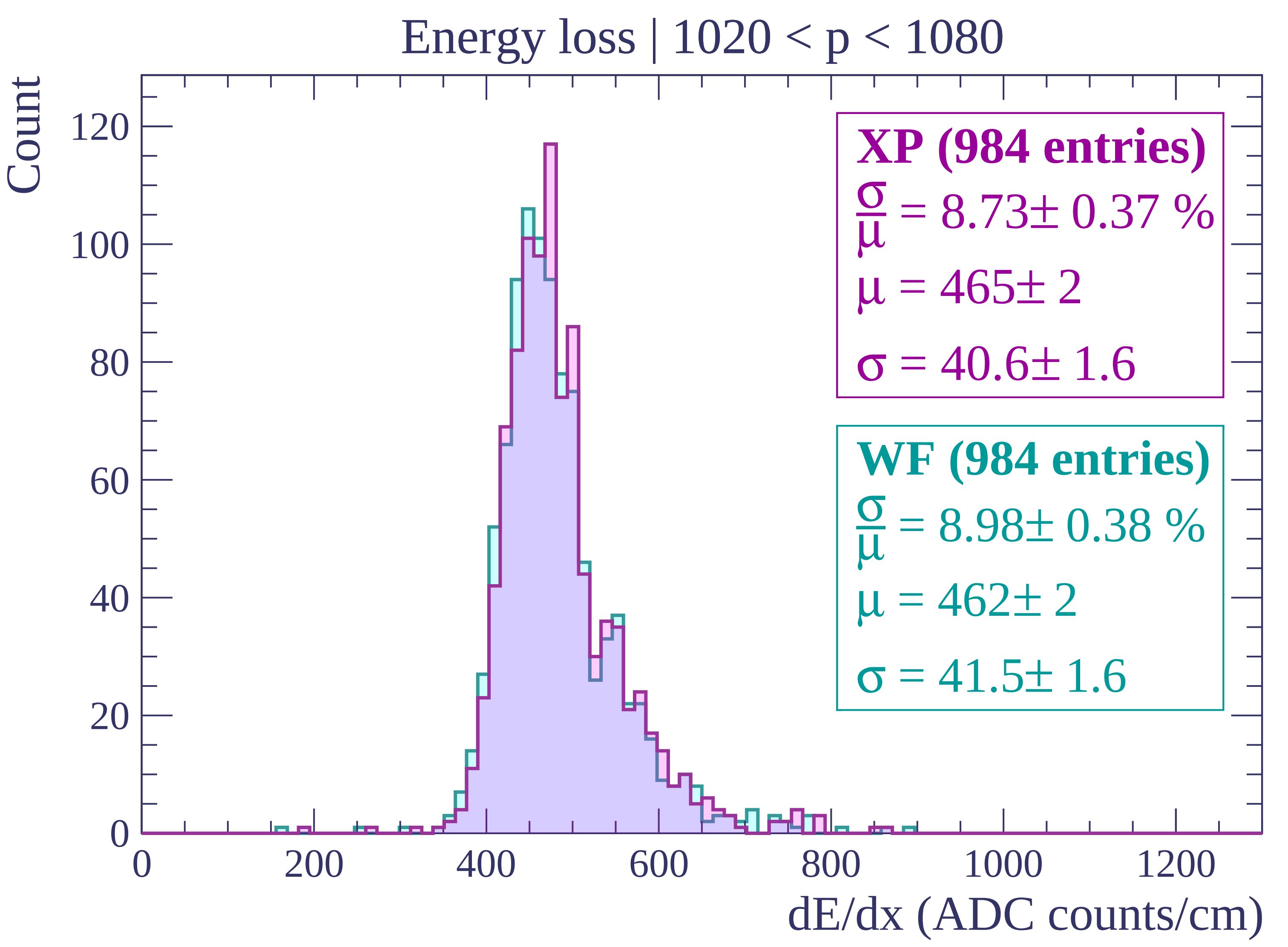
Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (930 entries)
$\frac{\sigma}{\mu}$ = 9.40± 0.49 %
μ = 452± 2
σ = 42.5± 2.1
WF (930 entries)
$\frac{\sigma}{\mu}$ = 9.82± 0.47 %
μ = 452± 2
σ = 44.4± 1.9

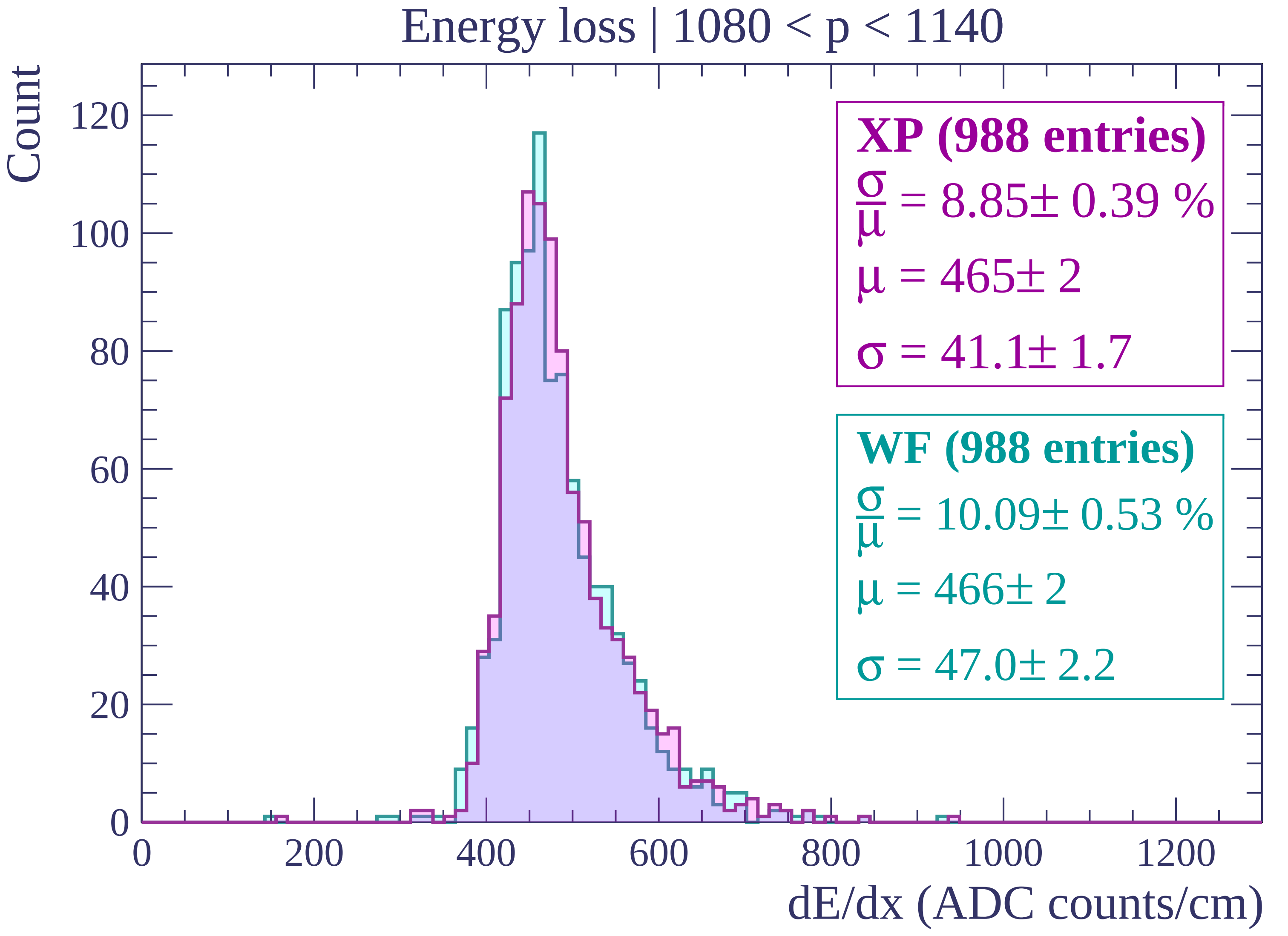

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (950 entries)
σ/μ = 9.80± 0.42 %
μ = 458± 2
σ = 44.8± 1.8
WF (950 entries)
σ/μ = 10.51± 0.46 %
μ = 456± 2
σ = 48.0± 1.9

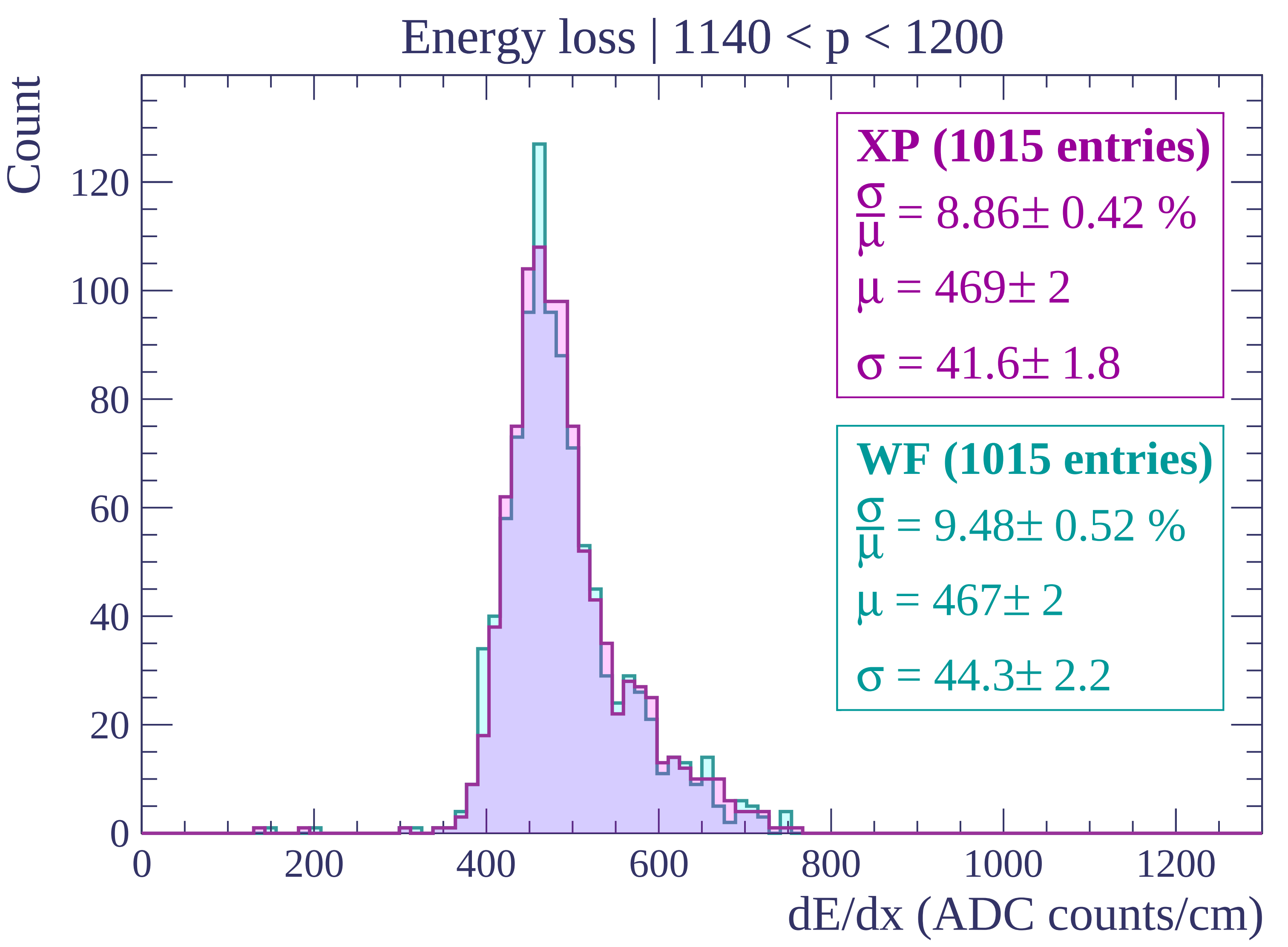

Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1015 entries)
$\frac{\sigma}{\mu}$ = 8.34± 0.36 %
μ = 458± 2
σ = 38.2± 1.5
WF (1015 entries)
$\frac{\sigma}{\mu}$ = 8.94± 0.42 %
μ = 455± 2
σ = 40.7± 1.8

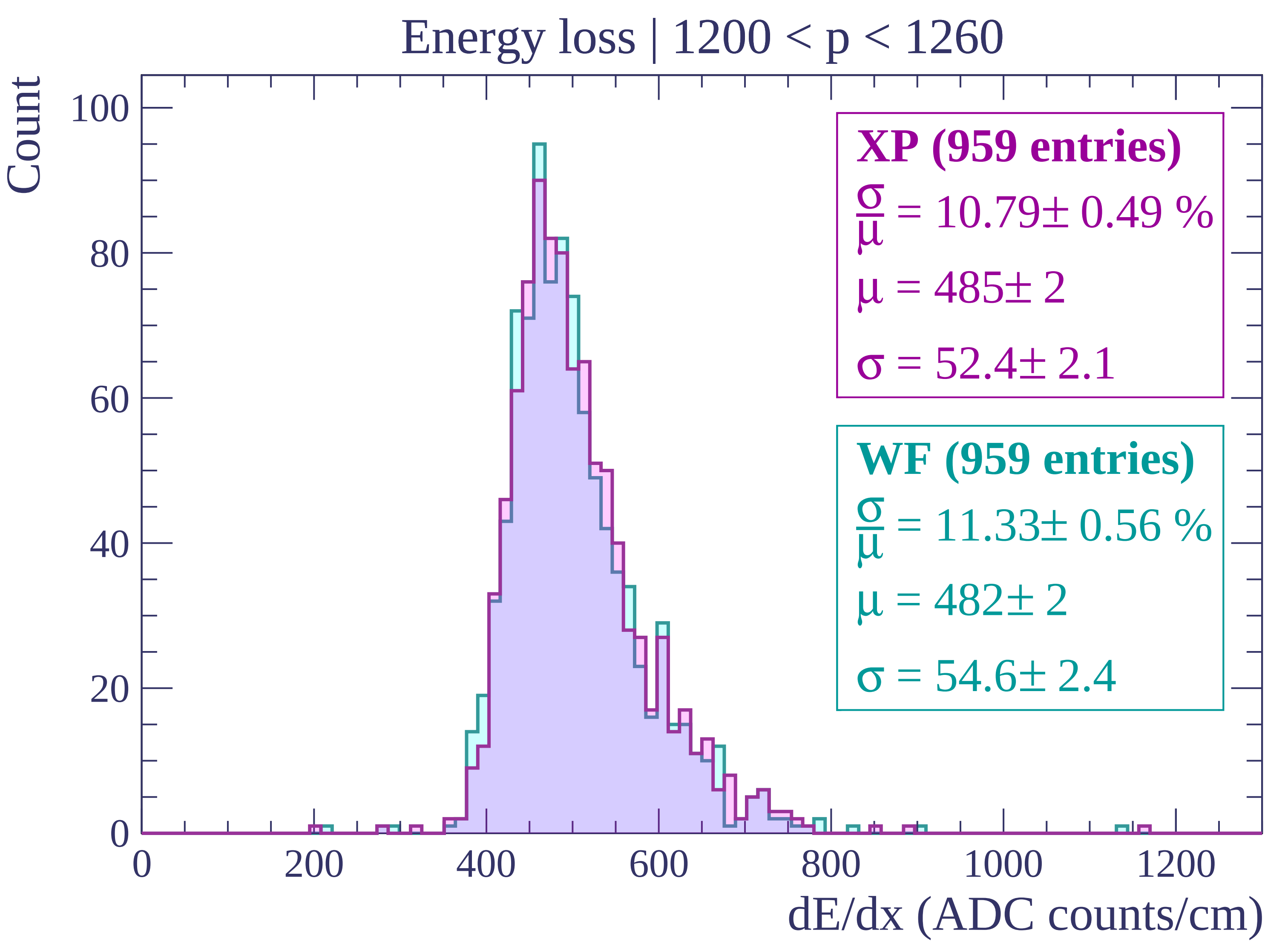


Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1008 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.41 %
μ = 464± 2
σ = 43.2± 1.7
WF (1008 entries)
$\frac{\sigma}{\mu}$ = 9.24± 0.36 %
μ = 461± 2
σ = 42.6± 1.5

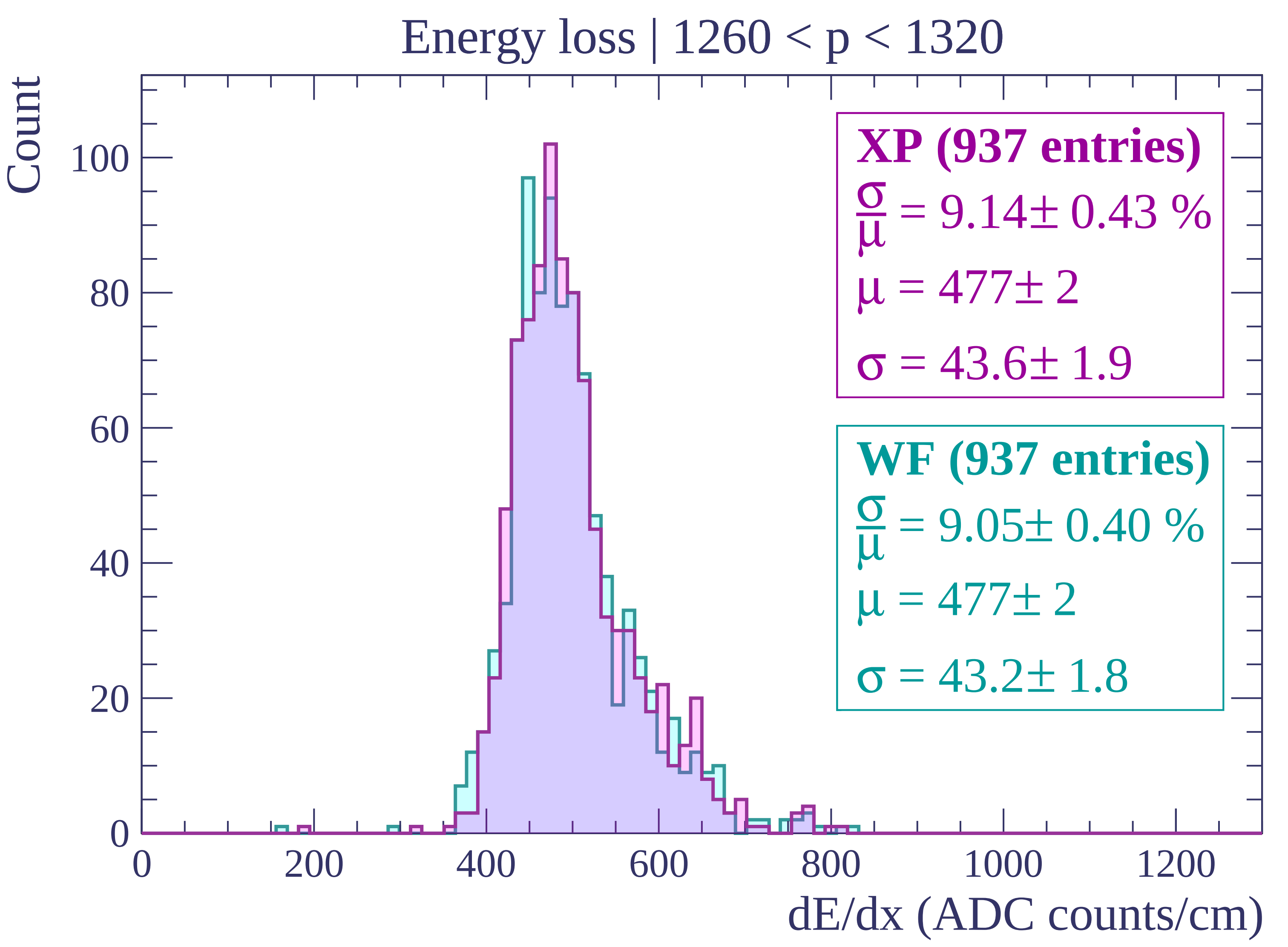

Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (984 entries)
σ/μ = 8.73± 0.37 %
μ = 465± 2
σ = 40.6± 1.6
WF (984 entries)
σ/μ = 8.98± 0.38 %
μ = 462± 2
σ = 41.5± 1.6


Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
XP (988 entries)
$\frac{\sigma}{\mu}$ = 8.85± 0.39 %
μ = 465± 2
σ = 41.1± 1.7
WF (988 entries)
$\frac{\sigma}{\mu}$ = 10.09± 0.53 %
μ = 466± 2
σ = 47.0± 2.2
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

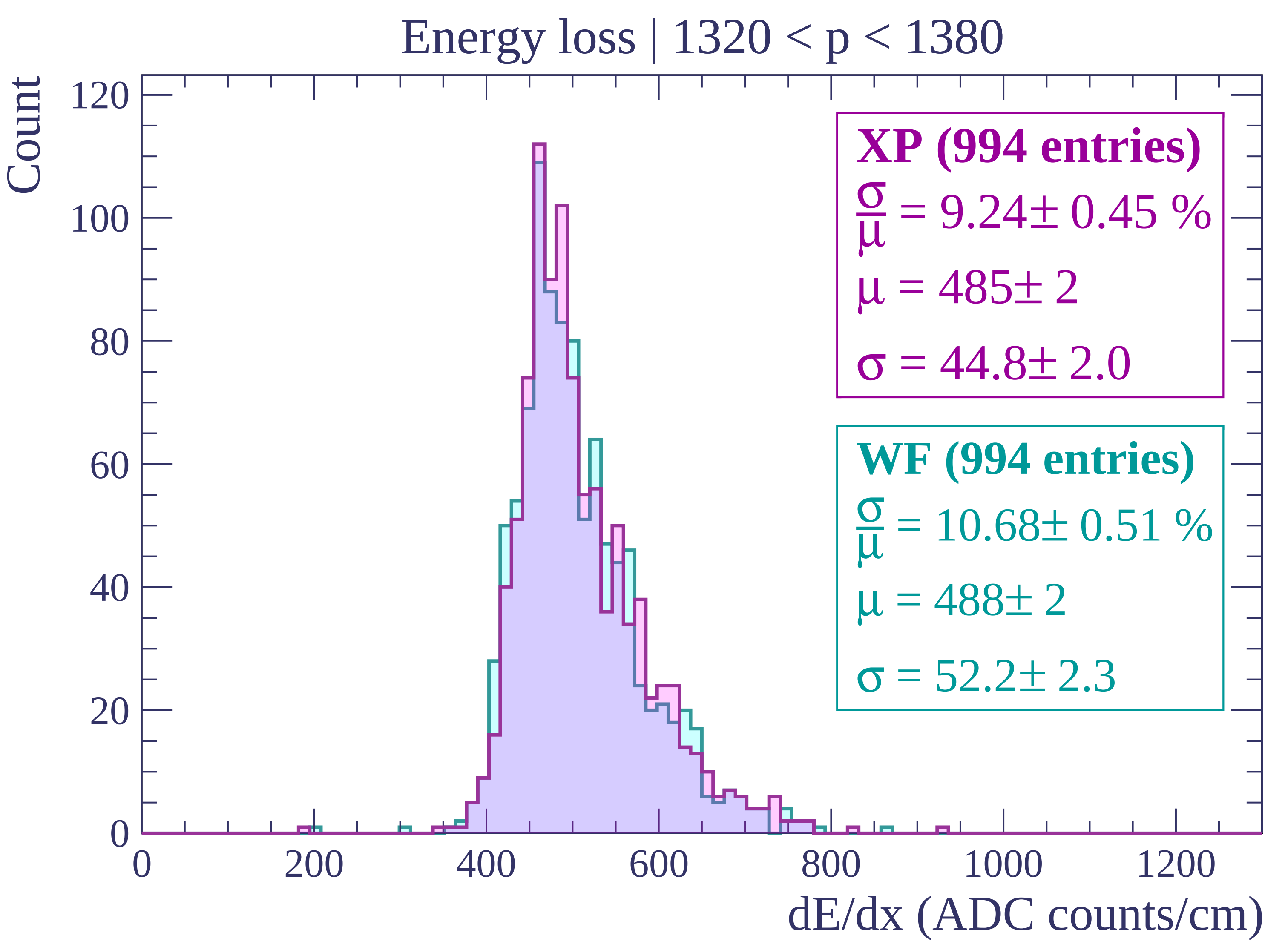

Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (1015 entries)
σ/μ = 8.86± 0.42 %
μ = 469± 2
σ = 41.6± 1.8
WF (1015 entries)
σ/μ = 9.48± 0.52 %
μ = 467± 2
σ = 44.3± 2.2
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200



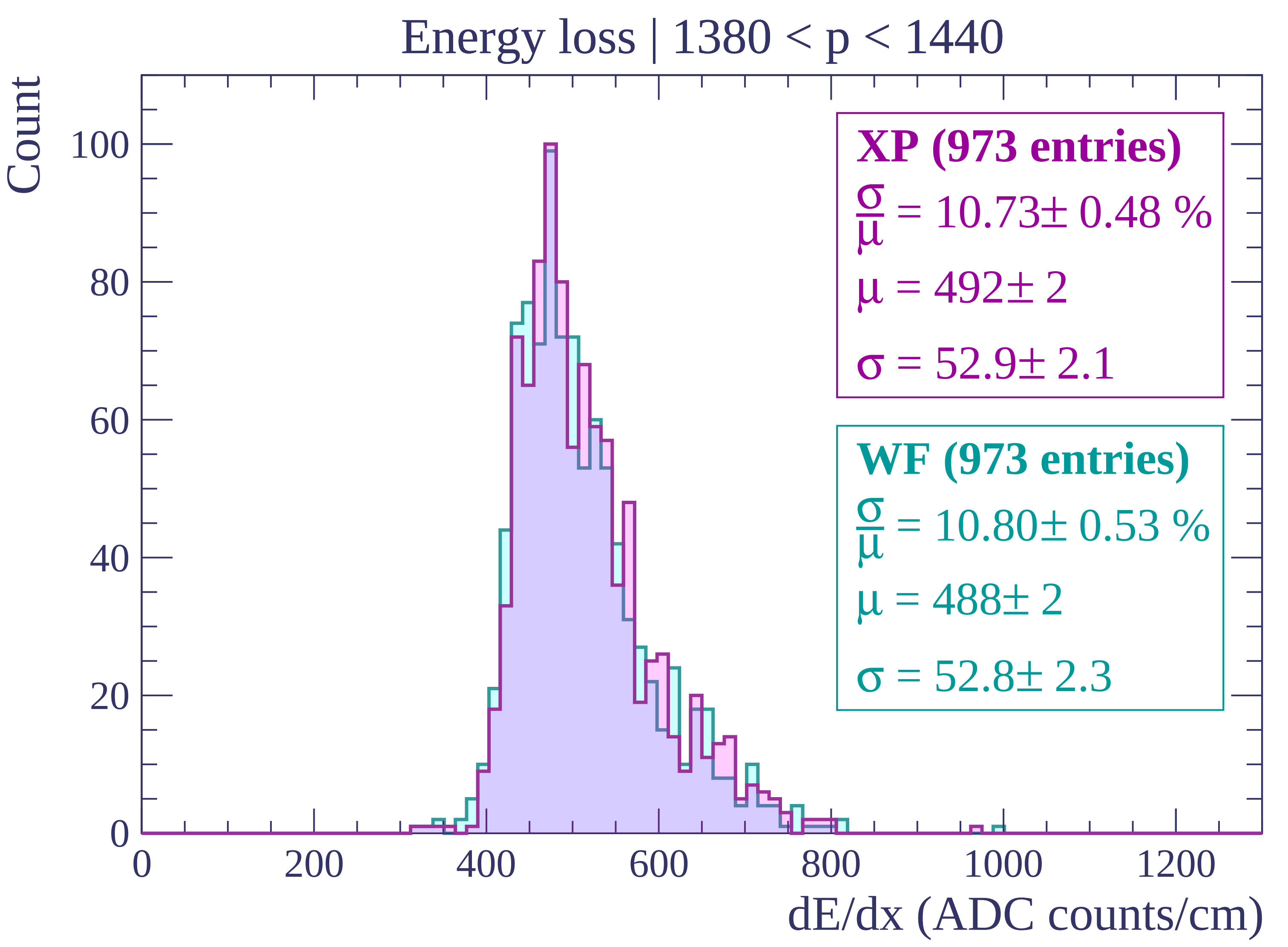
Energy loss | 1200 < p < 1260
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (959 entries)
σ/μ = 10.79± 0.49 %
μ = 485± 2
σ = 52.4± 2.1
WF (959 entries)
σ/μ = 11.33± 0.56 %
μ = 482± 2
σ = 54.6± 2.4

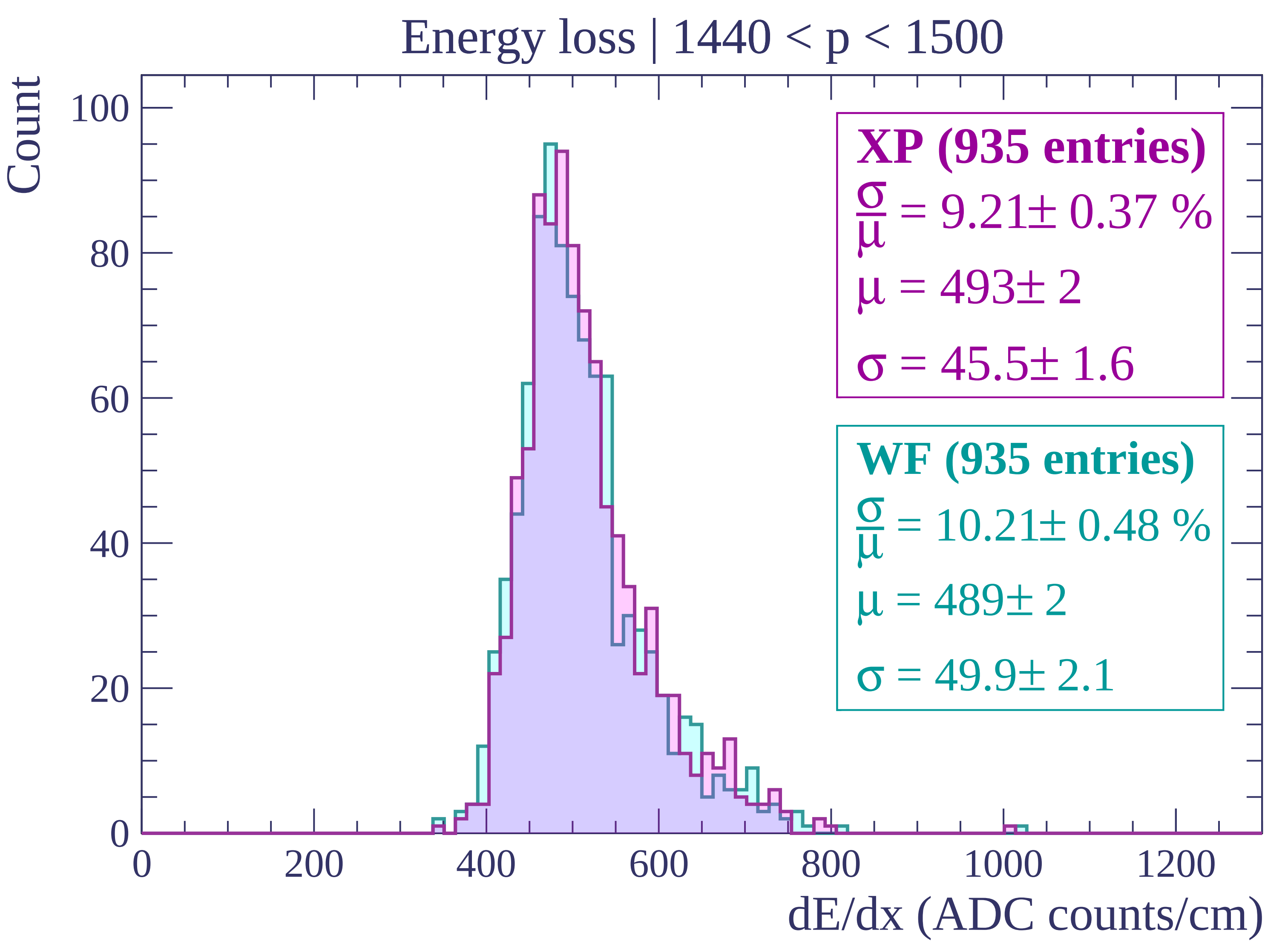


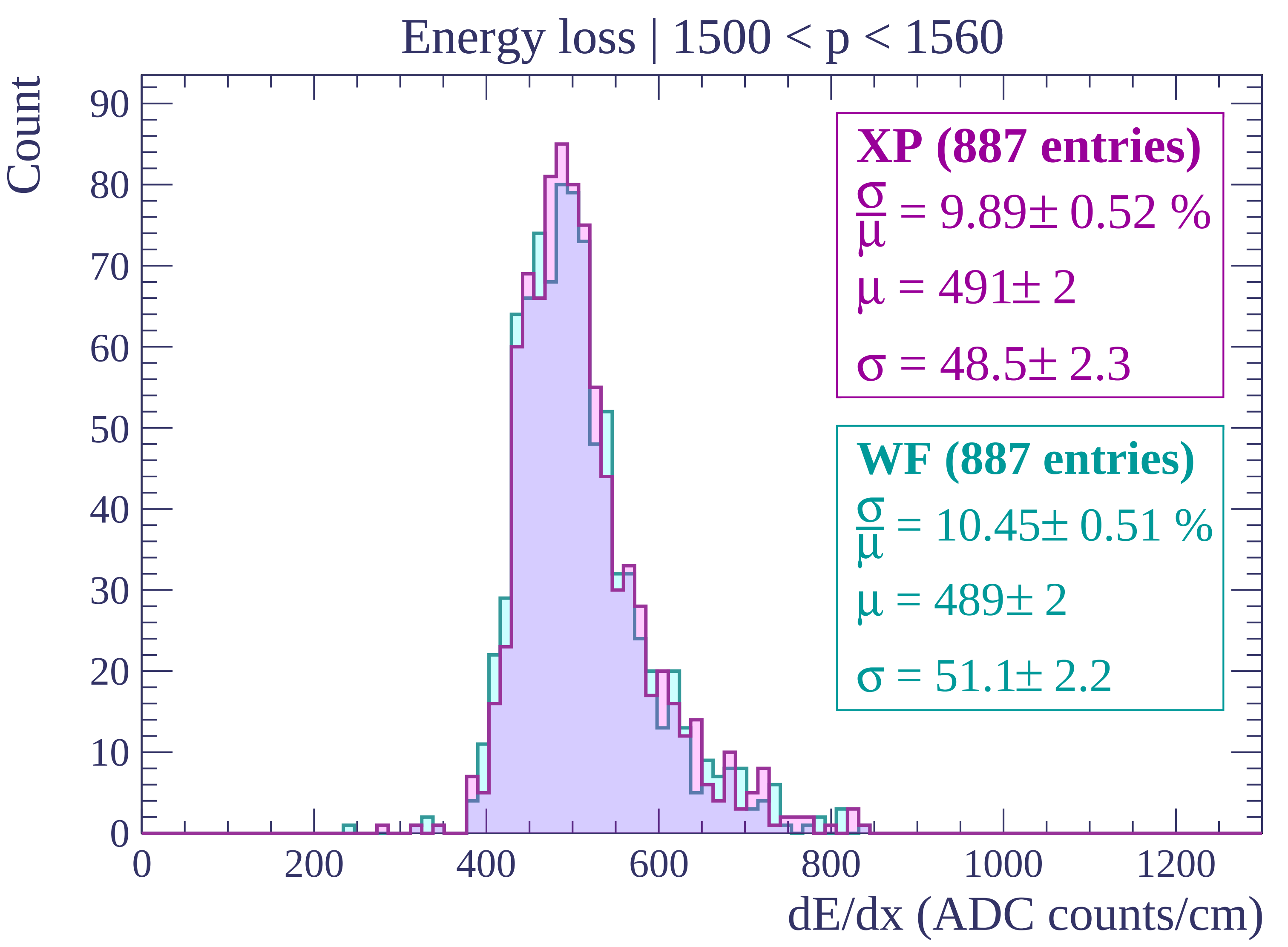
Energy loss | 1260 < p < 1320
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (937 entries)
$\frac{\sigma}{\mu}$ = 9.14± 0.43 %
μ = 477± 2
σ = 43.6± 1.9
WF (937 entries)
$\frac{\sigma}{\mu}$ = 9.05± 0.40 %
μ = 477± 2
σ = 43.2± 1.8

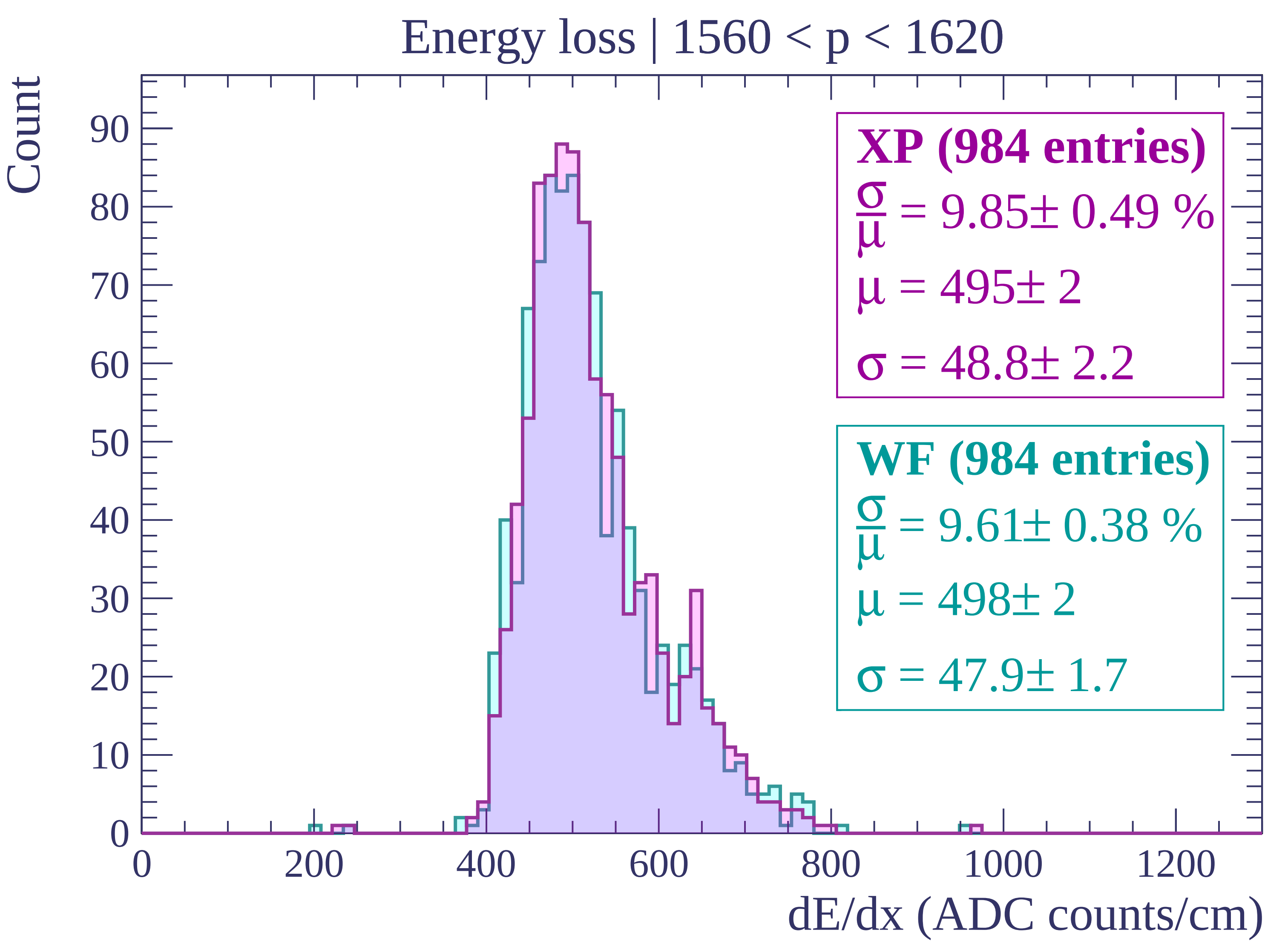


Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
XP (994 entries)
σ/μ = 9.24± 0.45 %
μ = 485± 2
σ = 44.8± 2.0
WF (994 entries)
σ/μ = 10.68± 0.51 %
μ = 488± 2
σ = 52.2± 2.3

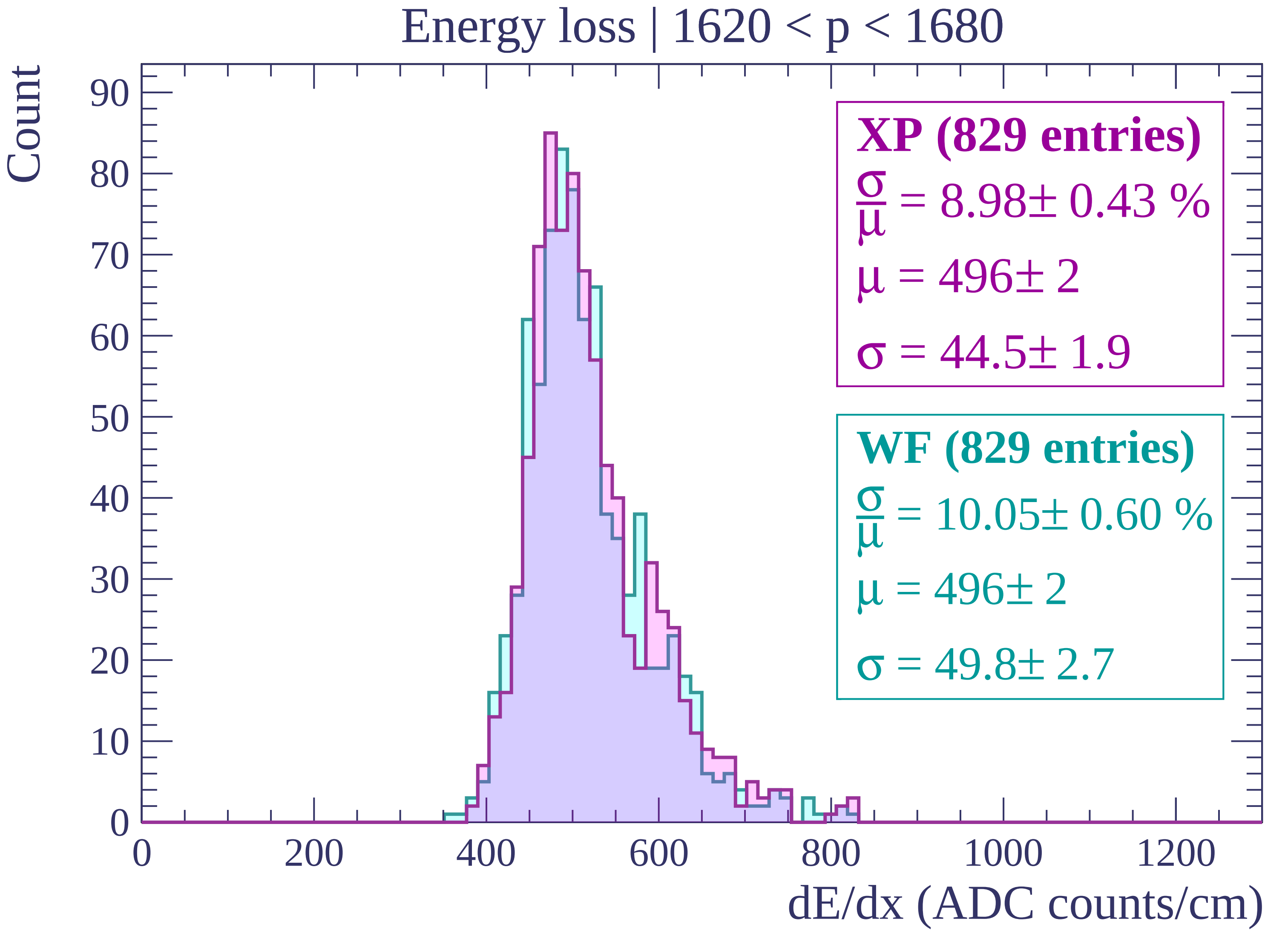

Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (973 entries)
σ/μ = 10.73± 0.48 %
μ = 492± 2
σ = 52.9± 2.1
WF (973 entries)
σ/μ = 10.80± 0.53 %
μ = 488± 2
σ = 52.8± 2.3


Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (935 entries)
$\frac{\sigma}{\mu}$ = 9.21± 0.37 %
μ = 493± 2
σ = 45.5± 1.6
WF (935 entries)
$\frac{\sigma}{\mu}$ = 10.21± 0.48 %
μ = 489± 2
σ = 49.9± 2.1

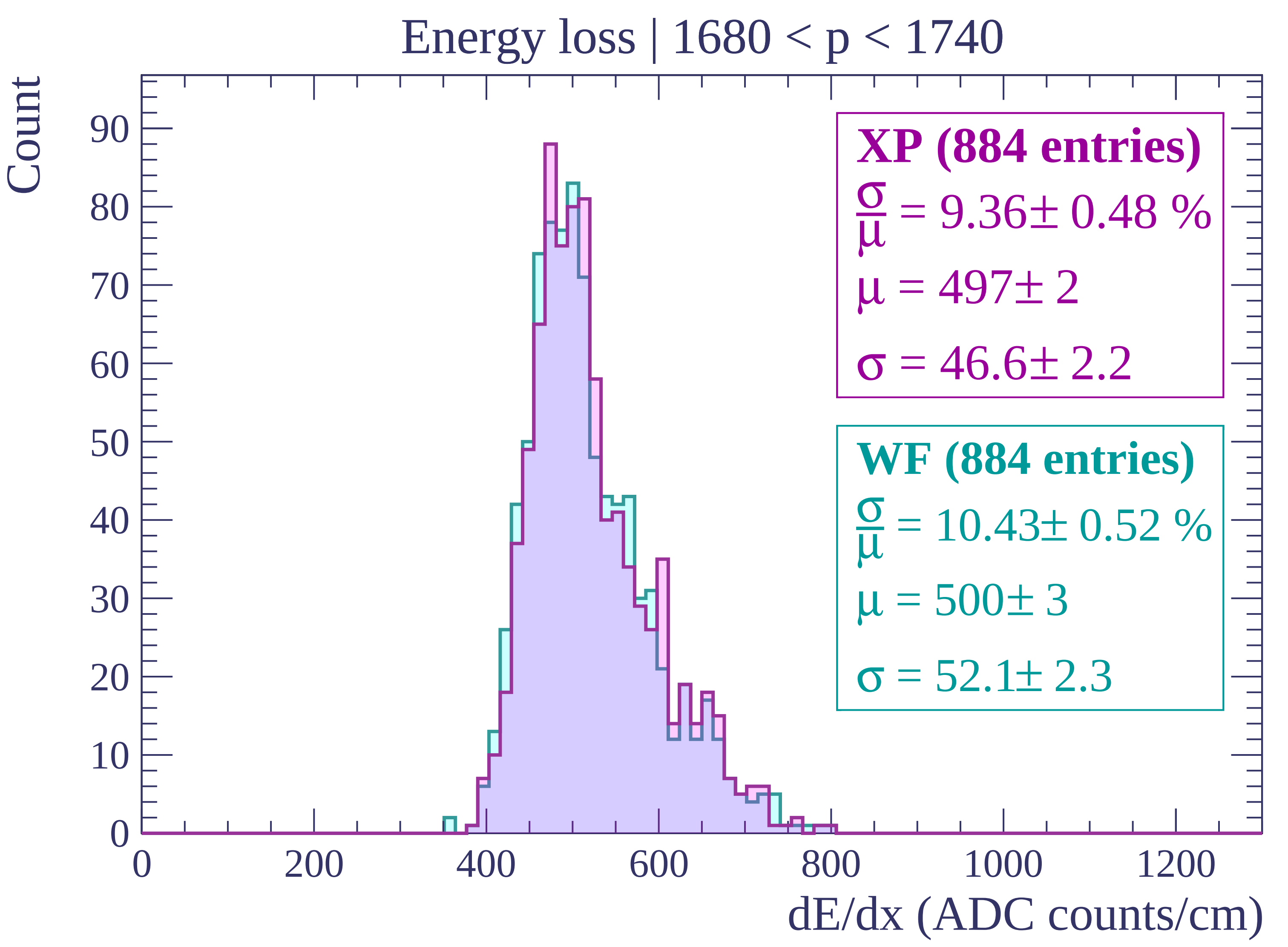

Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
XP (887 entries)
$\frac{\sigma}{\mu}$ = 9.89± 0.52 %
μ = 491± 2
σ = 48.5± 2.3
WF (887 entries)
$\frac{\sigma}{\mu}$ = 10.45± 0.51 %
μ = 489± 2
σ = 51.1± 2.2
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90

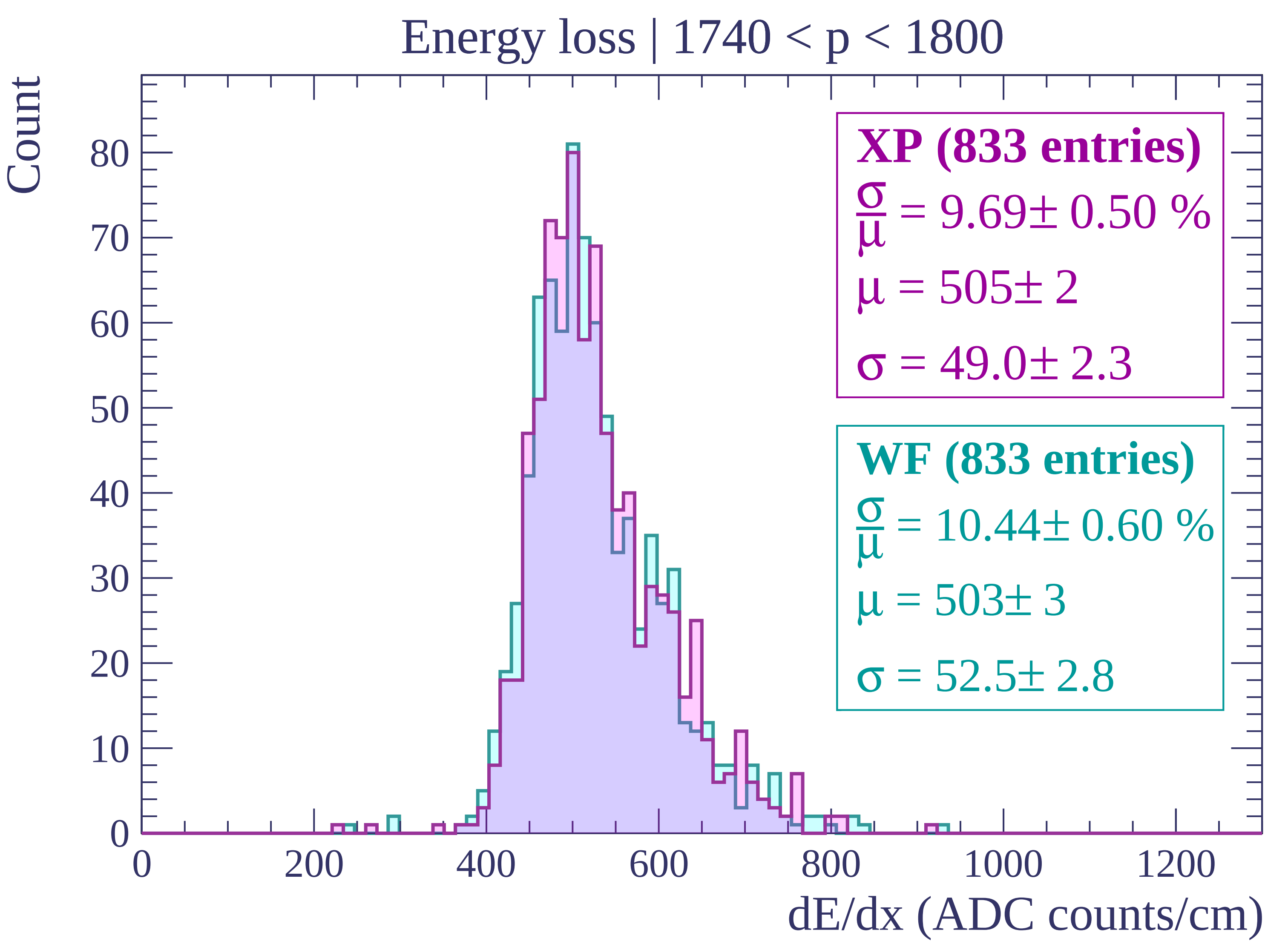


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (984 entries)
$\frac{\sigma}{\mu}$ = 9.85± 0.49 %
μ = 495± 2
σ = 48.8± 2.2
WF (984 entries)
$\frac{\sigma}{\mu}$ = 9.61± 0.38 %
μ = 498± 2
σ = 47.9± 1.7

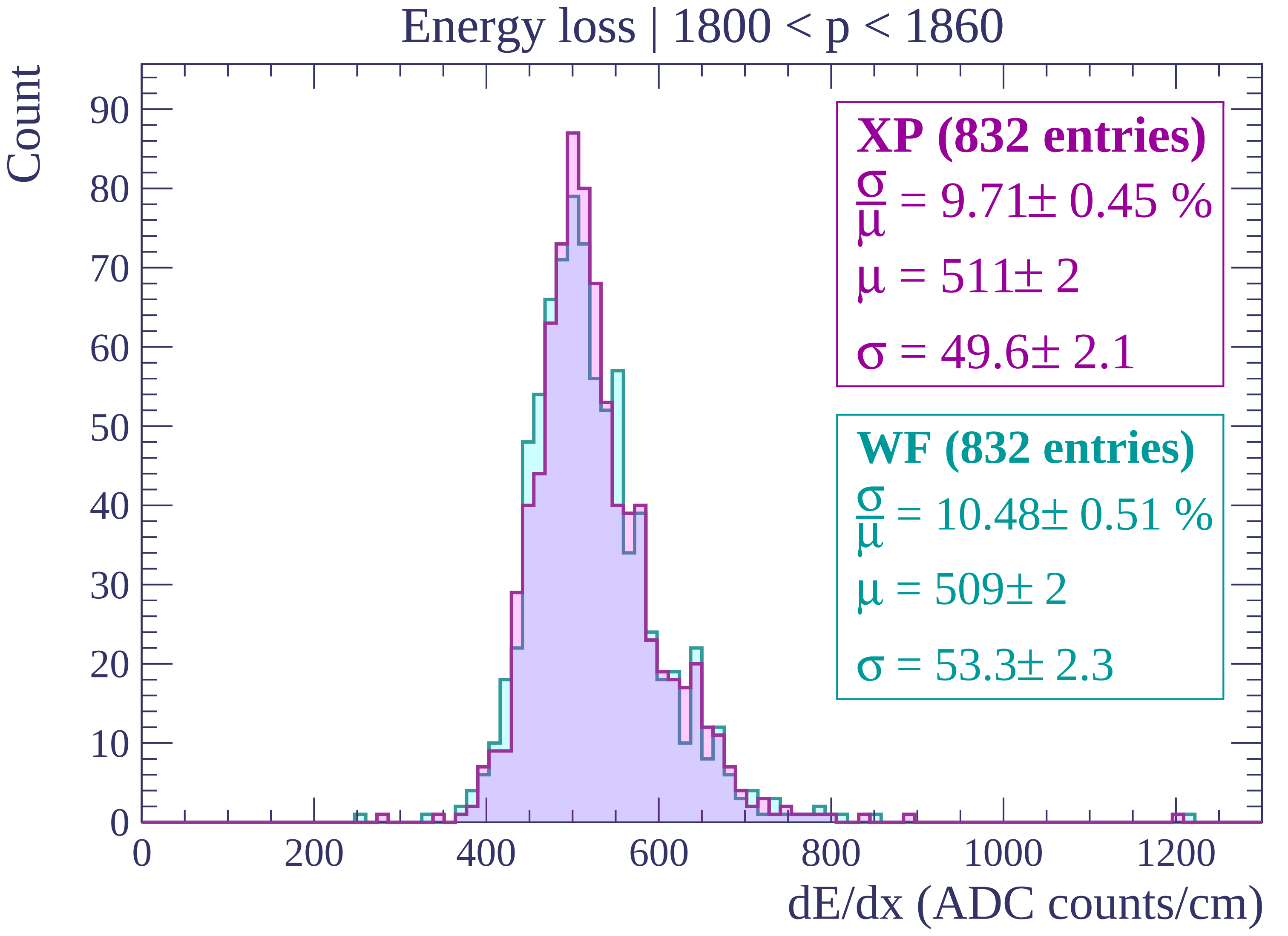


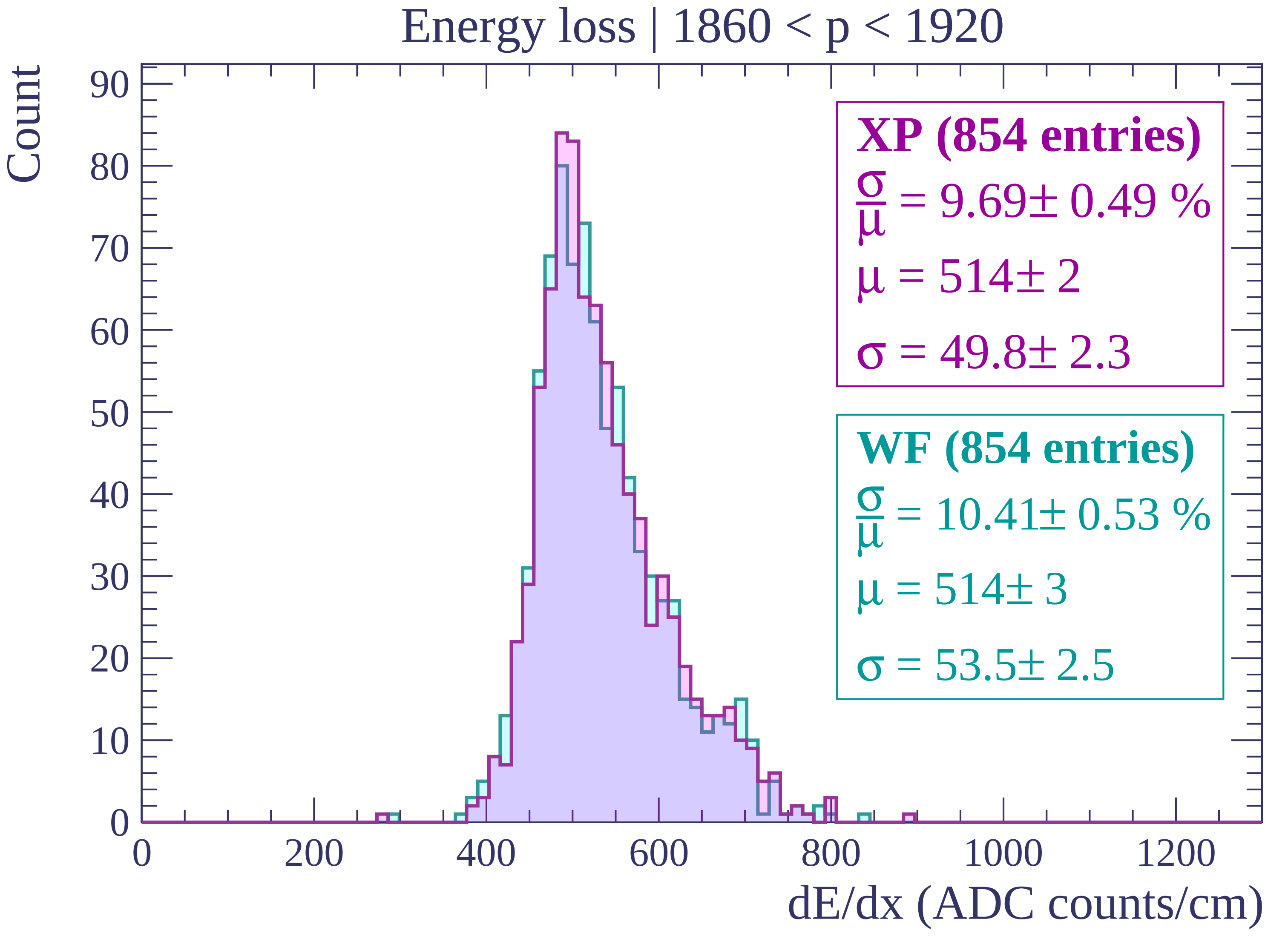
Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (829 entries)
$\frac{\sigma}{\mu}$ = 8.98± 0.43 %
μ = 496± 2
σ = 44.5± 1.9
WF (829 entries)
$\frac{\sigma}{\mu}$ = 10.05± 0.60 %
μ = 496± 2
σ = 49.8± 2.7

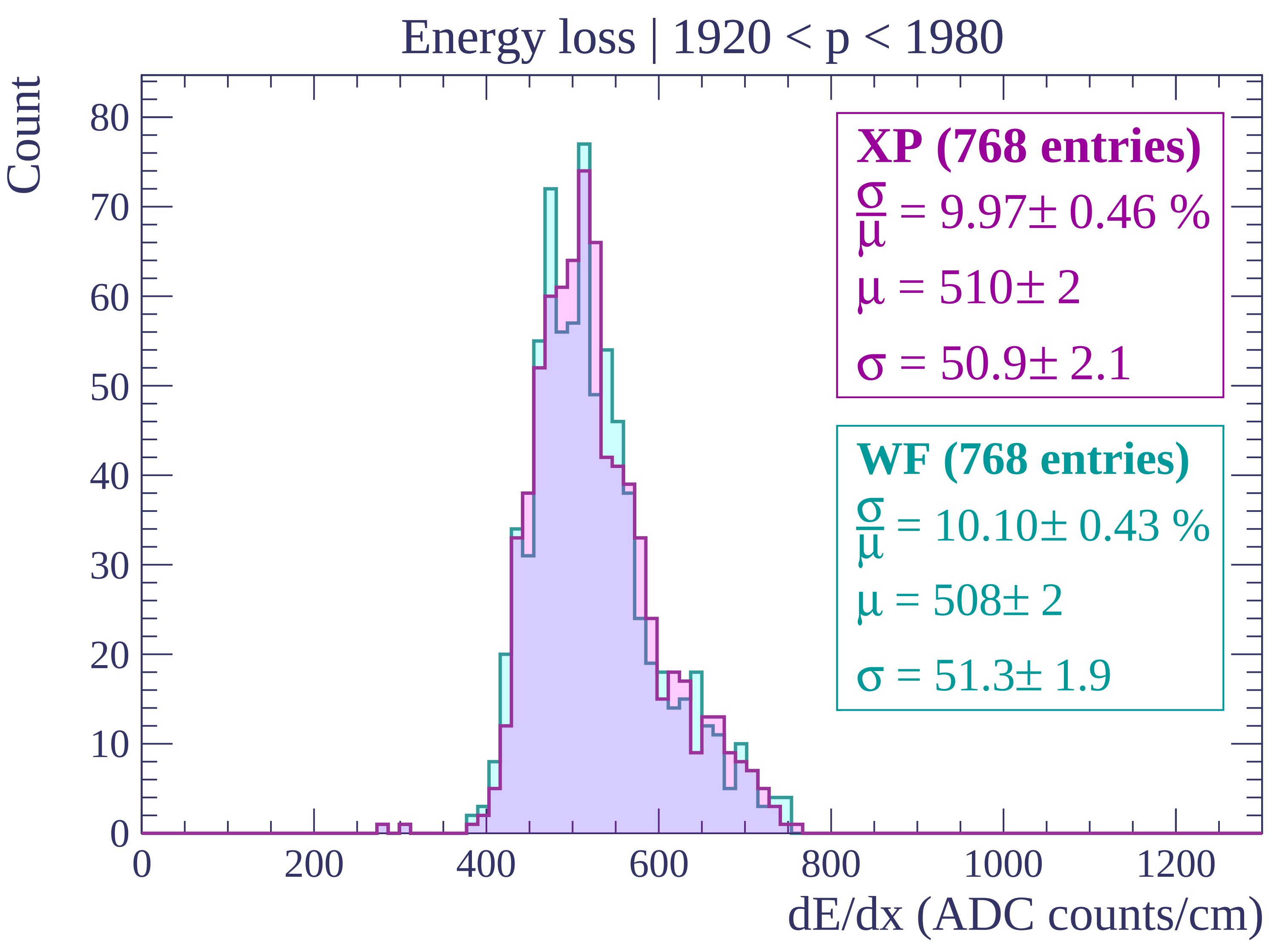

Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (884 entries)
σ/μ = 9.36± 0.48 %
μ = 497± 2
σ = 46.6± 2.2
WF (884 entries)
σ/μ = 10.43± 0.52 %
μ = 500± 3
σ = 52.1± 2.3

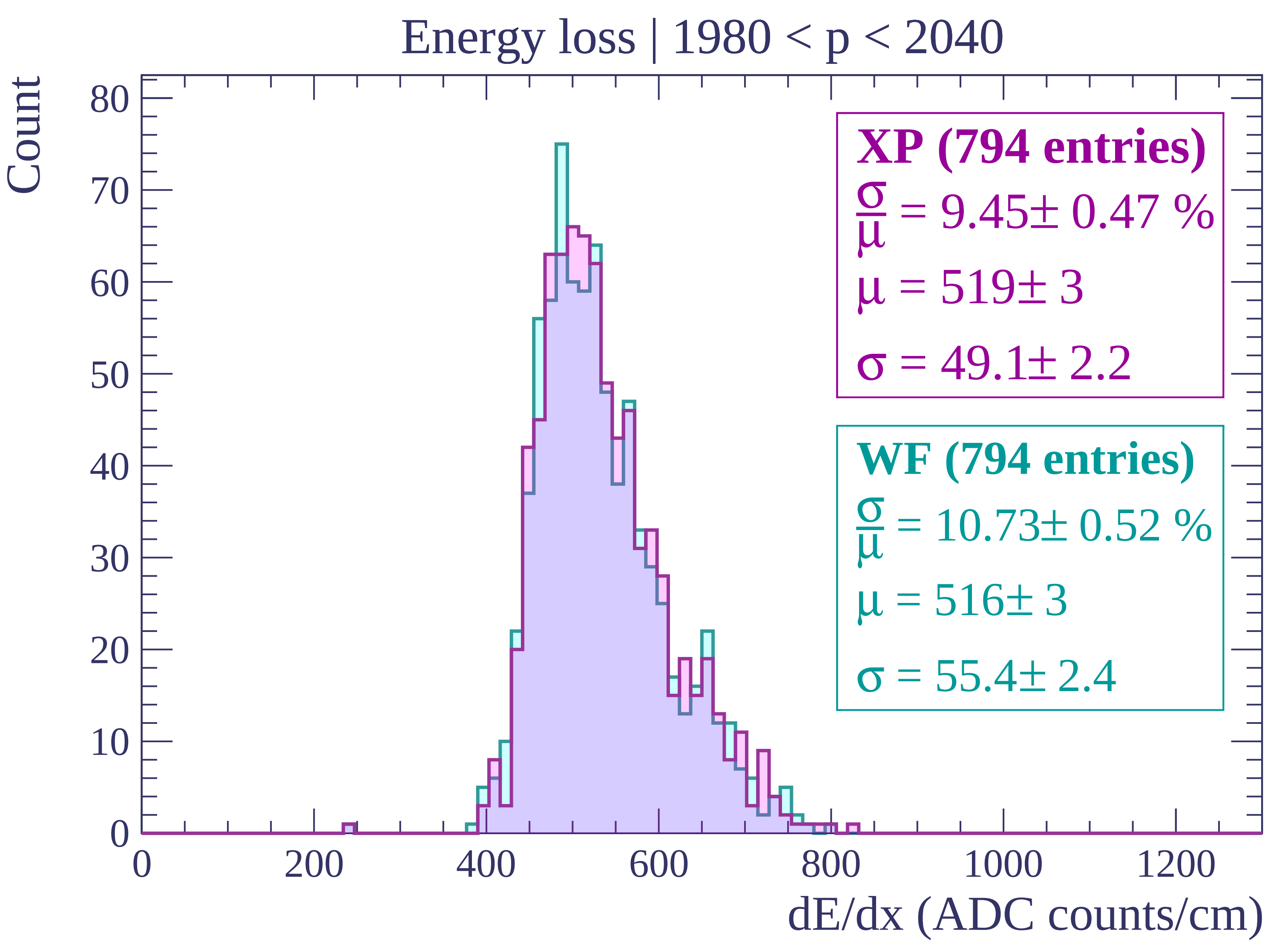


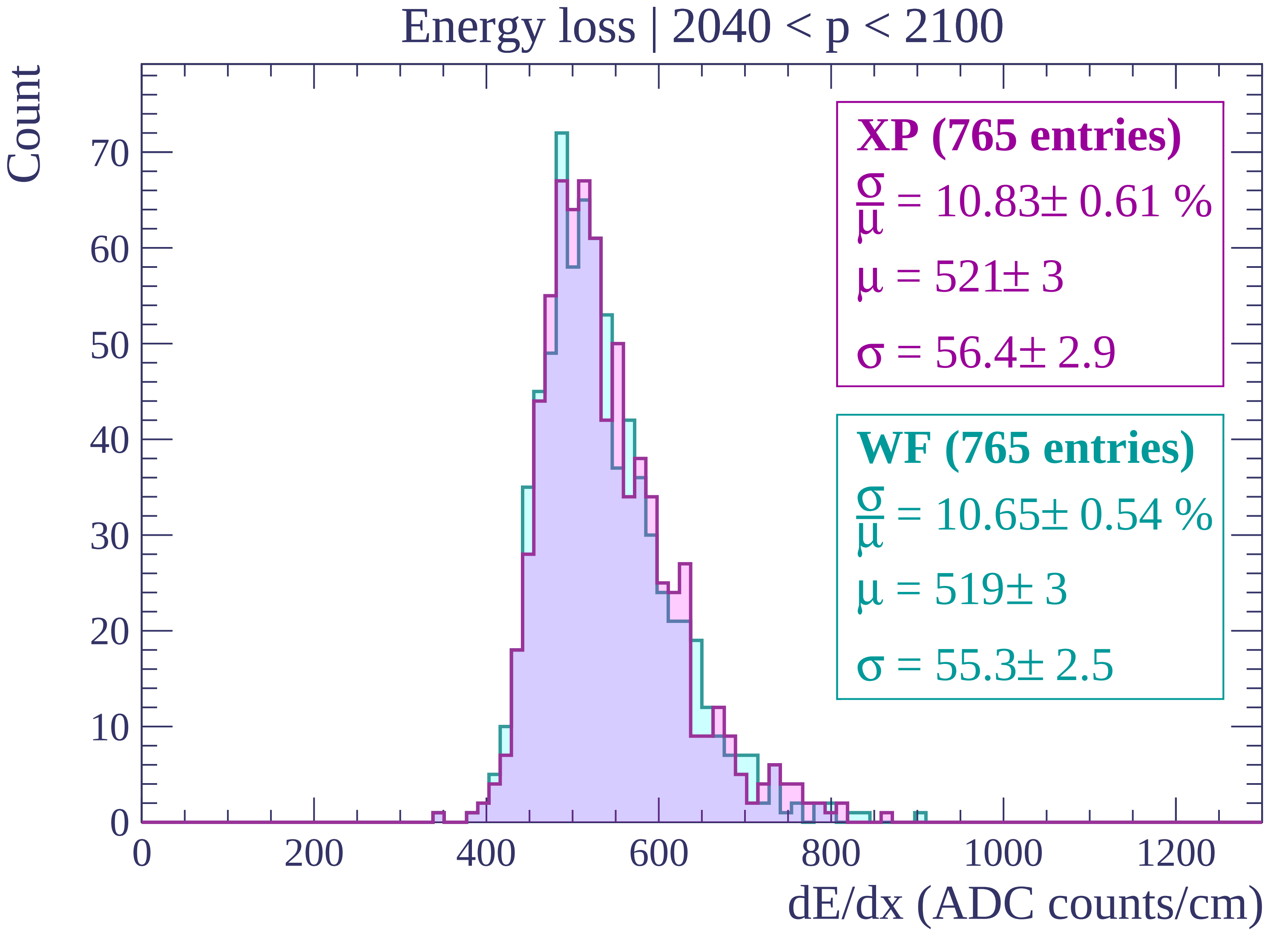
Energy loss | 1740 < p < 1800
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (833 entries)
$\frac{\sigma}{\mu}$ = 9.69± 0.50 %
μ = 505± 2
σ = 49.0± 2.3
WF (833 entries)
$\frac{\sigma}{\mu}$ = 10.44± 0.60 %
μ = 503± 3
σ = 52.5± 2.8


Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (832 entries)
σ/μ = 9.71± 0.45 %
μ = 511± 2
σ = 49.6± 2.1
WF (832 entries)
σ/μ = 10.48± 0.51 %
μ = 509± 2
σ = 53.3± 2.3

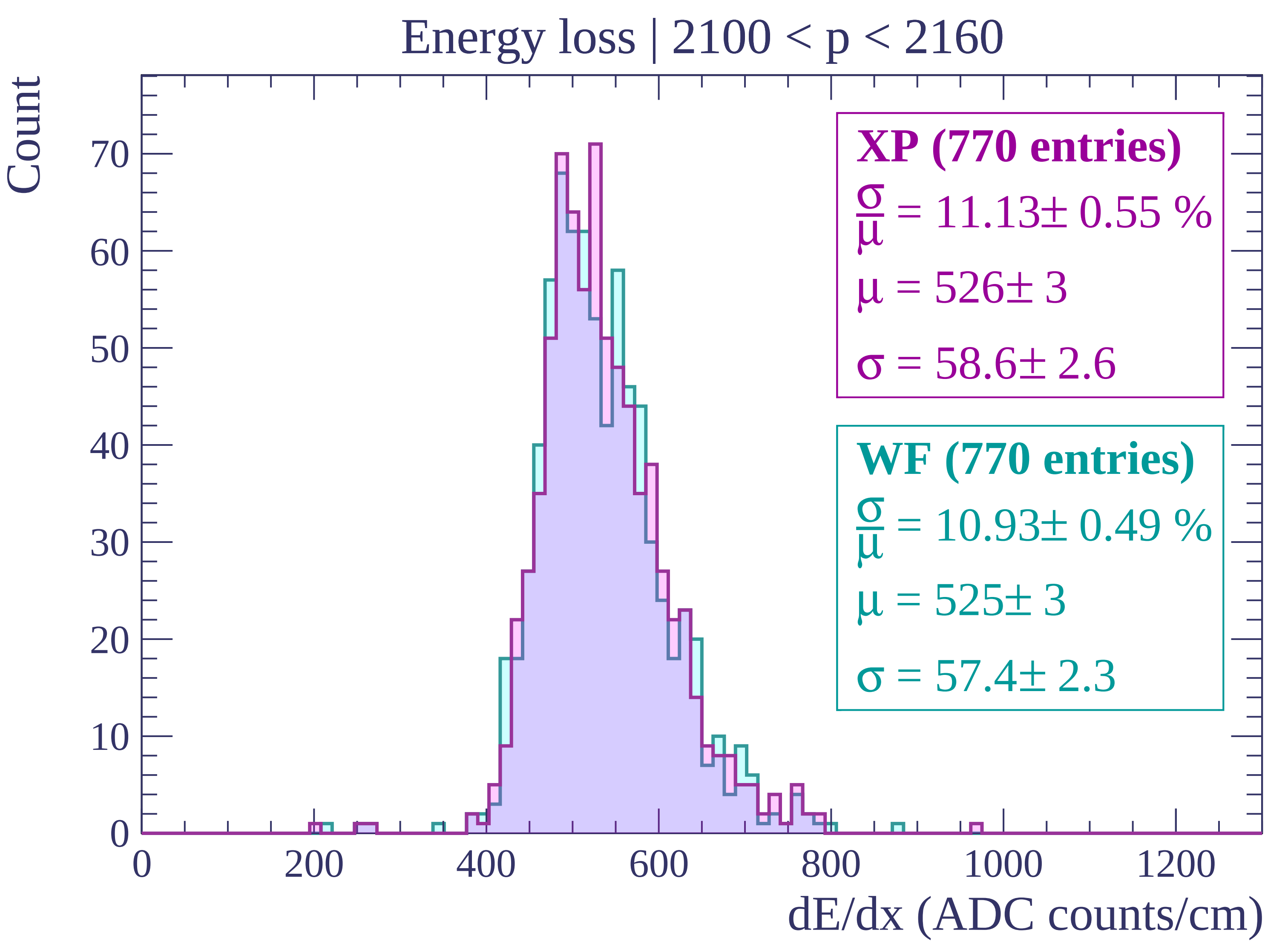

Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (854 entries)
σ/μ = 9.69± 0.49 %
μ = 514± 2
σ = 49.8± 2.3
WF (854 entries)
σ/μ = 10.41± 0.53 %
μ = 514± 3
σ = 53.5± 2.5


Energy loss | 1920 < p < 1980
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (768 entries)
σ/μ = 9.97± 0.46 %
μ = 510± 2
σ = 50.9± 2.1
WF (768 entries)
σ/μ = 10.10± 0.43 %
μ = 508± 2
σ = 51.3± 1.9

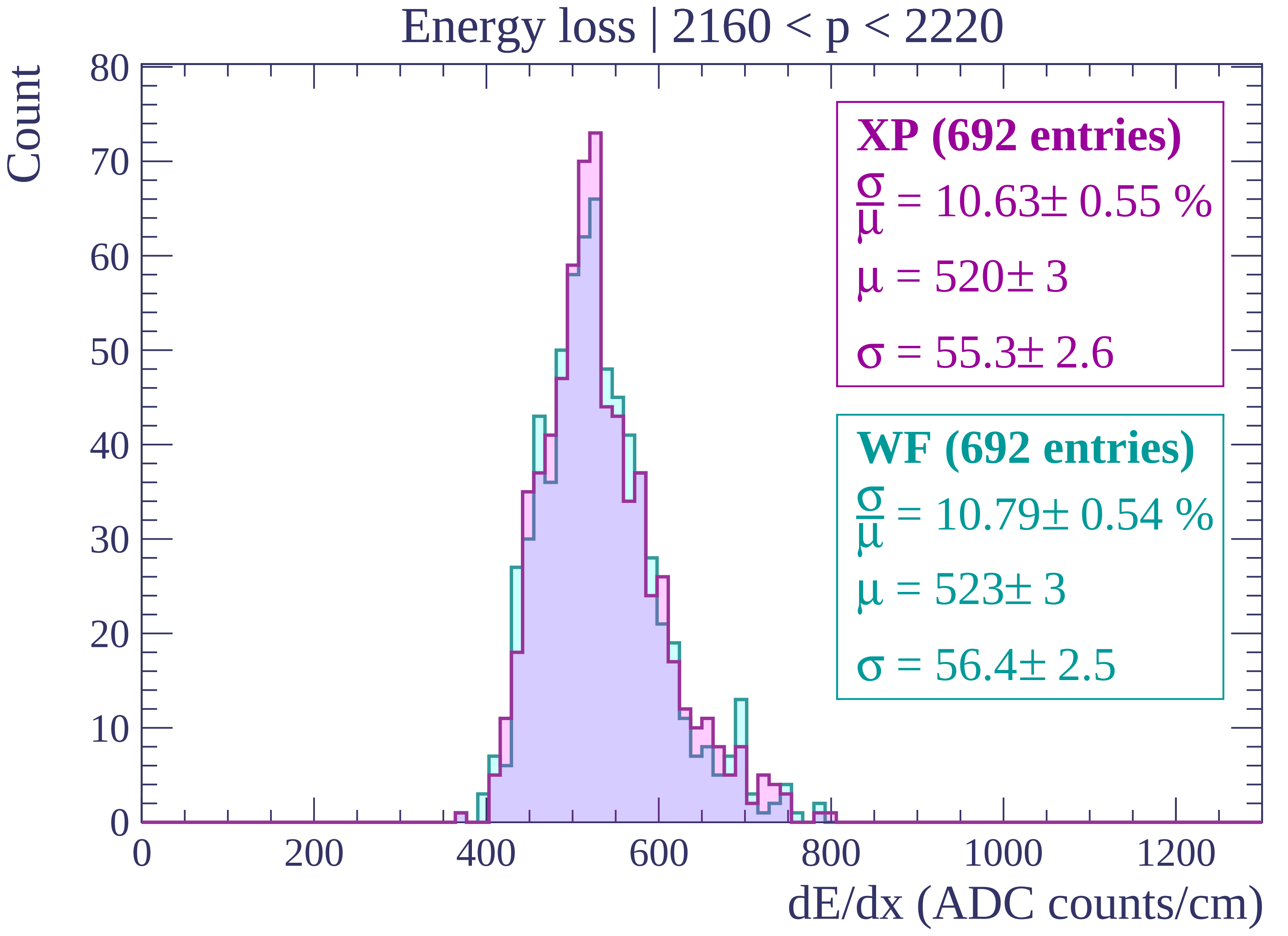

Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (794 entries)
$\frac{\sigma}{\mu}$ = 9.45± 0.47 %
μ = 519± 3
σ = 49.1± 2.2
WF (794 entries)
$\frac{\sigma}{\mu}$ = 10.73± 0.52 %
μ = 516± 3
σ = 55.4± 2.4

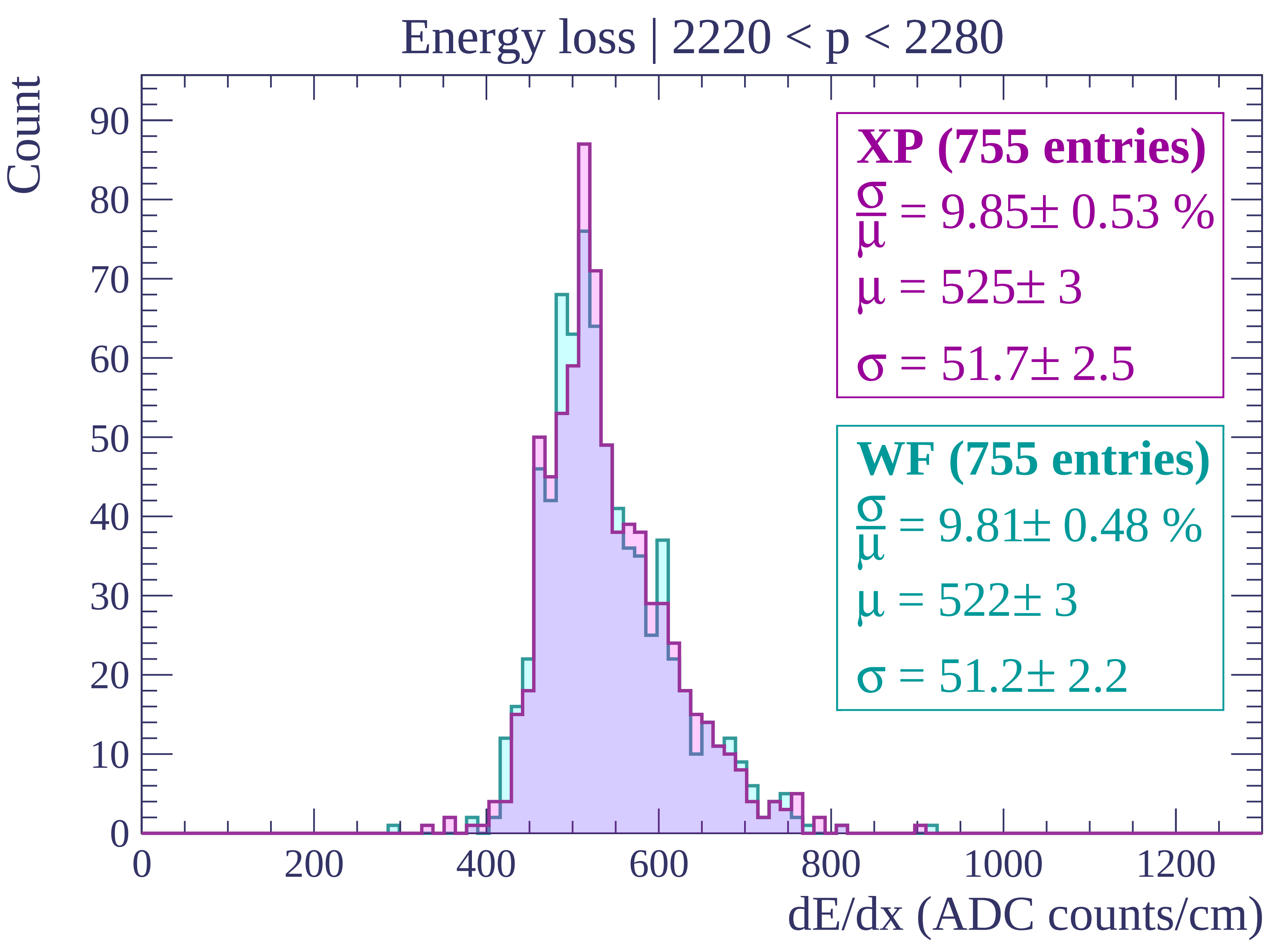


Energy loss | 2040 < p < 2100
Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (765 entries)
σ/μ = 10.83± 0.61 %
μ = 521± 3
σ = 56.4± 2.9
WF (765 entries)
σ/μ = 10.65± 0.54 %
μ = 519± 3
σ = 55.3± 2.5

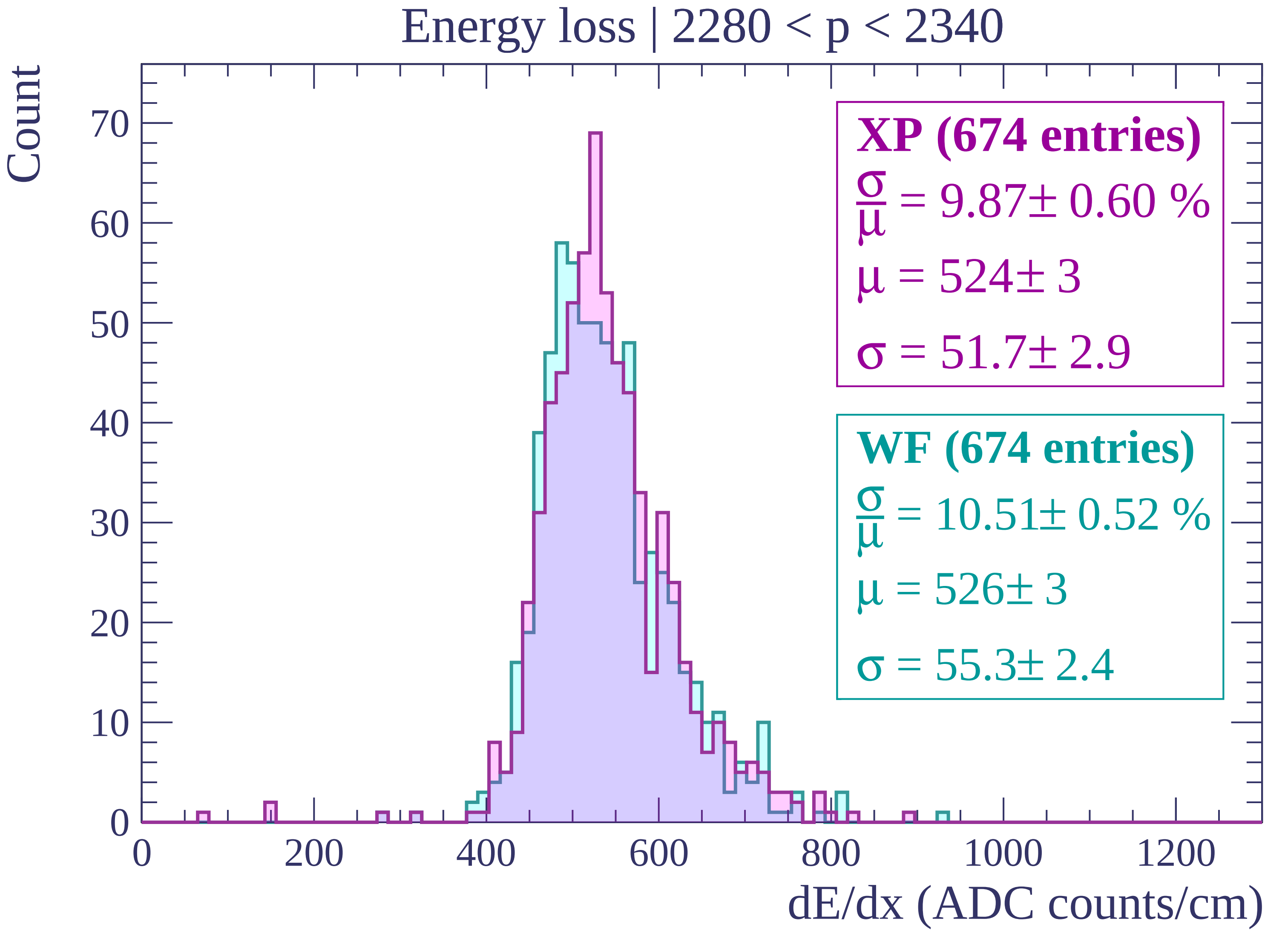


Energy loss | 2100 < p < 2160
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (770 entries)
σ/μ = 11.13± 0.55 %
μ = 526± 3
σ = 58.6± 2.6
WF (770 entries)
σ/μ = 10.93± 0.49 %
μ = 525± 3
σ = 57.4± 2.3

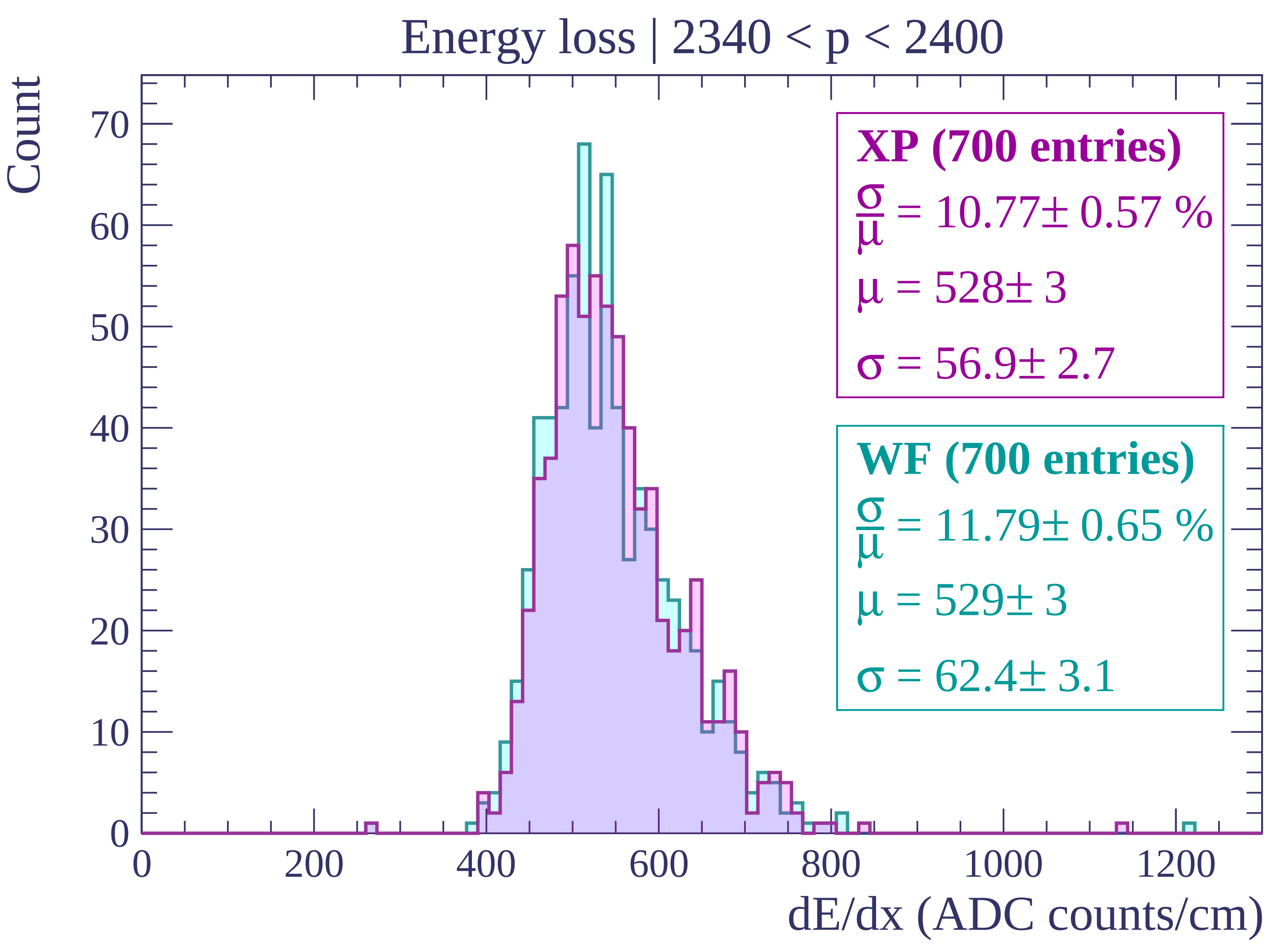

Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP (692 entries)
σ/μ = 10.63± 0.55 %
μ = 520± 3
σ = 55.3± 2.6
WF (692 entries)
σ/μ = 10.79± 0.54 %
μ = 523± 3
σ = 56.4± 2.5



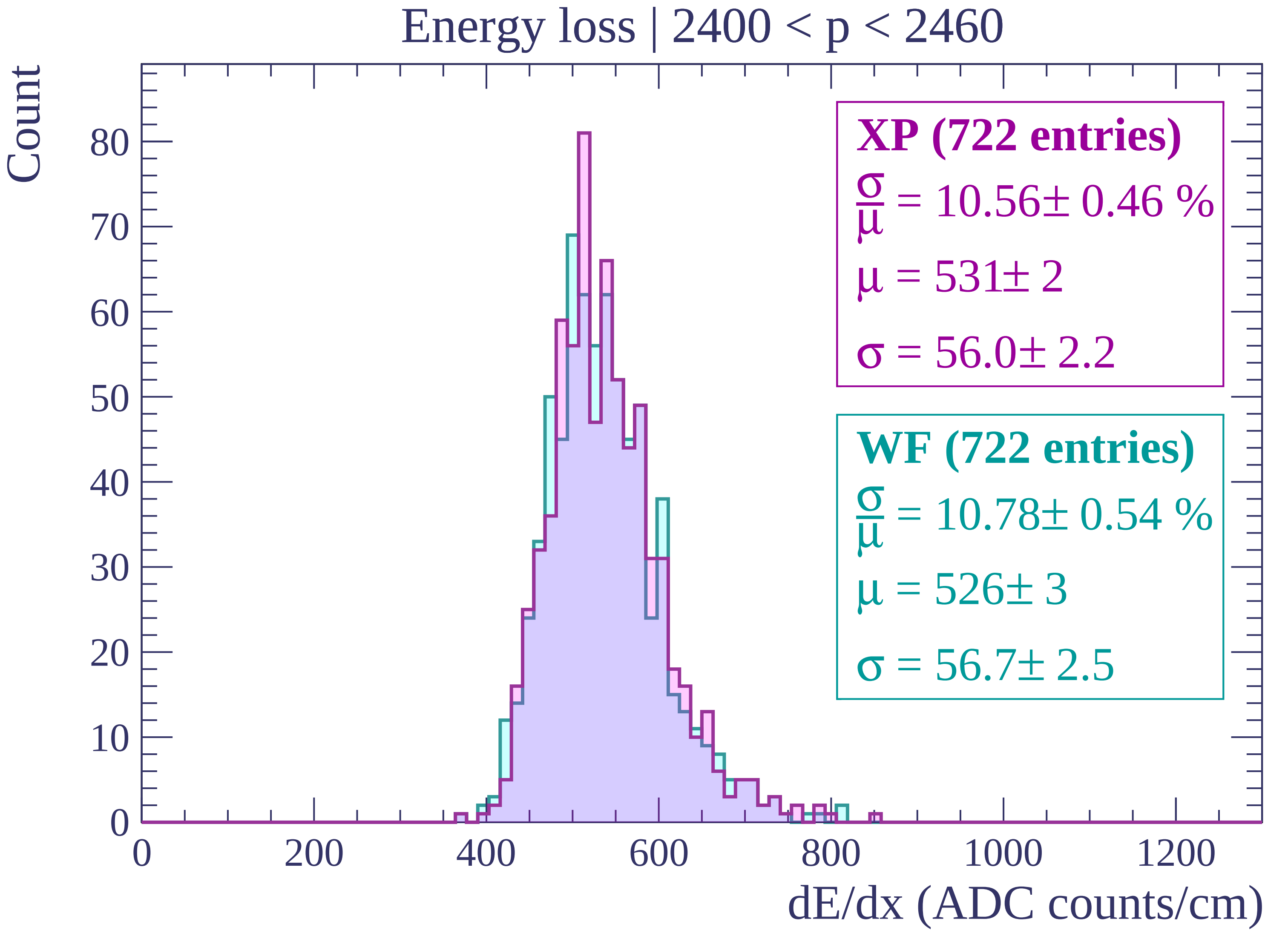
Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (755 entries)
$\frac{\sigma}{\mu}$ = 9.85± 0.53 %
μ = 525± 3
σ = 51.7± 2.5
WF (755 entries)
$\frac{\sigma}{\mu}$ = 9.81± 0.48 %
μ = 522± 3
σ = 51.2± 2.2



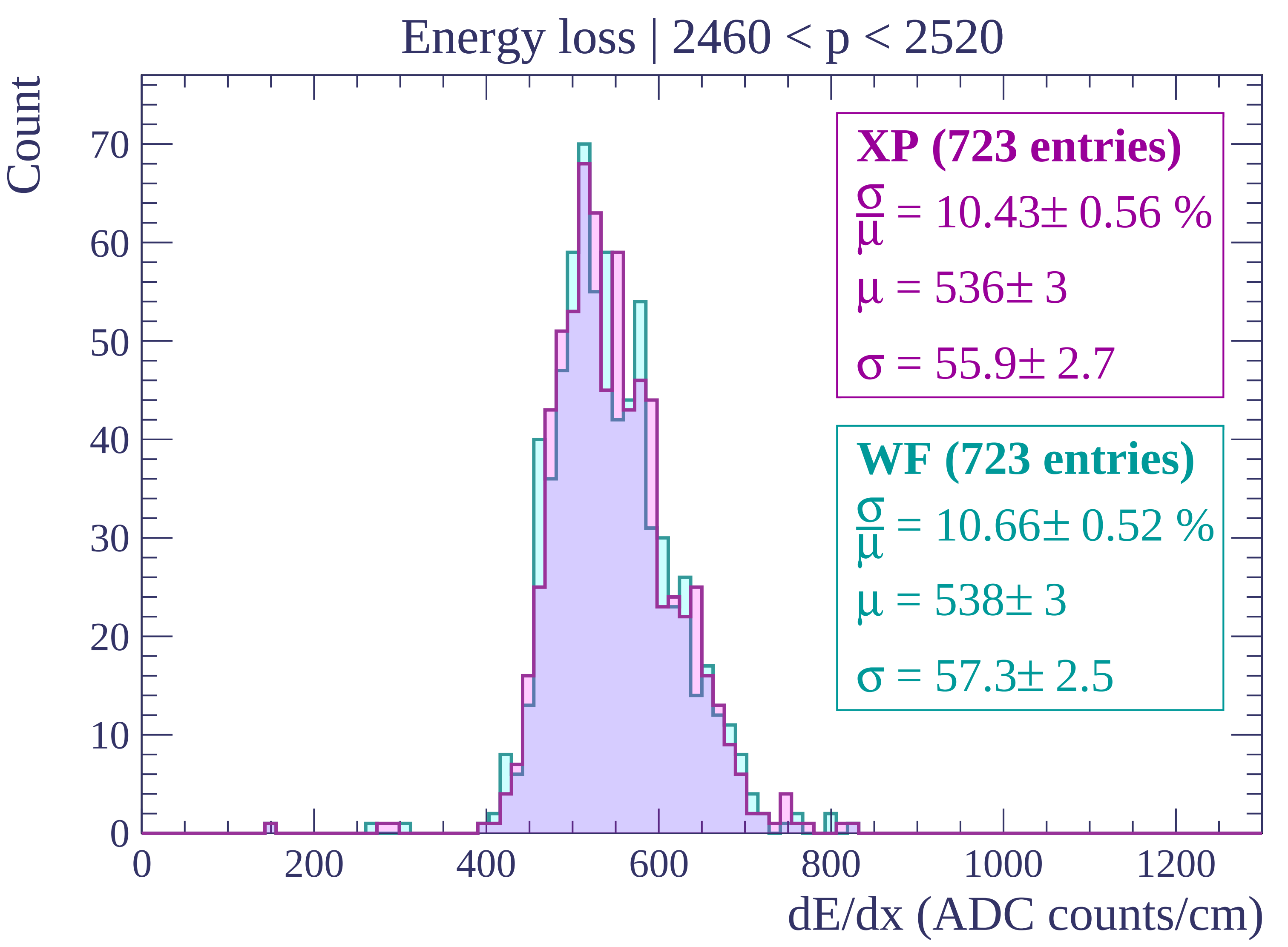
Energy loss | 2280 < p < 2340
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (674 entries)
σ/μ = 9.87± 0.60 %
μ = 524± 3
σ = 51.7± 2.9
WF (674 entries)
σ/μ = 10.51± 0.52 %
μ = 526± 3
σ = 55.3± 2.4



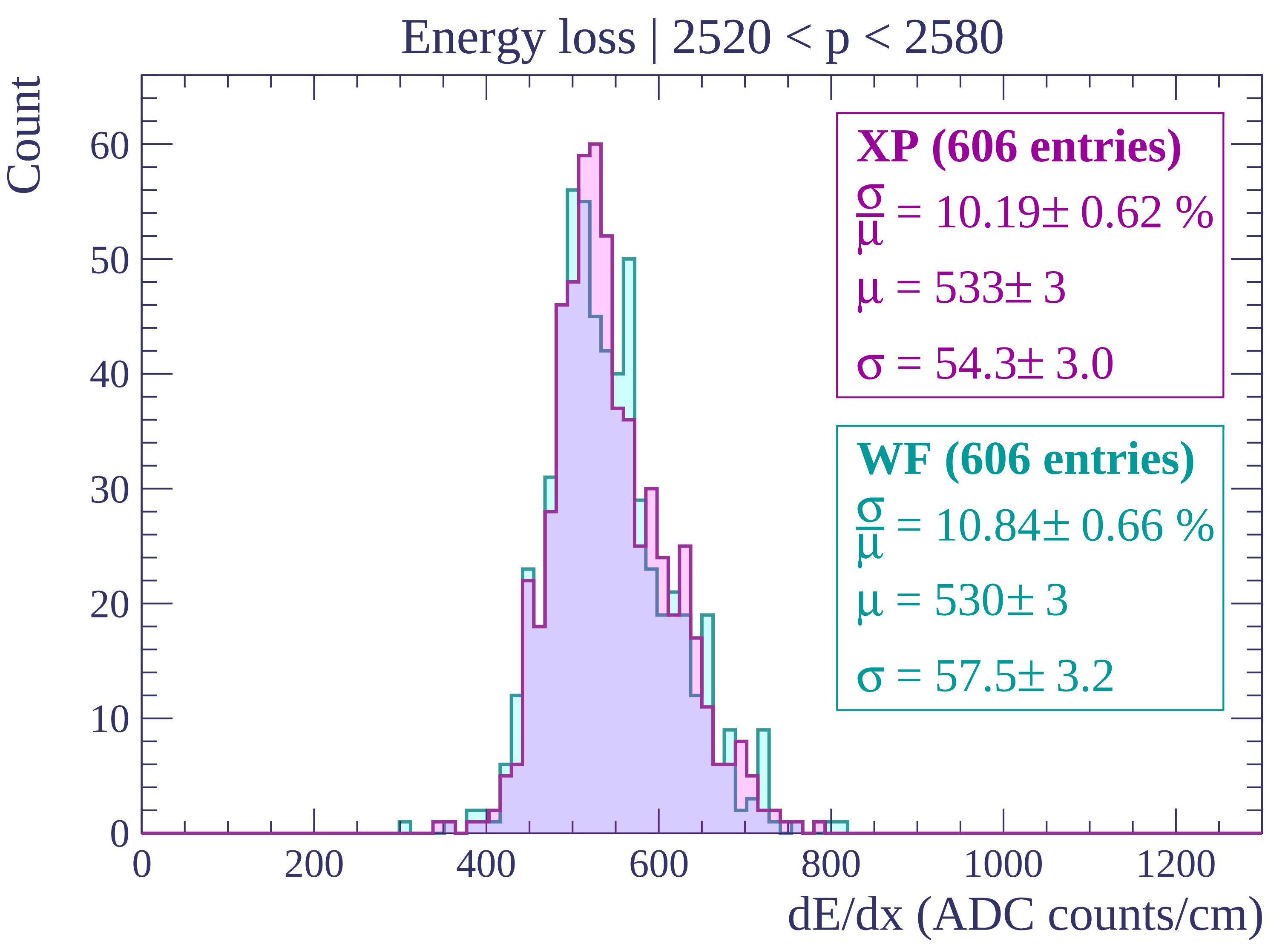
Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (700 entries)
σ/μ = 10.77± 0.57 %
μ = 528± 3
σ = 56.9± 2.7
WF (700 entries)
σ/μ = 11.79± 0.65 %
μ = 529± 3
σ = 62.4± 3.1

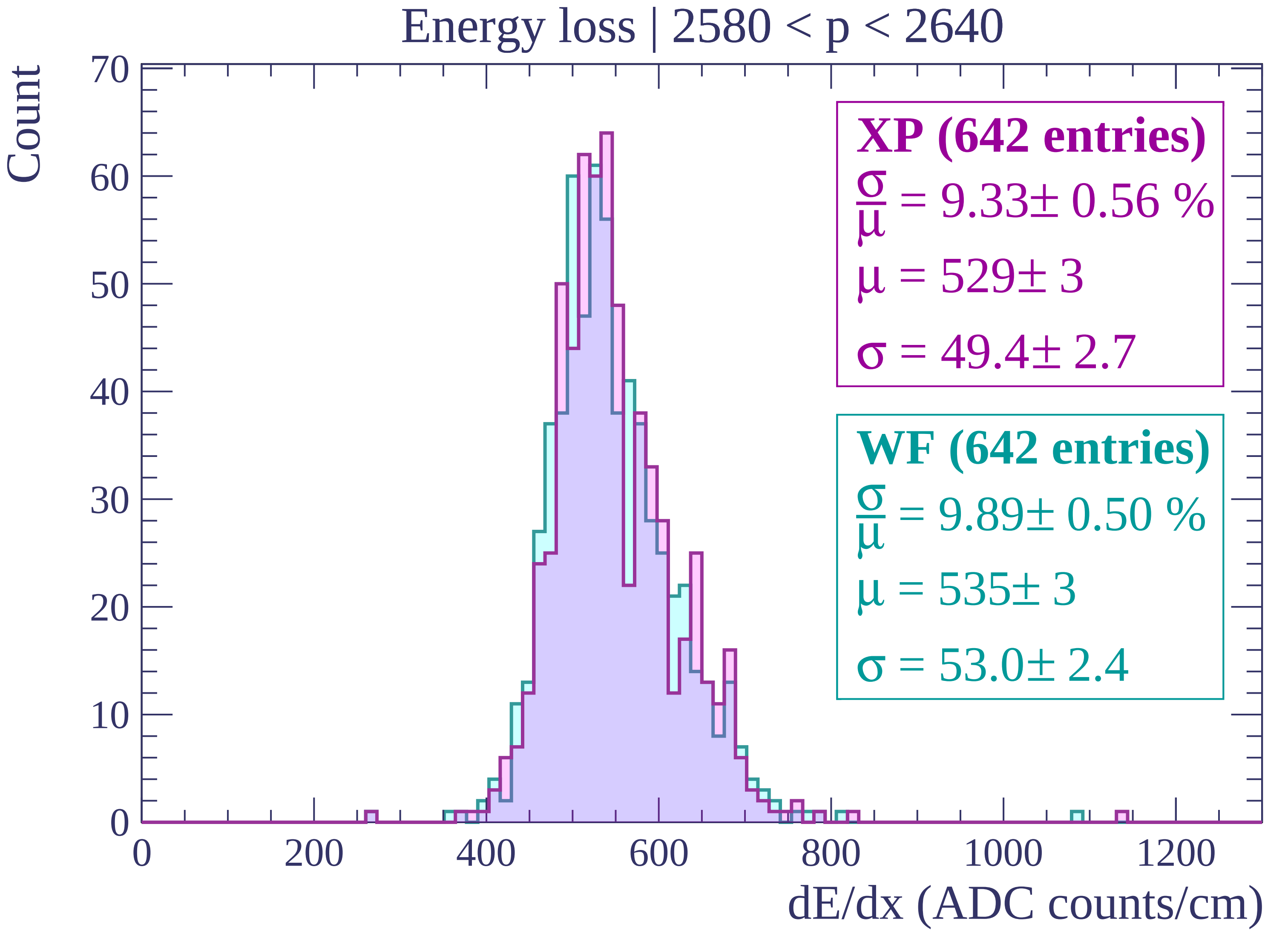


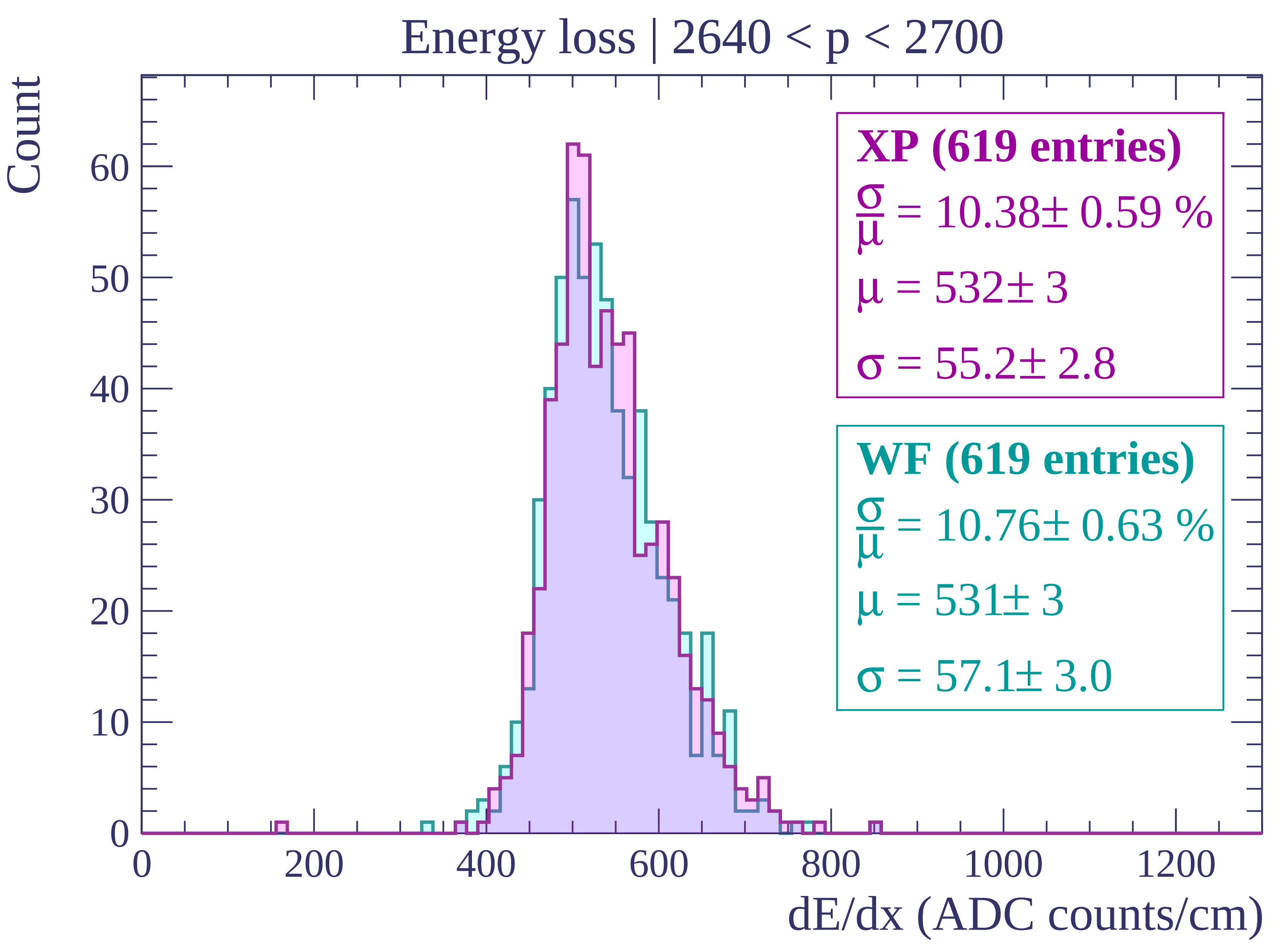
Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
XP (722 entries)
$\frac{\sigma}{\mu}$ = 10.56± 0.46 %
μ = 531± 2
σ = 56.0± 2.2
WF (722 entries)
$\frac{\sigma}{\mu}$ = 10.78± 0.54 %
μ = 526± 3
σ = 56.7± 2.5
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



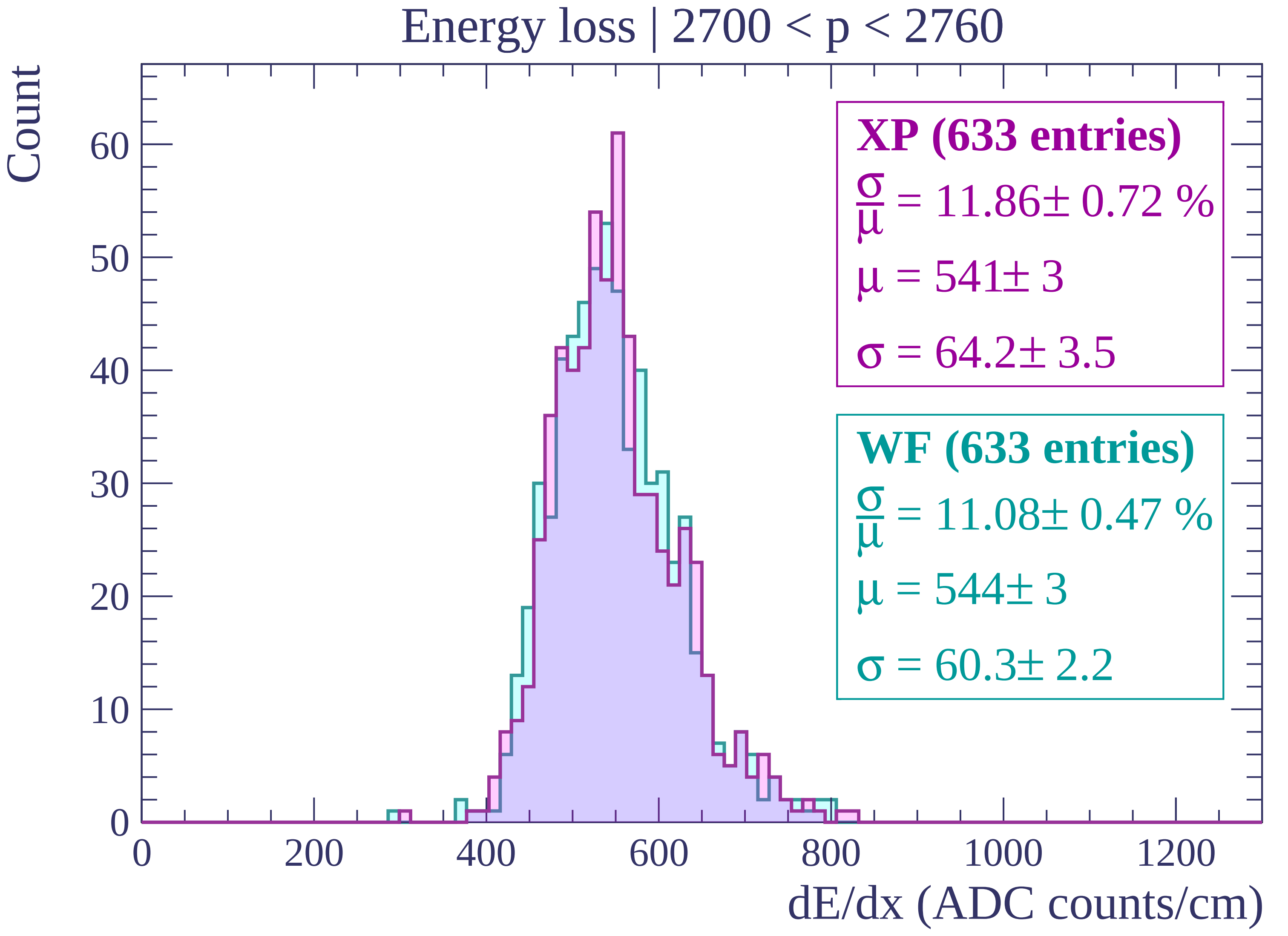
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (723 entries)
$\frac{\sigma}{\mu}$ = 10.43± 0.56 %
μ = 536± 3
σ = 55.9± 2.7
WF (723 entries)
$\frac{\sigma}{\mu}$ = 10.66± 0.52 %
μ = 538± 3
σ = 57.3± 2.5


Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (606 entries)
σ/μ = 10.19± 0.62 %
μ = 533± 3
σ = 54.3± 3.0
WF (606 entries)
σ/μ = 10.84± 0.66 %
μ = 530± 3
σ = 57.5± 3.2

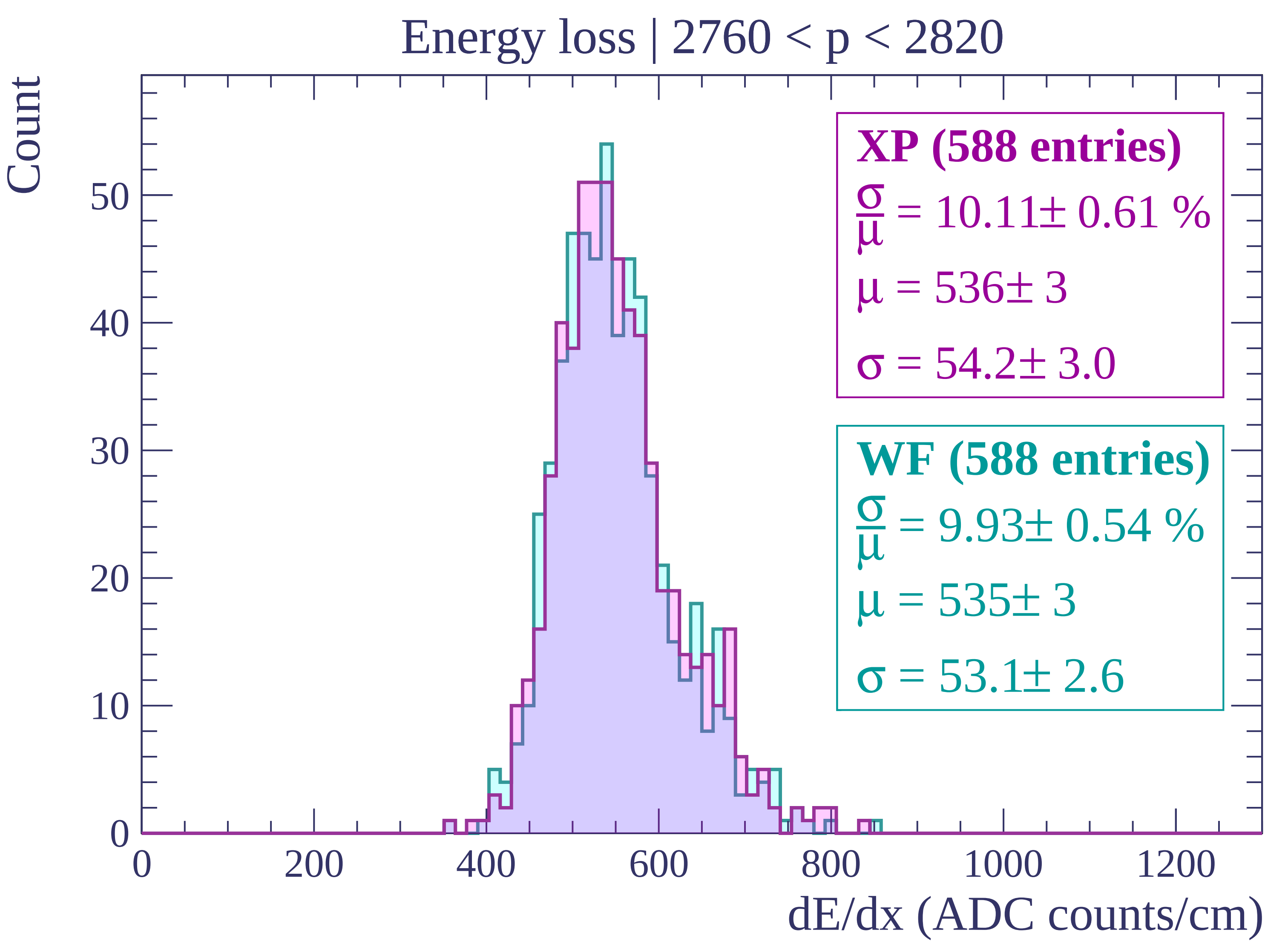

Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (642 entries)
σ/μ = 9.33± 0.56 %
μ = 529± 3
σ = 49.4± 2.7
WF (642 entries)
σ/μ = 9.89± 0.50 %
μ = 535± 3
σ = 53.0± 2.4



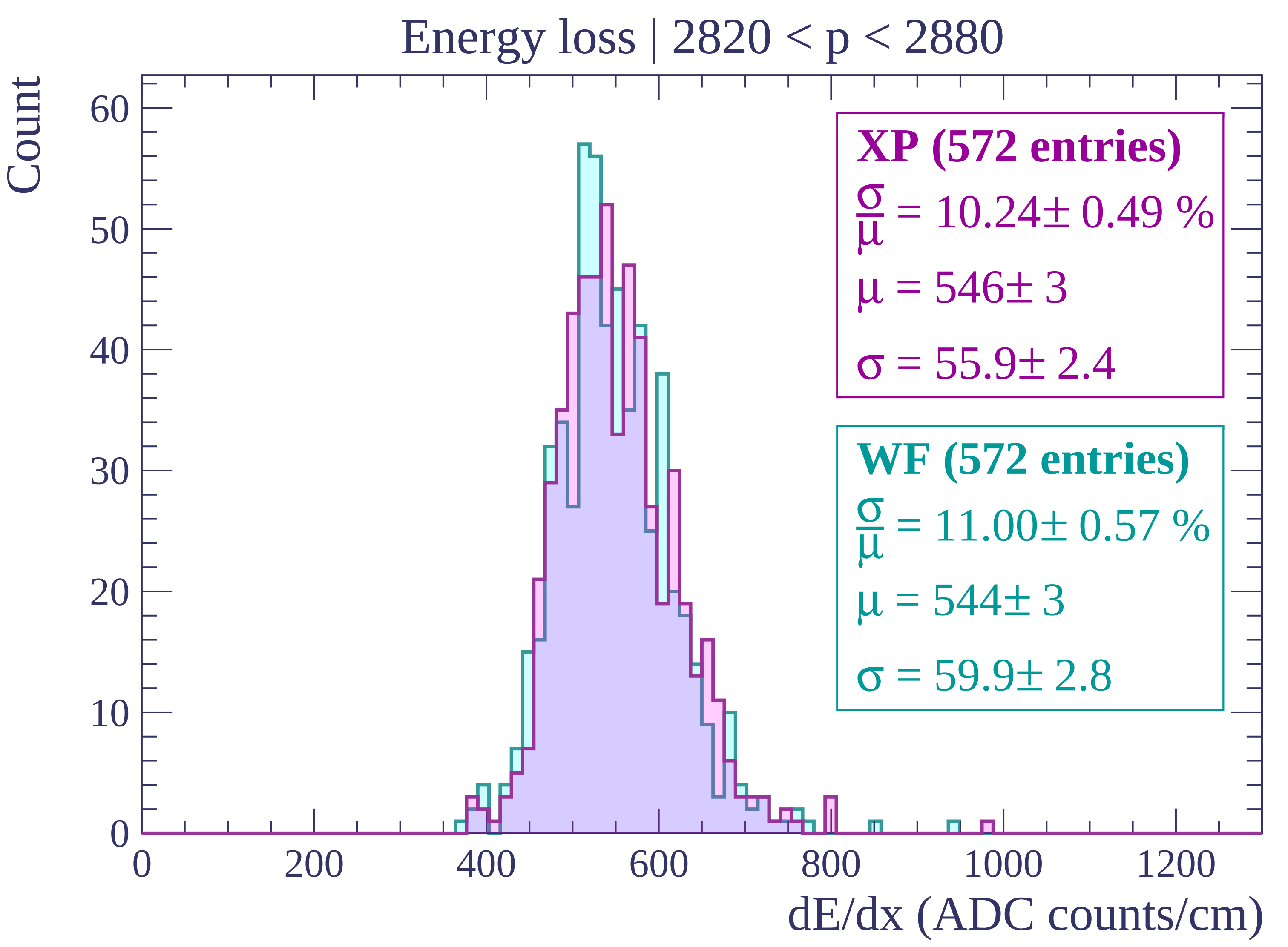
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
XP (619 entries)
σ/μ = 10.38± 0.59 %
μ = 532± 3
σ = 55.2± 2.8
WF (619 entries)
σ/μ = 10.76± 0.63 %
μ = 531± 3
σ = 57.1± 3.0

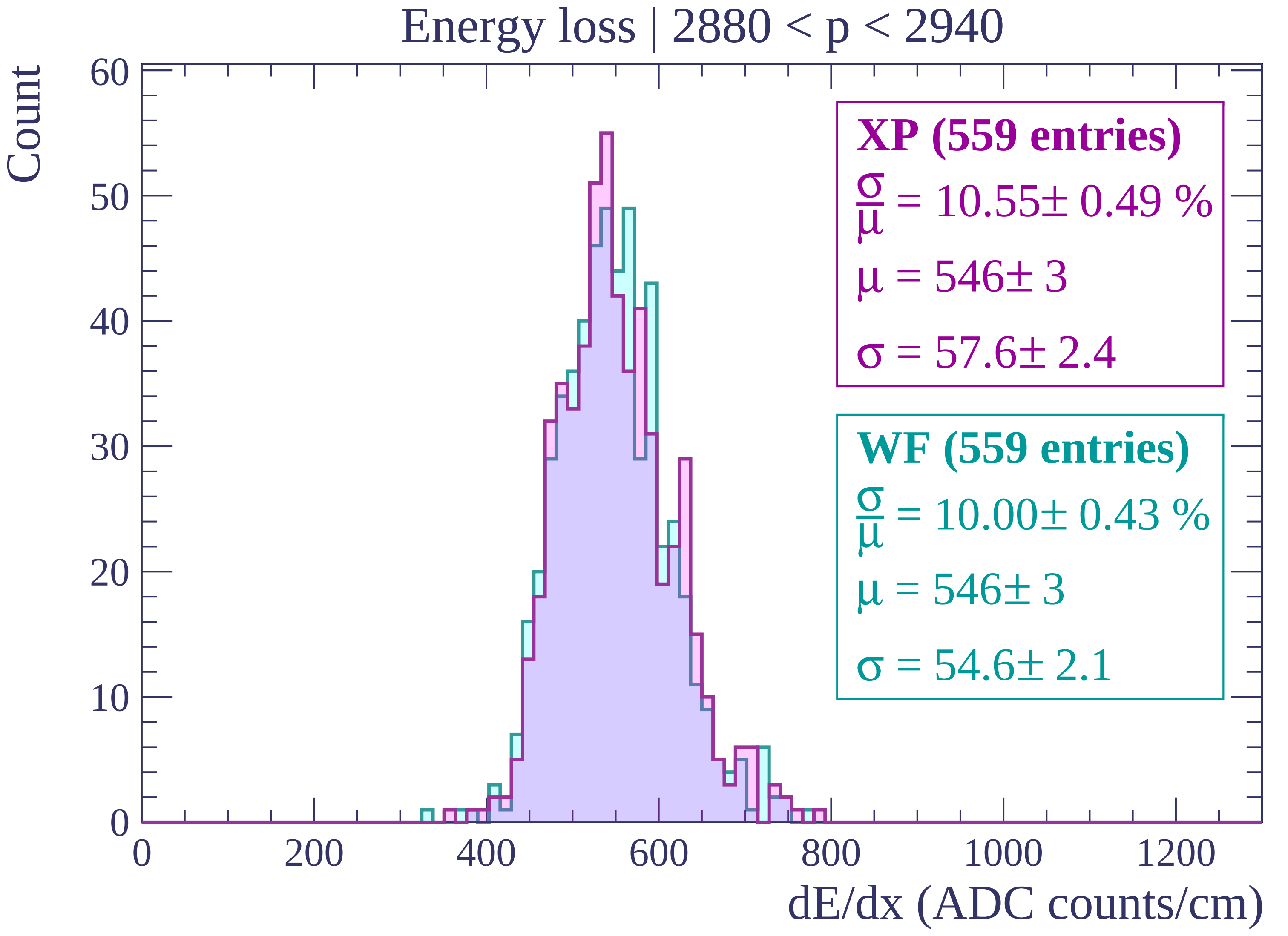


Energy loss | 2700 < p < 2760
Count
XP (633 entries)
$\frac{\sigma}{\mu}$ = 11.86± 0.72 %
μ = 541± 3
σ = 64.2± 3.5
WF (633 entries)
$\frac{\sigma}{\mu}$ = 11.08± 0.47 %
μ = 544± 3
σ = 60.3± 2.2
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200

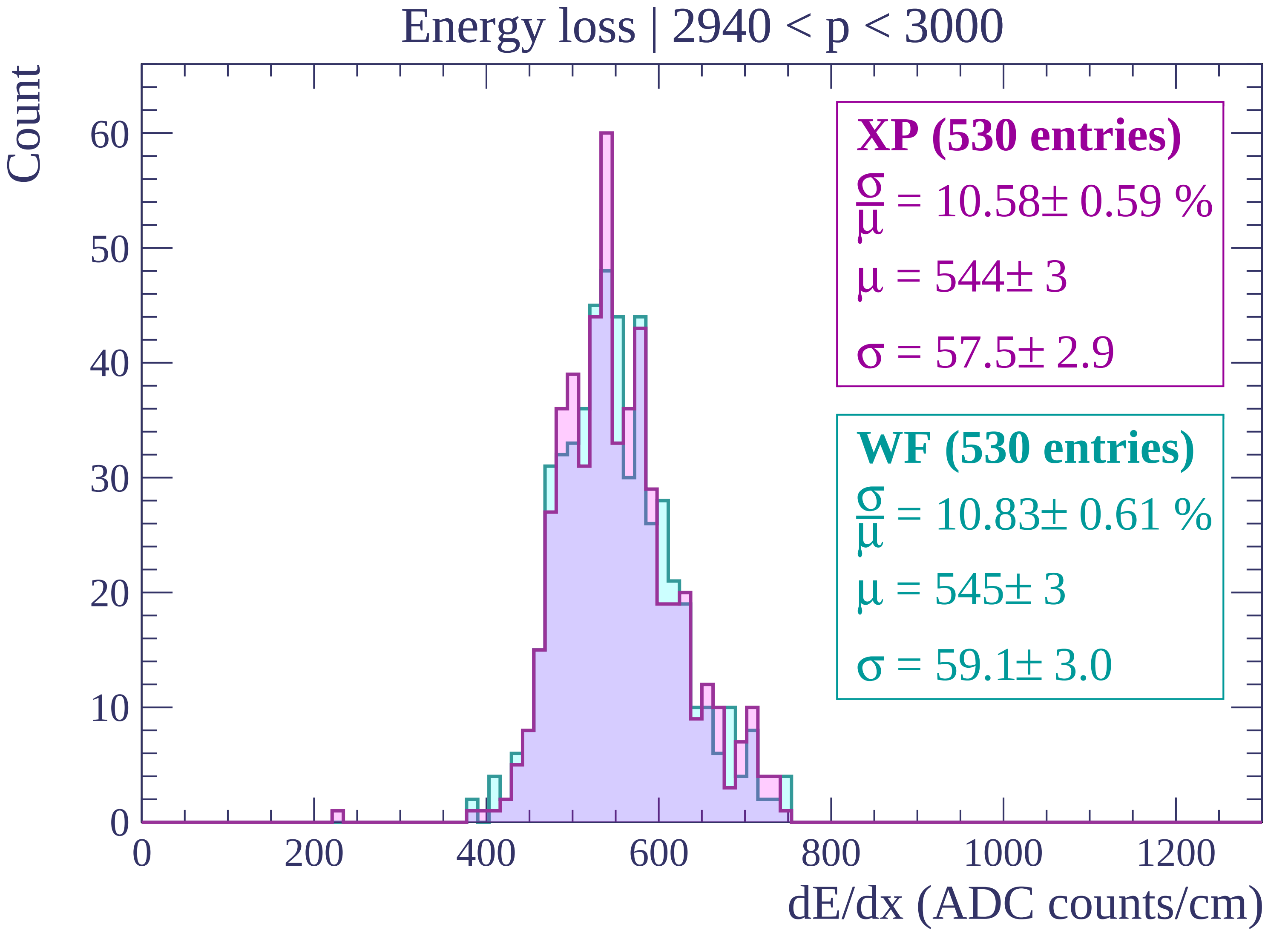


Energy loss | 2760 < p < 2820
Count
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (588 entries)
σ/μ = 10.11± 0.61 %
μ = 536± 3
σ = 54.2± 3.0
WF (588 entries)
σ/μ = 9.93± 0.54 %
μ = 535± 3
σ = 53.1± 2.6

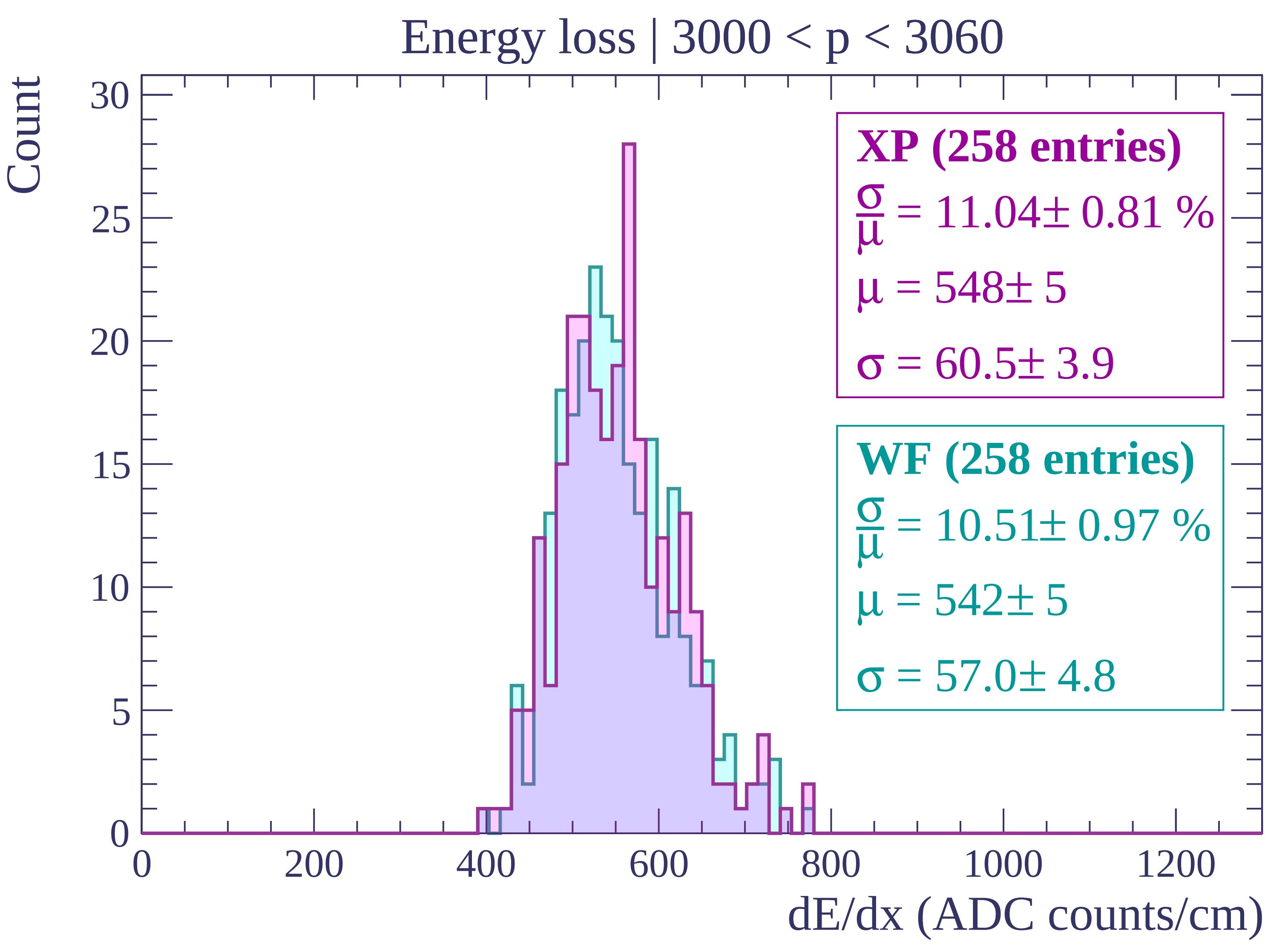

Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (572 entries)
σ/μ = 10.24± 0.49 %
μ = 546± 3
σ = 55.9± 2.4
WF (572 entries)
σ/μ = 11.00± 0.57 %
μ = 544± 3
σ = 59.9± 2.8

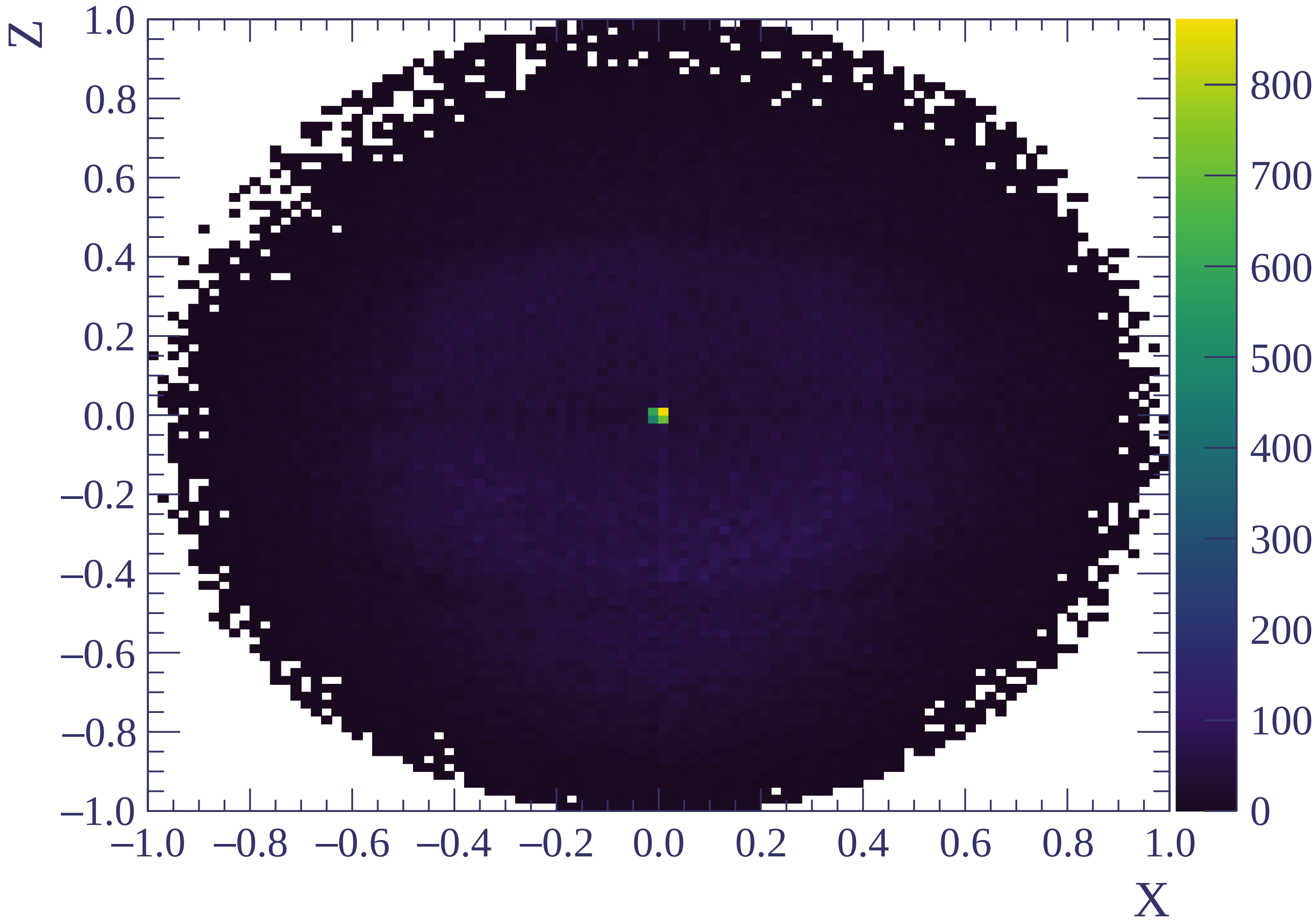


Energy loss | 2880 < p < 2940
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (559 entries)
σ/μ = 10.55± 0.49 %
μ = 546± 3
σ = 57.6± 2.4
WF (559 entries)
σ/μ = 10.00± 0.43 %
μ = 546± 3
σ = 54.6± 2.1

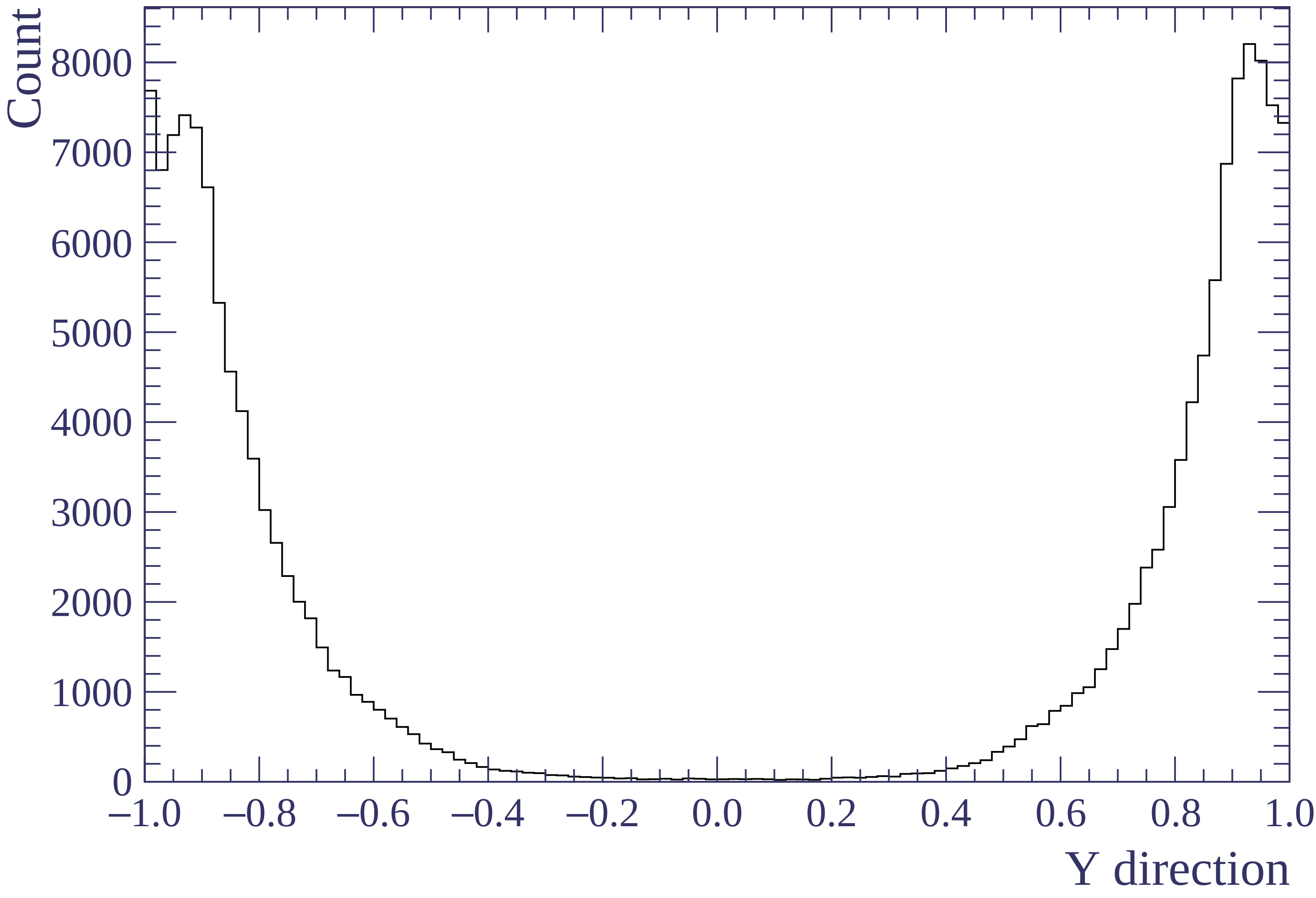


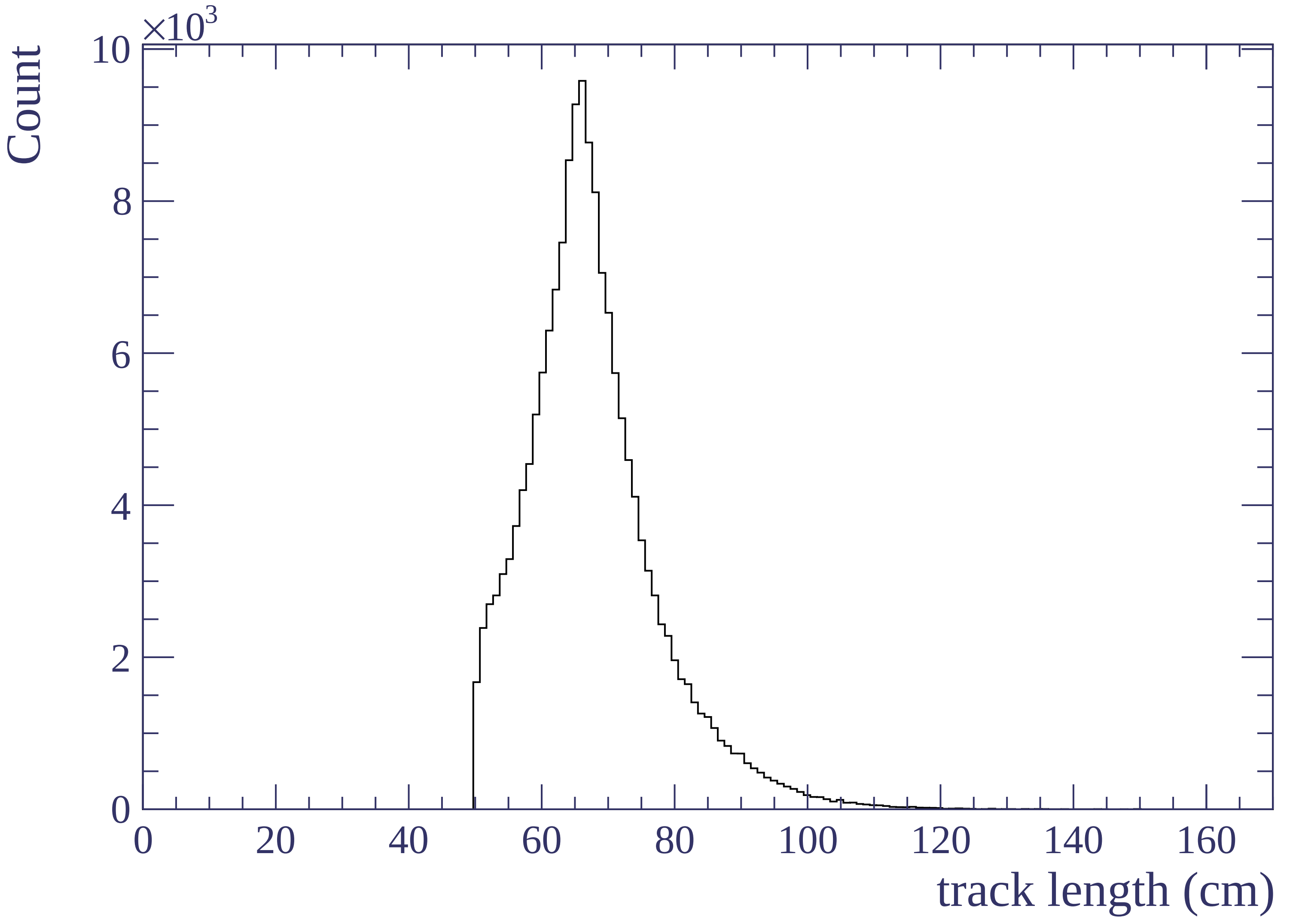
Energy loss | 2940 < p < 3000
Count
XP (530 entries)
σ/μ = 10.58± 0.59 %
μ = 544± 3
σ = 57.5± 2.9
WF (530 entries)
σ/μ = 10.83± 0.61 %
μ = 545± 3
σ = 59.1± 3.0
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)


Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
XP (258 entries)
σ/μ = 11.04± 0.81 %
μ = 548± 5
σ = 60.5± 3.9
WF (258 entries)
σ/μ = 10.51± 0.97 %
μ = 542± 5
σ = 57.0± 4.8
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

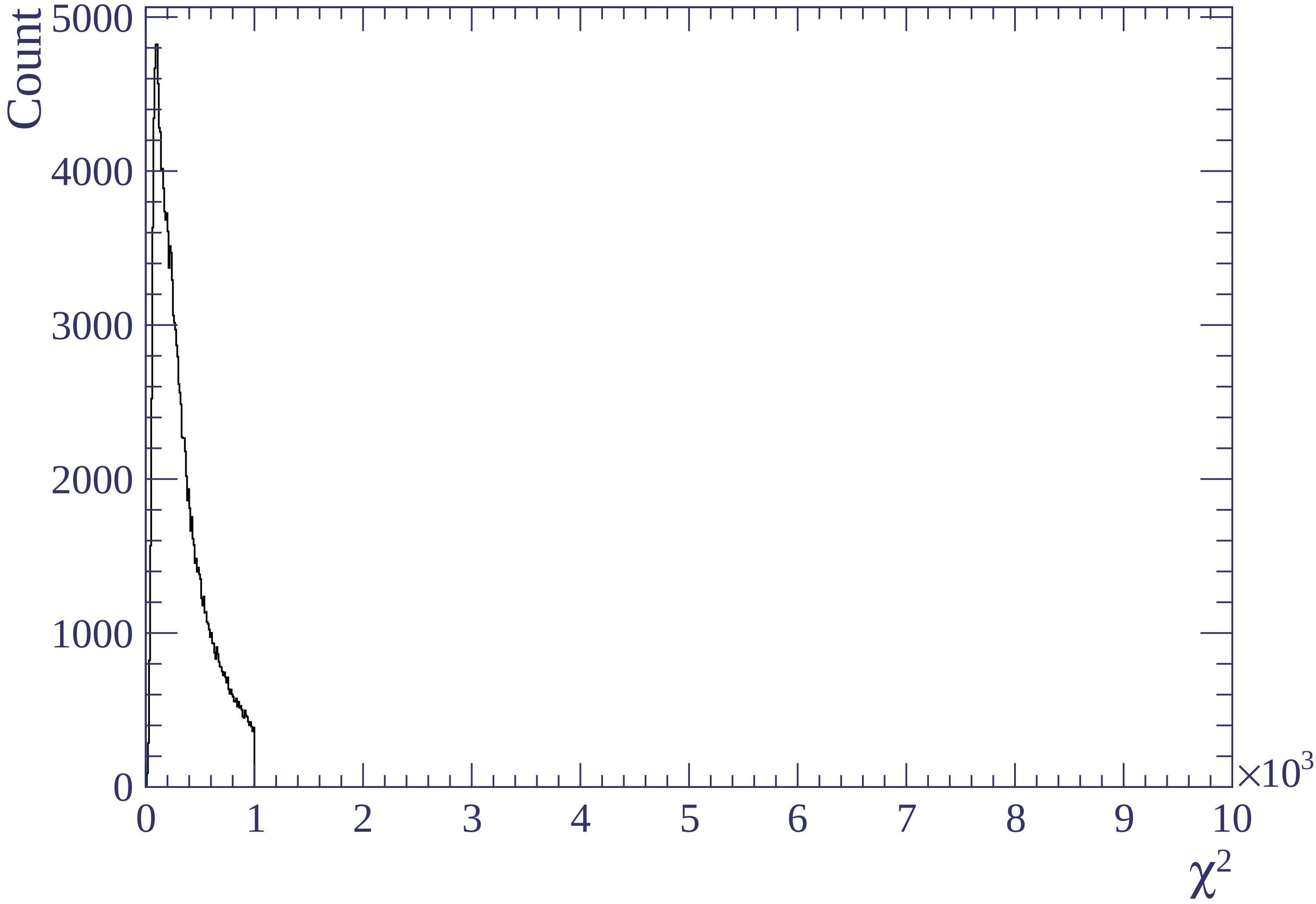


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
800
700
600
500
400
300
200
100
0

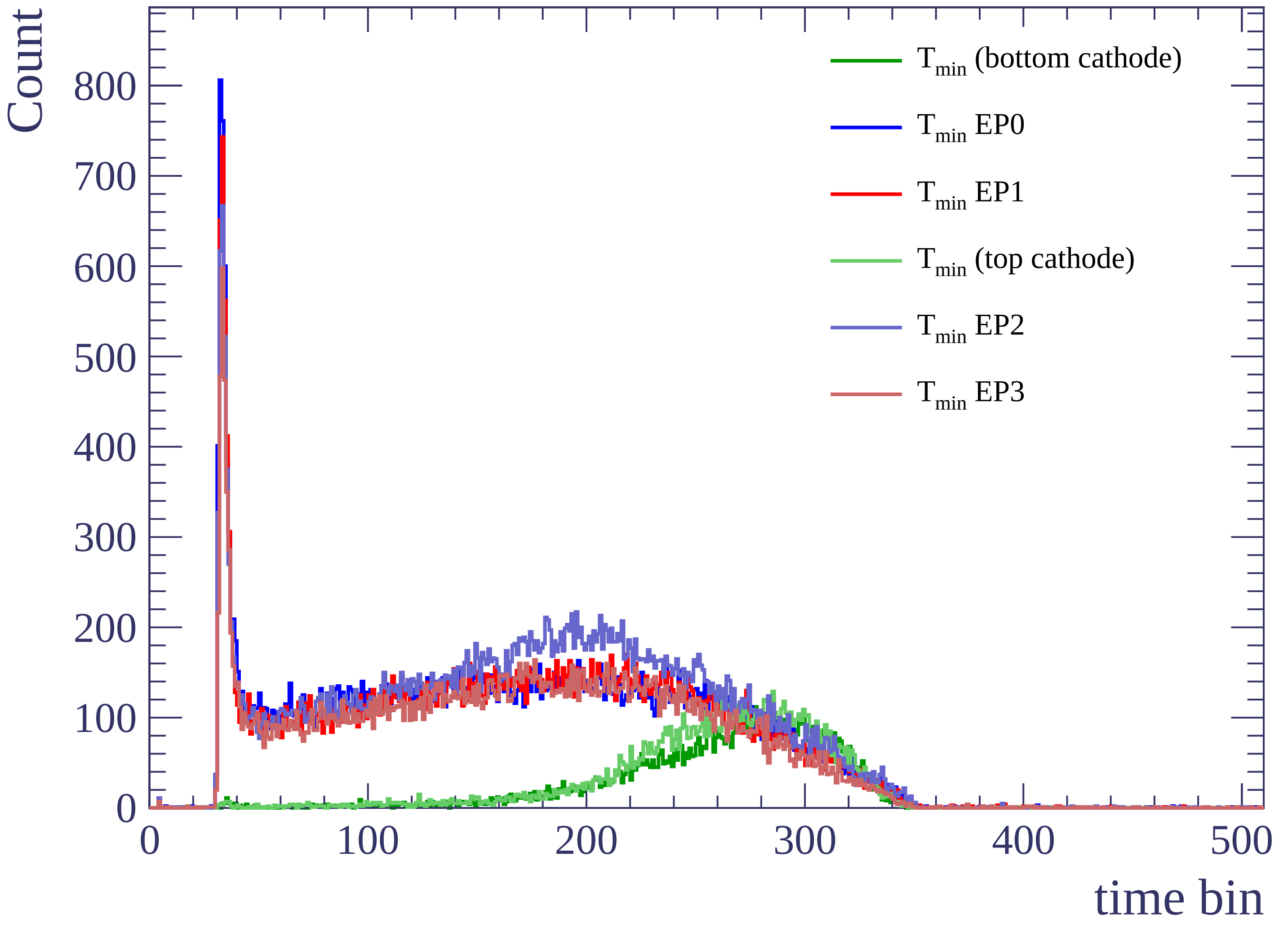

Count
8000
7000
6000
5000
4000
3000
2000
1000
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



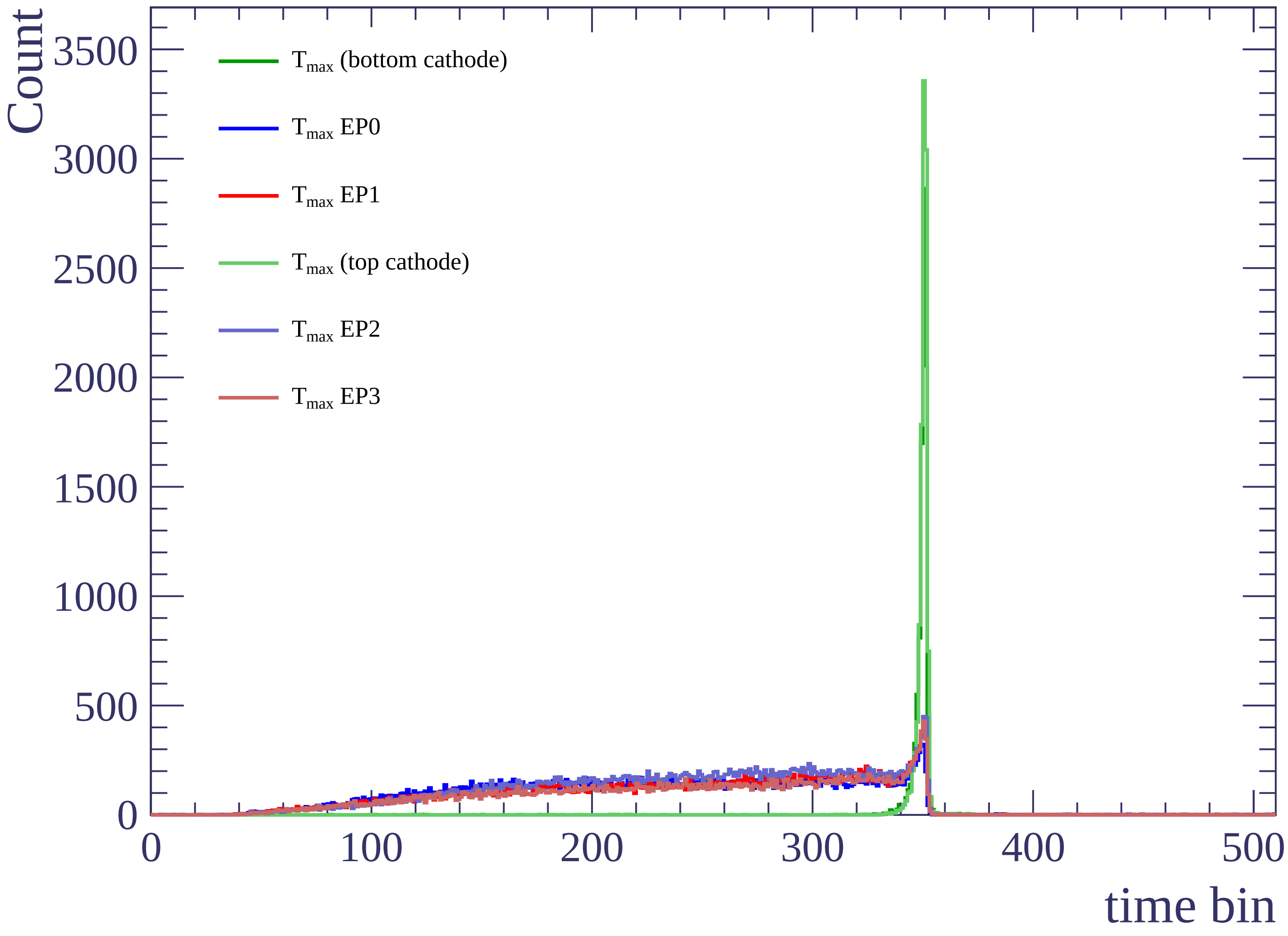
$\times 10^3$
Count
0
2
4
6
8
10
0
20
40
60
80
100
120
140
160
track length (cm)



Count
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



Count
800
700
600
500
400
300
200
100
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
3500
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3